ORACLE NetSuite

SuiteScript Developer Guide

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Setting Up Your SuiteScript Environment

Applies to: SuiteScript 2.x | SuiteCloud Developer

Before working with SuiteScript, you must configure both your NetSuite account and SuiteScript development environment. To do so, see the following topics.

- Enabling SuiteScript
- Showing Record and Field IDs in Your Account
- Setting Roles and Permissions for SuiteScript
- Setting Up Your SuiteScript Development Environment

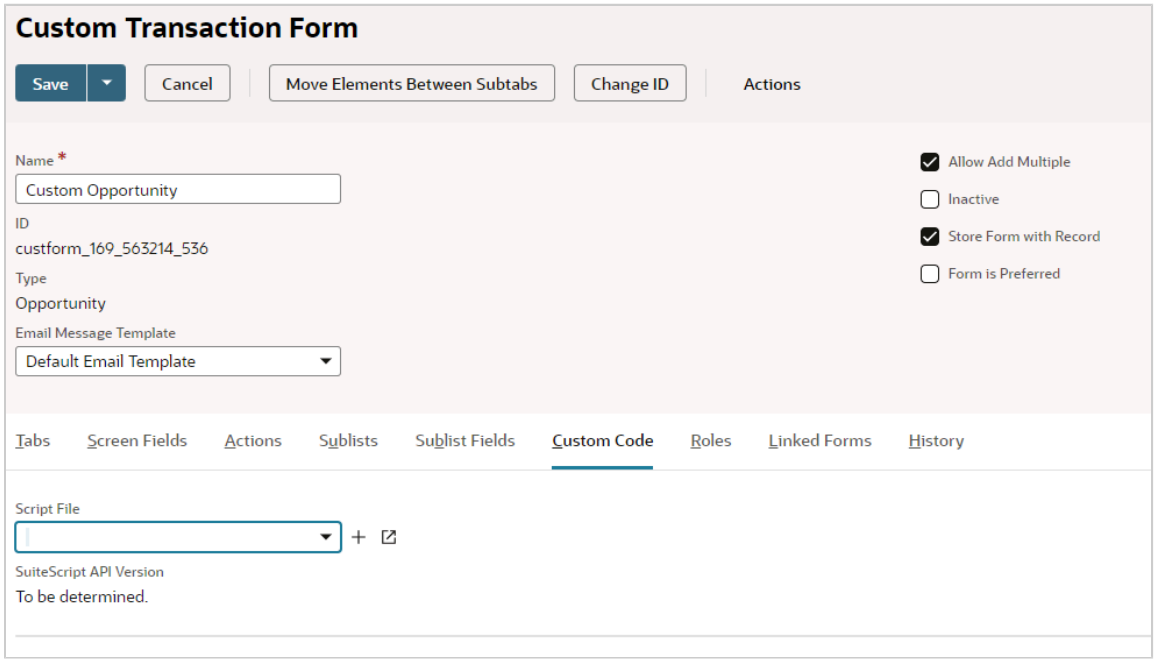
Enabling SuiteScript

Before you can use SuiteScript, a user with the Administrator role must enable the SuiteScript features that you plan to use.

To enable SuiteScript:

1. Go to Setup > Company > Enable Features.
2. Click the **SuiteCloud** subtab.
3. Under SuiteScript, check the **Client SuiteScript** or **Server SuiteScript** box (or both, depending on the scripts you want to run).
4. Click **Save**.

Note: If the Client SuiteScript feature is enabled, the **Custom Code** subtab becomes available on entry and transaction forms (see the following screenshot). On this subtab, you select the client script that you want to associate with the current form. For information about attaching client scripts to NetSuite forms, see the help topic Associating Custom Code (Client SuiteScript) Files With Custom Forms.



After enabling SuiteScript, continue configuring your NetSuite account:

- **Set up roles and permissions for SuiteScript** – Roles and permissions determine a user's level of access to specific areas in NetSuite. For information about SuiteScript permissions, see Setting Roles and Permissions for SuiteScript.
- **Set the preference to show internal IDs** – Internal record and field IDs are used as parameters in SuiteScript code. For information about the preference and how to view internal IDs, see Showing Record and Field IDs in Your Account,
- **Set up your development environment** – NetSuite provides a SuiteCloud plug-in or extension for specific IDEs. You can also use other development tools to create SuiteScript files. For more information, see Setting Up Your SuiteScript Development Environment

After setting up your SuiteScript environment, see the tutorial in SuiteScript 2.x Hello World for a sample SuiteScript implementation.

Showing Record and Field IDs in Your Account

Applies to: SuiteScript 2.x | SuiteCloud Developer

Records and fields are referenced in SuiteScript through their internal IDs. Internal IDs are unique identifiers that are associated to a record or field, and remain constant even if the record or field value is changed.

To configure your NetSuite account to display record and field IDs, see Show Internal IDs Preference.

If the **Show Internal IDs** preference is already checked, read the following sections for information about viewing internal IDs from within NetSuite:

- Finding Internal IDs of Records
- Finding Internal IDs of Record Fields

Note: The maximum number of instances (MAXINT) permitted for a record type is 2,000,000,000. When the internal ID limit for a custom record type is reached, NetSuite reports an error if you attempt to create a new instance of the record type. Existing records are not affected, but you won't be able to add new records of that type.

Show Internal IDs Preference

Before you can view the record or field IDs in your NetSuite account, you must configure your user preference to display internal IDs.

To show internal NetSuite IDs:

1. Go to Home > Set Preferences.
2. On the **General** subtab, check the **Show Internal IDs** box.
3. Click **Save**.

For examples of how internal IDs are referenced in the SuiteScript API, see the help topic record.load(options) or search.load(options).

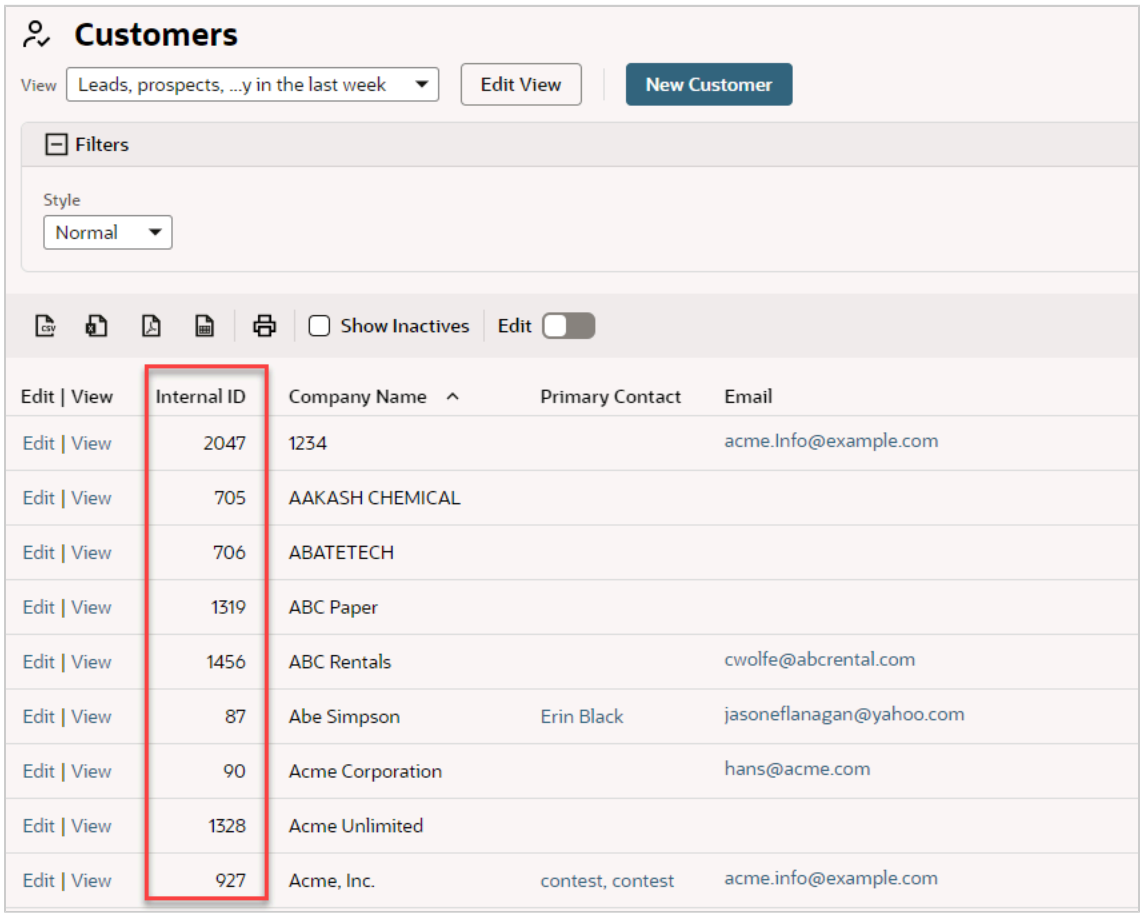
Important: When writing SuiteScript, all internal IDs must be specified in lowercase.

Finding Internal IDs of Records

Applies to: SuiteScript 2.x | SuiteCloud Developer

In NetSuite, each record is assigned an internal ID when it is created. If the **Show Internal IDs** preference box is checked, an Internal ID column is shown when viewing record lists.

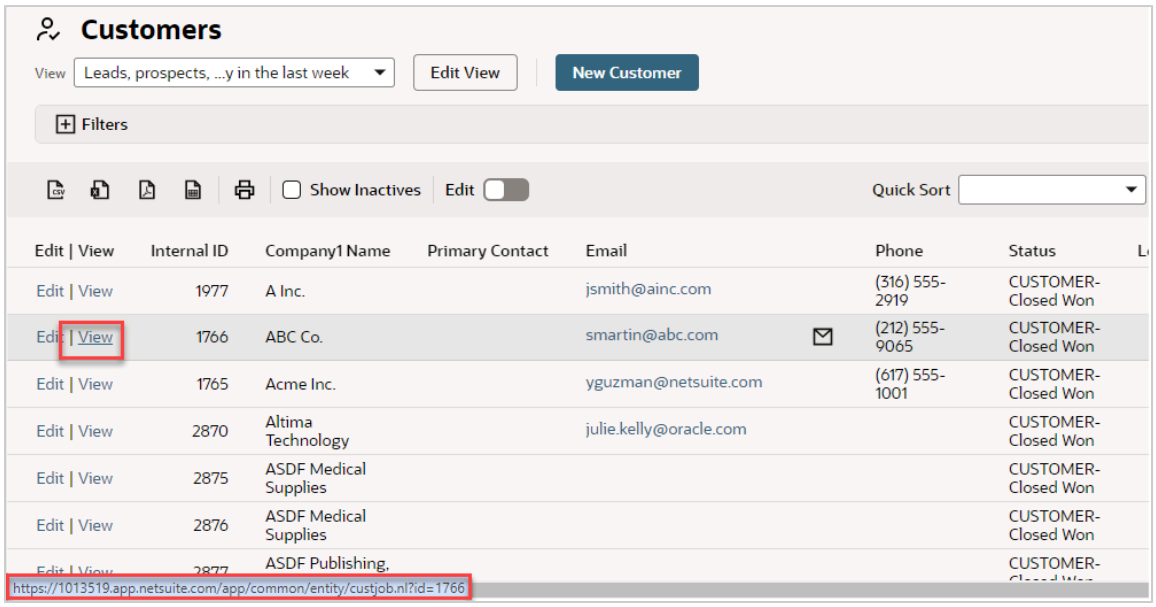
For example, to see the internal IDs of customer records, go to Lists > Relationships > Customers (Administrator). The internal ID of each record in the customer record list is found under the Internal ID column (see the following screenshot).



Edit | View	Internal ID	Company Name	Primary Contact	Email
Edit | View	2047	1234		acme.Info@example.com
Edit | View	705	AAKASH CHEMICAL		
Edit | View	706	ABATETECH		
Edit | View	1319	ABC Paper		
Edit | View	1456	ABC Rentals		cwolfe@abcrental.com
Edit | View	87	Abe Simpson	Erin Black	jasoneflanagan@yahoo.com
Edit | View	90	Acme Corporation		hans@acme.com
Edit | View	1328	Acme Unlimited		
Edit | View	927	Acme, Inc.	contest, contest	acme.info@example.com

When the **Show Internal IDs** preference is cleared, or if the internal IDs aren't shown on a particular page within NetSuite, you can see the record's internal ID by hovering over a link to that record. The internal ID is shown as a parameter in the URL in the browser status bar.

The following screenshot shows that if you hover over a link to the ABC Co. customer record, the internal record ID (1766) appears in the browser status bar.



Tip: You can also get the internal ID of a record type (for example, 'salesorder') by going to the SuiteScript Records Browser. For information about using the SuiteScript Records Browser, see Working with the SuiteScript Records Browser in the NetSuite Help Center.

Finding Internal IDs of Record Fields

Applies to: SuiteScript 2.x | SuiteCloud Developer

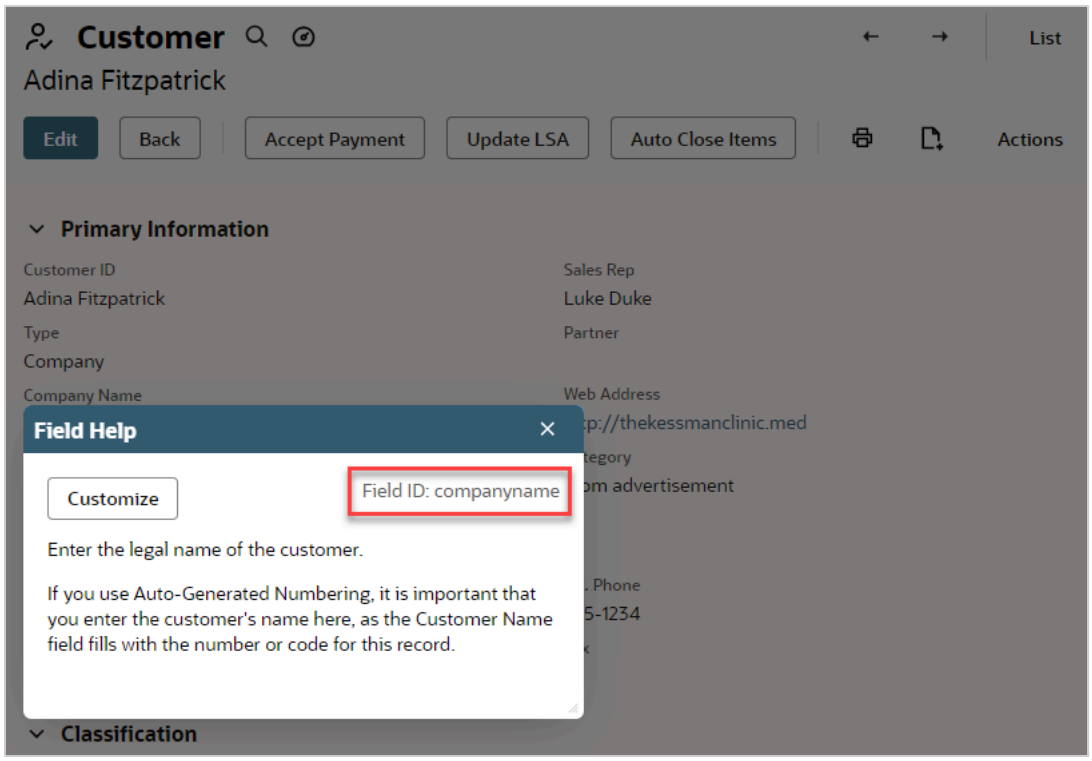
Internal field IDs must be used when referring to a field using SuiteScript APIs. The following sections show different ways to view a field's internal ID:

- Internal IDs for Standard and Custom Fields in Field Level Help Window
- Internal IDs for Custom Fields on List Pages
- Working with the SuiteScript Records Browser

Not all fields are visible to NetSuite users., You can use the SuiteScript Records Browser to find the internal ID of any field.

Internal IDs for Standard and Custom Fields in Field Level Help Window

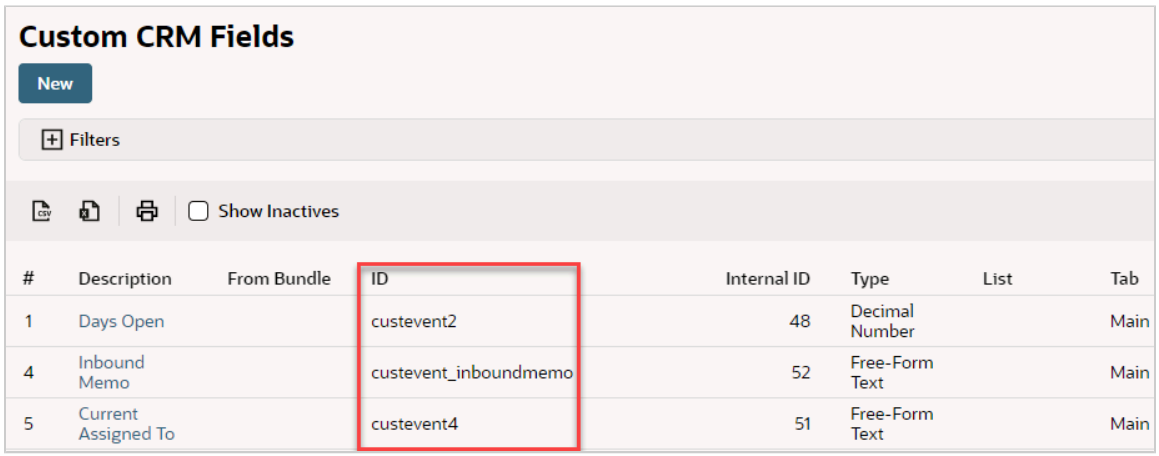
When the **Show Internal IDs** preference box is checked, you can view internal field IDs for both standard and custom fields by clicking the field label in the UI. The following screenshot shows the Field Level Help window that opens when you click a field label. In the example, the Company field label is clicked. The internal ID for the Company field is `companyname`, which appears in the bottom-right corner of the Field Level Help window.



Note: When creating custom fields, you can specify your own field ID or use the default ID assigned by NetSuite. To ensure that the field IDs make sense in the context of your business environment, you should define your own custom field IDs. For detailed information about creating custom fields and assigning custom field IDs, refer to Custom Fields.

Internal IDs for Custom Fields on List Pages

When the **Show Internal IDs** preference box is checked, internal IDs for each custom field are shown in the Internal ID column of a custom field page. For example, to see the internal IDs for custom CRM fields, go to Customization > Lists, Records, & Fields > CRM Fields (Administrator). The ID column appears in the list of custom fields as shown in the following screenshot.



#	Description	From Bundle	ID	Internal ID	Type	List	Tab
1	Days Open		custevent2	48	Decimal Number		Main
4	Inbound Memo		custevent_inboundmemo	52	Free-Form Text		Main
5	Current Assigned To		custevent4	51	Free-Form Text		Main

Setting Roles and Permissions for SuiteScript

Applies to: SuiteScript 2.x | SuiteCloud Developer

NetSuite provides standard roles with predefined permissions. A role (and its set of predefined permissions) lets customers, vendors, partners, and employees access specific areas of your data. Each role grants certain access levels for each permission. Use the Administrator role for complete access to all SuiteScript functionality, or use the permissions in the table to allow more granular access.

For more information about roles and permissions, see the help topic NetSuite Users & Roles.

To add SuiteScript permission:

1. Go to Setup > Users/Roles > Manage Roles.
2. Click **Edit** to make changes to a role.
3. On the **Permissions** subtab, click **Setup** and add the SuiteScript permission.

Note: If you customize a role to add SuiteScript permissions, you must also include the permissions to customize entry and transaction forms.

SuiteScript permissions

Action	Permission Required
View script records and script deployment records.	Role with SuiteScript permission (View level)
Create and view script records and script deployment records.	Role with SuiteScript permission (Create level)
Create, view, and edit script records and script deployment records.	Role with SuiteScript permission (Edit level)
Use **Save and Execute** from the script deployment record UI to run an on-demand scheduled script or an on-demand map/reduce script.	Role with SuiteScript permission (Full level)
Debug a SuiteScript 1.0, SuiteScript 2.0, or SuiteScript 2.x script.	Role with SuiteScript permission (Full level)
Use an API (for example, ScheduledScriptTask.submit()) to run an on-demand scheduled script or an on-demand map/reduce script.	Role with SuiteScript Scheduling permission (Full level)
Process a script with one of the following APIs: ■ SuiteScript 1.0 (see the help topic SuiteScript 1.0 Documentation) □ `nlobjContext.getPercentageComplete()` □ `nlobjContext.setPercentageComplete(pct)` ■ SuiteScript 2.x: Script.percentComplete	Role with SuiteScript Scheduling permission (Full level)

Setting Up Your SuiteScript Development Environment

Applies to: SuiteScript 2.x | SuiteCloud Developer

After configuring your NetSuite account for SuiteScript, you also need to set up your SuiteScript development environment. The following topics describe how to set up your development environment:

- Installing and Setting Up SuiteCloud Extension for Visual Studio Code
- Installing and Setting Up SuiteCloud IDE Plug-in for WebStorm
- Working with IDEs Other Than SuiteCloud IDEs

Working with IDEs Other Than SuiteCloud IDEs

Although SuiteCloud IDEs are preferred, you can use other development tools to create SuiteScript files. However, note that without the SuiteCloud IDEs, you will not be able to automatically upload SuiteScript files into the NetSuite File Cabinet.

If you use a development tool or IDE other than the SuiteCloud IDEs, see the following:

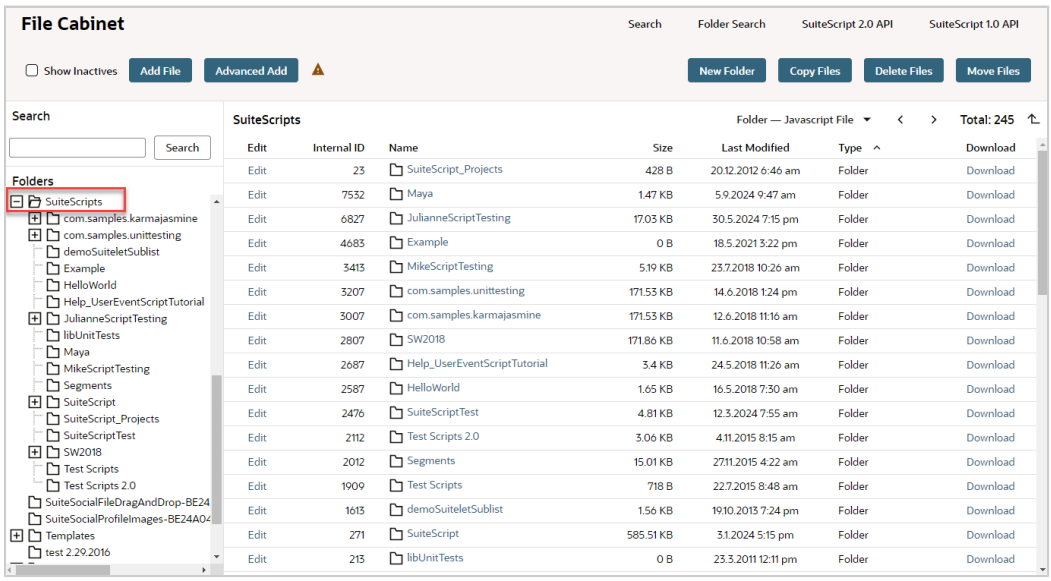
- Adding the SuiteScript Library File to Your IDE
- Uploading SuiteScript scripts to the File Cabinet without SuiteCloud IDEs

Adding the SuiteScript Library File to Your IDE

If you are working with an IDE other than the SuiteCloud IDEs, you should add the SuiteScript library file to your local SuiteScript project folder to take advantage of IDE features such as code completion.

To add the SuiteScript library file:

1. In NetSuite, go to Documents > Files > SuiteScripts.
2. Click the link for the **SuiteScript 2.0 API** file:



3. A .zip file of the API is downloaded to your browser Downloads folder.
4. Extract the API files as needed and add them to your IDE.

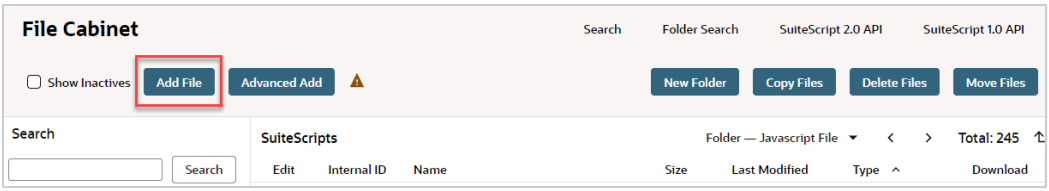
Uploading SuiteScript scripts to the File Cabinet without SuiteCloud IDEs

When the SuiteScript feature is enabled, a **SuiteScripts** folder is created in your NetSuite File Cabinet. However, you're not required to store your script files in the **SuiteScripts** folder. You can store script files in any location within the File Cabinet. Note that only scripts located in the File Cabinet can be executed in your NetSuite account.

Note: For scripts to run in the Customer, Vendor, Partner, or Employee Center, select the associated role on the Audience subtab of the script deployment record. Additionally, script files and any files referenced by the script must have either the Company-Wide Usage or Available Without Login preference enabled on the file record in the File Cabinet. For more information, see the help topic Permissions and File Cabinet Files.

To upload SuiteScript scripts to the File Cabinet:

1. Go to Documents > Files > SuiteScripts.
2. Click **Add File**.



3. In the popup window that appears, select the file you want to upload and click **Open**.

Note: If you make changes to a SuiteScript file that already exists in the File Cabinet, follow steps 1–3 to upload the changed file again. Click **OK** to overwrite the previous file and load your changes.

SuiteScript Governance and Limits

Applies to: SuiteScript 2.x | SuiteCloud Developer

NetSuite uses a SuiteScript governance model to optimize performance, based on usage units. If the number of allowable usage units is exceeded, script execution is terminated.

Usage units are tracked in two ways: by script type and by API. Each script type and each SuiteScript API has a set number of usage units.

In map/reduce scripts, keys are limited to 3,000 characters (specifically, in mapContext or reduceContext objects). You'll also see error messages if a key is longer than 3,000 characters or a value is larger than 10 MB. Keys longer than 3,000 characters will return the error KEY_LENGTH_IS_OVER_3000_BYTES. Values larger than 10 MB will return the error VALUE_LENGTH_IS_OVER_10_MB.

When using mapContext.write() or reduceContext.write(), keep your key strings under 3,000 characters and value strings under 10 MB. Also, consider the potential length of any dynamically generated strings, which might exceed these limits. It's also a good idea to avoid using keys to pass data - use values instead.

You can find governance limits for each SuiteScript 2.x API in their respective method topics. Methods are organized by modules, and all modules are listed in the help topic SuiteScript 2.x Modules. Also see SuiteScript 2.x API Governance.

Important: SuiteScript thresholds are based on the volume of activity that a company's users can manually generate. However, automated functions that generate excessive levels of activity may trigger metering of script execution as referenced in the NetSuite Main Terms of Service (TOS).

See the following help topics for more information:

- Script Type Usage Unit Limits
- SuiteScript 2.x API Governance
- Monitoring Script Usage
- Governance on Script Logging

Governance limits are also enforced on certain aspects of script execution. When you run a search using the N/search module, the number of search results you receive is limited. There are also limits on how long a script can run, based on the script type. See the following help topics to learn about these limits:

- Search Result Limits
- Script Execution Time Limits

Script Type Usage Unit Limits

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following table lists the maximum allowable usage units for each SuiteScript (1.0 and 2.x) script type. You can use the Script.getRemainingUsage() method to see how many usage units you have remaining for a particular script.

Script Type	Total usage units Allowed per Script	Notes
Bundle Installation Scripts (SuiteScript 1.0) SuiteScript 2.x Bundle Installation Script Type	10,000	The limit is 10,000 usage units per execution.
Client Scripts (SuiteScript 1.0) SuiteScript 2.x Client Script Type	1,000	Client scripts are metered on a per-script basis. If an account has one form-level client script attached to a form and one record-level client script deployed to a record (which may be associated with a form), each client script can total 1,000 usage units. Usage units are not shared by all the client scripts associated with a form or record. For information about record- and form-level client scripts, see the help topic Record-Level and Form-Level Script Deployments.
SuiteScript 2.1 Custom Tool Script Type	1,000	The limit is 1,000 usage units per execution of the tool.
SuiteScript 2.x Map/Reduce Script Type	—	Map/reduce scripts are available only in SuiteScript 2.x. There are no limits imposed on the full duration of a map/reduce script deployment instance. Instead, isolated components of the deployment, such as the usage units used by a single method invocation, are regulated. For more information, see the help topic Map/Reduce Governance.
Mass Update Scripts (SuiteScript 1.0) SuiteScript 2.x Mass Update Script Type	1,000	The limit is 1,000 usage units per record or execution of the script.
Portlet Scripts (SuiteScript 1.0) SuiteScript 2.x Portlet Script Type	1,000	—
RESTlets (SuiteScript 1.0) SuiteScript 2.x RESTlet Script Type	5,000	The SuiteScript governance model for RESTlets tracks usage units on the API level and the script level. For more information, see the help topic RESTlet Governance and Security
Scheduled Scripts (SuiteScript 1.0) SuiteScript 2.x Scheduled Script Type	10,000	Within one scheduled script, all actions combined cannot exceed 10,000 usage units. For example, a scheduled script that includes two calls to record.transform(options) and one call to email.send(options) consumes from 24 to 40 usage units (depending on the record type for record.transform(options)) out of a possible 10,000 usage units available. If you have a scheduled script with potentially long execution times, you should consider using a map/reduce script instead. SuiteScript 2.x does not have a method to allow you to set recovery points or provide a yield to avoid exceeding the allowed governance for a scheduled script. A map/reduce script has built-in yielding and can be submitted for processing in the same ways as a scheduled script.

Script Type	Total usage units Allowed per Script	Notes
SuiteScript 2.x SDF Installation Script Type	10,000	The limit is 10,000 usage units per execution.
Suitelets (SuiteScript 1.0) SuiteScript 2.x Suitelet Script Type	1,000	Within one Suitelet, all actions combined cannot exceed 1,000 usage units. For example, a Suitelet that calls record.create(options) and http.get(options) consumes from 12 to 20 usage units (depending on the record type for record.create(options)) out of a possible 1,000 usage units available. Regardless of the 1,000 unit limit for Suitelets, you should create your Suitelets to be responsive to users, otherwise user experience may be impacted.
User Event Scripts (SuiteScript 1.0) SuiteScript 2.x User Event Script Type	1,000	Regardless of the 1,000 usage unit limit for user event scripts, you should create your scripts so that they are responsive to users, otherwise user experience may be impacted.
Workflow Action Scripts (SuiteScript 1.0) SuiteScript 2.x Workflow Action Script Type (also referred to as Custom Action in SuiteFlow)	1,000	Within one workflow state, all actions combined cannot exceed 1,000 usage units. For example, if you have developed a custom action (using a workflow action script) that consumes 990 usage units, be aware of the unit consumption of the other actions within that state.
Core Plug-ins	Varies based on the plug-in	The default limit for core plug-ins that do not have more restrictive limits defined is 10,000. See the help topic for each core plug-in for any specific usage unit limits.
Custom Plug-ins	10,000	The limit is 10,000 usage units per plug-in.
SSP Application Scripts	1,000	For more information, see the help topic SSP Application Governance.

Note: There is also a limit of 1,000 usage units when using the SuiteScript Debugger. For more information, see Script Debugger Metering and Permissions.

SuiteScript 2.x API Governance

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following table shows the governance usage units used for each SuiteScript 2.x method. You can use the Script.getRemainingUsage() method to see how many usage units you have remaining for a particular script.

The governance model takes into account the NetSuite processing requirements for three categories of records: custom records, standard transaction records, and standard non-transaction records. For example, custom records require less processing than standard records, therefore, the usage unit cost for custom records is lower than for standard records. Similarly, standard non-transaction records require less processing than standard transaction records, therefore, the usage unit cost for standard non-

transaction records is lower than for standard transaction records. Standard transaction records include records such as cash refund, customer deposit, and item fulfillment. Standard non-transaction records include records such as activity, inventory item, and customer. You can see an example of this in the N/record module methods listed below.

Record categories are included in the SuiteScript Supported Records help topic. Record types categorized as activity, communication, customization, entity, file cabinet, item, list, marketing, subrecord, support, or website are considered to be standard non-transaction records.

For examples of API governance, see API Governance Examples.

For tips on how to monitor how many usage units your script is using, see Monitoring Script Usage.

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
N/action Module	Action(options) Action.promise(options)	None
	Action.execute(options) Action.execute.promise(options)	None
	Action.executeBulk(options)	50
	action.getBulkStatus(options)	None
	action.execute(options) action.execute.promise(options)	None
	action.executeBulk(options)	50
	action.find(options) action.find.promise(options)	None
	action.get(options) action.get.promise(options)	None
N/auth Module	auth.changeEmail(options)	10
	auth.changePassword(options)	10
N/cache Module	Cache.get(options)	1 if the value is present in the cache 2 if the loader function is used
	Cache.put(options)	1
	Cache.remove(options)	1
	cache.getCache(options)	None
N/certificateControl Module	Certificate.save()	10
	certificateControl.createCertificate(options)	10
	certificateControl.deleteCertificate(options)	10
	certificateControl.findCertificates(options)	10
	certificateControl.findUsages(options)	10
	certificateControl.loadCertificate(options)	10
N/commerce Modules	recordView.viewItems(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	recordView.viewWebsite(options)	None
N/compress Module	Archiver.add(options)	None
	Archiver.archive(options)	None
	compress.createArchiver()	None
	compress.gzip(options)	None
	compress.gunzip(options)	None
N/config Module	config.load(options)	10
N/crypto Module	Cipher.final(options)	None
	Cipher.update(options)	None
	Decipher.final(options)	None
	Decipher.update(options)	None
	Hash.digest(options)	None
	Hash.update(options)	None
	Hmac.digest(options)	None
	Hmac.update(options)	None
	crypto.createCipher(options)	None
	crypto.createDecipher(options)	None
	crypto.createHash(options)	None
	crypto.createHmac(options)	None
	crypto.createSecretKey(options)	None
N/crypto/certificate Module	SignedXml.asFile()	None
	SignedXml.asString()	None
	SignedXml.asXml()	None
	Signer.sign(options)	None
	Signer.update(options)	None
	Verifier.update(options)	None
	Verifier.verify(options)	None
	certificate.createSigner(options)	10
	certificate.createVerifier(options)	10
	certificate.verifyXmlSignature(options)	10
	certificate.signXml(options)	10
N/currency Module	currency.exchangeRate(options)	10
N/crypto/random Module	random.generateBytes(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	random.generateInt(options)	None
	random.generateUUID()	None
N/currentRecord Module	CurrentRecord.cancelLine(options)	None
	CurrentRecord.commitLine(options)	None
	CurrentRecord.findMatrixSublistLineWithValue(options)	None
	CurrentRecord.findSublistLineWithValue(options)	None
	CurrentRecord.getCurrentMatrixSublistValue(options)	None
	CurrentRecord.getCurrentSublistIndex(options)	None
	CurrentRecord.getCurrentSublistSubrecord(options)	None
	CurrentRecord.getCurrentSublistText(options)	None
	CurrentRecord.getCurrentSublistValue(options)	None
	CurrentRecord.getField(options)	None
	CurrentRecord.getLineCount(options)	None
	CurrentRecord.getMatrixHeaderCount(options)	None
	CurrentRecord.getMatrixHeaderField(options)	None
	CurrentRecord.getMatrixHeaderValue(options)	None
	CurrentRecord.getMatrixSublistField(options)	None
	CurrentRecord.getMatrixSublistValue(options)	None
	CurrentRecord.getSublist(options)	None
	CurrentRecord.getSublistField(options)	None
	CurrentRecord.getSublistText(options)	None
	CurrentRecord.getSublistValue(options)	None
	CurrentRecord.getSubrecord(options)	None
	CurrentRecord.getText(options)	None
	CurrentRecord.getValue(options)	None
	CurrentRecord.hasCurrentSublistSubrecord(options)	None
	CurrentRecord.hasSublistSubrecord(options)	None
	CurrentRecord.hasSubrecord(options)	None
	CurrentRecord.insertLine(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	CurrentRecord.removeCurrentSublist Subrecord(options)	None
	CurrentRecord.removeLine(options)	None
	CurrentRecord.removeSubrecord(options)	None
	CurrentRecord.selectLine(options)	None
	CurrentRecord.selectNewLine(options)	None
	CurrentRecord.setCurrentMatrixSublistValue (options)	None
	CurrentRecord.setCurrentSublistText(options)	None
	CurrentRecord.setCurrentSublistValue (options)	None
	CurrentRecord.setMatrixHeaderValue(options)	None
	CurrentRecord.setMatrixSublistValue(options)	None
	CurrentRecord.setText(options)	None
	CurrentRecord.setValue(options)	None
	Field.getSelectOptions(options)	None
	Field.insertSelectOption(options)	None
	Field.removeSelectOption(options)	None
	Sublist.getColumn(options)	None
	currentRecord.get() currentRecord.get.promise()	None
N/dataset Module	Dataset.getExpressionFromColumn(options)	None
	Dataset.run()	10
	Dataset.runPaged(options)	10
	Dataset.save(options)	10
	dataset.create(options)	None
	dataset.createColumn(options)	None
	dataset.createCondition(options)	None
	dataset.createJoin(options)	None
	dataset.createTranslation(options)	None
	dataset.describe(options)	10
	dataset.list()	10
	dataset.listPaged(options)	10
	dataset.loadDataset(options)	10
N/datasetLink Module	datasetLink.create(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
N/documentCapture Module	documentCapture.documentToStructure (options)	100
	documentCapture.documentToStructure. promise(options)	100
	documentCapture.documentToText(options)	100
	documentCapture.documentToText. promise(options)	100
	documentCapture. getRemainingConcurrency()	None
	documentCapture.getRemainingConcurrency. promise()	None
	documentCapture.getRemainingFreeUsage()	None
	documentCapture.getRemainingFreeUsage. promise()	None
	documentCapture.parseResult(options)	None
N/email Module	email.send(options) email.send.promise(options)	20
	email.sendBulk(options) email.sendBulk.promise(options)	10
	email.sendCampaignEvent(options) email.sendCampaignEvent.promise(options)	10
N/encode Module	encode.convert(options)	None
N/error Module	error.create(options)	None
N/file Module	File.appendLine(options)	None
	File.getContents()	None
	File.getReader()	None
	File.getSegments(options)	None
	File.appendLine(options)	None
	File.lines.iterator()	None
	File.resetStream()	None
	File.save()	20
	Reader.readChars(options)	None
	Reader.readUntil(options)	None
	file.create(options)	None
	file.delete(options)	20
	file.load(options)	10
N/format Module	format.format(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	format.parse(options)	None
N/format/i18n Module	CurrencyFormatter.format(options)	10
	NumberFormatter.format(options)	10
	PhoneNumberFormatter.format(options)	None
	PhoneNumberParser.parse(options)	None
	format.getCurrencyFormatter(options)	10
	format.getNumberFormatter(options)	10
	format.spellOut(options)	None
N/http Module	ServerRequest.getLineCount(options)	None
	ServerRequest.getSublistValue(options)	None
	ServerResponse.addHeader(options)	None
	ServerResponse.getHeader(options)	None
	ServerResponse.renderPdf(options)	10
	ServerResponse.sendRedirect(options)	None
	ServerResponse.setCdnCacheable(options)	None
	ServerResponse.setHeader(options)	None
	ServerResponse.write(options)	None
	ServerResponse.writeFile(options)	None
	ServerResponse.writeLine(options)	None
	ServerResponse.writePage(options)	None
	http.get(options) http.get.promise(options)	10
	http.delete(options) http.delete.promise(options)	10
	http.post(options) http.post.promise(options)	10
	http.put(options) http.put.promise(options)	10
	http.request(options) http.request.promise(options)	10
N/https Module	SecureString.appendSecureString(options)	None
	SecureString.appendString(options)	None
	SecureString.convertEncoding(options)	None
	SecureString.hash(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	SecureString.hmac(options)	None
	ServerRequest.getLineCount(options)	None
	ServerRequest.getSublistValue(options)	None
	ServerResponse.addHeader(options)	None
	ServerResponse.getHeader(options)	None
	ServerResponse.renderPdf(options)	10
	ServerResponse.sendRedirect(options)	None
	ServerResponse.setCdnCacheable(options)	None
	ServerResponse.setHeader(options)	None
	ServerResponse.write(options)	None
	ServerResponse.writeFile(options)	None
	ServerResponse.writeLine(options)	None
	ServerResponse.writePage(options)	10
	https.createSecretKey(options)	None
	https.createSecureString(options)	None
	https.get(options) https.get.promise(options)	10
	https.delete(options) https.delete.promise(options)	10
	https.post(options) https.post.promise(options)	10
	https.put(options) https.put.promise(options)	10
	https.request(options) https.request.promise(options)	10
	https.requestRestlet(options)	10
	https.requestSuiteTalkRest(options)	10
N/https/clientCertificate Module	clientCertificate.delete(options)	10
	clientCertificate.get(options)	10
	clientCertificate.post(options)	10
	clientCertificate.put(options)	10
	clientCertificate.request(options)	10
N/keyControl Module	Key.save()	10
	keyControl.createKey(options)	10

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	keyControl.deleteKey(options)	10
	keyControl.findKeys(options)	10
	keyControl.loadKey(options)	10
N/llm Module	llm.createChatMessage(options)	None
	llm.createDocument(options)	None
	llm.embed(options)	50
	llm.embed.promise(options)	50
	llm.evaluatePrompt(options)	100
	llm.evaluatePrompt.promise(options)	100
	llm.evaluatePromptStreamed(options)	100
	llm.evaluatePromptStreamed.promise(options)	100
	llm.generateText(options)	100
	llm.generateText.promise(options)	100
	llm.generateTextStreamed(options)	100
	llm.generateTextStreamed.promise(options)	100
	llm.getRemainingFreeUsage()	None
	llm.getRemainingFreeUsage.promise()	None
	llm.getRemainingFreeEmbedUsage()	None
	llm.getRemainingFreeEmbedUsage.promise()	None
N/log Module	log.audit(options)	Amount of logging in any 60–minute period is limited. See Governance on Script Logging.
	log.debug(options)	
	log.emergency(options)	
	log.error(options)	
N/machineTranslation Module	machineTranslation.createDocument(options)	None
	machineTranslation.translate(options)	100
	machineTranslation.translate.promise(options)	100
N/piremoval Module	PiRemovalTask.deleteTask()	20
	PiRemovalTask.run()	20
	PiRemovalTask.save()	20
	piremoval.createTask(options)	None
	piremoval.deleteTask(options)	20
	piremoval.getTaskStatus(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	piremoval.loadTask(options)	None
N/plugin Module	plugin.findImplementations(options)	None
	plugin.loadImplementation(options)	None
N/portlet Module	portlet.refresh()	None
	portlet.resize()	None
N/query Module	Component.autoJoin(options)	None
	Component.createColumn(options)	None
	Component.createCondition(options)	None
	Component.createSort(options)	None
	Component.join(options)	None
	Component.joinFrom(options)	None
	Component.joinTo(options)	None
	PagedData.iterator()	None
	Query.and(conditions)	None
	Query.autoJoin(options)	None
	Query.createColumn(options)	None
	Query.createCondition(options)	None
	Query.createSort(options)	None
	Query.join(options)	None
	Query.joinFrom(options)	None
	Query.joinTo(options)	None
	Query.run(options) Query.run.promise()	10
	Query.runPaged(options) Query.runPaged.promise(options)	10
	Query.not(condition)	None
	Query.or(conditions)	None
	Query.toSuiteQL()	None
	Result.asMap()	None
	ResultSet.asMappedResults()	None
	ResultSet.iterator()	None
	SuiteQL.run()	10
	SuiteQL.runPaged(options)	10
	query.create(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	query.createPeriod(options)	None
	query.createRelativeDate(options)	None
	query.delete(options)	5
	query.listTables(options)	None
	query.load(options) query.load.promise(options)	5
	query.runSuiteQL(options)	10
	query.runSuiteQLPaged(options)	10
N/record Module	Field.getSelectOptions(options)	None
	Macro(options) Macro.promise(options)	None
	Macro.execute(options) Macro.execute.promise(options)	None
	Record.cancelLine(options)	None
	Record.commitLine(options)	None
	Record.executeMacro(options) Record.executeMacro.promise(options)	None
	Record.findMatrixSublistLineWithValue(options)	None
	Record.findSublistLineWithValue(options)	None
	Record.getCurrentMatrixSublistValue(options)	None
	Record.getCurrentSublistField(options)	None
	Record.getCurrentSublistIndex(options)	None
	Record.getCurrentSublistSubrecord(options)	None
	Record.getCurrentSublistText(options)	None
	Record.getCurrentSublistValue(options)	None
	Record.getField(options)	None
	Record.getFields()	None
	Record.getLineCount(options)	None
	Record.getMacro(options)	None
	Record.getMacros()	None
	Record.getMatrixHeaderCount(options)	None
	Record.getMatrixHeaderField(options)	None
	Record.getMatrixHeaderValue(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Record.getMatrixSublistField(options)	None
	Record.getMatrixSublistValue(options)	None
	Record.getSublist(options)	None
	Record.getSublists()	None
	Record.getSublistField(options)	None
	Record.getSublistFields(options)	None
	Record.getSublistSubrecord(options)	None
	Record.getSublistText(options)	None
	Record.getSublistValue(options)	None
	Record.getSubrecord(options)	None
	Record.getText(options)	None
	Record.getValue(options)	None
	Record.hasCurrentSublistSubrecord(options)	None
	Record.hasSublistSubrecord(options)	None
	Record.hasSubrecord(options)	None
	Record.insertLine(options)	None
	Record.removeCurrentSublistSubrecord(options)	None
	Record.removeLine(options)	None
	Record.removeSublistSubrecord(options)	None
	Record.removeSubrecord(options)	None
	Record.save(options) Record.save.promise(options)	20 for transaction records 4 for custom records 10 for all other records
	Record.selectLine(options)	None
	Record.selectNewLine(options)	None
	Record.setCurrentMatrixSublistValue(options)	None
	Record.setCurrentSublistText(options)	None
	Record.setCurrentSublistValue(options)	None
	Record.setMatrixHeaderValue(options)	None
	Record.setMatrixSublistValue(options)	None
	Record.setSublistText(options)	None
	Record.setSublistValue(options)	None
	Record.setText(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Record.setValue(options)	None
	Sublist.getColumn(options)	None
	record.attach(options) record.attach.promise(options)	10
	record.copy(options) record.copy.promise(options)	10 for transaction records 2 for custom records 5 for all other records
	record.create(options) record.create.promise(options)	10 for transaction records 2 for custom records 5 for all other records
	record.delete(options) record.delete.promise(options)	20 for transaction records 4 for custom records 10 for all other records
	record.detach(options) record.detach.promise(options)	10
	record.load(options) record.load.promise(options)	10 for transaction records 2 for custom records 5 for all other records
	record.submitFields(options) record.submitFields.promise(options)	10 for transaction records 2 for custom records 5 for all other records
	record.transform(options) record.transform.promise(options)	10 for transaction records 2 for custom records 5 for all other records
N/recordContext Module	recordContext.getContext(options)	10
N/redirect Module	redirect.redirect(options)	None
	redirect.toRecord(options)	None
	redirect.toRecordTransform(options)	None
	redirect.toSavedSearch(options)	5
	redirect.toSavedSearchResult(options)	5
	redirect.toSearch(options)	None
	redirect.toSearchResult(options)	None
	redirect.toSuitelet(options)	None
	redirect.toTaskLink(options)	None
N/render Module	TemplateRenderer.addCustomDataSource (options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	TemplateRenderer.addQuery(options)	None
	TemplateRenderer.addRecord(options)	None
	TemplateRenderer.addSearchResults(options)	None
	TemplateRenderer.renderAsPdf()	None
	TemplateRenderer.renderPdfToResponse (options)	None
	TemplateRenderer.renderAsString()	None
	TemplateRenderer. renderToResponse(options)	None
	TemplateRenderer.setTemplateById(options)	None
	TemplateRenderer.setTemplateByScriptId (options)	None
	render.bom(options)	10
	render.create()	None
	render.mergeEmail(options)	None
	render.packingSlip(options)	10
	render.pickingTicket(options)	10
	render.statement(options)	10
	render.transaction(options)	10
	render.xmlToPdf(options)	10
N/runtime Module	Script.getParameter(options)	None
	Script.getRemainingUsage()	None
	Session.get(options)	None
	Session.set(options)	None
	User.getPermission(options)	None
	User.getPreference(options)	None
	runtime.getCurrentScript()	None
	runtime.getCurrentSession()	None
	runtime.getCurrentUser()	None
	runtime.isFeatureInEffect(options)	None
N/search Module (Search results are limited to 1000 records; see Search Result Limits)	Column.setWhenOrderedBy(options)	None
	Page.next() Page.next.promise()	5
	Page.prev() Page.prev.promise()	5

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	PagedData.fetch(options) PagedData.fetch.promise()	5
	Result.getValue(column) Result.getValue(options)	None
	Result.getText(column) Result.getText(options)	None
	ResultSet.each(callback) ResultSet.each.promise(callback)	10
	ResultSet.getRange(options) ResultSet.getRange.promise(options)	10
	Search.run()	None
	Search.runPaged(options) Search.runPaged.promise(options)	5
	Search.save() Search.save.promise()	5
	search.create(options) search.create.promise(options)	None
	search.createColumn(options)	None
	search.createFilter(options)	None
	search.createSetting(options)	None
	search.delete(options) search.delete.promise(options)	5
	search.duplicates(options) search.duplicates.promise(options)	10
	search.global(options) search.global.promise(options)	10
	search.load(options) search.load.promise(options)	5
	search.lookupFields(options) search.lookupFields.promise(options)	1
N/sftp Module	Connection.download(options)	100
	Connection.list(options)	10
	Connection.makeDirectory(options)	10
	Connection.move(options)	10
	Connection.removeDirectory(options)	10

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Connection.removeFile(options)	10
	Connection.upload(options)	100
	sftp.createConnection(options)	None
N/suiteAppInfo Module	suiteAppInfo.isBundleInstalled(options)	5
	suiteAppInfo.isSuiteAppInstalled(options)	5
	suiteAppInfo.listBundlesContainingScripts (options)	10
	suiteAppInfo.listInstalledBundles()	10
	suiteAppInfo.listInstalledSuiteApps()	10
	suiteAppInfo.listSuiteAppsContainingScripts (options)	10
N/task Module	CsvImportTask.submit()	100
	EntityDeduplicationTask.submit()	100
	MapReduceScriptTask.submit()	20
	MapReduceScriptTaskStatus. getCurrentTotalSize()	25
	MapReduceScriptTaskStatus. getPendingMapCount()	10
	MapReduceScriptTaskStatus. getPendingMapSize()	25
	MapReduceScriptTaskStatus. getPendingOutputCount()	10
	MapReduceScriptTaskStatus. getPendingOutputSize()	25
	MapReduceScriptTaskStatus. getPendingReduceCount()	10
	MapReduceScriptTaskStatus. getPendingReduceSize()	25
	MapReduceScriptTaskStatus. getPercentageCompleted()	10
	MapReduceScriptTaskStatus. getTotalMapCount()	10
	MapReduceScriptTaskStatus. getTotalOutputCount()	10
	MapReduceScriptTaskStatus. getTotalReduceCount()	10
	QueryTask.addInboundDependency(options)	None
	QueryTask.submit()	100
	RecordActionTask.submit()	50
	RecordActionTask.paramCallback	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	ScheduledScriptTask.submit()	20
	SearchTask.addInboundDependency()	None
	SearchTask.submit()	100
	SuiteQLTask.addInboundDependency (options)	None
	SuiteQLTask.submit()	100
	WorkflowTriggerTask.submit()	20
	task.checkStatus(options)	None
	task.create(options)	None
N/task/accounting/recognition Module	MergeArrangementsTask.submit()	20
	MergeElementsTask.submit()	20
	recognition.checkStatus(options)	50
	recognition.create(options)	None
N/transaction Module	transaction.void(options) transaction.void.promise(options)	10
N/translation Module	translation.get(options)	1
	translation.load(options)	1
	translation.selectLocale(options)	None
N/ui/dialog Module	dialog.alert(options)	None
	dialog.confirm(options)	None
	dialog.create(options)	None
N/ui/message Module	Message.hide()	None
	Message.show(options)	None
	message.create(options)	None
N/ui/serverWidget Module	Assistant.addField(options)	None
	Assistant.addFieldGroup(options)	None
	Assistant.addStep(options)	None
	Assistant.addSublist(options)	None
	Assistant.getField(options)	None
	Assistant.getFieldGroup(options)	None
	Assistant.getFieldGroupIds()	None
	Assistant.getFieldIds()	None
	Assistant.getFieldIdsByFieldGroup(fieldGroup)	None
	Assistant.getLastAction()	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Assistant.getLastStep()	None
	Assistant.getNextStep()	None
	Assistant.getStep(options)	None
	Assistant.getStepCount()	None
	Assistant.getSteps()	None
	Assistant.getSublist(options)	None
	Assistant.getSublistIds()	None
	Assistant.hasErrorHtml()	None
	Assistant.isFinished()	None
	Assistant.sendRedirect(options)	None
	Assistant.setSplash(options)	None
	Assistant.updateDefaultValues(values)	None
	AssistantStep.getFieldIds()	None
	AssistantStep.getLineCount(options)	None
	AssistantStep.getSublistFieldIds(options)	None
	AssistantStep.getSublistValue(options)	None
	AssistantStep.getSubmittedSublistIds()	None
	AssistantStep.getValue(options)	None
	Field.addSelectOption(options)	None
	Field.getSelectOptions(options)	None
	Field.setHelpText(options)	None
	Field.updateBreakType(options)	None
	Field.updateDisplaySize(options)	None
	Field.updateDisplayType(options)	None
	Field.updateLayoutType(options)	None
	Form.addButton(options)	None
	Form.addCredentialField(options)	None
	Form.addField(options)	None
	Form.addFieldGroup(options)	None
	Form.addPageInitMessage(options)	None
	Form.addPageLink(options)	None
	Form.addSecretKeyField(options)	None
	Form.addSublist(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Form.addSubmitButton(options)	None
	Form.addSubtab(options)	None
	Form.addTab(options)	None
	Form.getButton(options)	None
	Form.getField(options)	None
	Form.getSublist(options)	None
	Form.getSubtab(options)	None
	Form.getTab(options)	None
	Form.getTabs()	None
	Form.insertField(options)	None
	Form.insertSublist(options)	None
	Form.insertSubtab(options)	None
	Form.insertTab(options)	None
	Form.removeButton(options)	None
	Form.updateDefaultValues(options)	None
	List.addButton(options)	None
	List.addColumn(options)	None
	List.addEditColumn(options)	None
	List.addPageLink(options)	None
	List.addRow(options)	None
	List.addRows(options)	None
	ListColumn.addParamToURL(options)	None
	ListColumn.setURL(options)	None
	Sublist.addButton(options)	None
	Sublist.addField(options)	None
	Sublist.addMarkAllButtons()	None
	Sublist.addRefreshButton()	None
	Sublist.getField(options)	None
	Sublist.getSublistValue(options)	None
	Sublist.setSublistValue(options)	None
	Sublist.updateTotallingFieldId(options)	None
	Sublist.updateUniqueFieldId(options)	None
	serverWidget.createAssistant(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	serverWidget.createForm(options)	None
	serverWidget.createList(options)	None
N/url Module	url.format(options)	None
	url.resolveDomain(options)	None
	url.resolveRecord(options)	None
	url.resolveScript(options)	None
	url.resolveTaskLink(options)	None
N/util Module	util.each(iterable, callback)	None
	util.extend(receiver, contributor)	None
	util.isAsyncFunction(obj)	None
	util.isArray(obj)	None
	util.isBoolean(obj)	None
	util.isDate(obj)	None
	util.isFunction(obj)	None
	util.isNumber(obj)	None
	util.isObject(obj)	None
	util.isRegExp(obj)	None
	util.isString(obj)	None
N/workbook Module	Workbook.runPivot(options)	10 per intersection returned
	workbook.create(options)	None
	workbook.createCalculatedMeasure(options)	None
	workbook.createColor(options)	None
	workbook.createConditionalFilter(options)	None
	workbook.createConditionalFormat(options)	None
	workbook.createConditionalFormatRule (options)	None
	workbook.createConstant(options)	None
	workbook.createDataDimension(options)	None
	workbook.createDataDimensionItem(options)	None
	workbook.createDataMeasure(options)	None
	workbook.createDimensionSelector(options)	None
	workbook.createExpression(options)	None
	workbook.createFieldContext(options)	None
	workbook.createFontSize(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	workbook.createLimitingFilter(options)	None
	workbook.createMeasureSelector(options)	None
	workbook.createMeasureValueSelector (options)	None
	workbook.createPathSelector(options)	None
	workbook.createPivotAxis(options)	None
	workbook.createPivot(options)	None
	workbook.createPositionPercent(options)	None
	workbook.createPositionUnits(options)	None
	workbook.createPositionValues(options)	None
	workbook.createReportStyle(options)	None
	workbook.createReportStyleRule(options)	None
	workbook.createSection(options)	None
	workbook.createSort(options)	None
	workbook.createSortByDataDimensionItem(options)	None
	workbook.createSortByMeasure(options)	None
	workbook.createSortDefinition(options)	None
	workbook.createStyle(options)	None
	workbook.createTable(options)	None
	workbook.createTableColumn(options)	None
	workbook.createTableColumnFilter(options)	None
	workbook.list()	10
	workbook.loadWorkbook(options)	10
N/workflow Module	workflow.initiate(options)	20
	workflow.trigger(options)	20
N/xml Module	Document.adoptNode(options)	None
	Document.createAttribute(options)	None
	Document.createAttributeNS(options)	None
	Document.createCDATASection(options)	None
	Document.createComment(options)	None
	Document.createDocumentFragment()	None
	Document.createElement(options)	None
	Document.createElementNS(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Document.createProcessingInstruction(options)	None
	Document.createTextNode(options)	None
	Document.getElementById(options)	None
	Document.getElementsByTagName(options)	None
	Document.getElementsByTagNameNS(options)	None
	Document.importNode(options)	None
	Element.getAttribute(options)	None
	Element.getAttributeNode(options)	None
	Element.getAttributeNodeNS(options)	None
	Element.getAttributeNS(options)	None
	Element.getElementsByTagName(options)	None
	Element.getElementsByTagNameNS(options)	None
	Element.hasAttribute(options)	None
	Element.hasAttributeNS(options)	None
	Element.removeAttribute(options)	None
	Element.removeAttributeNode(options)	None
	Element.removeAttributeNS(options)	None
	Element.setAttribute(options)	None
	Element.setAttributeNode(options)	None
	Element.setAttributeNodeNS(options)	None
	Element.setAttributeNS(options)	None
	Node.appendChild(options)	None
	Node.cloneNode(options)	None
	Node.compareDocumentPosition(options)	None
	Node.hasAttributes()	None
	Node.hasChildNodes()	None
	Node.insertBefore(options)	None
	Node.isDefaultNamespace(options)	None
	Node.isEqualNode(options)	None
	Node.isSameNode(options)	None
	Node.lookupNamespaceURI(options)	None
	Node.lookupPrefix(options)	None

SuiteScript 2.x Module	API(s)	Governance Usage Units Used
	Node.normalize()	None
	Node.removeChild(options)	None
	Node.replaceChild(options)	None
	Parser.fromString(options)	None
	Parser.toString(options)	None
	XPath.select(options)	None
	xml.escape(options)	None
	xml.validate(options)	None

API Governance Examples

The following examples show how governance units are calculated in a user event script and in a scheduled script:

Script Details	Usage Units Calculated
A user event script on a standard transaction record type (such as invoice) that includes: ■ one call to record.delete(options) ■ one call to Record.save(options)	This script uses a total of 40 usage units: ■ record.delete(options) uses 20 usage units for a transaction record ■ Record.save(options) uses 20 usage units for a transaction record Each user event script can use a maximum of 1,000 usage units, so in this case, this script has plenty of room to be expanded.
A scheduled script on a standard non-transaction record type (such as customer) that includes: ■ one call to record.load(options) ■ one call to record.transform(options) ■ one call to email.send(options)	This script uses a total of 30 usage units: ■ record.load(options) uses 5 usage units for a standard non-transaction record ■ record.transform(options) uses 5 usage units for a non-standard transaction record ■ email.send(options) uses 20 usage units Each scheduled script can use a maximum of 10,000 usage units, so in this case, this script has plenty of room to be expanded.

For more examples, see Monitoring Script Usage.

SuiteScript Statement and Instruction Limits

Applies to: SuiteScript 2.x | SuiteCloud Developer

NetSuite enforces execution limits to help prevent scripts from running indefinitely or consuming excessive resources. If a script exceeds these limits, NetSuite throws an error and terminates execution. This error varies depending on which script version is used.

- For **SuiteScript 1.0, 2.0, and 2.x scripts**, the `SSS_INSTRUCTION_COUNT_EXCEEDED` error is thrown based on the instruction count metric, which measures the number of low-level operations (instructions)

executed by the script engine interpreter. The instruction count may vary depending on how JavaScript constructs are implemented internally. For example, a single JavaScript statement such as *Array.sort* may correspond to a single operation in user code but multiple instructions within the engine.

- For **SuiteScript 2.1 scripts**, the `SSS_STATEMENT_COUNT_EXCEEDED` error is thrown based on statement metric in the script engine interpreter. This metric counts JavaScript-level statements executed, providing a higher-level measurement than the instruction count.

If you encounter these errors, you should examine all of the execution loops in your script to ensure that they contain either a terminating condition or a condition that can be met.

Monitoring Script Usage

Applies to: SuiteScript 2.x | SuiteCloud Developer

You can monitor SuiteScript unit usage using the Script.getRemainingUsage() method as shown in the following examples.

Example 1

This example shows how to instantiate the current script object and call Script.getRemainingUsage() to write the script's remaining usage units to the execution log.

```
var scriptObj = runtime.getCurrentScript();
log.debug({
    title: "Remaining usage units: ",
    details: scriptObj.getRemainingUsage()
});
```

Example 2

This example shows how to instantiate the current script object and checks the remaining usage units. If there are more than 50 usage units remaining, the script will execute a certain set of instructions.

```
var scriptObj = runtime.getCurrentScript();
var unitsRemaining = scriptObj.getRemainingUsage();
if (unitsRemaining > 50)
{
    //execute code here
}
```

You can also monitor script unit usage by running the script in the SuiteScript Debugger. After a script completes execution, you can view unit usage details on the Execution Log subtab in the SuiteScript Debugger console. If usage units are exceeded, usage limit error messages are emailed to the script owner indicating the number of usage units that were executed in the script before the usage limit error occurred.

For more information about the SuiteScript Debugger, see the help topic SuiteScript Debugger.

Governance on Script Logging

Applies to: SuiteScript 2.x | SuiteCloud Developer

NetSuite governs the amount of script execution logging that can be done. The governance model is:

- A company can make up to 100,000 N/log calls across all scripts within a 60–minute period.
- System error logs are purged after 60 days. User-generated logs are purged after 30 days.

Use the N/log Module methods to log script execution details.

Script owners are notified if a script is logging excessively and NetSuite automatically adjusts the log level.This keeps the script running and prevents it from logging too much. For example, say one company has 10 scripts running during a 60–minute period. If one script makes 70,000 log.debug(options) calls within each 20–minute period, NetSuite will automatically raise the script's log level.

You can see the new log level in the Log Level field on the script's deployment page:

For example, if the offending script's log level was originally set to Debug, NetSuite will increase the level to Audit. This means that the `log.debug` line of code will continue to execute, however, nothing will be logged, because the log level for the script has been raised to Audit and the line of code is using the Debug level. For more information about log levels, see the help topic Using Log Levels. For more information about viewing script execution logs, see the help topic Script Execution Logs.

For more information, see the help topic N/log Module Guidelines.

Script Execution Logs

All script execution logs in NetSuite are stored for up to 30 days. You can view script execution logs using the Execution Log subtab, using a Server Script Log search, or using the Script Execution log page. Details are shown in the following table:

Access to Script Execution Logs	Details
Execution Log subtab on script and script deployment records	The capacity for script execution logs on the Execution Log subtab is shared by customers on the same NetSuite database. To prevent excessive logging, there's a 5 million log limit per database instance. If this limit is reached, all logs for all customers on that database are purged. If you need to view all logs up to 30 days on the Execution Log subtab, consider creating a custom solution using custom records to store log information from logs. For more information, see the help topic Using the Script Execution Log Subtab.
Server Script Log search	The capacity for script execution logs returned by the Server Script Log search is shared by all customers on the same NetSuite database.

Access to Script Execution Logs	Details
	To prevent excessive logging, script execution logs are governed by a total storage limit of 5 million log limit per database instance for this search. On each NetSuite server, if this limit is reached, logs across all customers on that server are purged. If you need to search all logs up to 30 days, consider creating a custom solution using custom records to store log information from logs. If a Server Script Log search is not returning logs less than 30 days old, go to Customization > Scripting > Script Execution Logs and use the filters at the top of the Script Execution Log page to search for logs. You can also check script deployments that have large amounts of logging and consider lowering the Log Level to Error or Emergency or limiting the number of log lines in your script.
Script Execution log page	The execution logs shown on the Script Execution log page are stored for 30 days regardless of volume. For more information about the Script Execution log, see the help topic Viewing a List of Script Execution Logs.

Script Owner Notifications

If NetSuite detects that a script is logging excessively, the owner of the script is notified that the script is logging excessively and could possibly exceed the 100,000 logging threshold (for a 60–minute time period).

NetSuite sends email notifications and adds a log entry to the script's Execution Log indicating that a script's log level has been increased. For more information about script execution log levels, see the help topic Setting Script Execution Log Levels.

Search Result Limits

Applies to: SuiteScript 2.x | SuiteCloud Developer

- Search results are limited to 1,000 records when you execute SuiteScript searches using the N/search Module. For information about working with SuiteScript searches in NetSuite, see the help topic N/search Module.
- If you load a saved search using search.load(options), and then call Search.run() to return a result set of search.ResultSet objects, you may get up to 4,000 results returned. For more information, see the help topic ResultSet.each(callback).
- Text columns in search results have a 4000-byte limit, or about 4000 characters in English. Using intricate character sets can lower this limit, since they need more bytes per character.

Script Execution Time Limits

Applies to: SuiteScript 2.x | SuiteCloud Developer

Each server script type and plug-in type has a limit on the amount of time it can run in a single execution. If the time limit is exceeded, an `SSS_TIME_LIMIT_EXCEEDED` error is thrown and script execution stops. However, a script may run for a long time, and you may not see the `SSS_TIME_LIMIT_EXCEEDED` error under these conditions:

- The script performs a large number of record operations without exceeding the usage unit limit. See Script Type Usage Unit Limits for usage unit limits.
- The script processes a large number of transactions for the same records, such as items or lot numbers, without exceeding the usage unit limit.
- The script causes a large number of user event scripts or workflows to execute.
- The script performs database queries or updates that collectively take a long time to finish.

Additionally, there is a 300 second (5-minute) time limit when scripts are executed in the SuiteScript 2.1 Debugger. For more information about metering in the debugger, see Script Debugger Metering and Permissions.

The following tables list the time limit (in seconds) for each script type, core plug-in type, and custom plug-in type.

Script Type	Time Limit (in Seconds)
Bundle Installation Scripts (SuiteScript 1.0) SuiteScript 2.x Bundle Installation Script Type	3,600
Client Scripts (SuiteScript 1.0) SuiteScript 2.x Client Script Type	300
Custom record action	300
SuiteScript 2.1 Custom Tool Script Type	300
Map/reduce (input stage) SuiteScript 2.x Map/Reduce Script Type	3,600
Map/reduce (map stage) SuiteScript 2.x Map/Reduce Script Type	300
Map/reduce (reduce stage) SuiteScript 2.x Map/Reduce Script Type	900
Map/reduce (summarize stage) SuiteScript 2.x Map/Reduce Script Type	3,600
Mass Update Scripts (SuiteScript 1.0) SuiteScript 2.x Mass Update Script Type	300
Portlet Scripts (SuiteScript 1.0) SuiteScript 2.x Portlet Script Type	300
RESTlets (SuiteScript 1.0) SuiteScript 2.x RESTlet Script Type	300
SuiteScript 2.x SDF Installation Script Type	3,600
Single-page application (SPA)	300
Scheduled Scripts (SuiteScript 1.0) SuiteScript 2.x Scheduled Script Type	3,600
Suitelets (SuiteScript 1.0)	300

Script Type	Time Limit (in Seconds)
SuiteScript 2.x Suitelet Script Type	
SuiteScript Server Pages (SSP) application	300
User Event Scripts (SuiteScript 1.0) SuiteScript 2.x User Event Script Type	300
Workflow Action Scripts (SuiteScript 1.0) SuiteScript 2.x Workflow Action Script Type	300

Core Plug-in Type	Time Limit (in Seconds)
Adjusted Consolidated Rates	300
Bank Connectivity Plug-in	300
Bank Statement Parser Plug-in	1,800
Custom GL Lines Plug-in	30
Dataset Builder Plug-in	300
Email Capture Plug-in	300
Financial Institution Connectivity Plug-in	3,600
Financial Institution Parser Plug-in	1,800
Payment Processing	300
Platform Extension	300
Promotions	300
Revenue Management	300
Shipping Partners	300
Tax Engine	60
Workbook Builder Plug-in	300

Custom Plug-in Type	Time Limit (in Seconds)
Plug-in Type	3,600
Plug-in Type Implementation	3,600

SuiteScript Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

General Development Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

Important: If you're using SuiteScript 1.0 or SuiteScript 2.0 for your scripts, consider converting these scripts to SuiteScript 2.1. Use SuiteScript 2.1 to take advantage of a new runtime engine which may improve performance, new features, and functionality enhancements, including newer ECMAScript language features. SuiteScript 2.1 server supports ES2023 and SuiteScript 2.1 client supports the ES version supported by the browser. For more information, see the help topics SuiteScript Versioning Guidelines and SuiteScript 2.1.

Here are some general best practices for writing SuiteScript scripts:

Using SuiteScript modules	■ When you specify the modules that your script uses, make sure that the order of the modules matches the order of the parameters in your main function. For example, if your script uses the N/error, N/record, and N/file modules and you specify them in this order, make sure that your function parameters are in the same order, as follows:
	`require(['N/error', 'N/record', 'N/file'],`
	`  function(error, record, file) {`
	`  ...`
	`});`

Code organization and structure	■ Good code is well organized. Data and operations in each function or class fit together. Organize your code into reusable chunks. Many functions can be used in a variety of forms. Any reusable functions should be stored in a common library file and then called into specific event functions for the required forms as needed. ■ Good code is written using small, readable, reusable functions. Including too many lines of code in one function makes it harder to understand and debug. Consider refactoring your code to keep all functions reasonable in size. ■ Use nesting judiciously. Extensive nesting may make your script unreadable. Try to keep your scripts as readable as possible. ■ During script development, use components, load them individually and then test each one -- inactivating all but the one you are testing when multiple components are tied to a single user event.
Naming conventions	Good code uses meaningful naming conventions. A well written script containing descriptive names for functions, variables, IDs, provides a good structure for building comments around. Functions:

- Although you can use any desired naming conventions for functions within your code, you should use custom name spaces or unique prefixes for all your function names.
- When working with the top-level NetSuite functions in Client SuiteScript (for example the pageInit function), you should name the new function to correspond to the NetSuite name. For example, a pageInit function can be named pageInit or formAPageInit. If your code is already established, you can wrap it with a top-level function that has the appropriate naming convention.
- Function Names: All function names should be written in lower camel case which means the first letter of each word (except the first) is capitalized.
- For IDs, enter all record, field, sublist, tab, and subtab IDs in lower case in your SuiteScript code. Prefix all custom script IDs and deployment IDs with an underscore (_).

Variables and Constants:

- Avoid using a single character as a variable name.
- Name your variables to identify the data type and describe the data stored in the variable. For example,
  - String variables – prefix with "st", such as stTitle
  - Integer variables – prefix with "int", such as intTotalCount
  - Float variables – prefix with "fl", such as flPrice
  - Boolean variables – prefix with "b", such as bIsDone
  - Array variables – prefix with "arr", such as arrPhoneCalls
  - Object variables – prefix with "Obj", such as ObjNewPet
  - Record variables – prefix with "rec", such as recCustomer
  - Date variables – prefix with "dt", such as dtFirstBillingDate
- Local variable scope is only within the function. Use standard camelCase naming.
- Global variable scope is within the whole file or more files if declared under included library. Use a variable in upper case spaced by underscore.
- To represent a pseudo constant, use a variable in upper case spaced by underscore. Constants are to be used to make code more readable.

IDs:

- For IDs, enter all record, field, sublist, tab, and subtab IDs in lower case. Prefix all custom script IDs and deployment IDs with an underscore (_).

File names:

- If you write SuiteScript code for multiple accounts, consider using the following file name convention: <Company Name/Abbreviation>_<Script Type>_<Requirement Description>.js. For example, MyCompany_CS_SetTaxable.js.

  The following are suggested Script Types:
  - CS – client scripts
  - UE – user events
  - SL – Suitelet
  - RL – RESTlet
  - PL – Portlet
  - SC – Scheduled
  - MR – Map/Reduce
  - Gl – SuiteGL
  - WA – Workflow Action
  - MU – Mass Update
  - BI – Bundle Installation

Commenting	■ Thoroughly comment your code. Include comments about what the script is doing and who authored it. This practice helps with debugging and development and assists NetSuite Customer Support in locating problems, if necessary. It also helps anyone else who works with the script. ■ Write useful comments. Good code comments explain what is being done and why it's being done. ■ When documenting functions try to provide a good abstract of the actions of the function without restating the function name. ■ A good way to write relevant comments is to plan your scripts using pseudo code and replace the pseudo code with a relevant comment where necessary. ■ At a minimum, you should include comments: (1) for the file level disclaimer, (2) for version control, (3) at the function level, and (4) at the logic level. ■ Note that in SuiteScript 2.x, inline comments aren't allowed within an object. You may not see a specific error, however, your script won't function properly if you include inline comments within an object.
White space and formatting	■ Use white space and formatting to enhance readability rather than to minimize typing effort. The following rules should always apply: □ All line returns within a function should be terminated with a ; (semicolon). □ Code blocks should be tab indented.
Using IDs	■ Use **static** ID values in your API calls where applicable because name values can change.
Passwords	■ Don't hard-code any passwords in scripts. The password and password2 fields are supported for scripting. For additional information, see SuiteScript Security Considerations.
Error and exception handling	■ Include proper error handling sequences in your script wherever data may be inconsistent, not available, or invalid for certain functions. For example, if your script requires a field value to validate another, ensure that the field value is available. ■ Use try-catch blocks. Try-catch blocks provide a way to “catch” an exception object thrown with the try block. As soon as an exception is thrown the catch block starts. An optional finally clause can also be used to run (guaranteed) if any exception occurs or not.
URLs	■ Don't hard-code URL links. If the domain name changes, hard-coded URLs won't work. To prevent issues with links, use the N/url Samples.
Date and Currency fields	■ Use the built-in library functions whenever possible for reading/writing Date/Currency fields and for querying XML documents.

Script parameters	■ Use script parameters or configuration files wherever possible, instead of hard coding values, to ensure that scripts can be configured easily
	■ You can use the runtime.getCurrentScript() function in the runtime module to reference script parameters. For example, use the following code to obtain the value of a script parameter named custscript_case_field:
	`define(['N/runtime'], function(runtime) {`
	`  function pageInit(context) {`
	`  var strField = runtime.getCurrentScript().getParameter('SCRIPT', 'custscript_case_field');`
	`  ...`
	`  });`
	■ For security reasons, don't include confidential information in script parameters. Information saved in script parameters can be indexed by search engines and therefore be viewable by the public. This means, for example, that the information could be found in Google searches. For more information, see Creating Script Parameters Overview.

Reserved words	■ Ensure that your scripts don't use any reserved words in SuiteScript. For more information, see the help topic SuiteScript Reserved Words.

Execution contexts	■ Consider the most appropriate script execution context for your script. The script execution context specifies how and when a script is triggered to run. If you don't use the appropriate context, blocks of scripts run when they shouldn't and result in unexpected errors in the data and the script. For more information, see the help topic Execution Contexts.
Loading records	■ For record Create or Edit operations, you should load records in dynamic mode. This is required to make updates on record object. Load records in a non-dynamic mode if you want to only get field values from the record and update on record isn't needed. ■ Don't use record.load(options) to load a whole record when you only need to get the value of a few fields. The performance of the load record API is similar to loading the record in the user interface without field rendering since it loads all configuration fields and sublists for the record. To avoid slowing down the execution of the script, use search.lookupFields(options) to get the value of a few fields.
Invoking methods	■ You should use parameter id's instead of parameter sequence. ■ You should use defined SuiteScript 2.0 native enums. You may find the list of valid enums on Help and SuiteScript 2.0 API pdf.
Asynchronous programming	■ When using promises in your SuiteScript scripts to implement asynchronous programming, you should consider the best practices listed in the Best Practices for Asynchronous Programming with SuiteScript help topic.
Custom code	■ Place all custom code and markup, including third party libraries, in your own namespace. **Important:** Custom code must not be used to access the NetSuite DOM. Developers must use SuiteScript APIs to access NetSuite UI components. ■ Use Subresource Integrity when you include content from external sources in Inline HTML fields. For more information, see Subresource Integrity.
Testing	■ Good code is well-tested. Testing serves as an executable specification of the code and examples of its use. ■ Always thoroughly test your code before using it on your live NetSuite data. ■ If the same script is used across multiple forms, ensure that you test any changes to the code for **each** form that the code is associated with. ■ During testing, the Deployment Status should be Testing to limit the script processing to the script owner only. If several users are testing a script (or if a Sandbox Account is being used), the Deployment Status can be set to Released, but you must choose the appropriate Audience so that the script doesn't run for users who aren't involved in the testing process. For production, the Deployment Status should set be Released. Again, it's critical to set the appropriate Audience so that the script doesn't run for unintended users. ■ During testing, the Log Level should be set to Debug.
Logging	■ Log messages in the script should make business sense; include the record id, record type and other important details in your log messages. Each log message should provide a clear picture of the functionality that the script is designed to achieve. ■ Key steps in each script should be logged as Audit. ■ Proper logging adds to the readability of the code and also significant on issue/error debugging. ■ During testing, the Log Level should be set to Debug. For a Production Deployment, the Log Level should be set to Audit. Note that if there are production issues that require diagnosis/debugging, the Deployment Log Level can be set back to Debug.
Other	■ Consider the time zone that your scripts must use. By default, server scripts use the time zone that is specified in your NetSuite account. Be sure to convert your time values to the correct time zone before you use them, if necessary.

	■ Consider field limits and translation when creating your scripts. Fields that use special characters used in some languages may support fewer characters than the field limit specified. For example, the Reference No. field (tranid) supports 45 English characters, but supports fewer Chinese characters. ■ Be sure to use BigInt values in your scripts to avoid potential errors. Integers larger than MAX_SAFE_INTEGER and smaller than MIN_SAFE_INTEGER must be enclosed in quotations or appended with '*n*' at the end of the integer literal. For more information about BigInt values, see BigInt.

You can also refer to the following Internet resources for JavaScript coding best practices:

- JavaScript Best Practices (w3)
- JavaScript Best Practices (w3 schools)

Client Script Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following are best practices for both form-level and record-level client SuiteScript development.

General	■ Use global (record-level) client scripts for more flexible deployment models (include in a bundle or SuiteCloud project) than client scripts attached to forms.
Using Record.setValue and Record.setCurrentSublistValue methods	■ Record.setValue(options) and Record.setCurrentSublistValue(options) execution is multi-threaded whenever child field values need to be sourced in. To synchronize your logic, use the `postSourcing` function or set the `forceSyncSourcing` synchronous parameter to `true`.
Advanced Employee Permissions	■ When the Advanced Employee Permissions feature is enabled, the search columns available to users is also dependent on the permissions assigned to the role. ■ When the Advanced Employee Permissions feature is enabled, the fields and sublists a user has access to can change depending on which employee is being viewed or edited. This is different from other records in NetSuite, where permissions granted to a role determines what the role can see for every instance of that record. In general, scripts should always verify that a field exists before trying to do something with it. Simply calling functions and methods that interact with fields before checking whether the field is there may result in inconsistent behavior. For example, the department field is permitted on the employee record. When you verify that the field exists and you don't have access, a null value is returned, and if the field is empty, an empty string is returned. Unsafe Practice Example `var employeeRecord = record.load ({ type: record.Type.EMPLOYEE, id: 115 }); employeeRecord.setValue({ fieldId: 'department', value: 'marketing' }); employeeRecord.save();` Safe Practice Example To detect if your role has access to a field for a specific employee, load the employee record object and call `getFields()`. If the field exists and you do have access a `true` value is returned. In the below example, the user has access to the department field for the employee with ID: 115.

	``
	var employeeRecord = record.load ({
	    type: record.Type.EMPLOYEE,
	    id: 115
	});
	var accessToDepartment = employeeRecord.getFields().includes('department');
	``

Editing sublists	■ When you are editing the current line of a sublist and a nested client script is triggered (for example, when a `fieldChanged` script is triggered by changing a value in the current sublist line), make sure that the nested script doesn't select a different line of the same machine as the current line. Otherwise, the changes made in the parent script may be lost.
Execution contexts	■ Use execution context filtering to specify how and when a client script is executed. Execution contexts provide information about how a script is triggered to run. For example, a script can be triggered in response to an action in the NetSuite application, or an action occurring in another context, such as a web services integration. You can use execution context filtering to ensure that your scripts are triggered only when necessary. For more information, see the help topic Execution Contexts.
setValue and setText methods	■ Methods that set values accept raw data of a specific type and don't require formatting or parsing. Methods that set text accept strings but can conform to a user-specified format. For example, when setting a numerical field type, `setValue()` accepts any number, and `setText()` accepts a string in a specified format, such as “2,000.39” or “2.00,39”. When setting a date field type, `setValue()` accepts any `Date` object, and `setText()` accepts a string in a specified format, such as “6/9/2016” or “9/6/2016”.

Debugging	■ When debugging client scripts, you can insert a `debugger;` statement in your script, and execution will stop when this statement is reached:
	``
	function runMe() {
	    var k = 1;
	    k *= (k + 9);
	    debugger;
	    console.log(k);
	}
	``
	When your script stops at the `debugger;` statement, you can examine your script properties and variables using the debugging tools in your browser.
	■ When debugging client scripts, some scripts might be minified. Minified scripts have all unnecessary characters removed, including white space characters, new line characters, and so on. Minifying scripts reduces the amount of data that needs to be processed and reduces file size, but it can also make scripts difficult to read. You can use your browser to de-minify scripts so that they’re more readable. To learn how to de-minify scripts, see the documentation for your browser.

Testing	■ When testing form-level client scripts, use Ctrl-Refresh to clear your browser cache and ensure that the latest scripts are being executed.
Client Scripts in SuiteApps and Bundles	■ If you need to include client scripts in a SuiteBundle or a SuiteApp, don't enable the Hide in SuiteBundle preference on the client script's file record. ■ If a client script included in a SuiteApp or SuiteBundle references another script file, such as a library file: 1. First, upload the library file to the File Cabinet. Don't enable the Hide In SuiteBundle preference. 2. Next, set your installation preferences, being sure not to hide any client script files or files referenced from client script files. 3. Finally, link the library file to the client script file.

Map/Reduce Script Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following are best practices for working with map/reduce scripts.

General	As described in Hard Limits on Function Invocations, NetSuite imposes governance limits on single invocations of map, reduce, getInputData, and summarize functions: ■ **map**: 1,000 usage units (the same as mass update scripts) ■ **reduce**: 5,000 usage units ■ **getInputData**: 10,000 usage units ■ **summarize**: 10,000 usage units If you are concerned about potential issues with these limits, review your script to make sure that your map and reduce functions are relatively lightweight. Your map and reduce functions shouldn't include a long or complex series of actions. For example, consider a situation in which your map or reduce function loads and saves multiple records all at the same time. This approach might cause an issue with the limits described above. If your getInputData function returns a list of record IDs, a better approach might be to use the map function to load each record, update fields on the record, and save it. If you have a script that performs complex operations within a single function (such as loading and saving multiple records, or transforming multiple records), consider using a different script type, such as a scheduled script.

Passing search data to getInputData	In the getInputData stage, your script must return an object that can be transformed into a list of key-value pairs. A common approach is to use a search. If you decide to use this technique, note that you should have your function return either a search.Search object or an object reference to a saved search. By contrast, if you run a search within the getInputData function and return the results (for example, as an array), there is a greater risk that the search will time out.
	Instead, you should use one of the following approaches:
	■ Return a search object. That is, return an object created using search.create(options) or search.load(options).
	■ Return a search object reference. That is, return an inputContext.ObjectRef object that references a saved search.
	In both cases, the time limit available to the search is more generous than it would be for a search executed within the function.
	The following snippet shows how to return a search object:
	`function getInputData()`
	`{`
	`  return search.create({`
	`  type: record.Type.INVOICE,`
	`  filters: [['status', search.Operator.IS, 'open']],`
	`  columns: ['entity'],`
	`  title: 'Open Invoice Search'`
	`  });`
	`}`
	And the following snippet shows how to return a search object reference:

	``
	...
	function getInputData {
	{
	    // Reference a saved search with internal ID 1234.
	
	    return {
	        type: 'search',
	        id: 1234
	    };
	}
	...
	``
	For information about additional ways to return data, see the help topic getInputData(inputContext).

Minimizing risk of data duplication	A map/reduce script can be interrupted at any time by an application server disruption. When this happens, the script is restarted. Depending on how the script is configured, when a map or reduce job starts again, it may attempt to retry processing for the same key-value pairs it had flagged for processing when the interruption occurred. Similarly, if an uncaught error disrupts the job, the system may retry processing for the pair that was being processed when the error occurred. **Note:** For an overview of the system's behavior following an interruption, see the help topic System Response After a Map/Reduce Interruption.
Handling restarts	When a job is restarted, there is an inherent risk of data duplication. Every map/reduce script should be written in such a way that each entry point function checks to see whether the function has been previously invoked. To do this, use the context.isRestarted property, which exists for every map/reduce entry point. If the function has been restarted, the script should provide any logic needed to avoid duplicate processing. For examples, see the help topic Adding Logic to Handle Map/Reduce Restarts.
Buffer size	When you deploy a script, the deployment record includes a field called Buffer Size. The default value of this field is 1. In general, you should leave this value set to the default. The Buffer Size field controls how many key-value pairs are flagged for processing at one time, and how frequently a map or reduce job saves data about its progress. Setting this field to a higher value may have a small performance advantage. However, the disadvantage is that, if the job is interrupted by an application server restart, there is a greater likelihood of one or more key-value pairs being processed twice. For that reason, you should leave this value set to 1, particularly if the script is processing records. For more details on this field, see the help topic Buffer Size.

Scheduled Script Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following are best practices for scheduled scripts. Also see the help topic Scheduled Script Handling of Server Restarts.

General	■ **Scheduled** deployments and **Not Scheduled** deployments are executed from the same processor pool. Keep this in mind as you deploy multiple scheduled scripts because

	the amount of pending script instances is the ultimate bottleneck for scheduled script execution throughput. ■ Create **Not Scheduled** deployments based on the anticipated number of simultaneous calls to this script and approximate execution time of the script. ■ Although there is no restriction on the number of **Not Scheduled** scripts that can be placed into the processing pool, too many pending scripts may create a backlog and compromise system performance. Because only one script can be run in a processor at a time, you shouldn't overload the system. ■ If you want to deploy scheduled scripts that are **scheduled to run hourly on a 24 hour basis**, the following sample values should be set on the Script Deployment page: □ Deployed = checked □ Daily Event = [radio button enabled] □ Repeat every 1 day □ Start Date = [today's date] □ **Start Time = 12:00 am** □ Repeat = every hour □ End By = [blank] □ No End Date = checked □ Status = Scheduled □ Log Level = Error □ Execute as Role = Set to **Administrator** If the **Start Time** is set to any other time than 12:00 am (for example, set to 2:00 pm), the script will start at 2:00 pm, but then finish its hourly execution at 12:00 am. It won't resume until the next day at 2:00 pm. ■ When possible, schedule your deployments during the hours of 2 AM to 6 AM PST. Deployments scheduled for submission during the hours of 6 AM to 6 PM PST may not run as quickly due to high database activity.
Deployments that continue to use queues	■ If you don't have an account with SuiteCloud Plus, all **Scheduled** deployments and **Not Scheduled** deployments that continue to use queues are executed from the same queue. If you deploy multiple scheduled scripts that continue to use queues, your queue size is the ultimate bottleneck for scheduled script execution throughput. For additional information, see the help topic Scheduled Script Deployments that Continue to Use Queues. ■ Although there is no restriction on the number of **Not Scheduled** scripts that can be submitted to SuiteCloud Processors, too many waiting scripts may create a backlog and compromise system performance. Because only one script can be run at a time, you shouldn't overload the system. ■ Although there is no restriction on the number of **Not Scheduled** scripts that can be placed into the processing pool, too many pending scripts may create a backlog and compromise system performance. Because only one script can be run in a processor at a time, you shouldn't overload the system.
Scheduled script vs Map/Reduce script	■ In general, you should use a map/reduce script rather than a scheduled script for any scenario where you want to process multiple records, and where your logic can be separated into relatively lightweight segments. However, a map/reduce script isn't as well suited to situations where you want to enact a long, complex function for each part of your data set. ■ Map/reduce scripts can be run manually or on a schedule, the same as scheduled scripts. However, map/reduce scripts offer several advantages over scheduled scripts. One advantage is that, if a map/reduce job violates certain aspects of NetSuite governance, the map/reduce framework automatically causes the job to yield and its work to be rescheduled, without disruption to the script. However, be aware that some aspects of map/reduce governance can't be handled through automatic yielding.

	■ SuiteScript 2.x scheduled scripts don't support recovery points or yielding, so use map/reduce scripts for large data processing. If you need to process a large amount of data or a large number of records, use a map/reduce script instead. The map/reduce script type has built in yielding and can be submitted for processing in the same ways as scheduled scripts. For information about map/reduce scripts, see the help topic SuiteScript 2.x Map/Reduce Script Type.

Suitelets and UI Object Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following are best practices for Suitelet development using UI objects and custom UI.

General	■ Suitelets are ideal for generating NetSuite pages (forms, lists), returning data (XML, text), and redirecting requests. ■ Limit the number of UI objects on a page: less than 100 rows for sublists, less than 100 options for on demand select fields, and less than 200 rows for lists.
HTML	■ Try using inline HTML fields embedded on the form before implementing a full custom HTML page route. ■ Use Subresource Integrity when you include content from external sources in Inline HTML fields. For more information, see Subresource Integrity.
iFrames	■ Append "ifrmcntnr=T" to the external URL when embedding in iFrame especially if you are using the Firefox browser. (For more about NetSuite and iFrame, see the help topic Embedding an Online Form in your Website Page.)
User credentials	■ When building a custom UI outside of the standard NetSuite UI (such as building a custom mobile page using Suitelet), use the N/auth Module and N/crypto Module to help users manage their credentials within the custom UI.
Calling a Suitelet and redirection	■ When calling a Suitelet using its external URL, escape the parameter values to avoid cross-site scripting injections, for example, by converting the appropriate characters to HTML entities. ■ For access or redirect to a Suitelet from another script, use url.resolveDomain(options) to discover the URL instead of hard- coding the URL.
Advanced Employee Permissions	■ When the Advanced Employee Permissions feature is enabled, keep the following in mind: □ To avoid inadvertently exposing employee data, use caution when running Suitelets or Restlets as an administrator. A user with a role that has limited access to the employee record can access a Suitelet or Restlet that runs as an administrator. Depending on how the Suitelet or Restlet is written, the user may have access to employee information that they would otherwise not see. □ Use caution when setting up Suitelets and Restlets to give access to users without having to log in since it could potentially expose employee information in uncontrolled ways.
Deployment	■ Deploy Suitelets as "Available without Login" only if necessary (no user context, login performance overhead). (See Setting Available Without Login.)

User Event Script Best Practices

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following are best practices for developing user event scripts.

General	■ For critical business logic, use a scheduled script to clean up after user events in case of errors. ■ Don't try to run a user event script from another one – instead, create a module with common code. ■ Make sure that the user event script doesn't access any sensitive field values.
Context	■ Use the `type` argument, the `context` object, and `context.UserEventType` enum to define and limit the scope of your user event logic. See the help topic context.UserEventType.
Entry points	■ For operations that depend on on the submitted record being committed to the database should happen in an `afterSubmit` script. ■ When updating transaction line items in a `beforeSubmit` script, ensure that the line item totals, net taxes, and discounts are equal to the `summarytotal`, `discounttotal`, `shippingtotal`, and `taxtotal` amounts. ■ Avoid assigning too many functions to one record type, as it can slow things down. This could negatively affect the user experience with that record type. For example, if there are ten `beforeLoad` scripts that must complete their execution before the record loads into the browser, the time needed to load the record may increase significantly. Be aware of the number of user events scripts used, including bundled user event scripts. ■ Use beforeSubmit to update fields or change the record before it's submitted. ■ Perform all post-processing operations of the current record on an `afterSubmit` event.
Storing values	■ If you want to store a value during a `beforeLoad` operation and then read that value during an `afterSubmit` operation in the same script, consider using a hidden custom field to store the value. You can add a hidden custom field to the form and store your value during the `beforeLoad` operation, and you can retrieve the value from the same field during the `afterSubmit` operation.
Execution contexts	■ Use execution context filtering to control how and when a user event script is executed. Execution contexts provide information about how a script is triggered. For example, a script can be triggered in response to an action in the NetSuite application, or an action occurring in another context, such as a web services integration. You can use execution context filtering to ensure that your scripts are triggered only when necessary. For more information, see the help topic Execution Contexts.
Performance	■ Try to keep user event script execution under 5 seconds, as they run frequently. You can use the Application Performance Management (APM) SuiteApp to test the performance of your scripts deployed on a specific record type. See the help topic Application Performance Management (APM).
Debugging	■ To debug a client script, include a `debugger;` statement as the first line in the script. When execution reaches that statement, you can examine your script properties and variables using the debugging tools in your browser. You can also use the `debugger;` statement in the SuiteScript Debugger to help you debug server scripts. For more information about the SuiteScript Debugger, see SuiteScript Debugger.
Hosted websites	■ Activities (user events) on a hosted website can trigger server SuiteScripts. In addition to sales orders, scripts on case records and customer records also run in response to web activities.

Optimizing SuiteScript Performance

Applies to: SuiteScript 2.x | SuiteCloud Developer

Write your scripts with performance in mind to get the best results. This is especially important for your custom scripts. You can see if there are custom scripts in your account at Customization > Scripting > Scripts. The following guidelines are suggested to optimize script performance:

General Scripting Guidelines

- Consider using SuiteScript 2.1 language constructs and converting your SuiteScript 2.0 scripts to SuiteScript 2.1. Some SuiteScript 2.1 language constructs can be used to improve script performance. For more information about SuiteScript 2.1, see the help topic SuiteScript 2.1.
- Combine similar scripts/functions whenever possible.
- Inactivate scripts/deployments that are no longer needed.
- Use asynchronous processing for user events.
- Use built-in permissions (by leveraging Run as Role) instead of scripting permissions.
- As a general rule, develop your user event scripts to run in under 5 seconds, your Suitelets and Portlets in under 10 seconds, and your scheduled scripts in under 5 minutes. This gives you a large enough margin of error to handle the outlier use cases (where the volume of work is unusually large, or the overall system is slow due to high load).
- To minimize execution logging after your script is tested and released, set your script log level to ERROR or EMERGENCY. See Setting Script Execution Log Levels.
- Deploy scripts to run as administrator only if necessary to minimize security risk and to eliminate performance overhead. See Executing Scripts Using a Specific Role.
- Use execution context filtering to control when a client script or user event script is executed. Execution contexts provide information about how a script is triggered to process. For example, a script can be triggered in response to an action in the NetSuite application, or an action occurring in another context, such as a web services integration. You can use execution context filtering to ensure that your scripts are triggered only when necessary. For more information, see the help topic Execution Contexts.

Accessing Records in Scripts

- If applicable, change the trigger of a script to avoid reloading a record.
- Avoid saving records multiple times in an After Record Submit event.
- Avoid loading and submitting a record on a Before Record Submit trigger.
- Minimize use of API calls that perform load, search, or save record operations.
- Use inline editable child custom records whenever your use case calls for batch processing of multiple related/child records during user events on the parent record. (See the help topic Custom Child Record Sublists in the NetSuite Help Center.)

Scripting Searches

- Optimize your search filters and columns to get faster results by filtering inactive records, using shorter date ranges, and removing unnecessary columns.
- Use faster operators such as starts with/between/withi instead of contains/formulas.
- Remove unused columns from your search results. Place any lines that add columns to search results in a comment. Then the returned values aren't used by the succeeding script logic.

- Wherever possible, combine searches for the same record into one main search with merged filters to improve performance by minimizing search instances.

SuiteScript Security Considerations

Applies to: SuiteScript 2.x | SuiteCloud Developer

You should take certain measures to ensure your SuiteScript script runs safely and securely. This includes the use of user credentials and executing scripts using a specific role, particularly when working with sensitive data. In general, you should follow these guidelines:

Security Consideration	Security Consideration Guidance
User passwords	■ Don't hard-code any passwords in scripts. Using plain text or other unencrypted user credentials is unsafe and can pose a security threat. Whenever possible, use Token-based Authentication (TBA) or OAuth2.0 to specify user credentials. ■ Plaintext passwords can be used with SFTP, certificates, and other areas, however NetSuite always gives the customer the option to use encrypted credential (GUID) instead.
Script Deployment	Deploy scripts to run as administrator only if necessary to reduce security risks and performance issues. See Executing Scripts Using a Specific Role.
Script parameters	For security reasons, don't include confidential information in script parameters. Information saved in script parameters can be indexed by search engines and therefore be viewable by the public. This means, for example, that the information could be found in Google searches. For more information, see Creating Script Parameters Overview.
SuiteScript 2.x RESTlet Script Type	The URLS for accessing RESTlets are protected by TLS encryption. Only requests sent using TLS encryption are granted access. For more information, see the help topic Supported TLS Protocol and Cipher Suites.
SuiteScript 2.x Suitelet Script Type	■ Deploy Suitelets as "Available without Login" only if necessary, such as when there is no user context or due to login performance overhead. See Setting Available Without Login.) ■ When building custom UI objects outside of the standard UI, such as when building a custom mobile page using a Suitelet, you can use the N/auth Module to help users manage their credentials within the custom UI.
SuiteScript 2.x User Event Script Type	To prevent users from accessing sensitive information, such as password and credit card data, the following internal field IDs cannot be read in `beforeSubmit` entry point scripts for external role users: ■ password ■ password2 ■ ccunumber (on the sales order record) ■ ccsecuritycode (on the sales order record) External role users include shoppers, online form users and other anonymous users, customer center users, etc.
N/auth Module	Be extremely careful when using the N/auth module to change an email or user password. Both the auth.changeEmail(options) and auth.changePassword(options) methods allow you to include a plaintext string for the password.
N/http Module and N/https Module	For security purposes, NetSuite blocks some headers from use in HTTP and HTTPS calls. For a list of blocked headers, see the help topic HTTP Header Information and HTTPS Header Information
N/https Module	Plaintext user credentials can be included in HTTPS request parameters or in the body. Using plain text or other unencrypted user credentials is unsafe and can pose a security

Security Consideration	Security Consideration Guidance
	threat. Whenever possible, use Token-based Authentication (TBA) or OAuth2.0 to specify user credentials.
N/query Module	When working with credit cards, you can retrieve only the encrypted version of the credit card number, at most. You may not be able to retrieve and credit card number data.
Content from external sources	Use Subresource Integrity when you include content from external sources in Inline HTML fields. For more information, see Subresource Integrity.

SuiteScript Debugger

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

NetSuite offers built-in tools so you can debug your SuiteScript 1.0, 2.0, and 2.1 scripts.

In SuiteScript Debugger help topics, the debugger for SuiteScript 1.0 and 2.0 scripts is called the *Script Debugger*, while the debugger for SuiteScript 2.1 scripts is called the *2.1 Script Debugger*. To use either debugger, you must be using a role with SuiteScript permission (Full level).

Note: If you are using SuiteScript 1.0 for your scripts, consider converting these scripts to SuiteScript 2.0 or SuiteScript 2.1 scripts. Use SuiteScript 2.0 to take advantage of new features, APIs, and functionality enhancements. SuiteScript 2.1 uses the Graal runtime engine which may improve script performance and adds new language features, such as support for the spread operator, classes, and destructuring. For more information, see the help topics SuiteScript 2.x Advantages and SuiteScript 2.1.

To learn more about debugging SuiteScript scripts, see the following topics.

- SuiteScript Debugger Overview
- Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts
- Debugging SuiteScript 2.1 Scripts
- Script Debugger Metering and Permissions
- Debugging Client Scripts
- Debugging a RESTlet

SuiteScript Debugger Overview

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

SuiteScript offers a script debugger for SuiteScript 1.0, SuiteScript 2.0, and SuiteScript 2.1 server scripts, core plug-in implementations, and on-demand debugging.

Important: You can't use the SuiteScript Debugger to debug SuiteScript 2.1 scripts. You can still test critical parts of your script in the SuiteScript Debugger as a SuiteScript 2.0 script before you run the script as a SuiteScript 2.1 script. For more information about SuiteScript 2.1, see the help topic SuiteScript 2.1.

You can debug SuiteScript 2.1 scripts with the new 2.1 Script Debugger. This debugger uses Chrome DevTools directly within NetSuite so you can debug scripts much like you would in Google Chrome.

You can't debug client scripts with the SuiteScript Debugger, but you can use your browser's tools. To debug client scripts, use the Chrome DevTools for Chrome and the Firebug debugger for Firefox. For more information, see your browser's documentation.

Note: Currently, only one script can be debugged at a time in a given debug session, regardless of the version of the script.

The following table shows which server script types are supported in each debugger. On-demand debugging is also supported. Client scripts are debugged on the client browser.

Script Type	Script Debugger (SuiteScript 1.0)	Script Debugger (SuiteScript 2.0)	2.1 Script Debugger (SuiteScript 2.1)
Bundle Installation	#	#	—
Map/Reduce	—	—	—
Mass Update	#	#	—
Portlet	#	#	—
RESTlet	#	#	—
Scheduled	#	#	#
SDF Installation	—	This script type runs on the client browser.	—
Suitelet	#	#	#
User Event	#	#	#
Workflow Action	#	#	—
Custom Plug-in	#	#	—

The following table shows which custom plug-ins are supported in each debugger. Support for SuiteScript 2.1 scripts will increase in future releases.

Core Plug-in	Script Debugger (SuiteScript 1.0)	Script Debugger (SuiteScript 2.0)	2.1 Script Debugger (SuiteScript 2.1)
Platform Extension	—	#	Not available
GL Plugin	#	—	
Payment Gateway	#	—	
Consolidated Rate Adjustor	#	—	
Promotions	#	—	
Tax Calculation	#	—	
Shipping Partners	#	#	
Email Capture	#	—	
Advanced Rev Rec	#	#	
Bank Connectivity	#	—	
Test Plugin	#	#	

For more information about using each debugger, see the following help topics:

- Debugging Overview
- Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts
- Debugging SuiteScript 2.1 Scripts

Debugging Overview

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

This overview section introduces:

- Debugging Modes
- The Debugger Domain
- Terms Related to Debugging

Note: To use the Script Debugger and the 2.1 Script Debugger, you must be using a role with SuiteScript permission (Full level).

Debugging Modes

The Script Debugger and the 2.1 Script Debugger both provide two debugging modes:

- On-demand debugging of scripts and code samples: With on-demand debugging, you can debug a new script or a code sample that doesn't have a defined Script Deployment. Scripts that don't require any form/record-specific interaction are good candidates for on-demand debugging. For more information, see On-Demand Debugging of SuiteScript 1.0 and SuiteScript 2.0 Scripts and On-Demand Debugging of SuiteScript 2.1 Scripts.
- Debugging deployed scripts: With deployed debugging, you can select an existing script or core plug-in implementation that has a defined Script Deployment or Plug-in Implementation record. To debug deployed scripts, you must be the script owner and the status of your script or core plug-in implementation must be set to Testing. For more information, see Debugging Deployed SuiteScript 1.0 and SuiteScript 2.0 Server Scripts and Debugging Deployed SuiteScript 2.1 Server Scripts.

The Debugger Domain

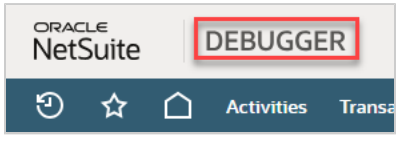
It's important to be aware that when you debug any version of SuiteScript scripts, you are operating in a separate Debugger domain.

Important: Any changes you make to your account when on this Debugger domain **will affect the data in your production account**. For example, if you execute a script in the Debugger that creates a new record, that record will appear in your production account.

Note: You may experience a decrease in performance when working on Debugger domains.

To access the debugger, go to Customization > Scripting > Script Debugger. The Debugger domain is available from a production account or a release preview account. If you're already logged in to your NetSuite account, you won't need to enter you account information again to access the Debugger domain.

When you are logged in to the Debugger domain, you see the Debugger logo at the top of the page:



When you're in the Debugger domain, you can debug SuiteScript 1.0, SuiteScript 2.0, and SuiteScript 2.1 scripts if the following requirements are met:

- You must have scripting permission. See Setting Roles and Permissions for SuiteScript for more information.
- You must be the assigned owner of the script (as indicated on the script record).
- If you are debugging a script that has a defined script deployment, the script must be in Testing mode before it can be loaded into the debugger. If you want to debug a script that has been released into production, you must change the script's status from Released to Testing on the Script Deployment page. See Setting Script Deployment Status for more information about the script deployment status.

Client scripts can be debugged using tools available in your browser. Both form-level and record-level client scripts should be tested on the form/record they run against.

If a bundled script has been installed into your account and the script has been marked as hidden, you will not be able to debug this script.

Be aware that the Script Debugger and the 2.1 Script Debugger are not any of the following:

- API test consoles
- Integrated development environments (IDE)
- Script deployment interfaces. To deploy a script to NetSuite, you must still create a script record and define the script's deployment parameters on the Script Deployment page.
- Script runner interfaces (for example, scheduled scripts still need to be placed INQUEUE for task completion)

Terms Related to Debugging

You may find it useful to learn the following terms related to debugging SuiteScript scripts:

Term	Definition
Call Stack	A stack (most recent on top) of all the active functions (and their local variables) called up to the current line of execution.
JavaScript Debugging Pane	The pane, normally on the right side of the Chrome DevTools tab (opened for SuiteScript 2.1 debugging), that displays breakpoints, watches, variables, and the call stack, and includes execution control buttons.
Line Break Point	A user-selected line in source code where program execution is stopped or paused.
Minify/de-minify	Minified code has had white space and other characters removed from display, so the entire script appears as one long line. This can occur when using the 2.1 Script Debugger. Minified code can be de-minified from within the Debugger. See Using Chrome DevTools.
User Event Break Point	A user event script where program execution is to be paused. The user event must be invoked during script execution for the break point to function.
Watch	A variable or expression that is monitored throughout the program's execution in the current scope.

Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

You debug SuiteScript 1.0 and 2.0 scripts in the Script Debugger. The Script Debugger lets you to debug SuiteScript 1.0 and 2.0 server scripts and core plug-in implementations, and can be used for on-demand debugging.

Note: To use the Script Debugger, you must be using a role with SuiteScript permission (Full level).

The following help topics teach you how to use the Script Debugger:

- SuiteScript Debugger Overview
- Script Debugger Interface
- On-Demand Debugging of SuiteScript 1.0 and SuiteScript 2.0 Scripts
- Debugging Deployed SuiteScript 1.0 and SuiteScript 2.0 Server Scripts
- Tips for Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts

Script Debugger Interface

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The Script Debugger includes several buttons for controlling the script execution and several tabs to view data during script execution. The Script Debugger UI includes a script area, buttons, and tabs.

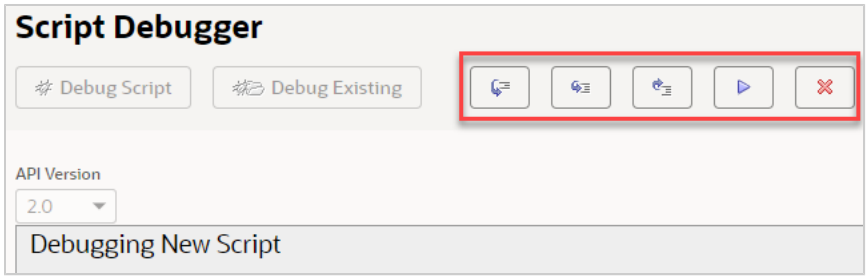
- Script Debugger Buttons
- Script Debugger Subtabs

For an example of how to use the Script Debugger buttons and tabs, see Example Use of the Script Debugger Buttons and Tabs.

Script Debugger Buttons

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The Script Debugger includes five buttons to let you to step through and run your script:



You can use these buttons to control script execution when the debugger stops at a particular line. Each button also has an associated keyboard shortcut.

- Step Over

  Resumes execution from the current line and stops at the next line (even if the current line is a function call). Keyboard shortcut is the space key.
- Step Into

  Resumes execution from the current line and stops at the first line in any function call made from the current line. Keyboard shortcut is the 'i' key.
- Step Out

Resumes execution from the current line until the end of the current function, and stops at the first line following the line from where this function was called, or until the next break point, or until the program terminates (either by error or by normal completion). Keyboard shortcut is the 'o' key.

- Continue

  Resumes program execution from the current line until the next break point -or- until the program terminates. Keyboard shortcut is the shift+space keys.

- Cancel

  Aborts execution of the program from the current line. Keyboard shortcut is the 'q' key.

Script Debugger Subtabs

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The Script Debugger page includes several subtabs to let you access information in your script as it runs:

- Execution Log Debugger Subtab
- Local Variables Debugger Subtab
- Watches Debugger Subtab
- Evaluate Expressions Debugger Subtab
- Break Points Debugger Subtab

Execution Log Debugger Subtab

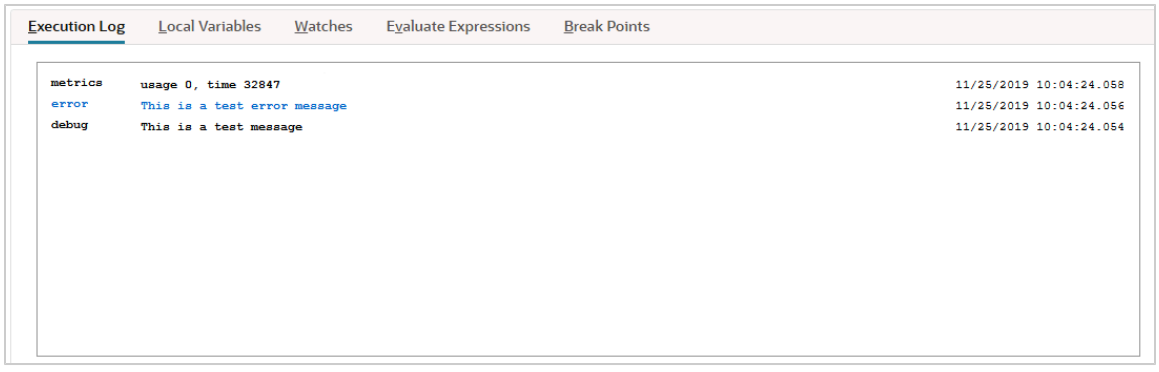
Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: The Execution Log subtab isn't used for SuiteScript 2.1 scripts. Logging of SuiteScript 2.1 scripts is done on the Chrome DevTools debugger console. For more information about debugging SuiteScript 2.1 scripts, see Chrome DevTools for SuiteScript 2.1 Script Debugging.

The Execution Log subtab shows all execution logs created by the currently executing program, including system errors. The execution log details that appear on this subtab are the same details that would normally appear in the Execution Log on the Script Deployment page. However, when you use the debugger, all script execution details appear on the Execution Log subtab on the Script Debugger page; these details will NOT appear on the Execution Log subtab of the Script Deployment page.

The type, subject, details, and timestamp are displayed on the Execution Log subtab. The timestamp is recorded on the server. It is converted to the current user's time zone for display.

Log details are collapsed by default, but you can see them by clicking the expand/collapse icon. Note that the subtab is automatically cleared at the start of each debugging session.



Local Variables Debugger Subtab

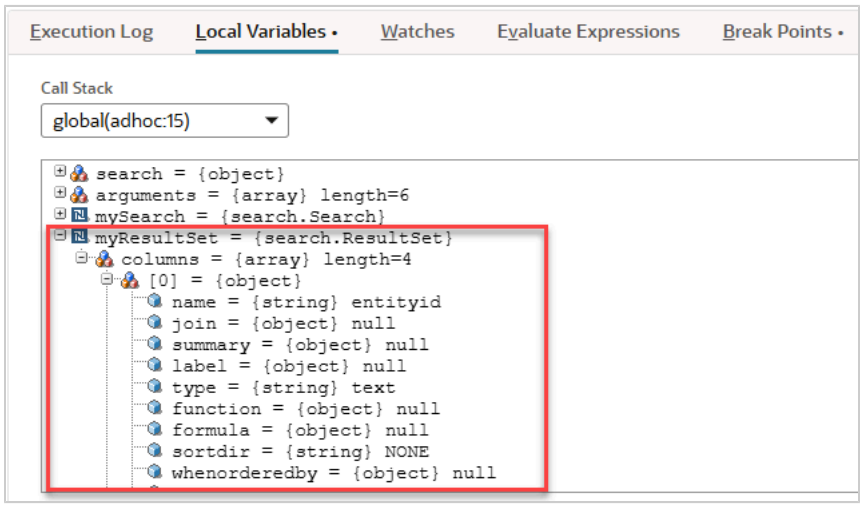
Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: The Local Variables subtab iisn't used for SuiteScript 2.1 scripts. Logging of SuiteScript 2.1 scripts is done on the Chrome DevTools debugger console. See Chrome DevTools for SuiteScript 2.1 Script Debugging for more information about debugging SuiteScript 2.1 scripts.

The Local Variables subtab shows a list of all local variables (primitives, objects, and NetSuite objects) currently in scope. Note that for NetSuite objects, all properties are private, even though they can be seen on the Local Variables subtab. Don't try to reference these properties directly in your script. Use the appropriate getter/setter functions instead.

The Local Variables subtab includes a call stack that shows the current execution stack of the script. The function call and current line number for that function are included in the list. Use the Call Stack list to switch between call stacks to view different local variables. In addition, watch expressions and expression evaluations are automatically performed in the context specified by this field.

Important: Due to performance considerations, the display limit for all variables is 1000. In addition, only the first 500 characters of a String are displayed. For large variables, use a watch to see the full member display (see Watches Debugger Subtab for additional information).



Watches Debugger Subtab

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: The Watches subtab isn't used for SuiteScript 2.1 scripts. Variable watches in SuiteScript 2.1 scripts are done on the Chrome DevTools debugger console. For more information about debugging SuiteScript 2.1 scripts, see Chrome DevTools for SuiteScript 2.1 Script Debugging.

The Watches subtab lets you add or remove variables and expressions to a list that's kept up-to-date ("watched") while your script executes.

The variables and expressions are always evaluated in the current call stack. This means that by default, they're evaluated at the current line of script execution. However, if you switch to a different function in the call stack, they're re-evaluated at that location.

- To add a variable or an expression, type it into the **Add Watch** field and press the Enter key.
- To remove a watched variable or expression, click on the red **x** icon to the left of the expression.
- To browse sub-properties of a user-defined object, an array, or a NetSuite object expression, click the expand/collapse icon next to the property name.

You can also use the Watches subtab to view object properties with a command-line interface. Any property that is viewable from the property browser can be added as a watch expression by referencing the property using dot (.) notation, even if the property is private in the script. For example, to watch the ID of a record object (referenced by a variable called *record*), type record.*id* in the **Add Watch** field.

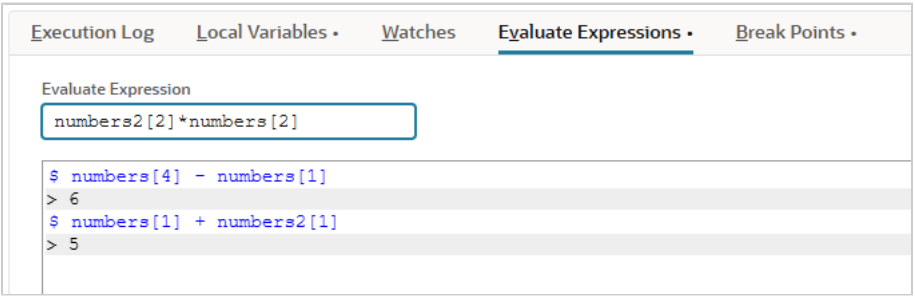
Evaluate Expressions Debugger Subtab

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: The Evaluate Expressions subtab isn't used for SuiteScript 2.1 scripts. SuiteScript 2.1 script logging is done on the Chrome DevTools debugger console. For more information about debugging SuiteScript 2.1 scripts, see Chrome DevTools for SuiteScript 2.1 Script Debugging.

Use the Evaluate Expressions subtab to run code at break points during the current program. This lets you access and even change the program's state.

Enter an expression in the Evaluate Expression field and hit the Enter key to evaluate the expression at the selected call stack. The results of an evaluated expression (if any) are displayed in the window below. Any change to the program's state is immediately reflected in the Local Variables and Watches subtabs.

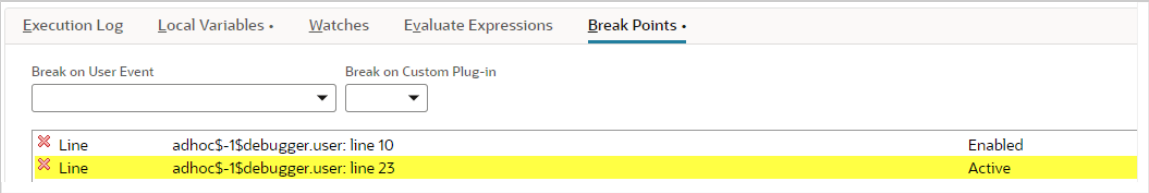


Break Points Debugger Subtab

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: The Break Points subtab isn't used for SuiteScript 2.1 scripts. SuiteScript 2.1 script logging is done on the Chrome DevTools debugger console. For more information about debugging SuiteScript 2.1 scripts, see Chrome DevTools for SuiteScript 2.1 Script Debugging.

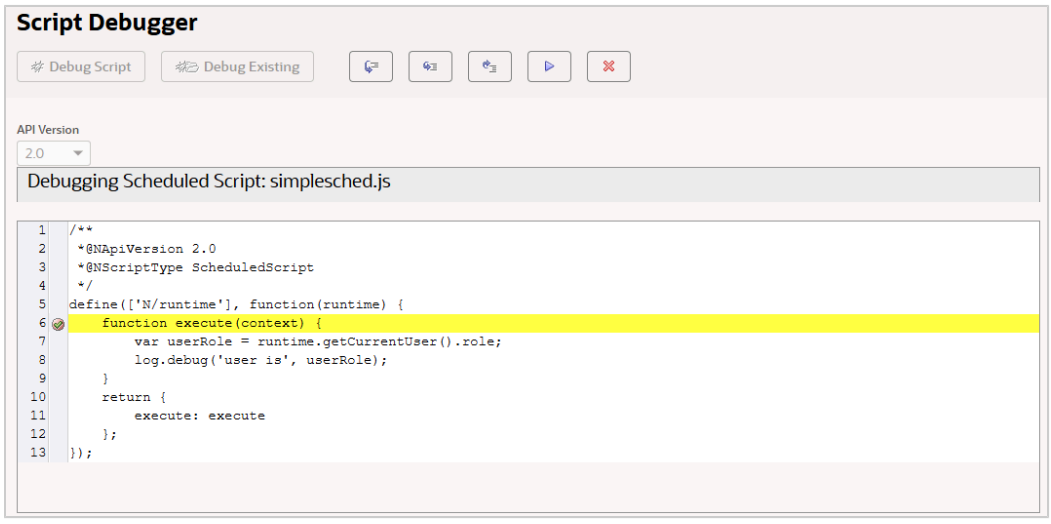
The Break Points subtab shows all your instruction-level (line) break points as well as your user event break points. You can add user event break points by selecting user events from the Break on User Event list.



You can set break points in your code using the Script Debugger code window. By setting breakpoints, you can run your code up to a certain point, and then halt the execution at the break point and examine the current state of the execution.

To add or remove break points in your code:

1. Click between the line number and the line of code to add a breakpoint:



2. To remove a break point, click the break point icon as it appears in the code. You can also remove a break point by clicking the red **x** icon next to the break point, as it appears on the **Break Points** tab.

   When you debug deployed scripts, you can set break points at each user event in your script. User events possibly invokable during script execution where the program halts execution.

Example Use of the Script Debugger Buttons and Tabs

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

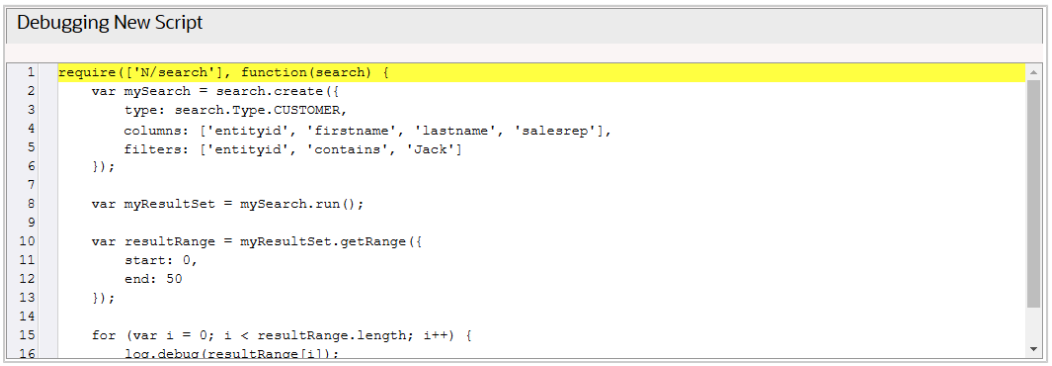
The following example shows how to debug a script using the execute, step over, and step into buttons. This example also shows how to use the Execution Log, Local Variables, and Watches tabs.

Note: The code in this example assumes you have at least two customers with an entityId that includes 'Jack'.

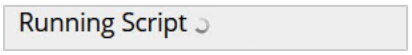
1. Go to Customization > Scripting > Script Debugger.
2. Enter the following code into the debugger window:

```
/** * @NApiVersion 2.x */ require (['N/search'], function (search) { var mySearch = search.create({ type: search.Type.CUS
TOMER, columns: ['entityid', 'firstname', 'lastname', 'salesrep'], filters: ['entityid', 'contains', 'Jack'] }); var
 myResultSet = mySearch.run(); var resultRange = myResultSet.getRange({ start: 0, end: 50 }); for (var i = 0; i < resul
tRange.length; i++) { log.debug(resultRange[i]); }
});
```

3. In the **API Version** dropdown list, select 2.0.
4. Click **Debug Script**.
5. Wait until the script is shown in the window with the first line highlighted. At this point, the script hasn't started to run yet.



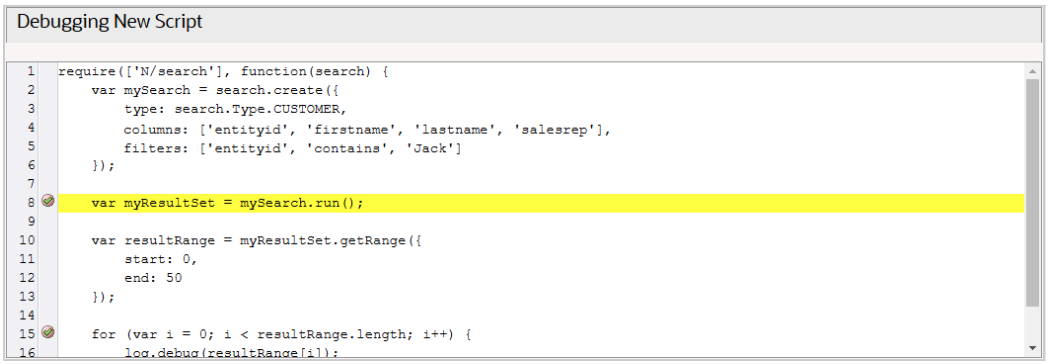
6. Set a breakpoint at the `var myResultSet = mySearch.run()` line by clicking between the line number (11) and the code line. When you add or remove a breakpoint, "Running Script" is briefly displayed:



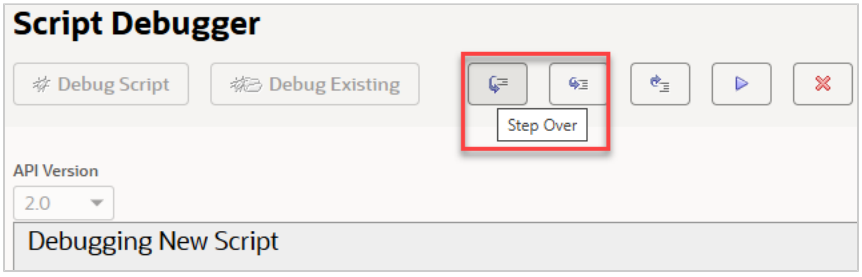
The script is not running, but the breakpoint is being inserted into the proper location.

7. Set another breakpoint at the `for` line (line 18). The two breakpoints are shown like this:

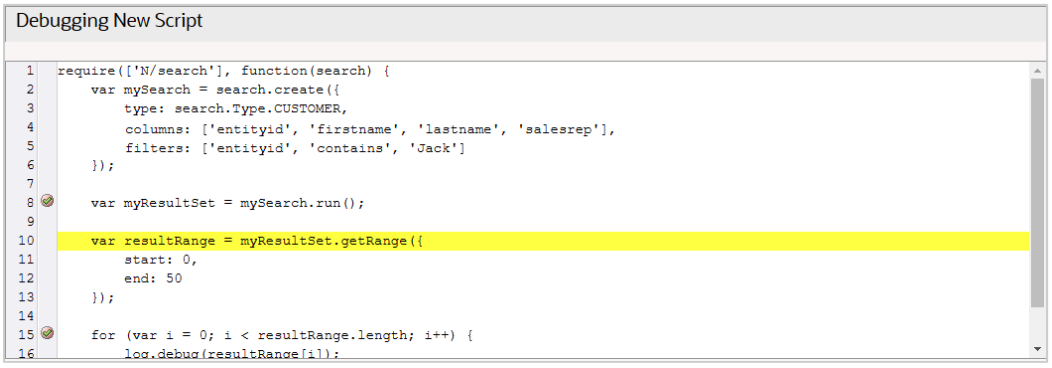
8. Click the Continue button to execute the script to the first breakpoint.
9. When the breakpoint is reached, the debugger pauses execution and highlights the line:



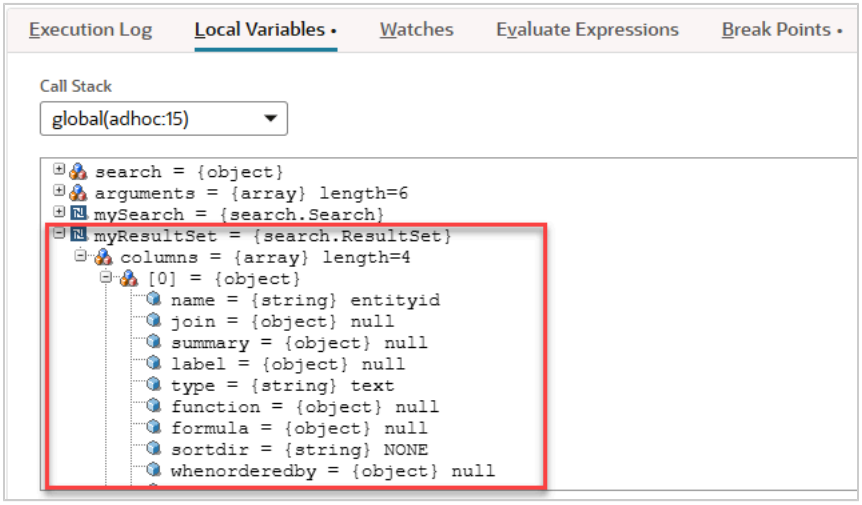
10. Click the Step Over button to execute the `mySearch.run()` line to get your search results:



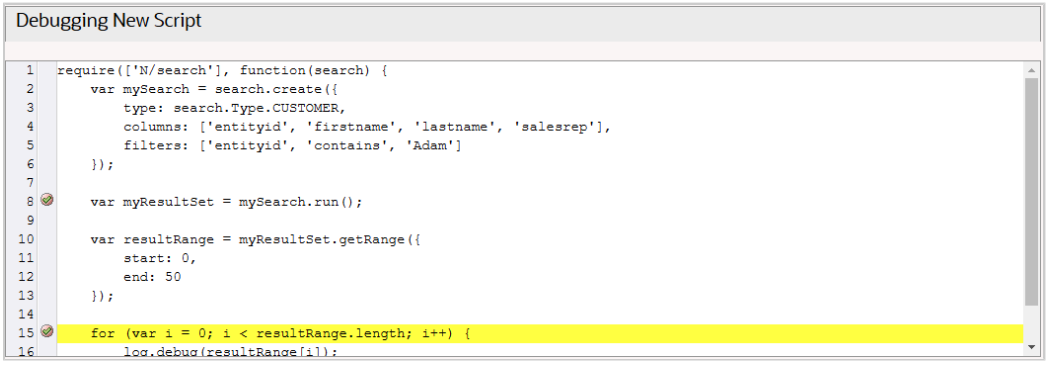
The debugger runs the search and pause at the next line of code:



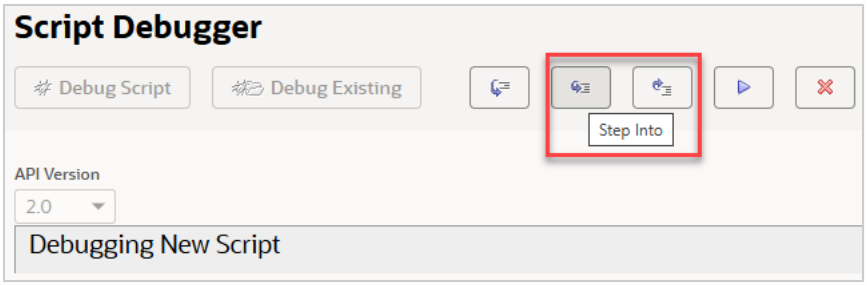
11. While the debugger is paused, click the Local Variables subtab to view the value of the `myResultSet` variable:



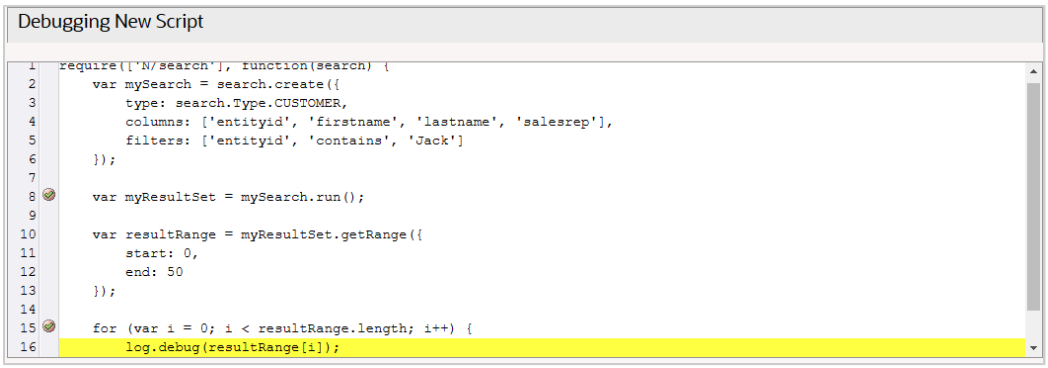
12. Click the Continue button to continue executing the script to the next breakpoint (line 15):



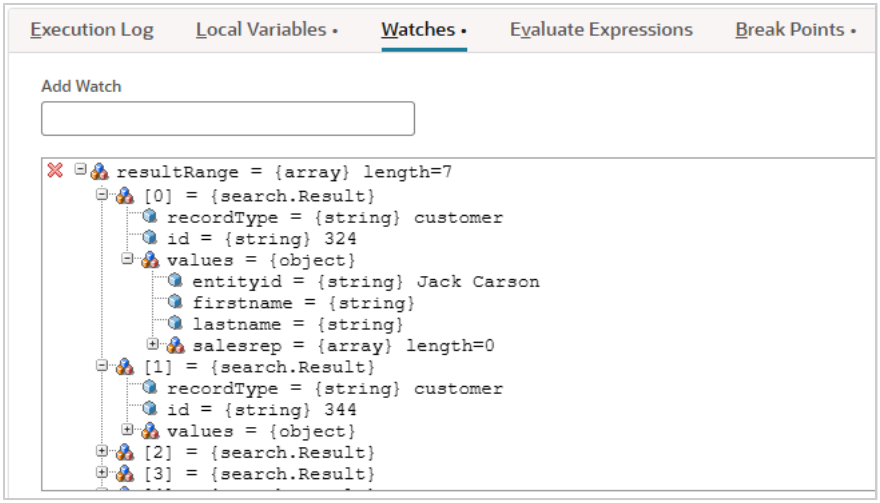
13. When the debugger pauses on line 15, click the Step Into button:



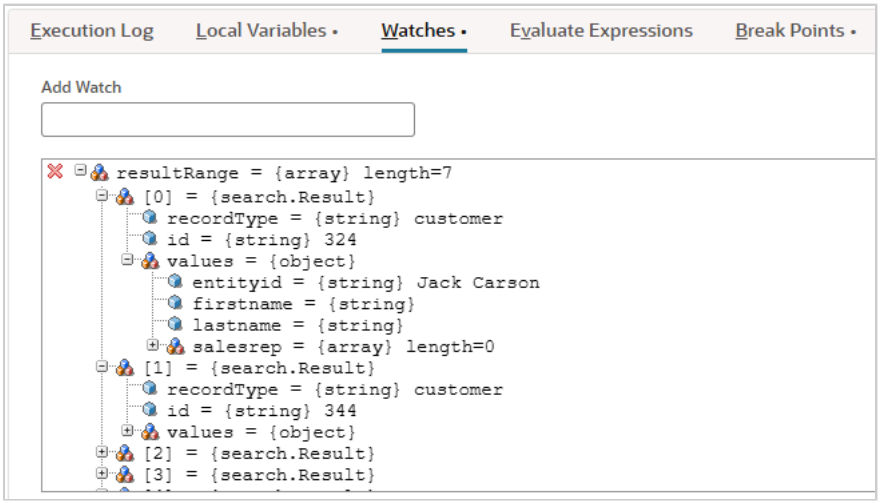
The debugger will execute the `for` line and pause at the first line in the `for` loop block of code, which is the `log.debug` statement:



14. While the debugger is paused, add a watch for the `resultRange` variable to watch the variable value update as the `for` loop. Enter resultRange into the Add Watch box on the Watches subtab and press the Enter key to see the value.



15. Click the Continue button. The debugger executes one iteration of the `for` loop and will pause again.
16. On the Watches tab, expand the `resultRange` variable to view its current value.



17. Click the second breakpoint to remove it. You must remove the breakpoint so the debugger does not pause at every iteration of the `for` loop. You can remove any breakpoint at anytime while the debugger is paused.
18. Click the Continue button to finish executing the script.

When the script completes execution, the Execution Log subtab shows all your log.debug statements:

On-Demand Debugging of SuiteScript 1.0 and SuiteScript 2.0 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

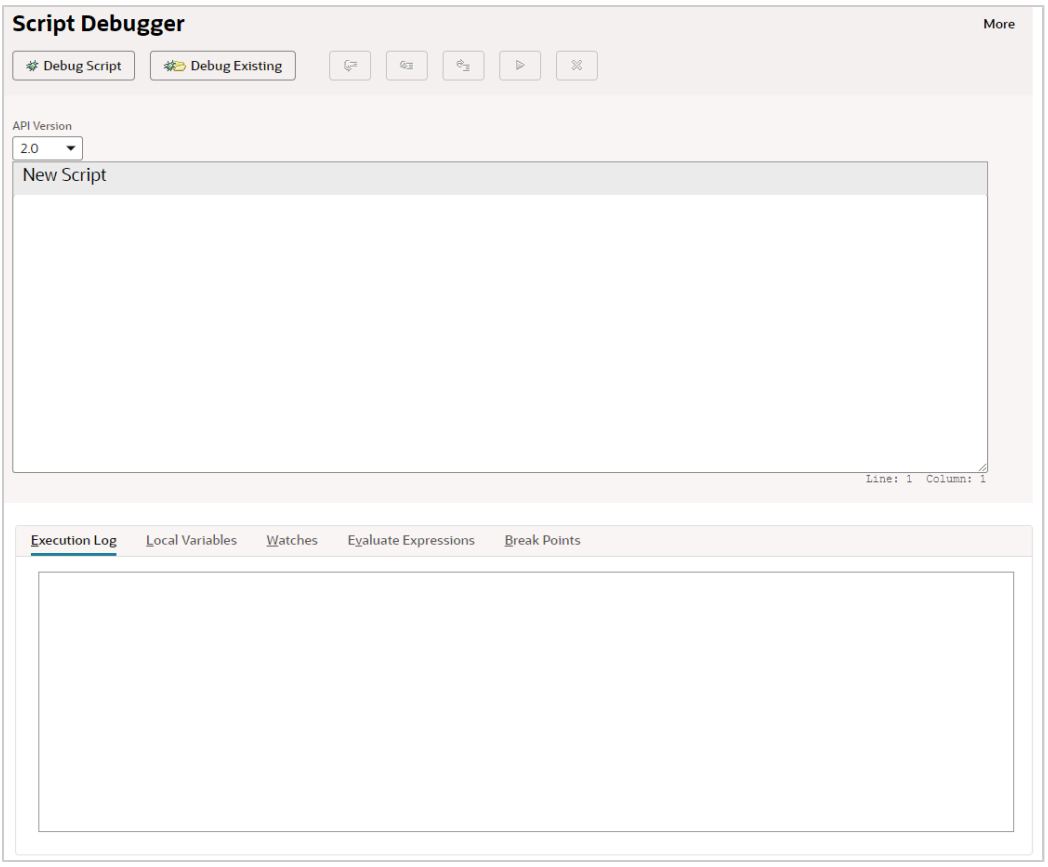
On-demand debugging is used for testing SuiteScript 1.0 or 2.0 scripts or code samples that don't have a defined script deployment. On-demand debugging is **not** used for scripts that have a Script record or defined deployment parameters set on the Script Deployment page. For information about debugging deployed SuiteScript 1.0 and 2.0 scripts, see Debugging Deployed SuiteScript 1.0 and SuiteScript 2.0 Server Scripts.

To use the Script Debugger in on-demand mode:

1. Go to Customization > Scripting > Script Debugger.

   Important: Any changes you make to your account when on this Debugger domain **will affect the data in your production account**. For example, if you execute a script in the Debugger that creates a new record, that record will appear in your production account.

   The following figure shows the Script Debugger page.

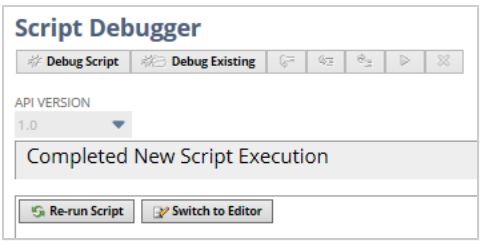


2. Select the **API Version** — 1.0 or 2.0. (If you want to debug a SuiteScript 2.1 script, see On-Demand Debugging of SuiteScript 2.1 Scripts).
3. Enter your code sample or script in the **New Script** text area. If your script is a 2.0 script and includes a `define` statement, you need to change that to a `require` statement to run the script in the debugger. For more information, see the help topic SuiteScript 2.x Global Objects.

   If you have written your code in an IDE, copy and paste the code into the **New Script** text area. If you modify your script in the debugger and you intend to save the changes, you must copy the updated script from the debugger and paste it into your IDE.
4. Click **Debug Script**.

   The script is immediately loaded into the debugger and the program's execution stops before running the first line of executable code.

   > **Important:** Make sure you always have an executable call in your script; otherwise, the script has nothing to execute and return.
5. With the script loaded into the debugger, you can step through each line to inspect local variables and object properties. You can also add watches, evaluate expressions, and set break points. See Script Debugger Interface for information about stepping into/out of functions, adding watches, setting and removing break points, and evaluating expressions.
6. When execution of your script is complete, a Completed New Script Execution message appears. You can either re-run the script (by clicking **Re-run Script**), or place the script into edit mode (by clicking **Switch to Editor**) and continue debugging.



Script execution details are logged on the Execution Log subtab of the Script Debugger. You can also use the N/log Module within your script to access methods for logging script execution details for all scripts. See Using the Script Execution Log Subtab to learn how SuiteScript 1.0 and 2.0 script execution details are logged.

Debugging Deployed SuiteScript 1.0 and SuiteScript 2.0 Server Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Deployed-mode debugging is for testing scripts or core plug-in implementations that have a defined Script Deployment or Core Plug-in Implementation record.

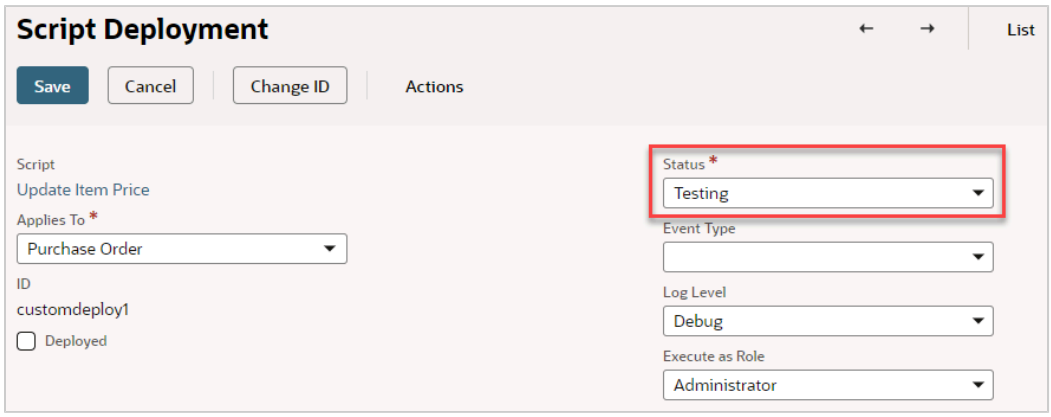
Note: SuiteScript does not support read-only sublists. If you are debugging a script that loads a record with a custom child record as a sublist, make sure the **Allow Child Record Editing** setting is checked for the child record in Customization. If this box is not checked, the sublist is read-only and will not load in the parent record. See the help topic Creating Custom Record Types for additional information about creating custom records.

To use the Script Debugger in deployed mode:

1. Go to Customization > Scripting > Script Debugger.

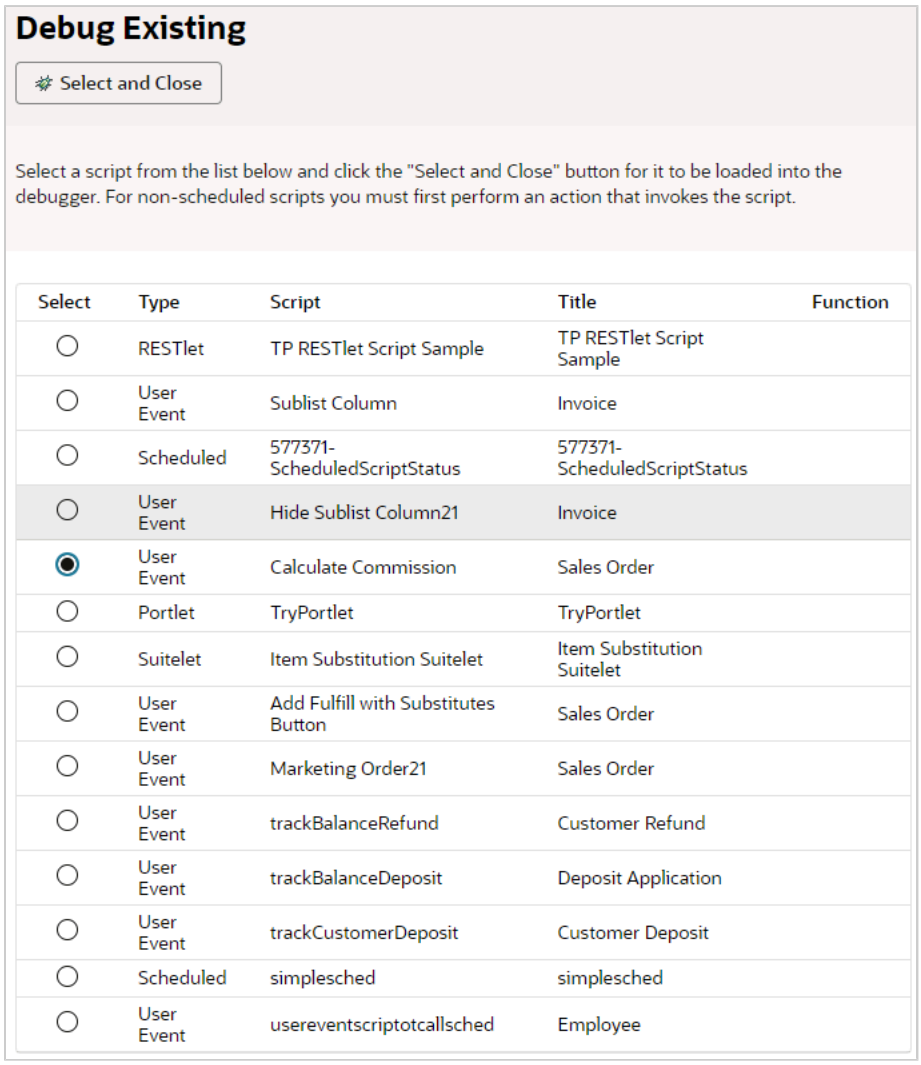
   Important: Any changes you make to your account when on this Debugger domain **will affect the data in your production account**. For example, if you execute a script in the Debugger that creates a new record, that record will appear in your production account.

2. When you are on the Debugger domain, go to the applicable Script Deployment or Core Plug-in Implementation record and verify that **Status** is set to Testing. If you want to debug a script or core plug-in implementation that has been released into production, you must change the status from **Released** to **Testing**.

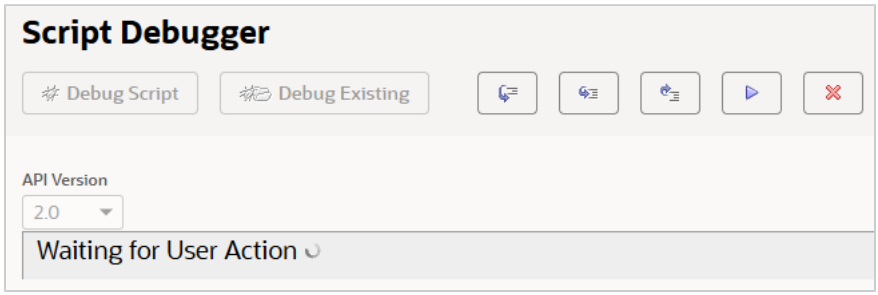


3. Verify you are the assigned owner of the script or core plug-in implementation. You can only debug a script or core plug-in implementation if you are the assigned owner.
4. To access the Script Debugger start page, use the appropriate option:
   - If you are on the Script Deployment page in View mode, click **Debug**.
   - If you are on the Script Deployment page in Edit mode, click **Save and Debug**.
   - If you are not on the Script Deployment page or you are debugging a core plug-in implementation, you can access the debugger by going to Customization > Scripting > Script Debugger.
5. On the Script Debugger page, select your **API Version** and then click **Debug Existing**. The list of scripts will be filtered based on the selected API version.

The Debug Existing popup window opens and shows all server scripts and core plug-in implementations available for debugging (see figure below) based on the selected API version. Note that only scripts and core plug-in implementations whose statuses are set to Testing are listed.



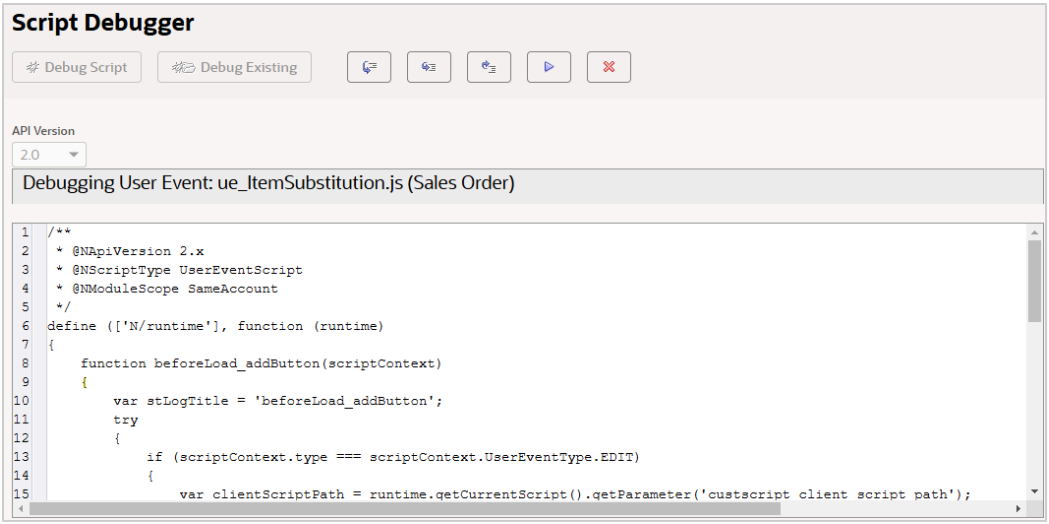
6. Select the script you want to debug and click **Select and Close**.
7. The Script Debugger page is updated with a Waiting for User Action status message. The debugger is waiting for you to perform the action that is associated with the selected script or plug-in implementation.



Note: Adding or updating records using CSV import may not trigger the debugging of scripts because the CSV import may create or update records using a different user role or entity. Remember, that you can only debug scripts if you are the assigned owner.

Note: You do not need to perform any user actions to load scheduled scripts into the Debugger. Simply select your scheduled script from the Script Debugger popup window, and then click Save and Close. The scheduled script automatically loads.

8. When the script is loaded into the debugger, it is shown on the Script Debugger page:



Note: The Debugger does not caption the user action during bulk processing, but the script is executed and a log is created.

Important: You should perform user actions in another window or another tab on your browser. This lets you keep the Script Debugger page open so that you can see the script or core plug-in implementation as it loads. To do this, right-click on the menu item (or other UI element) and select Open Link in New Tab or Open Link in New Window.

After the script is loaded into the Debugger, you can step through each line to inspect local variables and object properties. You can also add watches, evaluate expressions, and set break points. See Script Debugger Interface for details about stepping into/out of functions, adding watches, setting and removing break points, and evaluating expressions.

When a script finishes running in the Debugger, you can either re-run it (by clicking **Re-run Script**), or place it into edit mode (by clicking **Switch to Editor**) and continue debugging.

Important: If you modify any script or core plug-in implementation in the SuiteScript Debugger and you intend to save the changes, you must copy the updated code from the Script Debugger page and paste it into your IDE (or other editor). You must then re-load the updated file back into the NetSuite File Cabinet.

Script execution details are logged on the Execution Log subtab of the Script Debugger. You can also use the N/log Module in your script to write log messages during script execution. See Using the Script Execution Log Subtab to learn how SuiteScript 1.0 and 2.0 script execution details are logged.

Tips for Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The following tips may help you debug SuiteScript 1.0 and 2.0 scripts using the Script Debugger:

- Review the Execution Log subtab on the Script Debugger page to view all script messages and alerts, including those indicating any script errors.
- Execution metrics are shown on the Execution Log subtab of the Script Debugger page.
- Use breakpoints to pause script execution at predetermined lines of code. Breakpoints are managed on the Break Points subtab on the Script Debugger page. See Break Points Debugger Subtab for more information.
- Use the Watch subtab on the Script Debugger page to see values of variables within your script and to evaluate expressions instantly. See Watches Debugger Subtab for more information.
- Use the Evaluated Expressions subtab on the Script Debugger page to manually enter and evaluate expressions. See Evaluate Expressions Debugger Subtab for more information.
- To cancel a debugging session without finishing script execution, click the **X** button on the Script Debugger page.

Debugging SuiteScript 2.1 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

You can debug SuiteScript 2.1 scripts with the new 2.1 Script Debugger. This debugger uses Chrome DevTools directly from NetSuite to allow you to debug scripts similar to the way you debug JavaScript in the Google Chrome browser.

Note: To use 2.1 Script Debugger, you must be using a role with SuiteScript permission (Full level).

Note: The 2.1 Script Debugger is fully supported when using NetSuite in the Chrome browser. Other browsers, such as Mozilla Firefox and Apple Safari, have limited or no support. Functionality in those browsers is not guaranteed.

The following help topics teach you how to use the Script Debugger:

- 2.1 Script Debugger Overview
- Chrome DevTools for SuiteScript 2.1 Script Debugging
- Using Chrome DevTools
- On-Demand Debugging of SuiteScript 2.1 Scripts
- Debugging Deployed SuiteScript 2.1 Server Scripts
- Debugging Deployed SuiteScript 2.1 Scheduled Scripts and Suitelets
- Debugging Deployed SuiteScript 2.1 User Event Scripts

2.1 Script Debugger Overview

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Script debugging is an important part of the holistic developer experience. Debugging with Chrome DevTools is the most common way to debug JavaScript. To give NetSuite developers a modern coding experience, we built the 2.1 Script Debugger with Chrome DevTools so you can debug scripts like you would with JavaScript in Google Chrome.

You access the 2.1 Script Debugger the same way as the legacy Script Debugger for SuiteScript 1.0 and 2.0 scripts. To use the 2.1 Script Debugger, select 2.1 as the API Version on the Script Debugger page. When you click Debug Script, a new browser tab opens with the Chrome DevTools interface for debugging your script. Your script will be displayed on the new tab and you can pause/resume execution, set breakpoints, and step through your code line by line.

The 2.1 Script Debugger is available for all SuiteScript developers.

Note: Only SuiteScript 2.1 scripts can be debugged using the 2.1 Script Debugger. You can't use the 2.1 Script Debugger to debug SuiteScript 1.0 or 2.0 scripts.

The following SuiteScript 2.1 server script types can be debugged:

- Scheduled scripts
- Suitelets
- User event scripts

NetSuite plans to support debugging SuiteScript 2.1 versions of other script types and core plug-in implementations in a future release. And debugging "Hidden in SuiteBundle" files is not currently supported. See the help topic Add the Bundle Installation Script File to the File Cabinet for information about the Hide in SuiteBundle option for Bundle Installation scripts.

Note: Only one script can be debugged at a time in a given debug session, regardless of the version of the script. You cannot debug a SuiteScript 2.0 script at the same time as a SuiteScript 2.1 script.

The following table lists the Chrome DevTools debugging features and shows whether each is supported in the 2.1 Script Debugger. See Using Chrome DevTools for instructions on each capability.

Most common debugging capabilities in Chrome DevTools for JavaScript code	Supported in 2.1 Script Debugger
Pause your code with breakpoints	#
Find unused script code	# (see Chrome DevTools documentation)
Map preprocesses code to source	—
Change thread context	—

Most common debugging capabilities in Chrome DevTools for JavaScript code	Supported in 2.1 Script Debugger
Step through code: ■ step over a line of code ■ step into/out of a line of code ■ run all code up to a certain selected line ■ use breakpoints ■ restart at the top function of the call stack ■ pause and resume execution	#
Set and remove breakpoints	#
Edit code	—
View and edit local, closure, and global properties	#
View current call stack	#
Copy stack trace	# (see Chrome DevTools documentation)
Ignore a script or pattern of scripts	# (see Chrome DevTools documentation)
Blackbox a script from the Editor pane, the Call Stack pane, the Settings pane	# (see Chrome DevTools documentation)
Display the console	#
Use the Memory tab	—
Use the Profiler tab	—
Searching	#
Code coverage	#
Hover over code components shows info and values	#
De-minify the code	#
Set up a workspace	—
Keyboard shortcuts	# See https://developers.google.com/web/tools/chrome-devtools/shortcuts

Chrome DevTools for SuiteScript 2.1 Script Debugging

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The 2.1 Script Debugger offers a subset of the full Chrome DevTools features found in Chrome.

Note: Only SuiteScript 2.1 scripts can be debugged using the 2.1 Script Debugger. You cannot use the 2.1 Script Debugger to debug SuiteScript 1.0 or 2.0 scripts.

Note: Some of the following information, provided as a summary of the Chrome DevTools as it can be used in debugging JavaScript, is from Chrome DevTools. Refer to that site for complete information.

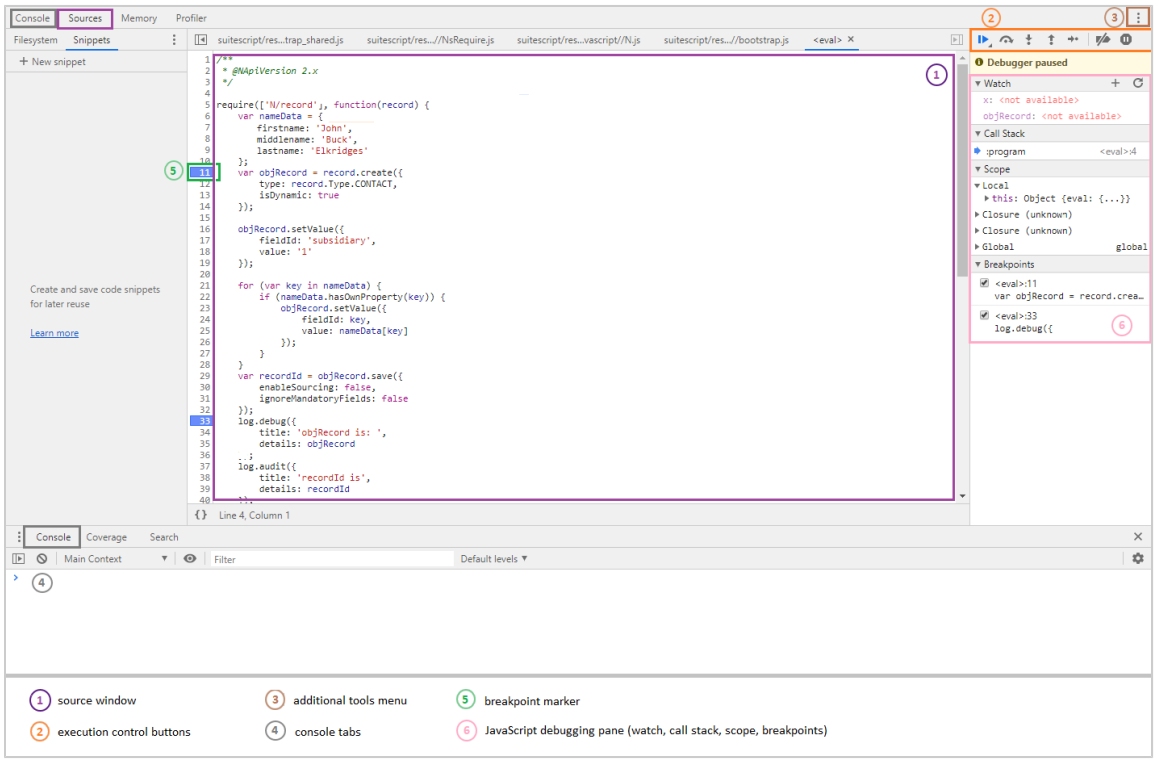
Also refer to Debug JavaScript and JavaScript Reference for additional help specific to debugging JavaScript code.

Chrome DevTools is a set of web developer tools built directly into the Google Chrome browser. Chrome DevTools can help you edit pages instantly and diagnose problems quickly. The JavaScript section of the Chrome DevTools site (JavaScript) has tutorials and videos showing you how to debug your JavaScript code. Most of these are included in the 2.1 Script Debugger. Se 2.1 Script Debugger Overview for a list of capabilities provided in the 2.1 Script Debugger.

Tip: Use Ctrl+Shift+I to access Chrome DevTools from the Chrome browser. Note that you do not have to do this when debugging SuiteScript2.1 scripts because NetSuite automatically provides the Chrome DevTools capability in a separate browser tab.

The following figure shows a sample Chrome DevTools Sources tab opened for debugging SuiteScript 2.1 scripts.

Note: Chrome DevTools for SuiteScript 2.1 scripts supports the Console and Sources tabs only. Currently, there is no support for the Memory or Profiler tabs when debugging SuiteScript 2.1 scripts.



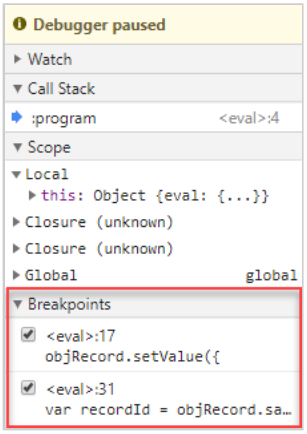
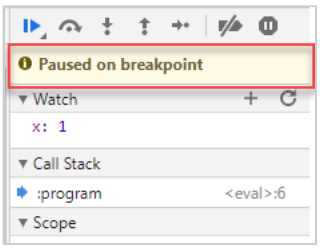
Note: Use of Chrome DevTools for NetSuite SuiteScript 2.1 script debugging adheres to all license parameters as described at Chrome DevTools License.

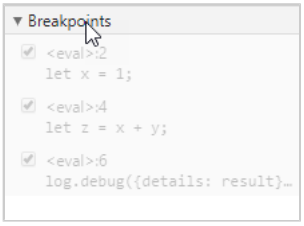
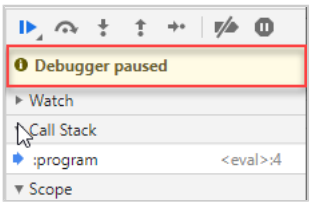
To use Chrome DevTools, see Using Chrome DevTools.

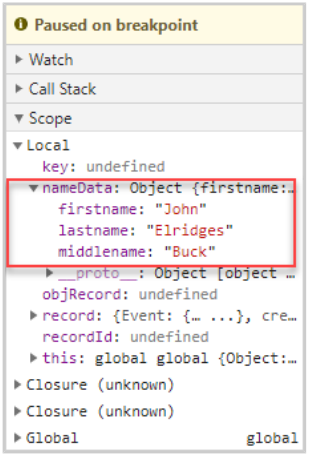
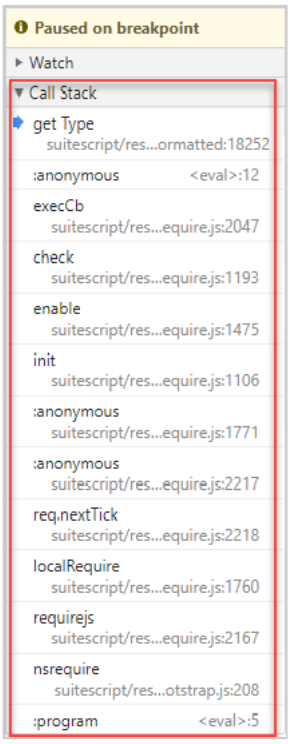
Using Chrome DevTools

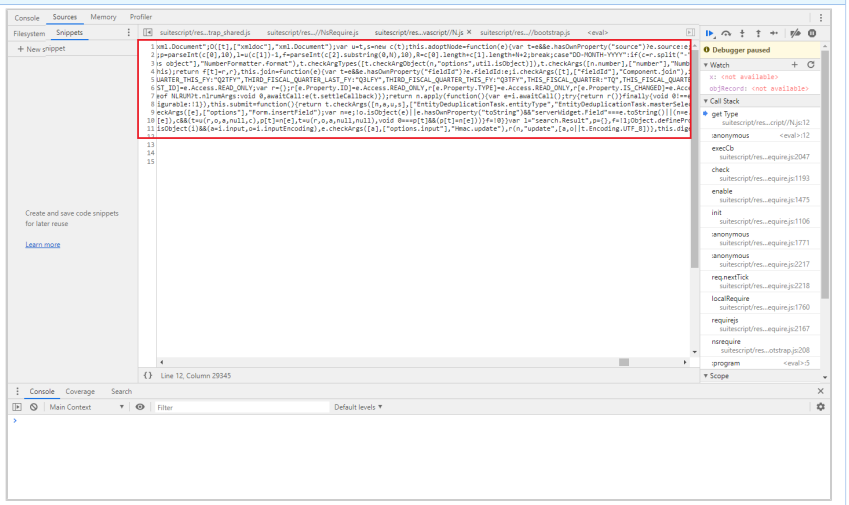
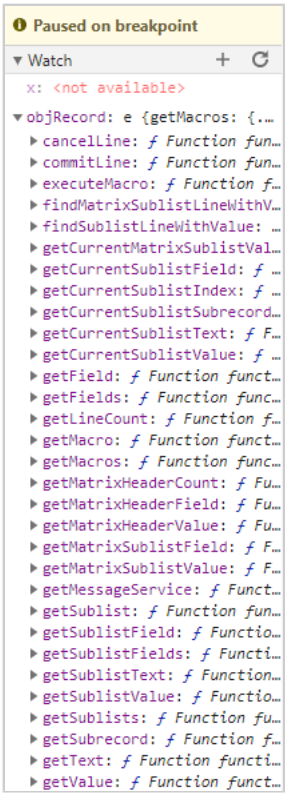
Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

When you start debugging SuiteScript 2.1 scripts using Chrome DevTools, you can set breakpoints, add watches, view the call stack, and start, pause, and resume script execution. Most of these tasks are performed in the 2.1 Script Debugger in the same way they are performed in Chrome DevTools. The following table provides quick instructions for different debug actions.

Debug action	Steps
Set and remove breakpoints	Breakpoints are shown in the Breakpoints section of the JavaScript debugging pane:  To set a breakpoint, click to the left of the line of code in the Source window where you want the break to be set. A breakpoint marker is displayed: When execution is paused due to a breakpoint, a Paused on breakpoint message is displayed:  To remove a breakpoint, click the breakpoint marker. **Note:** Breakpoints may "stick" between debugging sessions. If this happens, simply remove the breakpoints that you no longer need.
Disable/re-enable breakpoints	To disable all breakpoints, click the disable breakpoints button: . A disabled breakpoint marker appears grayed-out: 2

Debug action	Steps
	To re-enable all breakpoints, click the re-enable breakpoints button: . Breakpoints must first be set to disable/re-enable them. When all breakpoints are disabled, the breakpoints list is grayed out: 
Step through code	To step over a line of code, click the step over button: . The code pointer moves to the next line. To step into a line of code, click the step into button: . The code pointer moves into the function. Note that you cannot step into a SuiteScript module. Clicking the step into button when the code pointer is at a SuiteScript module will move the code pointer to the next line, effectively skipping over the SuiteScript module. To step out of a line of code, click the step out button: . The code pointer moves out of the function to the line after the function call.
Pause and resume execution	To pause execution, click the pause button: . When script debugging is paused, you will see the Debugger paused message:  To resume execution, click the resume execution button:
Restart execution from a line within the call stack	To restart execution from a point in the call stack: 1. Expand the Call Stack section of the JavaScript Debugging pane. 2. Click the line of code in the call stack where you want to resume execution. 3. Click the resume execution button:
View and edit local, closure, and global properties	SuiteScript and JavaScript objects can be viewed by expanding the Local, Closure, or Global sections of the JavaScript debugging pane:

Debug action	Steps
	
View current call stack	To view the call stack, click the Call Stack header on the JavaScript debugging pane:  You can also select a point in the call stack and resume execution from that point.
Display the console	You can display the console in the main debugger window or on the bottom pane of the debugger by clicking the corresponding Console tab. However, if you use the Console tab on the left side of the debugger, the source window closes.
De-minify display of the script code	Source code may be displayed minified when it is displayed on the Sources tab:

Debug action	Steps
	 This is often unreadable. You can de-minify the source (that is, add appropriate white space and new lines) by using the **{}** button located at the bottom of the source window.
Watch variable and expression values	Use the Watch window on the JavaScript debugging pane to see values of variables within your script and to evaluate expressions on-the-fly:  To add a variable or enter an expression to evaluate, click the plus (+) button on the Watch window header and enter the variable name or expression. Notice in the above example, the variable being watched is a SuiteScript object so all available members of that object are displayed.

Chrome DevTools supports keyboard shortcuts for nearly all actions available in the debugger. See Chrome DevTools Keyboard Shorcuts.

Your Chrome DevTools environment is retained between debugging session. For instance, if you open the Console window the first time you debug a script, Chrome DevTools will open with the Console window open in each subsequent debugging session.

On-Demand Debugging of SuiteScript 2.1 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

On-demand debugging is for testing SuiteScript 2.1 scripts or code snippets that do not have a defined script deployment. On-demand debugging is **not** for scripts that have a Script record or defined deployment parameters (set on the Script Deployment page). See Debugging Deployed SuiteScript 2.1 Server Scripts for information about debugging deployed SuiteScript 2.1 scripts.

To debug SuiteScript 2.1 scripts on demand:

1. Go to Customization > Scripting > Script Debugger.
2. Select 2.1 in the API Version field.
3. Enter your script code into the Debugger window. You may need to include a 'debugger;' statement for the script execution to pause in Chrome DevTools and allow you to gain control of the script execution. Note that if you use `@NApiVersion` JSDoc tag in your script code, it is essentially ignored. Regardless of the value of the `@NApiVersion` JSDoc tag, your script code will be debugged as a 2.1 script.
4. Click **Debug Script**. A new browser tab opens with the Chrome DevTools debugger and "Waiting for devtools to attach" is displayed on the Script Debugger page above the script. You might also notice that the font of the script in the Debugger window changes slightly, and that the legacy debugger buttons are grayed-out, with the exception of the red X button (that can be used to cancel debugging a SuiteScript 2.1 script). This is an added indication that the Chrome DevTools tab is being activated.

   Note: The first time you use the 2.1 Script Debugger, you may get notification that the popup window was blocked. To use the 2.1 Script Debugger, you must allow this popup window. To do this, click the button on the right side and select to always allow popup windows. You only need to do this one time. All future access to the 2.1 Script Debugger will be immediate.

   Note: When the Chrome DevTools tab opens, it may not show your code. Click the resume script execution button. to access your code. If your code is still not displayed in the Source window, click the :program line under the Call Stack section on the JavaScript debugging pane.
5. See Using Chrome DevTools for information about debug actions you can perform. You can use the console to view all log messages. If your code is displayed minified, use the de-minify code button. `{}`) to display your code in a readable manner.

   See Tips for Debugging SuiteScript 2.1 Scripts for additional help when using the Debugger in on demand mode for SuiteScript 2.1 scripts.
6. When execution of your code is complete, "Completed New Script Execution" is displayed on the Script Debugger page above the script. You can close the Chrome DevTools tab at any time after script execution is complete.

Note: When script execution is complete or when an error occurs during script execution, you will see the message: Debugging connection was closed. Reason: Websocket disconnected. This is a normal message. You can close the browser tab and return to the Script Debugger where you can re-run your script or enter a new script for debugging.

Debugging Deployed SuiteScript 2.1 Server Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Deployed-mode debugging is for testing scripts that have a defined Script Deployment record. Note that plug-in implementations are not supported.

Note: Only SuiteScript 2.1 scripts can be debugged using the 2.1 Script Debugger. You cannot use the 2.1 Script Debugger to debug SuiteScript 1.0 or 2.0 scripts.

You can debug SuiteScript 2.1 scheduled scripts, Suitelets, and user event scripts using the 2.1 Debugger. See the following help topics for more information:

- Debugging Deployed SuiteScript 2.1 Scheduled Scripts and Suitelets
- Debugging Deployed SuiteScript 2.1 User Event Scripts

Debugging Deployed SuiteScript 2.1 Scheduled Scripts and Suitelets

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

You can debug SuiteScript 2.1 scheduled scripts and Suitelets that are in File Cabinet and have a Script Deployment record.

Note: SuiteScript does not support read-only sublists. If you are debugging a script that loads a record with a custom child record as a sublist, make sure the **Allow Child Record Editing** setting is checked for the child record in Customization. If this box is not checked, the sublist is read-only and will not load in the parent record. See the help topic Creating Custom Record Types for additional information about creating custom records.

To debug SuiteScript 2.1 scheduled scripts and Suitelets:

1. Create your SuiteScript 2.1 scheduled script or Suitelet, upload it to the File Cabinet, and create a Script Deployment record for it.
2. Go to Customization > Scripting > Script Debugger.
3. Select 2.1 in the API Version field.
4. Click **Debug Existing Script**. The Debug Existing popup window opens showing all deployed SuiteScript 2.1 scripts.
5. Select your desired deployed script and click **Select and Close**.

6. The Script Debugger page updates displaying "Waiting for User Action" message above the script. This means that it is waiting for you to perform the action associated with how the script is deployed.
7. When you perform the necessary action, the "Waiting for devtools to attach" message is displayed above the script on the Script Debugger page and a new browser tab opens for Chrome DevTools.

   Note: The first time you use the 2.1 Script Debugger, you may get notification that the popup window was blocked. To use the 2.1 Script Debugger, you must allow this popup window. To do this, click the button on the right side and select to always allow popup windows. You only need to do this one time. All future access to the 2.1 Script Debugger will be immediate.
8. Your deployed script code is displayed on the new tab where you can debug as desired. A "Debugging <script>.js" message is displayed on the Script Debugger page indicating that the Chrome DevTools tab is ready for use.
9. The Debugger will automatically stop at the first line of your script, which is normally the `define` or `require` statement. You can now debug your code as desired. See Using Chrome DevTools for information about debug actions you can perform. Use the console to view all log messages. See Tips for Debugging SuiteScript 2.1 Scripts for additional help when debugging scripts.
10. When execution of your code is complete, "Completed Deployed Script Execution" is displayed on the Script Debugger page above the script. You can close the Chrome DevTools tab at any time after script execution is complete.

Note: When script execution is complete or when an error occurs during script execution, you will see the message: Debugging connection was closed. Reason: Websocket disconnected. This is a normal message. You can close the browser tab and return to the Script Debugger where you can re-run your script or enter a new script for debugging.

Debugging Deployed SuiteScript 2.1 User Event Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

SuiteScript 2.1 user event scripts can be debugged using the 2.1 Debugger similar to other supported server script types (scheduled scripts and Suitelets) as described in Debugging Deployed SuiteScript 2.1 Scheduled Scripts and Suitelets. However, because a user event script can have multiple entry points, there are some special steps to take when debugging SuiteScript 2.1 user event scripts.

You can debug every entry point included in your SuiteScript 2.1 user event script within the same browser session, however, you must reattach to Chrome DevTools for each entry point included in the script.

Note: SuiteScript does not support read-only sublists. If you are debugging a script that loads a record with a custom child record as a sublist, make sure the **Allow Child Record Editing** setting is checked for the child record in Customization. If this box is not checked, the sublist is read-only and will not load in the parent record. See the help topic Creating Custom Record Types for additional information about creating custom records.

See the following topics for an example of debugging a user event script deployed on the Customer record.

- Creating the User Event Script and Starting the Debugger Example
- Debugging the beforeLoad Entry Point Example
- Debugging the beforeSubmit and afterSubmit Entry Points Example

Creating the User Event Script and Starting the Debugger Example

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

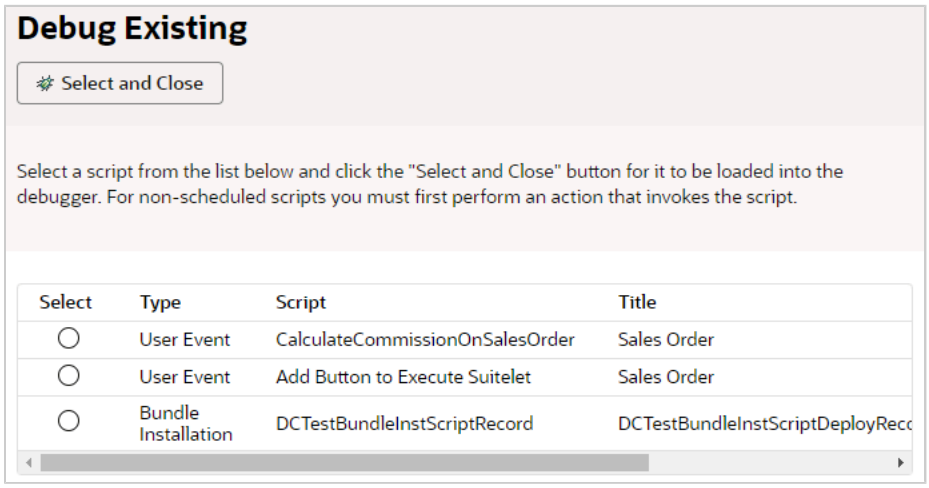
First, you create your user event script file and upload it.

To create the user event script and start the debugger:

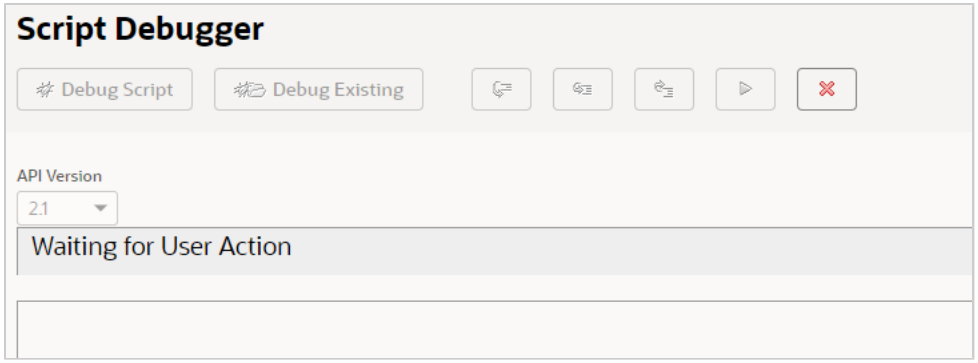
1. Create your user event script file and upload it to the File Cabinet. An example script is shown here:

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 */
define (['N/record'], function (record) {
    function beforeLoad(context) {
        if (context.type !== context.UserEventType.CREATE)
            return;

        var customerRecord = context.newRecord;

        customerRecord.setValue('phone', '555-555-5555');

        if (!customerRecord.getValue('salesrep')) {
            customerRecord.setValue('salesrep', 59); // replace 59 with a value specific to your account
        }
    }

    function beforeSubmit(context) {
        if (context.type !== context.UserEventType.CREATE)
            return;

        var customerRecord = context.newRecord;

        customerRecord.setValue('comments', 'Please follow up with this customer!');
    }

    function afterSubmit(context) {
        if (context.type !== context.UserEventType.CREATE)
            return;

        var customerRecord = context.newRecord,

        if (customerRecord.getValue('salesrep')) {
            var call = record.create({
                type: record.Type.PHONE_CALL,
                isDynamic: true
            });

            call.setValue('title', 'Make follow-up call to new customer');
            call.setValue('assigned', customerRecord.getValue('salesrep'));
            call.setValue('phone', customerRecord.getValue('phone'));

            try {
                var callId = call.save();
                log.debug('Call record created successfully', 'Id: ' + callId);
            } catch (e) {
                log.error(e.name);
            }
        }
    }

    return {
        beforeLoad: beforeLoad,
        beforeSubmit: beforeSubmit,
        afterSubmit: afterSubmit
    }
});
```

2. Create a script record for your script. For this example, set the Name field to Debug Customer w/ 2.1 UE and set the Applies To: field to Customer.
3. To have the script appear on the debugger tab, it must be executed by creating and saving a new Customer record (List > Relationships > Customers > New).
4. Go to Customization > Scripting > Script Debugger.
5. Select Debug Existing and select the Debug Customer w/ 2.1 UE user event script.



6. Click **Select and Close**. The Script Debugger waits for a user action to proceed:



7. Follow specific instructions for debugging each entry point.

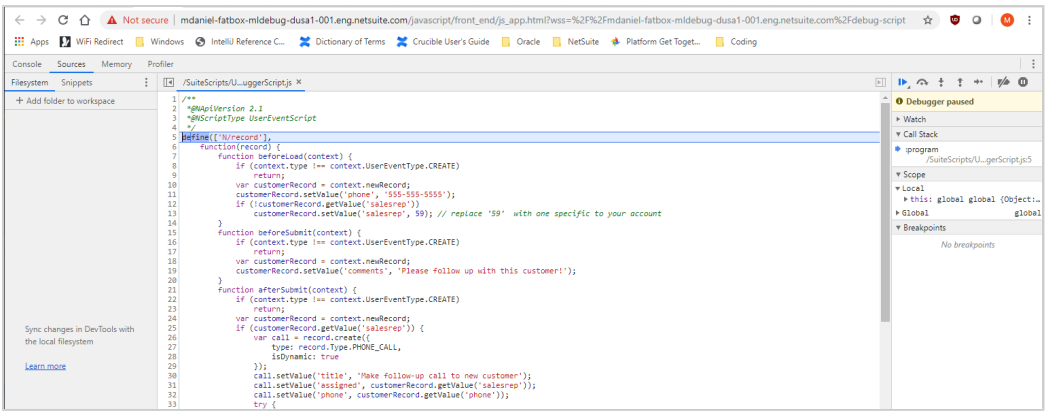
Debugging the beforeLoad Entry Point Example

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

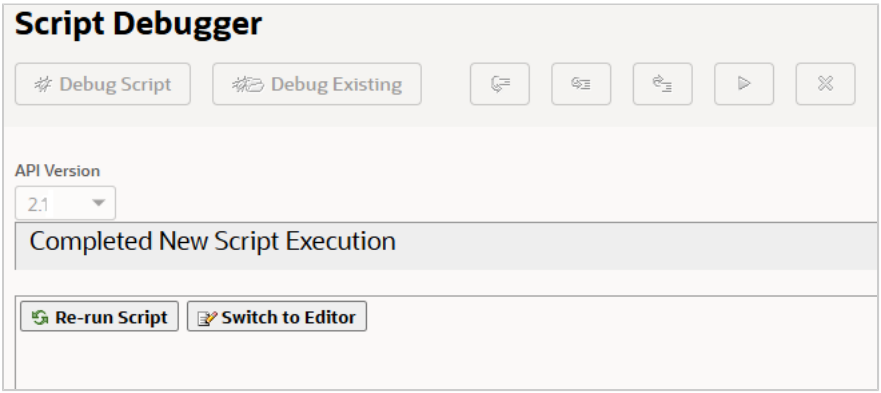
After you create the user event script and start the debugger, you debug the beforeLoad entry point. For more information, see Creating the User Event Script and Starting the Debugger Example.

To debug the beforeLoad entry point:

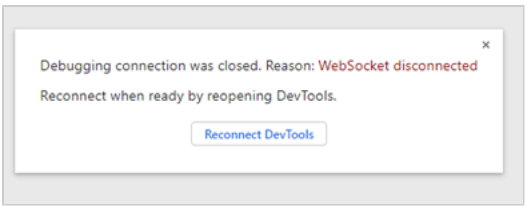
1. Follow the steps above to create the user event script and start the debugger.
2. To trigger the beforeLoad entry point, either create a new Customer or edit an existing Customer.
3. When the customer record is loaded, a new Chrome DevTools tab opens in your browser showing your user event script ready to be debugged using the 2.1 Debugger:



4. The execution of your script stops at the top of the script. You can now add breakpoints, watches, and so on, and begin debugging your script. You can also add a `debugger;` statement at the top of the beforeLoad entry point code to specifically stop the execution of the script at that entry point.
5. When you have finished stepping over/playing through the last line of the beforeLoad entry point code, the Script Debugger tab will show:



And you will see a message in the Chrome DevTools tab that indicates the debugging connection was closed:



You don't need to click the Reconnect DevTools button on this message. If you want to reexecute the beforeLoad entry point in your script, save the Customer record to restart the debugging session.

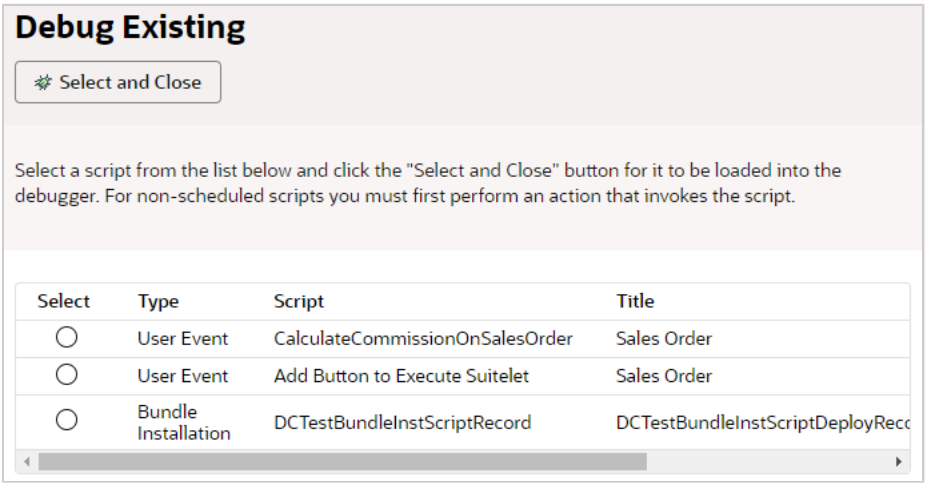
Debugging the beforeSubmit and afterSubmit Entry Points Example

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

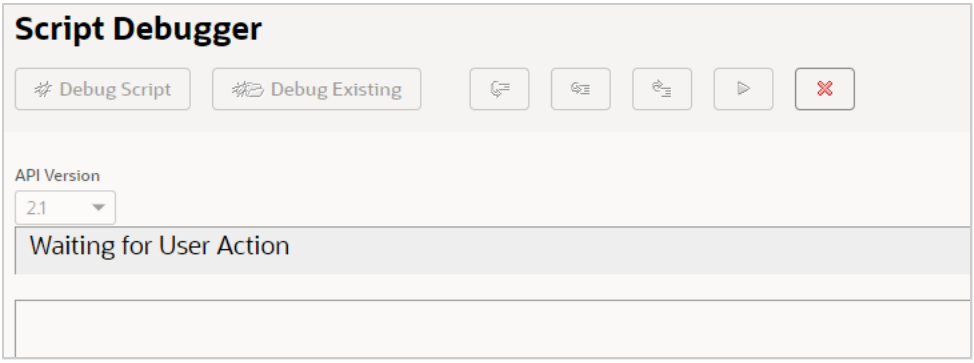
After you debug the beforeLoad entry point, you debug the beforeSubmit and afterSubmit entry points. For more information, see Creating the User Event Script and Starting the Debugger Example and Debugging the beforeLoad Entry Point Example.

To debug the beforeSubmit and afterSubmit entry points:

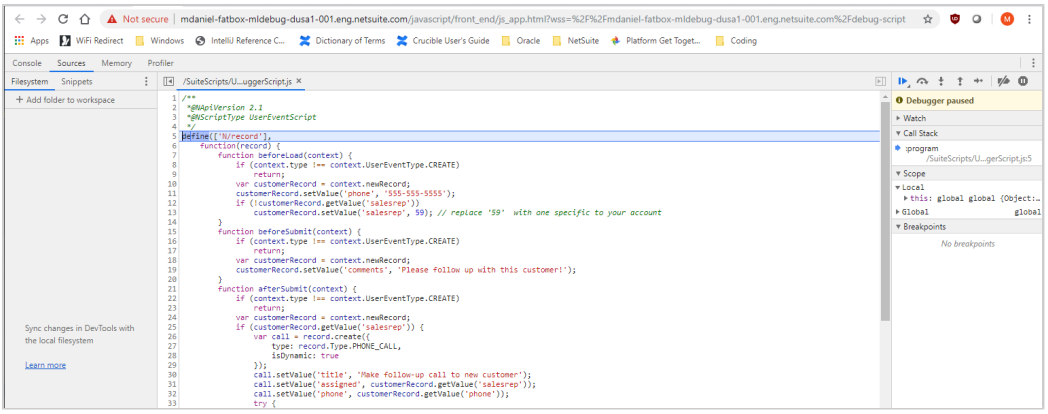
1. Go to Customization > Scripting > Script Debugger. Or, if you are already in the Script Debugger from debugging the beforeLoad entry point, execute the script again.
2. Create a new Customer or edit an existing Customer. **Do not save the record yet**. You first need to get ready to debug the script, as shown in the next steps, before clicking Save.
3. Select Debug Existing and select the Debug Customer w/ 2.1 UE user event script:



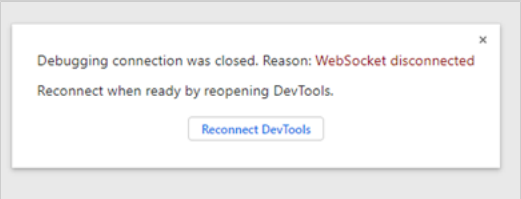
4. Click **Select and Close**. The Script Debugger waits for a user action to proceed:



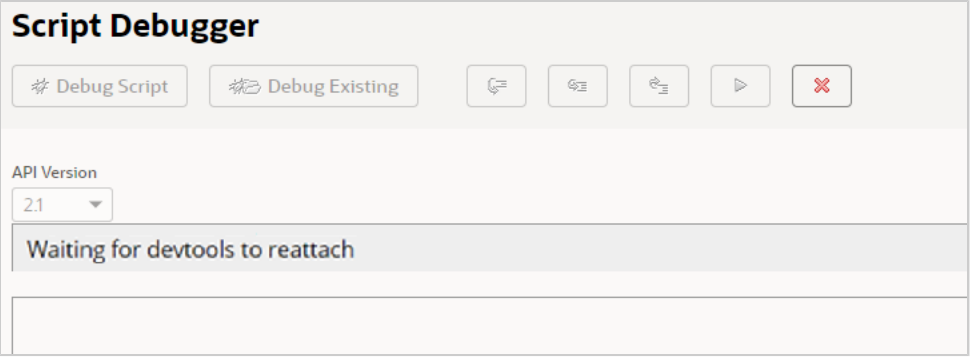
5. To trigger the beforeSubmit entry point, save the Customer record. A new Chrome DevTools tab opens in your browser showing your user event script ready to be debugged using the 2.1 Debugger:



6. The execution of your script stops at the top of the script. You can now add breakpoints, watches, and so on, within the beforeSubmit entry point and begin debugging your script. You can also add a 'debugger;' statement at the top of the beforeSubmit and afterSubmit entry point code to specifically stop the execution of the script at that entry point.
7. After the last line of the beforeSubmit function executes, the following message will appear in the Chrome DevTools tab:

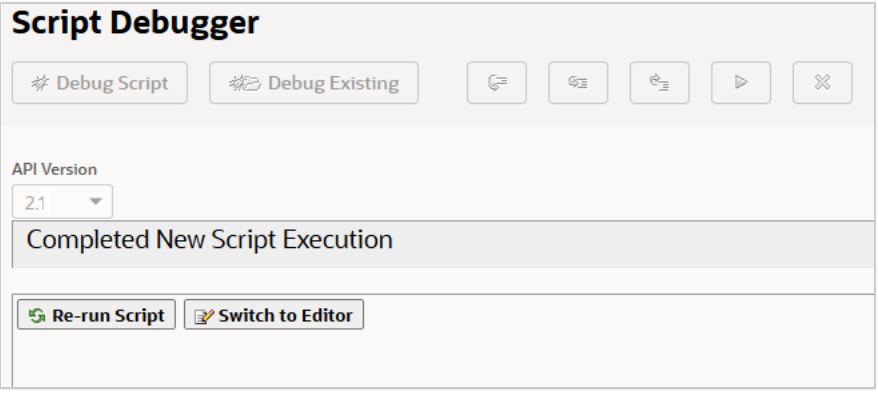


And the Script Debugger shows:

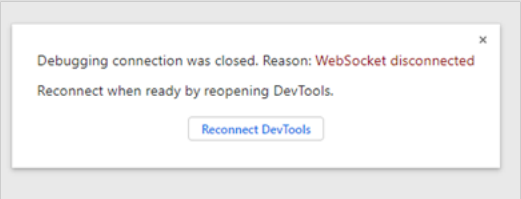


Note: This reattach phase only appears if you have both a beforeSubmit and an afterSubmit entry point in your user event script. If you have only the beforeSubmit entry point or the afterSubmit entry point, your debugging session will end at the completion of the entry point code.

8. To reattach the debugger and continue to debug your afterSubmit entry point code, click the Reconnect DevTools button on the message. The Chrome DevTools tab will reload your script and execution will stop at the beginning of the afterSubmit entry point code. You can now add breakpoints, watches, etc. and begin debugging your script.
9. When you are done stepping over/playing through the last line of the afterSubmit entry point code, the Script Debugger tab will show:



And you see a message in the Chrome DevTools tab that indicates the debugging connection was closed:



You do not need to click the Reconnect DevTools button on this message. If you want to re-execute any entry point in your script, you need to reload a customer record and follow the steps to debug the beforeLoad entry point or to debug the beforeSubmit and afterSubmit entry points.

Tips for Debugging SuiteScript 2.1 Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Note: Only SuiteScript 2.1 scripts can be debugged using the 2.1 Script Debugger. You cannot use the 2.1 Script Debugger to debug SuiteScript 1.0 or 2.0 scripts.

The following tips may help you debug SuiteScript 2.1 scripts using the Chrome DevTools debugger:

- If your script code appears minified (lacking white space and new line characters), you can use the de-minify button. `{}` to change the display of the code so that it is readable.
- Details normally displayed on the Execution Log of the Script Debugger page are shown on the Chrome DevTools console when debugging SuiteScript 2.1 scripts.
- Execution metrics for SuiteScript 2.1 scripts are shown in the Chrome DevTools console when the script completes execution. These are generally the same metrics displayed for SuiteScript 1.0 and 2.0 scripts in the Script Debugger page and include script messages and alerts, including script errors.
- Use breakpoints to pause script execution at predetermined lines of code. Active breakpoints are listed in the Breakpoints List in the JavaScript debugging pane (on the Chrome DevTools tab).
- Use the Watch window on the JavaScript debugging pane to see values of variables within your script and to evaluate expressions on-the-fly.
- Use the Call Stack window on the JavaScript debugging pane to view different areas of your script. You can also select a point in the call stack and resume execution from that point.
- To cancel a debugging session without completing script execution, click the **X** button on the Script Debugger page. The Chrome DevTools tab can be left opened or it can be closed. If you leave it open, it will be used for the next debugging session. Note that simply closing the tab does not fully cancel the current debugging session. You must use the red X button.
- Each 2.1 debugging session has a timeout. If you do not perform any actions in the Chrome DevTools debugger within the timeout period, your debug session will end. See Idle Timeouts.

Debugging Client Scripts

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

Client scripts are executed on the client browser based on how they are deployed. To debug a client script, use the debugging tools available on your browser. SuiteScript supports several browsers:

- Google Chrome (preferred)
- Mozilla Firefox (preferred)
- Microsoft Edge
- Apple Safari

Each browser includes its own debugging tools. The following table includes links to documentation for each browser.

Browser	Documentation Link
Google Chrome	https://developers.google.com/web/tools/chrome-devtools/javascript
Mozilla Firefox	https://developer.mozilla.org/en-US/docs/Tools/Debugger
Microsoft Edge	https://docs.microsoft.com/en-us/microsoft-edge/devtools-guide/debugger
Apple Safari	https://developer.apple.com/library/archive/documentation/AppleApplications/Conceptual/Safari_Developer_Guide/Debugger/Debugger.html and https://support.apple.com/guide/safari-developer/welcome/mac

Client scripts (both form- and record-level) should be tested on the form or record they are deployed on. See the help topic SuiteScript 2.x Client Script Entry Points and API for more information about triggering a client script. Also see the help topic Record-Level and Form-Level Script Deployments for more information about deploying a client script.

All browsers support common actions within their debugging environments. These actions including stepping through or over code, setting breakpoints, and pausing/resume execution.

Important: To debug a client script, the script MUST include a 'debugger;' statement. Place the 'debugger;' statement near the top of the script so that the debugger is invoked immediately when the script is triggered. Execution will stop when the statement is reached, allowing you to examine script properties and variables using the debugging tools in your browser.

To debug a SuiteScript client script:

1. Create, upload, and deploy your client script. Note that to debug a client script, it must include a 'debugger;' statement. For example:

```
debugger;
var k = 1;
k *= (k + 9);
console.log(k);
```

2. Get ready to perform the action to trigger your client script. For example, if your client script trigger (entry point) is saveRecord on the Purchase Order record, open a new purchase order in NetSuite, enter your data, but do not click **Save**.
3. Access the debugger in your browser (before triggering the script):
   - Enter CTRL+SHIFT+I in Chrome
   - Enter CTRL+SHIFT+I in Firefox
   - Enter CTRL+OPTION+I in Safari

   Note: If you access the browser debugger before you are ready to trigger your script, the debugger may pause at every action you perform in the browser causing unnecessary and premature interaction with the debugger.
4. When the debugger is open, click **Save** on the purchase order (in this example). The debugger will pause the execution of your script at the location of the ' debugger;' statement. You can now use

the debugger tools in your browser to step through the code, look at values, set breakpoints, etc. Refer to the documentation for your browser for more information.

When debugging client scripts, some scripts might be minified. Minified scripts have all unnecessary characters removed, including white space and new line characters. You can use your browser debug tools to de-minify the script. This is typically done using the { } button. See the documentation for your browser for more information.

5. When you are done using the debugger, you can close it. Your NetSuite page will remain displayed and you can continue using NetSuite.

For additional information about working with client scripts, see the help topics SuiteScript 2.x Client Script Entry Points and API and Record-Level and Form-Level Script Deployments.

Debugging a RESTlet

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

You can debug RESTlet code in the same way that you debug other types of SuiteScript code.

If you have installed and set up the SuiteCloud IDE, a debug client is available for your use. The RESTlet/Suitelet Debug Client enables you to debug deployed RESTlet and Suitelet SuiteScripts with the SuiteCloud IDE Debugger. The client is only accessible after a debug session is started. For information about SuiteCloud IDE, see SuiteCloud IDE Plug-in for Eclipse Guide.

Note: In addition to debugging RESTlet script code, you should test the HTTP request to be sent to the RESTlet. Free tools are available for this purpose, such as Send HTTP Tool (https://www.softpedia.com/get/Internet/Servers/Server-Tools/WIN-HTTP-Sender.shtml).

To debug a deployed RESTlet:

1. Before you deploy a RESTlet to be debugged, ensure that the script does not include the HTTP authorization header, as this header can prevent the debugger from working.
2. Ensure that on the script deployment record, the **Status** value is set to **Testing**.
3. Go to Customization > Scripting > Script Debugger.
4. Click the **Debug Existing** button in the main Script Debugger page.

   Note: This button only appears when you have deployed scripts with the status is set to **Testing**.
5. Select the RESTlet script that you want to debug in the Script Debugger popup.

   After you click the **Select** option button, the RESTlet's cookies display in a banner.
6. Copy the cookies and paste them into a text file so that you have them available.
7. Click the **Select and Close** button in the Script Debugger popup.

   The Script Debugger page displays a message that it is waiting for user action.
8. Set the cookies in your client application to the values you copied in step 6, and send the RESTlet request.

   The Script Debugger page displays the script execution as pending at the NetSuite function `restletwrapper(request)`.
9. With the script loaded into the debugger, you can now step through each line to inspect local variables and object properties. You can also add watches, evaluate expressions, and set break

points. See Script Debugger Interface for information about stepping into/out of functions, adding watches, setting and removing break points, and evaluating expressions.

Script Debugger Metering and Permissions

Applies to: SuiteScript 1.0 | SuiteScript 2.x | SuiteCloud Developer

The debugger adheres to the following metering and permission restrictions for all SuiteScript 1.0, 2.0, and 2.1 script types:

- You can only debug one script at a time. Attempting to debug multiple scripts simultaneously (for example, by opening two different browser windows) will result in the same script/debugging session appearing in both windows.
- You can debug only your own scripts in their current login session.
- There is a 1000 unit usage limit on all scripts being debugged. This is important to note, particularly for script types such as scheduled scripts, which are permitted 10,000 units when running in NetSuite. If, for example, you load a 2,000 unit scheduled script into the Debugger and attempt to step through or execute your code, the Debugger will throw a usage limit error when it reaches 1000 units.
- Email error notification is disabled for scripts being debugged.
- Execution log details are displayed on the Execution Log subtab in the Debugger rather than in the execution log on the Script Deployment page for all SuiteScript 1.0 and 2.0 scripts. For SuiteScript 2.1 scripts, execution log details are displayed on the Console tab of the Chrome DevTools debugging window.

Also see SuiteScript Governance and Limits for governance limits on script types and API modules.

Idle Timeouts

The SuiteScript Debugger and the 2.1 Script Debugger for all script types have idle timeouts. Your debug session will end if you stop interacting with the debugger, or if you have exceeded the maximum allowed amount for a debug session.

SuiteScript Debugger Timeouts

There is a two-minute time limit on SuiteScript 1.0 and 2.0 scripts sitting idle in the debugger. If you do not perform some user action in the debugger within the two minutes, the following error is thrown:

- You have exceeded the maximum allowable idle time for debugging scripts. To debug another script, reload the script debugger page and start a new debugging session.

If you receive this message and you are debugging a deployed SuiteScript 1.0 or 2.0 script, click **Go Back**. You must then reload your script by clicking **Debug Existing**.

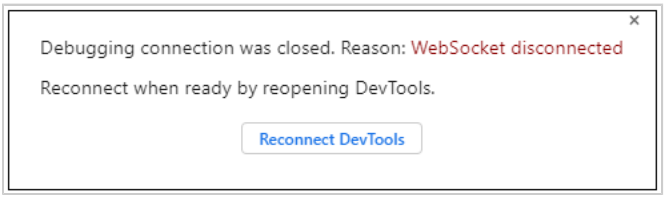
If you are debugging a SuiteScript 1.0 or 2.0 script in on-demand mode, click **Go Back**, click in the Debugger text area, and re-type (or copy and paste) your code snippet. See Debugging SuiteScript 1.0 and SuiteScript 2.0 Scripts for information about deployed and on-demand debugging modes for SuiteScript 1.0 and 2.0 scripts.

There is also a ten–minute global timeout on all scripts being debugged. This means that even if you are performing user actions within the debugger every two minutes, the debugger will still timeout after ten minutes.

SuiteScript 2.1 Debugger Timeouts

A timeout can occur in two ways when using the 2.1 Script Debugger:

- There has been no user action within the debugger for 5-minutes (300 seconds). If you don't perform some user action in the debugger within five minutes, Chrome DevTools will disconnect and you will see this message:

  Clicking **Reconnect DevTools** doesn't resume your debug session. You must return to the Script Debugger tab and click the **X** button. Then click **Switch To Editor** and click **Debug Script** to start a new debug session. If you leave the Chrome DevTools tab open, it will be used for the new debugging session. If you close the Chrome DevTools tab, a new one will open when you start a new debugging session.
- You have reached the overall time limit for a debugger session. This time limit is set using the IDLE_SESSION_TIMEOUT_IN_MINUTES general preference, which sets the maximum time for a 2.1 Script Debugger session, regardless of whether there is user action or not. You can set the IDLE_SESSION_TIMEOUT_IN_MINUTES by going to Setup > General Preferences. Although you can enter a value larger than 20 minutes, a 2.1 Script Debugger session is limited to a maximum of 20 minutes. Note that this preference also sets the timeout for a NetSuite user session (see the help topic Setting General Account Preferences).

There are several errors thrown when a 2.1 Script Debugger timeout occurs:

- An error message "You have exceeded the maximum allowable idle time for debugging scripts. To debug another script, reload the script debugger page and start a new debugging session" will be thrown.
- DEBUGGER_SESSION_TIMEOUT error will be thrown if the timeout occurred based on the IDLE_SESSION_TIMEOUT_IN_MINUTES general preference value.
- DEBUGGER_IDLE_TIMEOUT error will be thrown if there has been no user action for five minutes.

If any of these timeouts occur, you can return to the Script Debugger page and click the **X** button to stop debugging. You can then click **Switch to Editor** to reload the script and start a new debugging session.

SuiteCloud Processors

Applies to: SuiteScript 2.x | SuiteCloud Developer

SuiteCloud Processors is the current system used to process scheduled scripts and map/reduce scripts. Before SuiteCloud Processors was introduced, scheduled scripts and map/reduce scripts were exclusively processed by scheduling queues. All scheduled script and map/reduce script jobs submitted to the same queue were processed on a FIFO (first in, first out) basis, based on the queue submission time stamp.

This system had several limitations. The scheduling queues didn't provide automated load balancing or a way to prioritize specific jobs. Users with access to multiple queues (accounts with SuiteCloud Plus) were forced to manually determine the optimal configuration of jobs to queues. For jobs that needed to be processed in a certain order, this method was useful but it created unintended dependencies among many of the jobs submitted. If processing for one job was delayed it would create a bottleneck resulting in several jobs waiting in one queue when other queues were underutilized or not utilized at all.

SuiteCloud Processors solves many of those limitations. A scheduler now automatically determines the order in which jobs start to process. The scheduler uses algorithms that are based on user-defined priority levels, submission time, and user-defined preferences. The result is increased throughput, reduced wait times, and the elimination of most bottlenecks. In addition, SuiteCloud Processors requires less user intervention and enables scheduled scripts and map/reduce scripts to start sooner.

Note: Some features of SuiteCloud Processors are available only to accounts that have one or more SuiteCloud Plus licenses. For more information about SuiteCloud Plus, see the help topic SuiteCloud Plus Settings.

For additional information about SuiteCloud Processors, see:

- SuiteCloud Processors Terminology
- SuiteCloud Processors Basic Architecture
- SuiteCloud Processors Supported Task Types
- SuiteCloud Processors Processor Allotment Per Account
- SuiteCloud Processors Priority Levels
- SuiteCloud Processors Priority Elevation
- Monitoring SuiteCloud Processors Performance

SuiteCloud Processors Terminology

Applies to: SuiteScript 2.x | SuiteCloud Developer

To understand how SuiteCloud Processors work, familiarize yourself with these key terms.

Term	Definition
Job	A piece of work submitted to SuiteCloud Processors for processing. Each job is executed by a single processor.

Term	Definition
Priority	Priority of a job. The priority of submitted jobs determines the order in which the scheduler sends the jobs to the processor pool. Priorities are set on the deployment record or from the SuiteCloud Processors Priority Settings Page.
Processor	A virtual unit of processing power that executes a job. It's not a separate physical entity, but a single processing thread.
Processor Pool	Represents the number of processors available to a specific account. For accounts without SuiteCloud Plus, the processor pool contains two processors. For more information, see Accounts Without a SuiteCloud Plus License. For accounts with SuiteCloud Plus, the processor pool contains at least two and up to 60 processors depending on the number of SuiteCloud Plus licenses the account has. For more information, see the help topic SuiteCloud Plus Settings.
Queue	With SuiteCloud Processors, a queue is no longer a separate processing mechanism. On scheduled script deployments, the **Queue** field remains to accommodate deployments that rely on the first in, first out (FIFO) order imposed by an individual queue. However, all jobs that use queues are processed by the same processor pool that handles the jobs without queues. All jobs compete using the same common processing algorithm.
Scheduler	The scheduler determines the order in which jobs are sent to the processor pool. It uses algorithms that are based on user-defined priority levels, submission time, and user-defined preferences.
Task	A script instance that's submitted for processing. Each task is handled by one or more jobs.

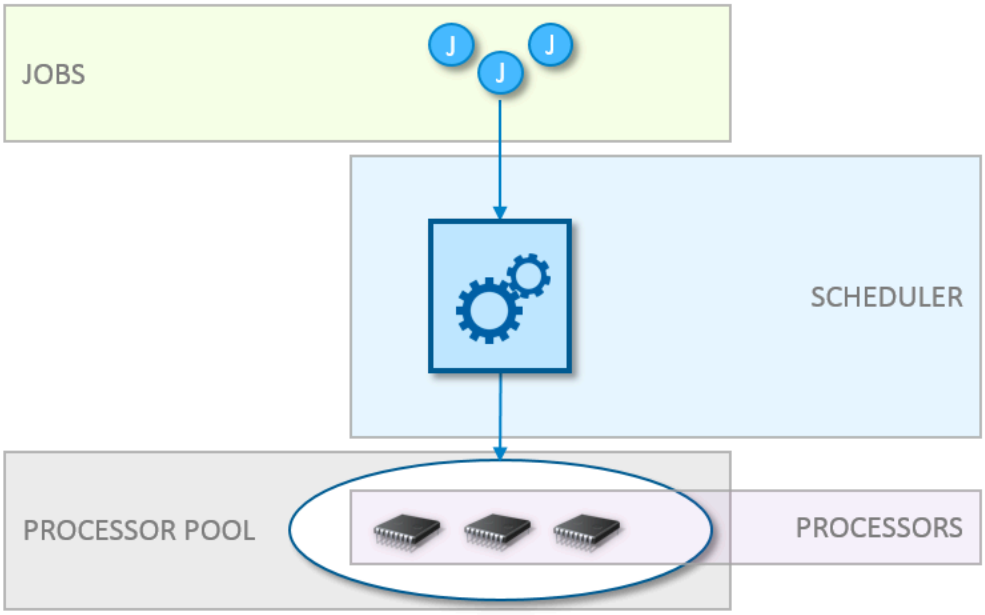
SuiteCloud Processors Basic Architecture

Applies to: SuiteScript 2.x | SuiteCloud Developer

SuiteCloud Processors currently supports the processing of scheduled scripts and map/reduce scripts. You can submit these scripts in several ways:

- You can set a one-time or recurring submission schedule from the script deployment record UI.
- You can select **Save and Execute** from the script deployment record UI to submit an on-demand instance of the script.
- You can use a SuiteScript API to submit a script on demand.

Each submitted script instance is called a "task". Each submitted scheduled script task is handled by one job. Each submitted map/reduce script task is handled by multiple jobs: one each for the getInput, shuffle, and summarize stages; and a minimum of one each for the map and reduce stages. The scheduler sends submitted jobs to the processor pool.



The scheduler sends the jobs to the processor pool in a specific order. This order is determined by priority and the order of submission. Jobs with higher priority are sent before jobs with lower priority. Jobs with the same priority are sent to the processor pool in the order of submission. For more information about priorities and examples of scheduling based on priorities, and to understand how scheduled scripts and map/reduce scripts are processed differently, see SuiteCloud Processors Priority Levels and SuiteCloud Processors Priority Scheduling Examples.

SuiteCloud Processors includes advanced settings for priority elevation and processor reservation which can also impact the order in which jobs are sent to the processor pool. For more information, see SuiteCloud Processors Priority Elevation.

Important: The role used to submit scheduled script tasks and map/reduce script tasks for processing must have the following permissions:

- Documents and Files: View, Create, Edit, or Full
- SuiteScript: View, Create, Edit, or Full
- SuiteScript Scheduling: Full

For more information about access levels for permissions, see the help topic Access Levels for Permissions.

For more information about submitting a script with the deployment record, see the following topics:

- SuiteScript 2.x Scheduled Script Type
- SuiteScript 2.x Map/Reduce Script Type

For more information about submitting a script with a SuiteScript API, see the following topics:

- task.ScheduledScriptTask
- task.MapReduceScriptTask

SuiteCloud Processors Supported Task Types

Applies to: SuiteScript 2.x | SuiteCloud Developer

SuiteCloud Processors currently supports processing for two script type tasks:

- SuiteScript 2.x Map/Reduce Script Type
- SuiteScript 2.x Scheduled Script Type

SuiteCloud Processors Impact on Map/Reduce Deployments

All map/reduce script deployments include a **Priority** field. For additional information, see SuiteCloud Processors Priority Levels.

For accounts with SuiteCloud Plus, the **Queues** multi-select field is replaced with the **Concurrency Limit** field. Instead of designating a specific queue, you set the maximum number of processors available to the deployment. This value equals the number of jobs submitted for the map and reduce stages.

For map/reduce deployments created prior to the introduction of SuiteCloud Processors, the **Concurrency Limit** value is set by default. The default value is equal to the value set on the **Queues** field. If **Queues** was previously set to 1, 3, 7, and 9, then **Concurrency Limit** is set to 4. If **Select All** was previously enabled, then **Concurrency Limit** is set to empty (the number of jobs initially created for the map and reduce stages is equivalent to the total number of processors in the processor pool).

For additional information, see the help topic SuiteScript 2.x Map/Reduce Script Type.

SuiteCloud Processors Impact on Scheduled Script Deployments

All scheduled script deployments include a **Priority** field. For additional information, see SuiteCloud Processors Priority Levels.

For all scheduled script deployments created **after** the introduction of SuiteCloud Processors, the **Queue** field is removed. These deployments can't use queues.

For all scheduled script deployments created **prior** to the introduction of SuiteCloud Processors, the **Queue** field remains by default. This applies to accounts with and without SuiteCloud Plus. You can decide whether to stop using queues. The deployment record includes a **Remove Queue** option. After you choose this option, the deployment stops using a queue and can't go back to using one.

The **Queue** field remains to accommodate deployments that rely on the FIFO order of processing imposed by an individual queue. However, all jobs that use queues are processed by the same processor pool that handles the jobs that don't use queues. All jobs compete with each other using the same common processing algorithm.

Note: For deployments that continue to use queues, all jobs assigned to the same queue should have the same priority. In most cases, you can keep the default (standard) priority of these jobs. However, in some cases, you may want to change these jobs to a higher or lower priority. One scenario is if you want to ensure that a specific queue always has a processor available. In that case, designate the jobs assigned to that queue as high priority. Alternatively, if you have a group of lower priority jobs, you can designate them as low priority and assign them to the same queue. That will ensure that only one is processed at a time.

Important: If your existing scheduled script deployments rely on implicit dependencies imposed by queues, you should update and test your scripts before you remove queues. Your scripts may be impacted if they rely on the sequence of FIFO (first in, first out).

One solution is to programmatically submit scripts in a certain order. To do this, use task.ScheduledScriptTask in the first script to submit the second script. This makes sure that the jobs are submitted to the processor pool in the correct order.

SuiteCloud Processors Processor Allotment Per Account

Applies to: SuiteScript 2.x | SuiteCloud Developer

Important: SuiteCloud Processors are used to process all scheduled script and map/reduce script instances. This topic doesn't apply to scheduled script deployments that continue to use queues. See the help topic SuiteCloud Plus Settings for more information.

SuiteCloud Processor Allotment

Service Tier	Maximum SuiteCloud Plus Licenses	Maximum SuiteCloud Processors
Standard	1	5
Premium	3	15
Enterprise	6	30
Ultimate	12	60

*The default number of SuiteCloud Processors is 2 if you haven't purchased any SuiteCloud Plus licenses.

For more information, see the help topics SuiteCloud Plus Settings and NetSuite Service Tiers.

Accounts Without a SuiteCloud Plus License

Accounts without a SuiteCloud Plus license have access to two processors. The extra processor doubles the processing bandwidth for scheduled scripts and map/ reduce scripts.

If any of your scripts depend on having one processor only, you may need to update them for this feature. When an account has access to only one processor, jobs are processed one at a time. If all jobs

have the same priority, the order of processing is always first in, first out (FIFO). Some scripts may depend on this behavior. With the addition of an extra processor, these scripts may no longer process in the correct order.

For example, if you have a script that processes data for use in a second script, the first script must complete processing before the second script begins. With one processor, you can submit the scripts in the appropriate order and know that the order of processing is as expected. With two processors, the second script submitted might start running before the first script completes.

To handle the above scenario, perform the following steps:

1. Check your scheduled and map/reduce scripts for ones that depend on a specific order of execution.
2. Use the following SuiteScript APIs in the first script to programmatically submit the dependent script. This makes sure that the scripts are always executed in the correct order.
   - For a scheduled script, call task.ScheduledScriptTask and place the code at the end of the script.
   - For a map/reduce script, call task.MapReduceScriptTask and place the code at the end of the summarize stage.

SuiteCloud Processors Priority Levels

Applies to: SuiteScript 2.x | SuiteCloud Developer

SuiteCloud Processors consider submission time and priority level when processing scripts. This means that higher priority jobs are sent to the processor pool first. Lower priority jobs are sent to the processor pool after all higher priority jobs are sent. The scheduler sends jobs with the same priority to the processor pool in the order of they're submitted.

Available priority options are:

- High: Use this priority level to mark critical jobs that require more immediate processing. The scheduler sends these jobs to the processor pool first.
- Standard: This is the default setting and is a medium priority level. The scheduler sends these jobs to the processor pool if there are no high priority jobs waiting.
- Low: Use this priority level to mark jobs that can tolerate a longer wait time. The scheduler sends these jobs to the processor pool if there are no high or standard priority jobs waiting.

For example, if jobs 1 and 4 are high priority, jobs 2 and 5 are standard priority, and job 3 is low priority, the scheduler sends the jobs to the processor pool in this order:

- Job 1
- Job 4
- Job 2
- Job 5
- Job 3

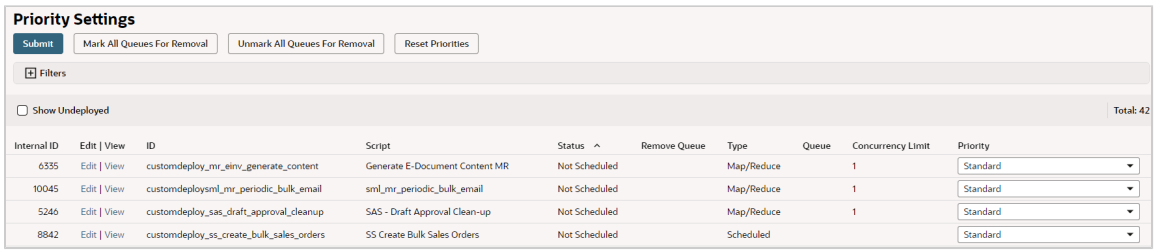
By default, all jobs have a standard priority, but you can change the priority on the deployment record or on the SuiteCloud Processors Priority Settings Page. For examples that demonstrate how changing the priority can change the order of processing, see SuiteCloud Processors Priority Scheduling Examples.

SuiteCloud Processors Priority Settings Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

The Priority Settings page lists each scheduled script deployment and map/reduce script deployment in your account. Each line item corresponds to one deployment record. To access the Priority Settings page, go to Customization > Scripting > Priority Settings.

The Priority Settings page lets you manage the SuiteCloud Processors Priority Levels for multiple deployments at one time. You can also use this page to remove queues for scheduled script deployments.



To save changes made to the page, such as changing priorities, use the Submit button. To reset all listed scripts to Standard priority, use the Reset Priorities button.

The Priority Settings page lists priority settings for scripts that include the following columns:

Column	Description
Internal ID	The internal ID for the script deployment record, as shown on the Script Deployments list page (for example, 345).
Edit | View	Links to the edit and view modes of the deployment record.
ID	The ID of the script deployment record (for example, customdeploy_testscript1).
Script	Corresponds to the Name field value on the script record associated with the deployment record.
API Version	Shows the SuiteScript version.
Status	Indicates how and when a script can be submitted for processing. This value is set with the Status field on the deployment record. Possible values are: ■ **Testing**: Indicates that you can test the script in the SuiteScript Debugger. The script deployment can be submitted for processing by the script owner only. ■ **Scheduled**: Indicates that you can schedule a single or recurring instance of the script on the deployment record. You can't submit an on demand instance of the script when it has this status. ■ **Not Scheduled**: Indicates that you can submit an on demand instance of the script with the **Save and Execute** button or a SuiteScript API. You can submit an on demand instance of the script only if there's no other unfinished instance of the same script.
Remove Queue	Applies only to scheduled script deployments. Check this box to remove queues from an existing scheduled script deployment. You can remove queues for multiple deployments at one time with this column. Queues aren't shown as removed until you submit the page.
Type	Indicates whether the deployment is for a scheduled script or map/reduce script.
Queue	Applies only to scheduled script deployments. Shows a value if the deployment is still using queues. After you remove queues, this value is empty.

Column	Description
Concurrency Limit	Applies only to map/reduce script deployments on accounts that use SuiteCloud Plus. The maximum number of processors available to a map/reduce script deployment. You set this value on the map/reduce deployment record.
Priority	The priority setting for the deployment. For additional information, see SuiteCloud Processors Priority Levels.

SuiteCloud Processors Priority Scheduling Examples

Applies to: SuiteScript 2.x | SuiteCloud Developer

The examples in this section, Default Priority Scheduling Example and Varying Priority Scheduling Example, show three diagrams. The diagrams compare how 18 jobs are handled by:

- Pre-2017.2 Scheduling Queues
- 2017.2 SuiteCloud Processors: All queues are removed
- 2017.2 SuiteCloud Processors: Some queues are removed; jobs 1 - 3, 15, and 18 still use queues

The following table shows the duration of each job submitted. As the examples show, the processing environment doesn't affect the time it takes to complete each job after processing starts.

Note: Each map/reduce stage is handled by a minimum of one job. So, the duration listed for map/reduce jobs is per stage. For example, the reduce stage jobs 11 and 12 have a combined duration of 5. This means that the sum of the duration of jobs 11 and 12 is 5. If one job is canceled or not created, the combined duration is only the time for the remaining job.

Job	Duration
1	5
2	2
3	2
4	6
5 +6	2
7 + 8	1
9 + 10	5
11 + 12	5
13 + 14	2
15	3
16	3
17	3
18	2

Default Priority Scheduling Example

Applies to: SuiteScript 2.x | SuiteCloud Developer

This example assumes default priority settings (standard priority) and default preferences. It also assumes that each job executes in full without yielding.

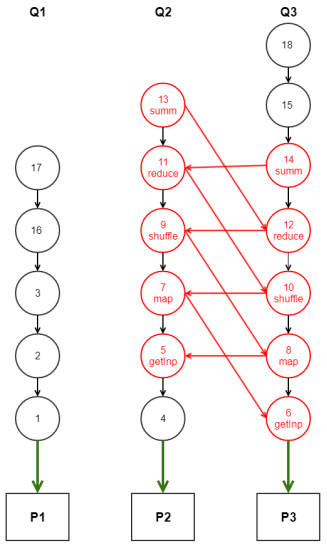
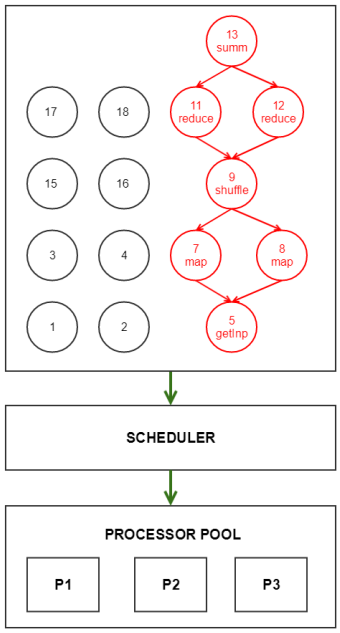
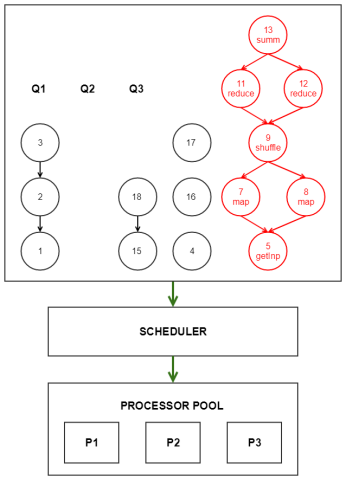
The map/reduce deployment in the first diagram has a **Queues** value of 2 and 3. The map/reduce deployment in the second and third diagrams has a **Concurrency Limit** value of 2.

Note: With the introduction of SuiteCloud Processors, the **Queues** field is replaced with the **Concurrency Limit** field on map/reduce deployments.

Diagram Key for Default Priority Scheduling Example

- Black circles are scheduled script jobs. Each scheduled script is processed by a single job.
- Red circles are map/reduce jobs. In this example, all of the map/reduce jobs belong to a single map/reduce task. The map/reduce jobs include a label that indicates the map/reduce stage being processed.
- Jobs in columns labeled Q1, Q2, and Q3 are for deployments that still use queues. The jobs in columns without a label are for deployments that no longer use queues.
- P1, P2, and P3 represent the processors that are available in the processor pool.

  Note: Although processors are labeled in this example, they aren't technically individual entities. Also, a real account would never have only three processors available to it. The minimum number of processors available to an account with SuiteCloud Plus is listed at SuiteCloud Plus Settings.
- Numbers within each circle represent the time stamp on the job submission, with 1 being the first job submitted. The example also uses this number as a unique identifier for each job.
- Arrows show dependencies. For example, in the Scheduling Queues diagram, job 2 can't start until job 1 is complete.
  - Black arrows represent queue dependencies. Jobs within a queue must be processed in the order they were submitted.
  - Red arrows represent dependencies imposed by the map/reduce algorithm.
  - Green arrows indicate access to processors.

Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
		

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
101	Jobs **1**, **4**, and **6** start.	Jobs **1**, **2**, and **3** start.	Jobs **1**, **4**, and **5** start.
102	Job **6** cancels job **5**. The getInput stage can't be processed by multiple jobs, so the first getInput job to start (job **6**) automatically cancels all others in the queues.	—	—
103	Job **6** completes. The next job in Q3, job **8**, starts.	Jobs **2** and **3** complete. Jobs **4** and **5** start.	Job **5** completes. There's no job **6** with SuiteCloud Processors. In the Scheduling Queues example, jobs **5** and **6** are both getInputstage jobs. The getInput stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple getInput stage jobs are no longer needed. Job **7** starts.
104	Job **8** completes. The map stage can be processed by multiple jobs. But since the other map stage job, job **7**, is unable to start (the job utilizing its processor, job **4**, isn't yet complete), job **8** completes the entire map	—	Job **7** completes. Since job **7** completes the entire map stage, job 8 is no longer needed. Job **8** is canceled. Job **9** starts.

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
	stage. Job **7** is no longer needed, so it is canceled. The next job in Q3, job **10**, starts.		
105	Job **10** cancels job **9**. The shuffle stage can't be processed by multiple jobs, so the first shuffle job to start (job **10**) automatically cancels all others in the queues.	Job **5** completes. There's no job **6** with SuiteCloud Processors. In the Scheduling Queues example, jobs **5** and **6** are both getInputstage jobs. The getInput stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple getInput stage jobs are no longer needed. Job **7** starts.	—
106	Job **1** completes. The next job in Q1, job **2**, starts.	Job **7** completes. Since job **7** completes the entire map stage, job **8** is no longer needed. Job **8** is canceled. Job **1** completes. Jobs **9** and **15** start. Jobs **11 - 13** are dependent on job **9**. There's no job **10** or job **14** with SuiteCloud Processors. In the Scheduling Queues example, jobs **9** and **10** are both shuffle stage jobs and jobs **13** and **14** are both summarize stage jobs. The shuffle and summarize stages can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple shuffle and summarize stage jobs are no longer needed.	Job **1** completes. The next job in Q1, job **2**, starts.
107	Job **4** completes. Q2 is blocked. The next job in Q2, job **11**, can't start. It is dependent on job **10**, and job **10** isn't complete.	—	Job **4** completes. The first job in Q3, Job **15**, starts.
108	Job **2** completes. The next job in Q1, job **3**, starts.	—	Job **2** completes. The next job in Q1, job **3**, starts.
109	Job **10** completes. Job **11** can now start and Q2 is no longer blocked. The next job in Q3, job **12**, starts. These are both reduce stage jobs. The reduce stage can be processed by multiple jobs.	Jobs **4** and **15** complete. Jobs **16** and **17** start.	Job **9** completes. There's no job **10** with SuiteCloud Processors. In the Scheduling Queues example, jobs **9** and **10** are both shuffle stage jobs. The shuffle stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple shuffle stage jobs are no longer needed. Job **11** starts.

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
110	Job **3** completes. The next job in Q1, job **16**, starts.	—	Jobs **3** and **15** complete. There are no additional jobs to process in Q1. Jobs **12** and **16** start.
111	Job **11** completes. Q2 is blocked. The next job in Q2, job **13**, can't start. It is dependent on job **12**, and job **12** isn't complete.	Job **9** completes. Job **11** starts.	—
112	Job **12** completes. Job **13** can now start and Q2 is no longer blocked. The next job in Q3, job **14**, can also start. But these are both summarize stage jobs, and the summarize stage can't be processed by multiple jobs. Job **13** starts first, so job **14** is canceled. Since job **14** is canceled, the next job in Q3, job **15**, starts	Jobs **16** and **17** complete. Jobs **12** and **18** start.	Jobs **11** and **12** complete. Jobs **13** and **17** start.
113	Job **16** completes. The next job in Q1, job **17**, starts.	—	Job **16** completes. The next job in Q3, Job **18**, starts.
114	Job **13** completes. There are no additional jobs to process in Q2.	Jobs **11**, **12**, and **18** complete. Job **13** starts	Job **13** completes. There's no job **14** with SuiteCloud Processors. In the Scheduling Queues example, jobs **13** and **14** are both summarize stage jobs. The summarize stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple summarize stage jobs are no longer needed.
115	Job **15** completes. The next job in Q3, job **18**, starts.	—	Jobs **17** and **18** complete. There are no additional jobs to process.
116	Job **17** completes. There are no additional jobs to process in Q1.	Job **13** completes. There are no additional jobs to process.	—
117	Job **18** completes. There are no additional jobs to process in Q3.	—	—

Varying Priority Scheduling Example

Applies to: SuiteScript 2.x | SuiteCloud Developer

This example assumes default preferences. It also assumes that each job executes in full without yielding.

The map/reduce deployment in the first diagram has a **Queues** value of 2 and 3. The map/reduce deployment in the second and third diagrams has a **Concurrency Limit** value of 2.

Note: With the introduction of SuiteCloud Processors, the **Queues** field is replaced with the **Concurrency Limit** field on map/reduce deployments.

The second and third diagrams vary from Default Priority Scheduling Example as follows:

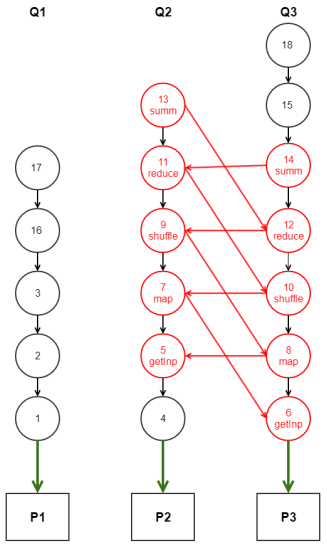
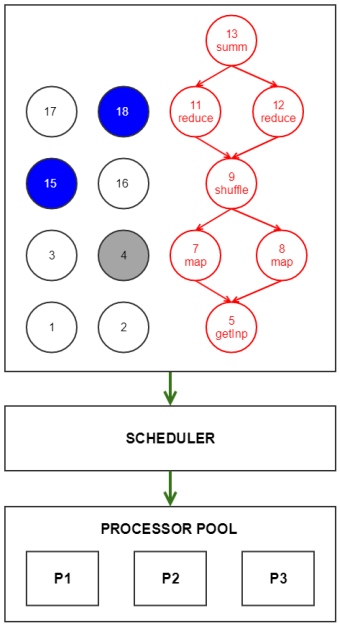
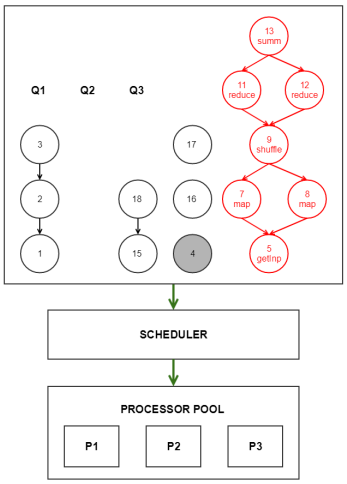
- In the second diagram, jobs 15 and 18 are high priority. Job 4 is low priority. The remaining jobs are standard priority.
- In the third diagram, job 4 is low priority. Jobs 15 and 18 in the third diagram are standard priority.

Diagram Key for Varying Priority Scheduling Example

- Black circles are scheduled script jobs. Each scheduled script is processed by a single job.
- Red circles are map/reduce jobs. In this example, all of the map/reduce jobs belong to a single map/reduce task. The map/reduce jobs include a label that indicates the map/reduce stage being processed.
- **Circles with blue fill are high priority. The circles with gray fill are low priority. The circles with white fill are standard priority.**
- Jobs in columns labeled Q1, Q2, and Q3 are for deployments that still use queues. The jobs in columns without a label are for deployments that no longer use queues.
- P1, P2, and P3 represent the processors that are available in the processor pool.

  Note: The processors are labeled for this example. Although technically, processors aren't distinguished as individual entities. Also, a real account would never have only three processors available to it. The minimum number of processors available to an account with SuiteCloud Plus is listed at SuiteCloud Plus Settings.

- Numbers within each circle represent the time stamp on the job submission, with 1 being the first job submitted. The example also uses this number as a unique identifier for each job.
- Arrows show dependencies. For example, in the Scheduling Queues diagram, job 2 can't start until job 1 is complete.
  - Black arrows represent queue dependencies. Jobs within a queue must be processed in the order they were submitted.
  - Red arrows represent dependencies imposed by the map/reduce algorithm.
  - Green arrows indicate access to processors.

Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
		

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
101	Jobs **1**, **4**, and **6** start.	Jobs **15**, **18**, and **1** start. Jobs **15** and **18** are designated as high priority so they are assigned to processors first (in the order of submission). Job **1** is the first standard priority job submitted so it is the next job assigned to a processer.	Jobs **1**, **5**, and **15** start. Job **4** is designated low priority, so it is processed after all other available standard priority jobs. All other standard priority jobs submitted before job **15** have dependencies that are blocking them.
102	Job **6** cancels job **5**. The getInput stage can't be processed by multiple jobs, so the first getInput job to start (job **6**) automatically cancels all others in the queues.	—	—
103	Job **6** completes. The next job in Q3, job **8**, starts.	Job **18** completes. Job **2** starts.	Job **5** completes. There's no job **6** with SuiteCloud Processors. In the Scheduling Queues example, jobs **5** and **6** are both getInputstage jobs. The getInput stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple getInput stage jobs are no longer needed. Job **7** starts.
104	Job **8** completes. The map stage can be processed by multiple jobs. But since the other map stage job, job **7**, is	Job **15** completes. Job **3** starts.	Jobs **7** and **15** complete. Since job **7** completes the entire map stage, job 8 is no longer needed. Job **8** is canceled.

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
	unable to start (the job utilizing its processor, job **4**, isn't yet complete), job **8** completes the entire map stage. Job **7** is no longer needed, so it is canceled. The next job in Q3, job **10**, starts.		Jobs **9** and **16** start.
105	Job **10** cancels job **9**. The shuffle stage can't be processed by multiple jobs, so the first shuffle job to start (job **10**) automatically cancels all others in the queues.	Job **2** completes. Job **5** starts. Job **4** is designated low priority, so it is processed after all other available standard priority jobs.	—
106	Job **1** completes. The next job in Q1, job **2**, starts.	Jobs **1** and **3** complete. Jobs **16** and **17** start.	Job **1** completes. The next job in Q1, job **2**, starts.
107	Job **4** completes. Q2 is blocked. The next job in Q2, job **11**, can't start. It is dependent on job **10**, and job **10** isn't complete.	Job **5** completes. There's no job **6** with SuiteCloud Processors. In the Scheduling Queues example, jobs **5** and **6** are both getInputstage jobs. The getInput stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple getInput stage jobs are no longer needed. Job **7** starts.	Job **16** completes. Job **17** starts.
108	Job **2** completes. The next job in Q1, job **3**, starts.	Job **7** completes. Since job **7** completes the entire map stage, job 8 is no longer needed. Job **8** is canceled. Job **9** starts.	Job **2** completes. The next job in Q1, job **3**, starts.
109	Job **10** completes. Job **11** can now start and Q2 is no longer blocked. The next job in Q3, job **12**, starts. These are both reduce stage jobs. The reduce stage can be processed by multiple jobs.	Jobs **16** and **17** complete. Job **4** starts. Job **4** is designated as low priority, but all the remaining standard priority jobs are dependent on job **9**.	Job **9** completes. There's no job **10** with SuiteCloud Processors. In the Scheduling Queues example, jobs **9** and **10** are both shuffle stage jobs. The shuffle stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple shuffle stage jobs are no longer needed. Job **11** starts.
110	Job **3** completes. The next job in Q1, job **16**, starts.	—	Jobs **3** and **17** complete. There are no additional jobs to process in Q1. Job **12** starts. The next job in Q3, Job **18**, starts.
111	Job **11** completes. Q2 is blocked. The next job in Q2, job **13**, can't start. It is	—	—

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
	dependent on job **12**, and job **12** isn't complete.		
112	Job **12** completes. Job **13** can now start and Q2 is no longer blocked. The next job in Q3, job **14**, can also start. But these are both summarize stage jobs, and the summarize stage can't be processed by multiple jobs. Job **13** starts first, so job **14** is canceled. Since job **14** is canceled, the next job in Q3, job **15**, starts	—	Jobs **11**, **12** and **18** complete. There are no additional jobs to process in Q3. Jobs **13** and **4** start.
113	Job **16** completes. The next job in Q1, job **17**, starts.	Job **9** completes. There's no job **10** with SuiteCloud Processors. In the Scheduling Queues example, jobs **9** and **10** are both shuffle stage jobs. The shuffle stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple shuffle jobs are no longer needed. Jobs **11** and **12** start.	—
114	Job **13** completes. There are no additional jobs to process in Q2.	—	Job **13** completes. There's no job **14** with SuiteCloud Processors. In the Scheduling Queues example, jobs **13** and **14** are both summarize stage jobs. The summarize stage can't be processed by multiple jobs. Since this map/reduce deployment isn't using queues, multiple summarize stage jobs are no longer needed.
115	Job **15** completes. The next job in Q3, job **18**, starts.	Jobs **4** and **12** complete.	—
116	Job **17** completes. There are no additional jobs to process in Q1.	Job **11** completes. Job **13** starts.	—
117	Job **18** completes. There are no additional jobs to process in Q3.	—	—
118	—	Job **13** completes. There's no job **14** with SuiteCloud Processors. In the Scheduling Queues example, jobs **13** and **14** are both summarize stage jobs. The summarize stage can't be processed by multiple jobs. Since this map/reduce	Job **4** completes. There are no additional jobs to process.

Time Slot	Scheduling Queues Only	SuiteCloud Processors – All Queues Removed	SuiteCloud Processors – Some Queues Removed
		deployment isn't using queues, multiple summarize stage jobs are no longer needed. There are no additional jobs to process.	

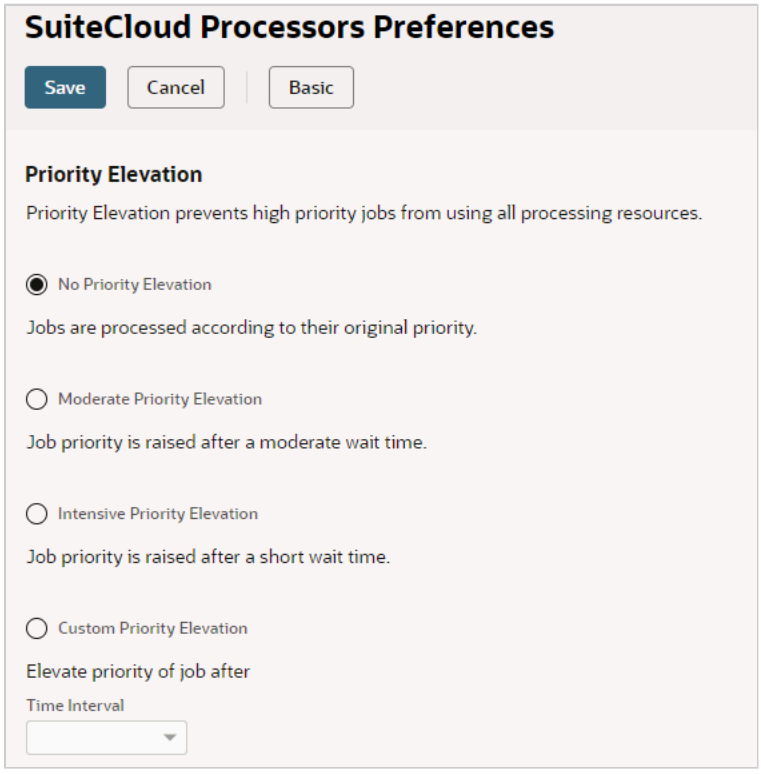
SuiteCloud Processors Priority Elevation

Applies to: SuiteScript 2.x | SuiteCloud Developer

Priority elevation lets you to automatically raise the priority of each low or standard priority job after a specific amount of time. The time interval starts when the job is submitted. You usually don't need to use these settings so priority elevation is disabled by default. However, you may need to use them if lower priority jobs experience excessive wait times.

Note: Priority elevation only affects lower priority jobs with a wait time greater than the time interval indicated. If a lower priority job is sent to the processor pool before the time interval ends, it is processed with its original priority.

To access the priority elevation settings, go to Setup > Preferences > SuiteCloud Processors.



There are three settings:

- **No Priority Elevation**: The default setting.
- **Moderate Priority Elevation**:

- If a low priority job is still waiting after four hours, it's elevated to standard priority.
- If a standard priority job is still waiting after four hours, it's elevated to high priority.

With this option, if priority elevation applies, the system elevates low priority jobs to high priority jobs after eight hours. Specifically, these jobs are elevated to standard priority after four hours and to high priority after another four hours.

- **Intensive Priority Elevation**:
  - If a low priority job is still waiting after one hour, it's elevated to standard priority.
  - If a standard priority job is still waiting after one hour, it's elevated to high priority.

  With this option, if priority elevation applies, the system elevates low priority jobs to high priority jobs after two hours. Specifically, these jobs are elevated to standard priority after one hour and to high priority after another hour.
- **Custom Priority Elevation**: This option lets you to specify a custom time interval for priority elevation using the **Time Interval** field.

The **Time Interval** field shows the time interval set for priority elevation. When you select **Custom Priority Elevation**, you can edit the field. Otherwise, the field shows a value that corresponds to the option selected, but you can't edit it..

Note: Click **Advanced** at the top of the page to access **Custom Priority Elevation** and **Time Interval**.

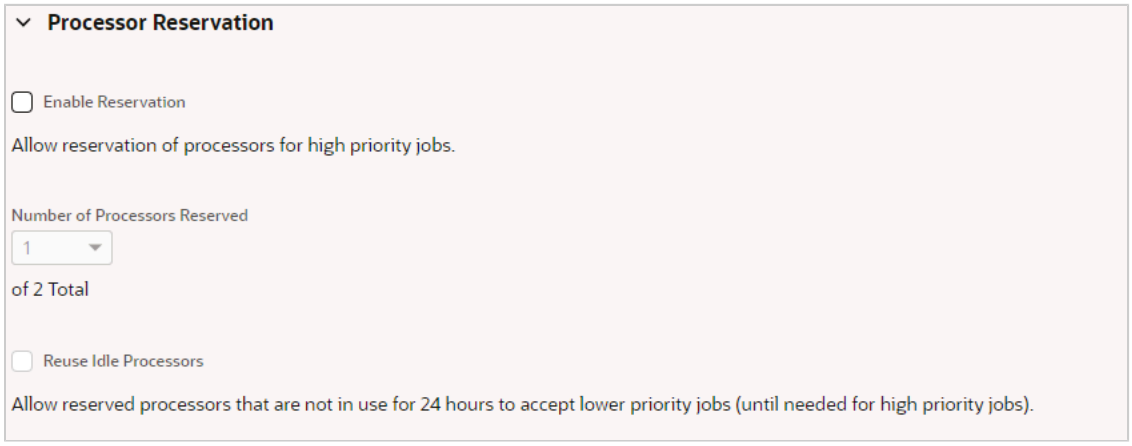
SuiteCloud Processors Processor Reservation Advanced Settings

Applies to: SuiteScript 2.x | SuiteCloud Developer

Processor reservation lets you to reserve processors for high priority jobs. You usually don't need to use these settings so processor reservation is disabled by default.

Important: Processor reservation is only available for SuiteCloud Plus accounts.

To access the priority reservation settings, go to Setup > Preferences > SuiteCloud Processors. Click **Advanced** at the top of the page.



When you select **Enable Reservation**, you can reserve all but one of your available processors from the **Number of Processors Reserved** list. If a high priority job is submitted, it's sent to the processor pool if there's at least one processor available. If a standard or low priority job is submitted, it's sent to the processor pool only if there are more processors available than the number reserved. For example, let's say you have 10 processors reserved out of 25 total processors. A standard or low priority job is sent to the processor pool if there are at least 11 processors available. If there are 10 processors or fewer available, the lower priority job has to wait.

Important: Changes to the Number of Processors Reserved apply to all jobs that haven't yet started. This can have immediate affect map/reduce scripts, since each stage is processed by at least one job. If a high priority map/reduce script task is executing and you change the setting, the new value applies to all jobs for the tasks that haven't started yet. This includes jobs created from yielding.

Processor reservation reduces the number of processors available for standard and low priority jobs. So, it can reduce the throughput of these jobs. The **Reuse Idle Processors** setting temporarily releases reserved processors that haven't been used in the past 24 hours. This increases the number of processors available for lower priority jobs.

When **Reuse Idle Processors** is enabled, it starts an hourly recurring audit. The system uses the data collected to determine whether to release reserved processors. After reserved processors are released, the system checks the audit data to decide if it needs to increase reserved processors.

Important: If the system decreases or increases the number of reserved processors, additional decreases aren't made for 24 hours. However, additional increases (up to the selected limit) can still be made after each hourly audit. This process continues if **Reuse Idle Processors** is enabled.

The system analyzes the following data points during this process:

- **a** = total number of processors available
- **b** = the value you set for **Number of Processors Reserved** (the maximum number of reserved processors; this number doesn't change)
- **c** = maximum number of jobs concurrently processed in the last 24 hours
- **d** = maximum number of high priority jobs that concurrently waited more than a minute in the last hour
- **e** = current number of reserved processors (this number can change if **Reuse Idle Processors** is enabled)

If	Then
c is less than **a** This means that some reserved processors weren't used.	The number of processors available to lower priority jobs is increased by **a – c** (up to the value of **a**).
e is less than **b AND d** is more than **0**	The number of reserved processors is increased by **d** (up to the value of **b**) The system can't increase the number of reserved processors over the limit set for **Number of Processors Reserved**. In other words, **e** can't be greater than **b**.

SuiteScript Monitoring, Auditing, and Logging

Applies to: SuiteScript 2.x | SuiteCloud Developer

SuiteScript execution can be monitored and audited using script logs. You can view Script execution details on the Script page, the Script Deployment page, and in the SuiteScript Debugger. You can also view a list of all records that have a user event script or a client script associated with a record on the Scripted Record page. And, you can also set runtime options to specify when a script is executed. The Application Performance Management (APM) SuiteApp can be used to view and manage the performance of NetSuite customizations and business critical operations. For more information, see the following help topics:

- Using the Script Execution Log Subtab
- Viewing a List of Script Execution Logs
- The Scripted Records Page
- SuiteScript Monitoring with the Application Performance Management (APM)
- Setting Runtime Options
- Governance on Script Logging
- Using the Context Filtering Subtab
- Reviewing Outbound HTTPS and SFTP Requests

Using the Script Execution Log Subtab

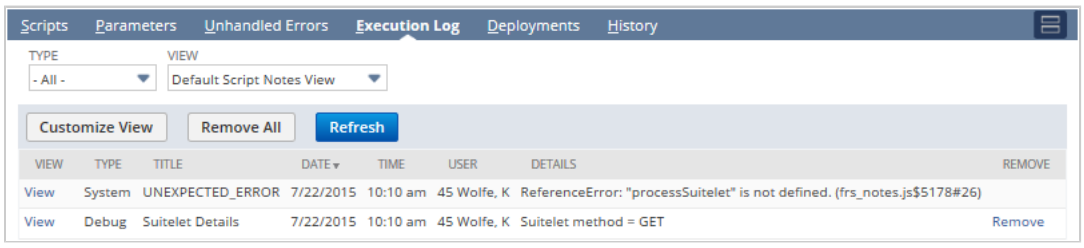
Applies to: SuiteScript 2.x | SuiteCloud Developer

Script execution details are logged on the Execution Log subtab included on the Script page, Script Deployment page, and on the SuiteScript Debugger. You can search and customize the logs to find what you need.

The Execution Log subtab displays all dates and times in the local time zone set in the user preferences.

Important: When you're using the SuiteScript Debugger to debug a script, all logging details appear on the Execution Log tab in the Debugger. You'll need to deploy the script first to see logging details on the Script Deployment page's Execution Log subtab.

The following figure shows two types of execution log messages for a Suitelet.

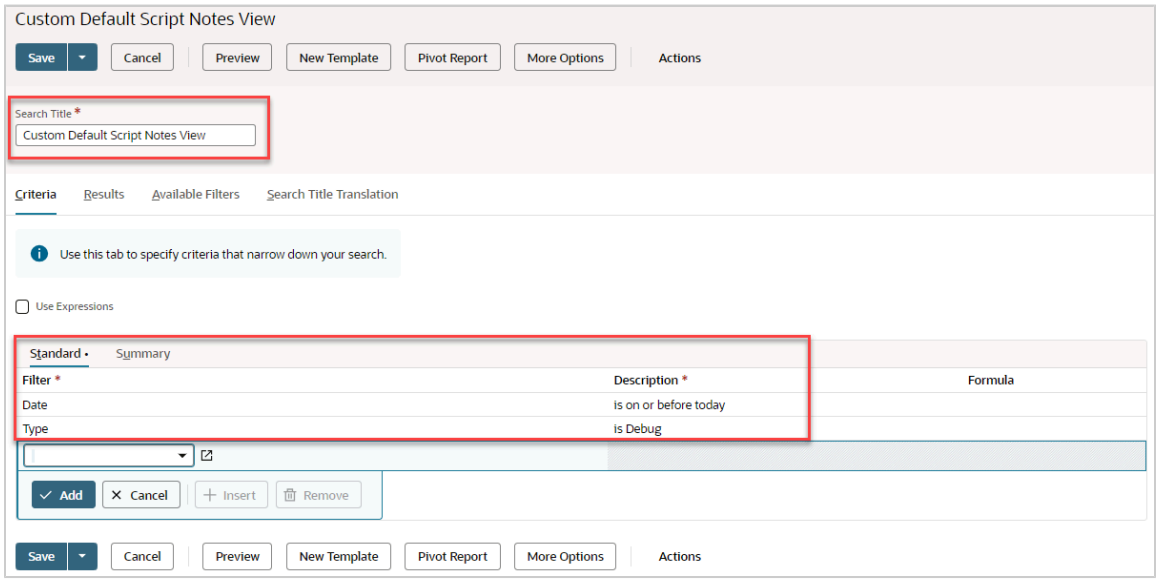


The first message is an unexpected error that is generated because a Suitelet script called an undefined method. The second message is user-generated by the following line(s) in this Suitelet code using the SuiteScript 2.x N/log Module:

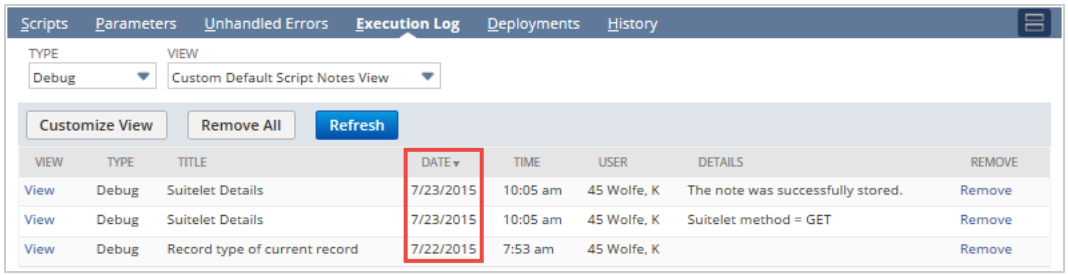
```
log.debug({ title: 'Suitelet Details', details: 'Suitelet method = ' + request.getMethod()
});
```

By default, the Execution Log subtab shows the Default Script Notes View. This default view shows all log types from the current day's script executions.

To view script executions for days other than the current day, click the Customize View button. Specify search criteria such as script execution dates, details, names, and script types, and then name your custom view in the Search Title field, as shown here:



The following figure shows a customized view of the execution log. The new view type has been selected from the View list. This view shows log details for days other than the current day.



For more information about viewing script execution logs, see Viewing a List of Script Execution Logs.

Important: The capacity for script execution logs on the Execution Log subtab is shared by customers on the same database. For further protection against excessive logging, script execution logs are governed by a total storage limit on each instance of the NetSuite database. On each NetSuite server, if the database table that stores logs reaches this limit, all logs (across all customers on that server) are purged. This means that you may have logs up to 30 days if the volume is low, but you may have logs of less than a week if the log volume is extremely large. Use the Script Execution Log page for logs up to 30 days, regardless of volume. For more information about governance of script logs, see Governance on Script Logging.

Viewing a List of Script Execution Logs

Applies to: SuiteScript 2.x | SuiteCloud Developer

The Script Execution Log page shows log details across all scripts for 30 days. You can access the Script Execution Log page by going to Customization > Scripting > Script Execution Logs.

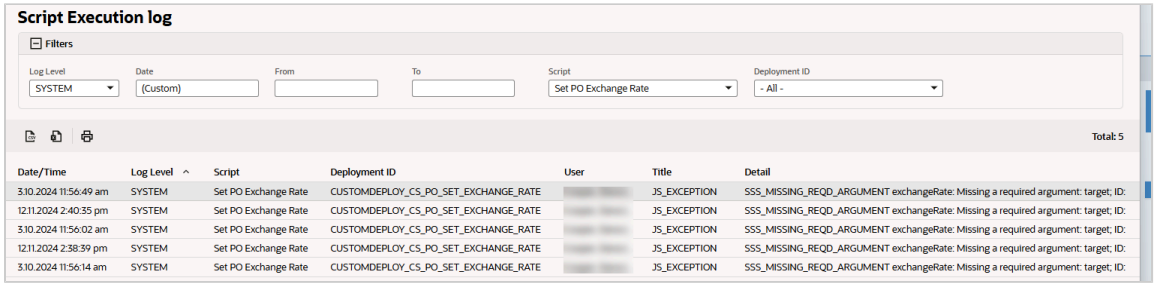
This page displays all dates and times in your local time zone set in the user preferences.

On this page, you can do the following:

- Search for specific logs using filter options, such as log level, execution date range, and script name. You can view other logs by using filter options.
- Download the list as a CSV file or an Excel spreadsheet. Downloads may be limited to 10,000 entries.
- Print the log list.

Note: These logs will not be moved with your NetSuite account to the new data center built on Oracle Cloud Infrastructure (OCI). For the first 29 days after the move, the displayed values will be calculated using data stored since the date of the move, rather than from the last 30 days.

The following figure shows a script execution log list filtered to show scripts that violated the web services and RESTlet account concurrency governance.



The Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

You can view a list of all records that have a script or workflow associated with a record on the Scripted Records page. By default, the list only shows records with at least one script or workflow attached.

Tip: To view a list of all records in your account, check the Show Undeployed box. You can also use the Script filter list to show only records associated with specific scripts.

The Scripted Records page is helpful for troubleshooting problematic scripts and workflows. You can use the page to:

- Specify the execution order of scripts associated with each specific record type
- Edit script deployment statuses
- Deploy or inactivate script deployments
- Change workflow statuses
- Set workflow initiation details

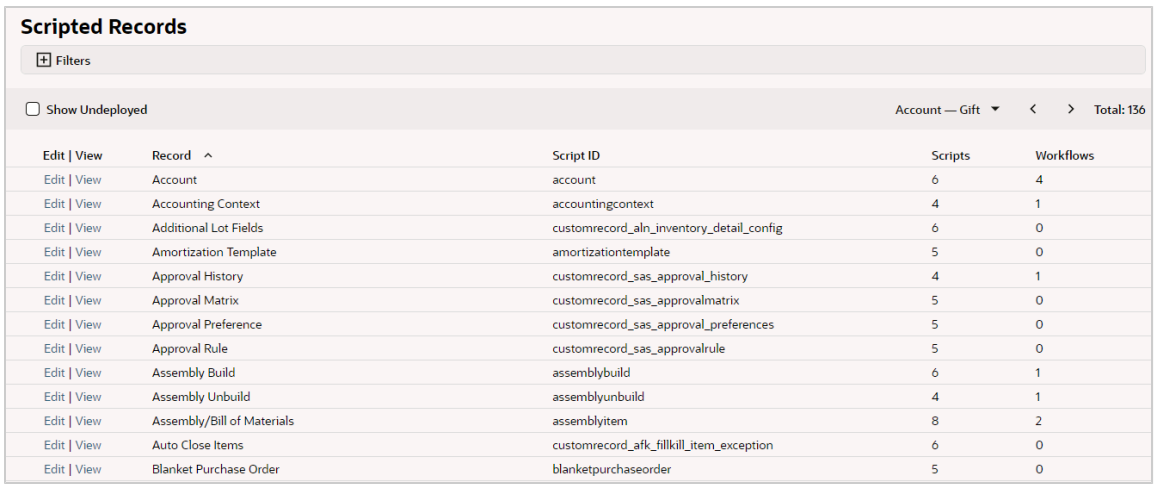
To learn more about the Scripted Records page, see the following topics:

- Accessing the Scripted Records Page
- Scripted Records Subtabs
- Execution Order of Deployed Scripts

Accessing the Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

Access the Scripted Records page by going to Customization > Scripting > Scripted Records.



Edit | View	Record	Script ID	Scripts	Workflows
Edit | View	Account	account	6	4
Edit | View	Accounting Context	accountingcontext	4	1
Edit | View	Additional Lot Fields	customrecord_aln_inventory_detail_config	6	0
Edit | View	Amortization Template	amortizationtemplate	5	0
Edit | View	Approval History	customrecord_sas_approval_history	4	1
Edit | View	Approval Matrix	customrecord_sas_approvalmatrix	5	0
Edit | View	Approval Preference	customrecord_sas_approval_preferences	5	0
Edit | View	Approval Rule	customrecord_sas_approvalrule	5	0
Edit | View	Assembly Build	assemblybuild	6	1
Edit | View	Assembly Unbuild	assemblyunbuild	4	1
Edit | View	Assembly/Bill of Materials	assemblyitem	8	2
Edit | View	Auto Close Items	customrecord_afk_fillkill_item_exception	6	0
Edit | View	Blanket Purchase Order	blanketpurchaseorder	5	0

Each record is shown on the Scripted Records page. To access a specific scripted record, do the following:

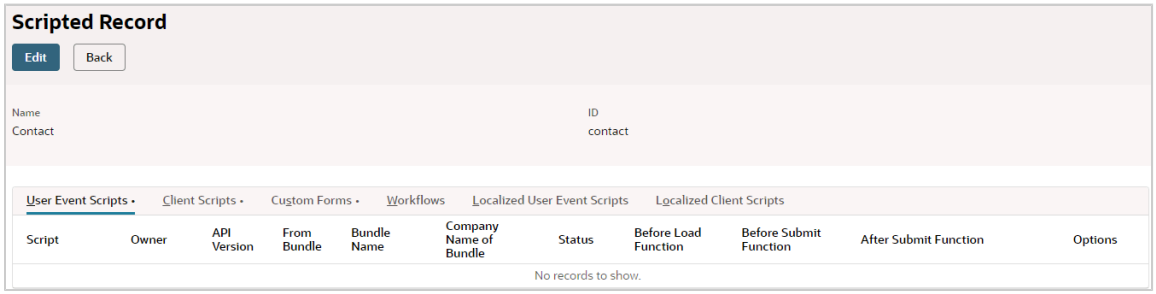
- Click **Edit** to change a record's script deployment order or to deploy or undeploy scripts for that record type, or
- Click **View** to view information about the scripts associated with a record type.

Scripted Records Subtabs

Applies to: SuiteScript 2.x | SuiteCloud Developer

When you open a record from the Scripted Records page, subtabs for each customization type is shown. The subtabs quickly show which scripts or workflows are deployed. Use these subtabs to view or update deployments.

Clicking on a script or workflow from a subtab opens the corresponding script or workflow record.



To learn more about each subtab, see:

- User Event Scripts and Localized User Event Scripts Subtabs on the Scripted Records Page
- Client Scripts and Localized Client Scripts Subtabs on the Scripted Records Page
- Custom Forms Subtab on the Scripted Records Page
- Workflows Subtab on the Scripted Records Page (if enabled)

User Event Scripts and Localized User Event Scripts Subtabs on the Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

Use the User Event Scripts and Localized User Event Scripts subtabs on the Scripted Records page to:

- Rearrange a script's execution order. You can drag scripts up or down the list to determine when each script executes. For more information, see Execution Order of Deployed Scripts.
- Change a script's deployment status.
- Deploy a script by checking the Deployed box.
- View information about the user event scripts, both localized and non-localized, for a record type:

Column Name	Description
Script	Shows the script name. Click the name to open its script deployment record. For more information, see the help topic Script Deployment.
Owner	Shows the script owner.
API Version	Shows the SuiteScript version of the script.
From Bundle	Lists the ID of the bundle that contains the script. Click the bundle ID to view bundle details.
Bundle Name	Shows the bundle name. Click the name to view bundle details.
Company Name of Bundle	Shows the bundle name assigned by the company.
Status	Shows whether the script status is Testing or Released. For details, see Setting Script Deployment Status.
Countries	Lists the countries where the script runs.

Column Name	Description
	Note: The Countries column appears only on the Localized User Event Scripts subtab.
Before Load Function	Lists the functions that execute at the `beforeLoad` entry point.
Before Submit Function	Lists the functions that execute at the `beforeSubmit` entry point.
After Submit Function	Lists the functions that execute at the `afterSubmit` entry point.
Options	Shows any script options and their current settings.

Important: Deploying many user event scripts for an entry point can impact performance. As a rule, keep each entry point to about ten scripts or fewer.

For information about user event script entry points, see the help topic SuiteScript 2.x User Event Script Entry Points and API.

For information about localized user event scripts and the localization context, see Localization Context.

Client Scripts and Localized Client Scripts Subtabs on the Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

Use the Client Scripts and Localized Client Scripts subtabs on the Scripted Records page to:

- Rearrange a script's execution order. You can drag scripts up or down the list to determine when each script executes. For more information, see Execution Order of Deployed Scripts.
- Change a script's deployment status.
- Deploy a script by checking the Deployed box.
- View information about the client scripts, both localized and non-localized, for a record type:

Column Name	Description
Script	Shows the script name. Click the name to open its script deployment record. For more information, see the help topic Script Deployment.
Owner	Shows the script's owner.
API Version	Shows which API version the script uses.
From Bundle	Lists the ID of the bundle that contains the script. Click the bundle ID to view bundle details.
Bundle Name	Shows the bundle name. Click the name to view bundle details.
Company Name of Bundle	Shows the bundle name assigned by the company.
Status	Shows whether the script status is Testing or Released. For details, see Setting Script Deployment Status.
Page Init Function	Lists the functions that execute at the `pageInit` entry point.
Save Record Function	Lists the functions that execute at the to the `saveRecord` entry point.
Field Changed Function	Lists the functions that execute at the `fieldChanged` entry point.

Column Name	Description
Validate Line Function	Lists the functions that execute at the `validateLine` entry point.

Note: You can deploy up to 10 client scripts (localized or not) per record. To go past that limit, ask the person in your company who manages NetSuite to check the Remove Client Script Deployment Limit per Record preference. For details, see the help topic Setting General Account Preferences.

For information about client script entry points, see the help topic SuiteScript 2.x Client Script Entry Points and API.

For more information about localized client scripts and the localization context, see Localization Context.

Custom Forms Subtab on the Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

Use the **Custom Forms** subtab on the Scripted Records page to:

- View information about the custom forms for a record type:

Column Name	Description
Form	Shows the custom form's name.
From Bundle	Lists the ID of the bundle that contains the script. Click the bundle ID to view bundle details.
Bundle Name	Shows the bundle's name. Click the name to view bundle details.
Company Name of Bundle	Shows the bundle name assigned by the company.
Script	Shows the script name. Click the name to open its script deployment record. For information, see the help topic Script Deployment.
Page Init Function	Lists the functions that execute at to the `pageInit` entry point.
Save Record Function	Lists the functions that execute at the `saveRecord` entry point.
Field Changed Function	Lists the functions that execute at the `fieldChanged` entry point.
Validate Line Function	Lists the functions that execute at the `validateLine` entry point.

Workflows Subtab on the Scripted Records Page

Applies to: SuiteScript 2.x | SuiteCloud Developer

Use the Workflows subtab on the Scripted Records page to:

- Update the workflow status using the **Status** list. You can set the workflow state to Suspended, Not Initiating, Testing, or Released.
- Set the initiation trigger type to All, Before Record Load, Before Record Submit, or After Record Submit. Use the **Trigger Type** list to set the type of trigger on which the workflow is to initiate.
- Choose the workflow event triggers:
  - Check the **On Create** box to run the workflow when a record is created.
  - Check the **On View or Update** box to run the workflow on view or update events.

To learn more about events that trigger workflows, see the help topic Initiating a Workflow on an Event.

- View information about the workflows for a record type:

Column Name	Description
Workflow	Shows the workflow name. Click the name to open the workflow. For more information, see the help topic Workflow Manager Interface.
Internal ID	Shows the workflow's internal ID. Click the ID to open the workflow. For more information, see the help topic Workflow Manager Interface.
Description	If available, shows the workflow description.
From Bundle	Shows the bundle ID. Click the bundle ID to view bundle details.
Bundle Name	Shows the bundle's name. Click the name to view bundle details.
Company Name of Bundle	Shows the bundle name assigned by the company.
Owner	Shows the workflow's owner.
Status	Shows the workflow's current state, whether it is in Testing, Released, Not Initiating, or Suspended. For details, see Setting Script Deployment Status.
Trigger Type	Shows which client event triggers the workflow: All, Before Record Load, Before Record Submit, or After Record Submit. For details, see the help topic Client Triggers Reference.
On Create	Shows whether the workflow starts when you create a record. For details, see the help topic Initiating a Workflow on an Event.
On View or Update	Shows whether the workflow starts when you view or update a record. For details, see the help topic Initiating a Workflow on an Event.

For information about SuiteFlow workflows, see the help topic SuiteFlow Overview. The SuiteFlow feature must be enabled to use workflows. See the help topic Enabling SuiteFlow.

Execution Order of Deployed Scripts

Applies to: SuiteScript 2.x | SuiteCloud Developer

NetSuite sets script execution order in two steps: it first groups deployments by entry point, then executes them in the order shown on the Scripted Records page.

Note: Besides the two-stage sequence, keep these behaviors in mind:

- NetSuite sets execution priority at the script level—not for individual entry points.
- In user event scripts, the `afterSubmit` phase starts with scripts that also have a `beforeSubmit` entry point, then follows the listed order.

The following example involves four deployed user event scripts on any record. The table shows their order on the User Event Scripts subtab, and the entry points each script defines.

List Position in the Subtab	Script	Defined Entry Points
1	Script A	`beforeLoad`

List Position in the Subtab	Script	Defined Entry Points
		`afterSubmit`
2	Script B	`beforeLoad`
		`beforeSubmit`
3	Script C	`afterSubmit`
4	Script D	`beforeSubmit`
		`afterSubmit`

NetSuite executes the entry point functions in this order:

- `beforeLoad` – Script A > Script B
- `beforeSubmit` – Script B > Script D
- `afterSubmit` – Script D > Script A > Script C

Although Script D is listed last on the subtab, its `afterSubmit` function runs before those in Scripts A and C because it also has a `beforeSubmit` entry point.

Note: The following scenarios can also affect execution order:

- If the record also has a workflow with an `afterSubmit` event, NetSuite finishes the user event script first.
- When both localized and non-localized scripts exist on a record, non-localized scripts run first. For more details on localized client scripts, see Localization Context.

SuiteScript Monitoring with the Application Performance Management (APM)

Applies to: SuiteScript 2.x | SuiteCloud Developer

The Application Performance Management (APM) SuiteApp helps you monitor and optimize your NetSuite customizations and key business operations with a performance dashboard.

With APM, you get data visualizations, page time summaries, and script analysis tools to help you boost the NetSuite UI's speed.

For additional information about APM, see the help topic Application Performance Management (APM).

Setting Runtime Options

Applies to: SuiteScript 2.x | SuiteCloud Developer

If you haven't already created a Script Deployment record for your SuiteScript file, see the help topic SuiteScript 2.x Entry Point Script Creation and Deployment.

After you've set up your script deployment, see these help topics to learn about more deployment and runtime options:

- Setting Script Execution Event Type from the UI
- Setting Script Execution Log Levels
- Executing Scripts Using a Specific Role
- Setting Available Without Login
- Setting Script Deployment Status
- Defining Script Audience
- Creating Script Parameters Overview

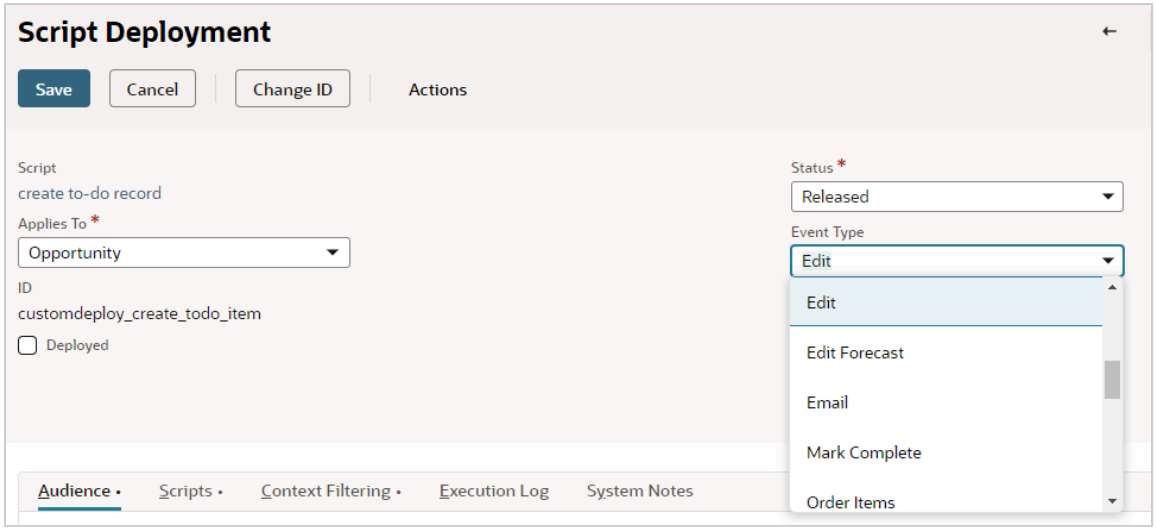
Also see these help topics for information related to script deployments, but not necessarily specific to any deployment/runtime options.

- Script Deployment
- Methods of Deploying a Script

Setting Script Execution Event Type from the UI

Applies to: SuiteScript 2.x | SuiteCloud Developer

On the Script Deployment page, use the Event Type field to choose the event that'll trigger your script (see following figure). If the Event Type field is left blank, the script will execute only on the events specified in the script file.



Note: The Event Type field is available on Script Deployment pages for Suitelet, user event, and record-level client scripts only.

Use the Event Type field to specify a script execution context at the time of script deployment, without having to modify your .js script file. After you select an event type and click Save, the deployed script will execute only on that event, regardless of the event types specified in the .js script file.

Note that event types selected on the Script Deployment page take precedence over the event types specified in the .js script file. For example, if the **create** event type is specified in the script, selecting **Edit** from the Event Type field on the Script Deployment page will restrict the script from running on any event other than Edit.

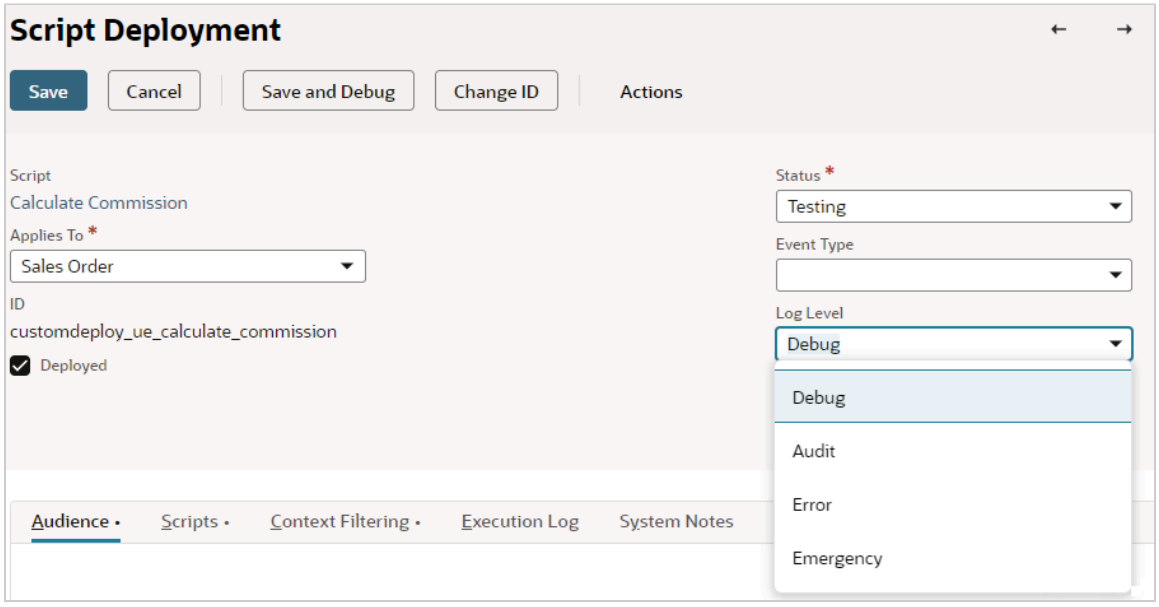
The following snippet is from a user event script. Notice that the event type specified in the code is create (`context.UserEvent.CREATE`). If the Edit event type is specified on the Script Deployment page, the script will execute only when the specified record is edited, not created.

```
function followUpCallAfterSubmit(type) { // execute the logic in this script onlyif a new customer is created if (context.type !== context.UserEvent Type.CREATE) return; // obtain a handle to the newly created customer record var customerRecord = context.newRecord; // remainder of script..... }
}
```

Setting Script Execution Log Levels

Applies to: SuiteScript 2.x | SuiteCloud Developer

In the Log Level field on the Script Deployment page, specify which log entries you want to appear on the Execution Log subtab:



The Log Level field is essentially used as a basic filtering mechanism. Each log entry written with N/log Module methods specifies a log level based on the specific method used: log.audit(options), log.debug(options), log.emergency(options), log.error(options).

Select one of the following log levels from the Log Level field:

- **Debug**: For scripts in testing mode. Selecting this level will show all log messages (debug, audit, error, and emergency).
- **Audit**: For scripts going into production. Selecting this level will show the events that have occurred during the processing of the script (for example, "A request was made to an external site.").
- **Error**: For scripts going into production. Selecting this level will show only unexpected script errors.

- **Emergency**: For scripts going into production. Selecting this level will show only the most critical errors in the script log.

Important: NetSuite governs the amount of logging that can be done by a company in any 60 minute time period. For complete details, see Governance on Script Logging.

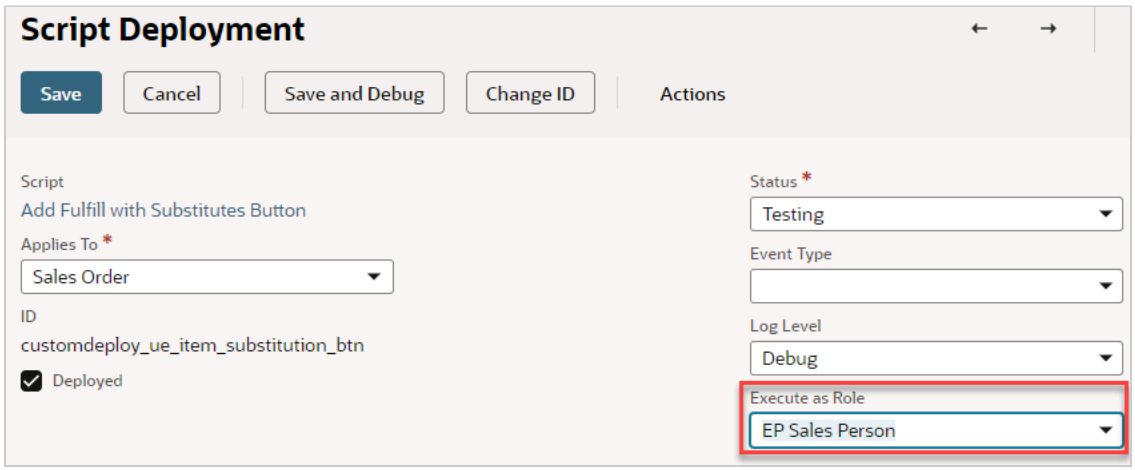
Note: The log level you specify on the Script Deployment page is independent of any error handling within your script.

See Using the Script Execution Log Subtab for details on how to further customize your view of all log entries.

Executing Scripts Using a Specific Role

Applies to: SuiteScript 2.x | SuiteCloud Developer

The **Execute as Role** field sets which role's permissions uses at runtime. It lists all standard and custom roles, plus Current User, which runs the script with the logged-in user's permissions. For example, if you choose Sales Person, the script always runs with that role's permissions—even if an Administrator triggers it. If no existing role fits, create a custom one with the needed permissions and choose it instead.



Important: Keep these behaviors in mind:

- When one script triggers another, the triggered script runs with the first script's role—not the role set on its own deployment.
- While you're testing, set the deployment status to **Testing** so the script runs only for you and the selected role; other users won't trigger it.

For more details on roles in script deployments, see the following topics:

- Script Types That Support Execute as Role

- Setting Your Script to Run With Administrative Privileges

For an overview of roles and permissions in SuiteScript, see Setting Roles and Permissions for SuiteScript. For limits that are specific to client scripts, see the help topic Client Script Role Restrictions.

Script Types That Support Execute as Role

Applies to: SuiteScript 2.x | SuiteCloud Developer

The **Execute as Role** field appears only on script types triggered by user actions that may need more permissions than the initiating user has. Choose the role NetSuite should apply at runtime when the current user doesn't have enough access.

You can set it for the following script types:

- Suitelet
- User Event
- Portlet
- Mass Update
- Workflow Action

To know more about script types, see the help topic SuiteScript 2.x Script Types.

Setting Your Script to Run With Administrative Privileges

Applies to: SuiteScript 2.x | SuiteCloud Developer

Sometimes a script needs to do something the initiating user's role can't. In these cases, you can execute the script as Administrator so it can read or change restricted records—or perform other high-privilege tasks—without failing.

Choose **Administrator** in the deployment's **Execute as Role** field only when the script needs full privileges, regardless of who's logged in.

For example:

- A script that creates follow-up tasks after a sales order is saved and needs to read employee records—data many sales roles can't access.
- Bundle installation scripts always need Administrator rights.
- Scripts that run without a NetSuite session—like web store scripts or Suitelets marked Available Without Login—when they need NetSuite data. For more information, see Setting Available Without Login.

To run a script as Administrator:

1. Go to Customization > Scripting > Script Deployments
2. Edit the deployment you want to change.
3. In **Execute as Role**, select **Administrator**.
4. Click Save.

Important: Because the Administrator role gives unrestricted access, use it sparingly and only when other roles can't meet the script's needs.

Setting Available Without Login

Applies to: SuiteScript 2.x | SuiteCloud Developer

You can allow Suitelet scripts to be executed without a login. Select the Available Without Login box on the Script Deployment page give users access to the Suitelet even if they're not logged in.

Warning: Suitelets configured as available without login should not be used in integration use cases, including SDN partner SuiteApps. Suitelets configured as available without login are a violation of Built for NetSuite (BFN) standards.

To ensure that users without an active NetSuite session can access the Suitelet, **Online Form User** must be selected in the External Roles field. All other roles that require access to the script should also be selected in the Internal Roles and External Roles fields. Values from the Departments, Groups, Employees, and Partners fields must be cleared.

Note: The Available Without Login box is available on the Script Deployment page for Suitelets only. When working with Client Scripts and externally available Suitelets, the Website feature must be enabled.

To be able to select the **Online Form User** role in the External Roles field, the Online Forms feature must be enabled. To enable the feature, go to Setup > Company > Enable Features. Click the CRM subtab, and check the **Online Forms** box in the Marketing section. If the role is still not available after enabling the feature, you may need to log out and log back in to see the updated roles list.

The following are some uses cases when you might want to make a Suitelet externally available (however, when considering these use cases, keep in mind that using Suitelets externally without login is forbidden in BFN):

- Hosting one-off online forms (for example, capturing partner conference registrations).
- Inbound partner communication (such as, listening for payment notification responses from PayPal or Google checkout, or for generating an unsubscribe request from email campaigns page, which requires access to account information but should not require a login or hosted website).
- For Facebook, Google, and Yahoo mashups in which the Suitelet lives in those websites but needs to communicate to NetSuite using POST requests.

Important: Note that the data contained within the Suitelet will be less secure when it is allowed to be accessed (using Suitelet execution) without login.

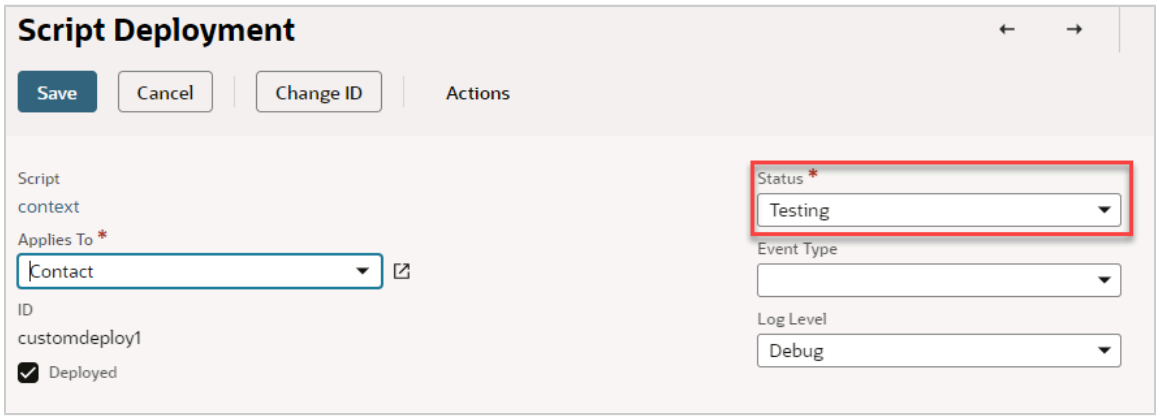
For information about errors when using the Available Without Login URL, see the help topic Errors Related to the Available Without Login URL.

Setting Script Deployment Status

Applies to: SuiteScript 2.x | SuiteCloud Developer

A script's deployment status can be set to Testing or Released. When the status is set to Testing, the script will execute for the script owner and specified audience. When the status is set to Released, the script will execute in the accounts of all specified audience members

To set the deployment status, select either Testing or Release from the Status field on the Script Deployment page:



Note: The Testing and Released statuses do not apply to scheduled scripts. To learn about scheduled script deployment statuses, see the help topic Scheduled Script Submission.

Status Set to Testing

When a script's deployment status is set to Testing, only the script owner can execute the script. The script owner can test the script functionality in a different role using the Audience subtab, but the script

will not be available for other users until the status is set to Released. For information, see Using the Audience Subtab to Test Scripts.

Note that when using the SuiteScript Debugger to test scripts, the script's deployment status must be set to Testing. You cannot debug a Deployed script if the status has been set to Released. (See Debugging Deployed SuiteScript 1.0 and SuiteScript 2.0 Server Scripts and Debugging Deployed SuiteScript 2.1 Server Scripts for more information about using the SuiteScript Debugger to test existing scripts.)

If you are working with Suitelet Script Deployment records, see the help topic Errors Related to the Available Without Login URL,to learn how Testing status affects internally and externally available Suitelets.

Note: A bundle installation script can't execute in target accounts if its deployment status is set to Testing.

Status Set to Released

When a script's deployment status is set to Release, the script executes in the accounts of all specified audience members. (See Defining Script Audience for information about defining script audiences.) When the deployment status is set to Released, the script is considered to be “production ready.”

Note that if you do not specify any values on the Audience subtab, the script will execute only for the script owner, even if the script deployment status is set to Released.

Note: Bundle installation scripts and scheduled scripts do not have an audience. If the deployment status is set to Released for a bundle installation script, the script will execute automatically in target accounts when the associated bundle is installed or updated.

If you are working with Suitelet Script Deployment records, see the help topic Errors Related to the Available Without Login URL, to learn how Released status affects internally and externally available Suitelets.

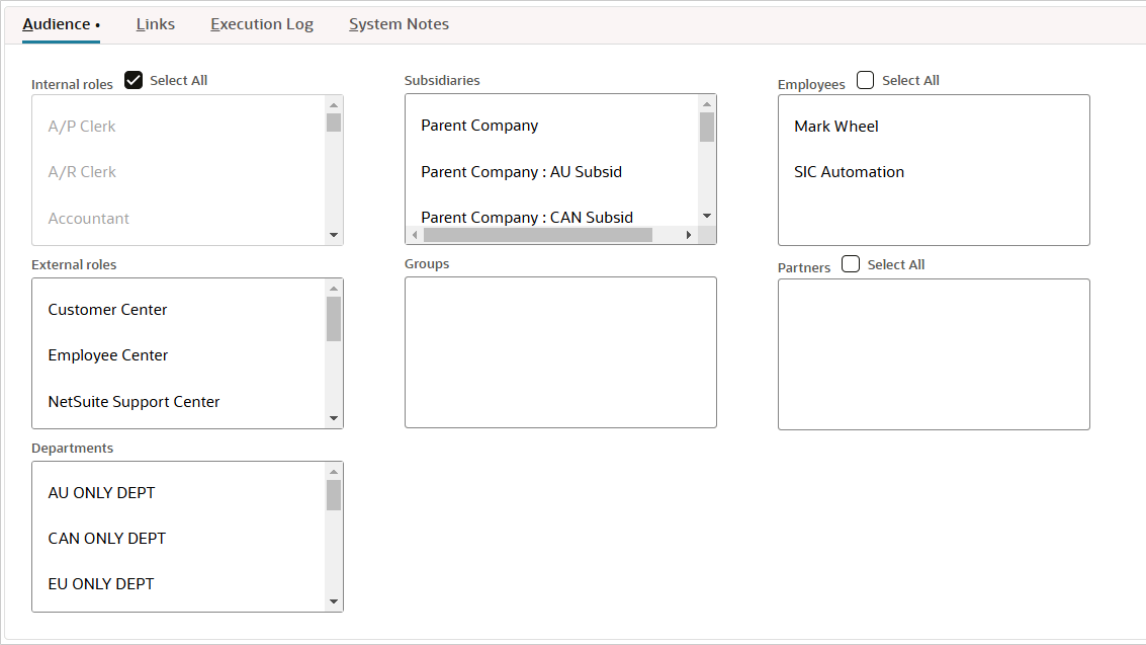
Defining Script Audience

Applies to: SuiteScript 2.x | SuiteCloud Developer

On the Audience subtab on the Script Deployment page, define the audience access for the script. When the script is deployed, it will only run in the roles specified in the Audience subtab.

The Audience subtab is included on the Script Deployment page for these script types: Suitelet, portlet, user event, global client (deployed at the record-level), RESTlet, mass update, and workflow action. You cannot specify an audience for scheduled scripts, map/reduce scripts, or bundle installation scripts.

Note: Mass update script deployments and mass updates can both be assigned an audience. It is the user's responsibility to make sure the two audiences are in sync. For information about working with the mass update script type, see the help topic SuiteScript 2.x Mass Update Script Type.



General guidelines for specifying an audience:

- If you don't specify any values on the Audience subtab, the script will execute only for the script owner even if the script deployment status is set to Released. For information about the differences between the Released and Testing deployment statuses, see Setting Script Deployment Status.
- If you choose both role and department options on the Audience subtab, a user must belong to one of the selected roles AND one of the selected departments to execute the script. If you choose options for any other combinations of types (groups, employees, and partners), a user need only belong to a selected option of one type OR of another.
- If you want the script to run only for specific users, select the appropriate roles, departments, groups, employees, or partners. If you want the script to run for all NetSuite users, do the following:
  - Check the Select All box beside the Internal Roles field
  - Select all roles in the External Roles field

  Note: When giving access to multiple roles, particularly external roles, ensure that each selected role has a valid need to access the script.

Be sure to save the Script Deployment record after all audience members have been defined.

If you are working with Suitelet Script Deployment records, see the help topic Errors Related to the Available Without Login URL, which discusses the relevance of the Select All box as it pertains to internally and externally available Suitelets.

Using the Audience Subtab to Test Scripts

When the Status field on the Script Deployment record is set to Testing, you can test scripts assigned to specific audiences if you are a member of that audience type. For information about the Testing deployment status, see Setting Script Deployment Status.

For example, if you have a script that you want to execute for everyone in the Support Management role, you can log into NetSuite and then switch to the Support Management role to test the script. If the deployment status for the script is set to Testing, the script will run only for the script owner. After you have determined that the script runs as expected for the Support Management role, you can change the script's deployment status to Released. After it is set to Released, it will execute in the account for all those assigned to the Support Management role.

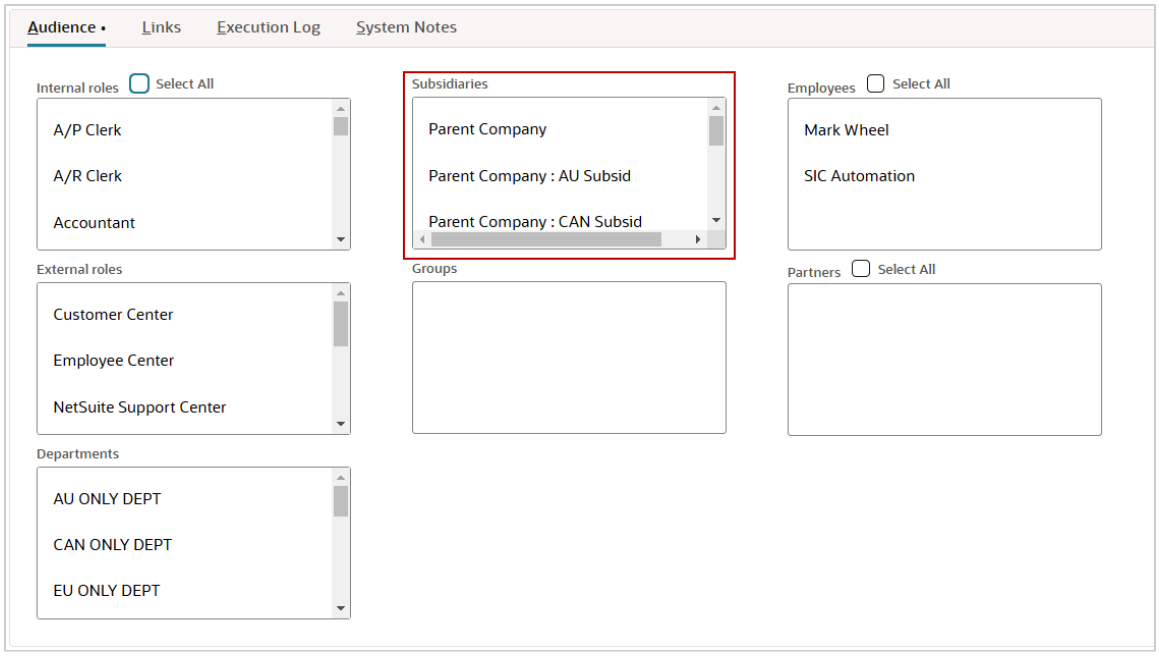
This is a good approach for script owners to verify that their script will run in the accounts for specified roles, departments, groups, employees, or partners.

Using the Audience Subtab in OneWorld Accounts

Script authors and owners who are developing scripts for NetSuite OneWorld accounts can specify a script audience based on subsidiary. In OneWorld accounts, the Audience subtab includes a Subsidiaries multiselect field. Don't forget to save your changes after selecting subsidiaries.

If the currently logged user is a member of a subsidiary which is selected in the multiselect field on the Script Deployment page, then the script is executed.

Script owners working in a OneWorld account that has multiple subsidiaries can select the subsidiaries they want their script to run in, and then log into an account of one of the specified subsidiaries. When a script's deployment status is set to Testing, the script will execute for the script owner and specified audience. This is a good way for script owners to verify that the script will run in accounts for specified subsidiaries.



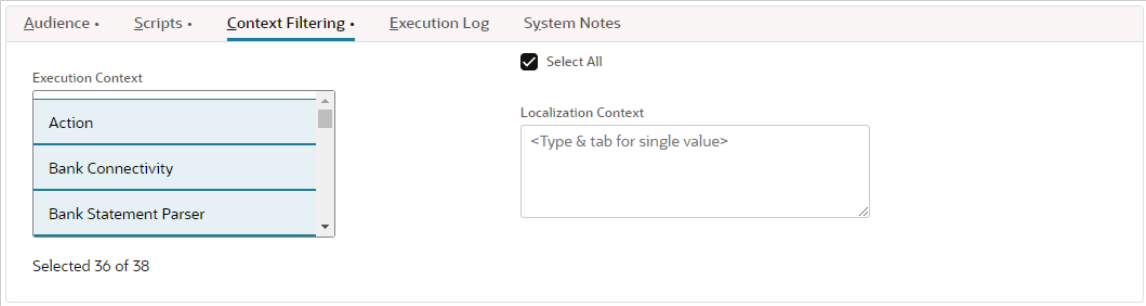
Using the Context Filtering Subtab

Applies to: SuiteScript 2.x | SuiteCloud Developer

In a script deployment record, the Context Filtering subtab lets you specify the contexts in which a script will run. You can use context filtering to ensure that a script runs only when necessary, which can improve performance.

You can specify the execution context and localization context for a script:

- Execution Context
- Localization Context



You can use the N/recordContext module to get the context type for a record. For more information, see the help topic N/recordContext Module.

Execution Context

Execution contexts provide information about how or when a client script or user event script is triggered to execute. For example, a script can be triggered in response to an action in the NetSuite application, or an action occurring in another context, such as a web services integration. You can use execution context filtering to ensure that your scripts are triggered only when necessary. You can specify that a script should execute only in certain contexts, and this filtering can improve performance in contexts where the script is not required. For more information, see the help topic Execution Contexts.

On the Context Filtering subtab of a script deployment record, you can select the contexts in which you want your script to execute. Your script will not execute if it is triggered in a context that is not selected. By default, all contexts are selected except for Web Application and Web Store. Often, it is not required to trigger scripts in these contexts, so they are disabled by default to improve performance. If you want your script to be triggered in these contexts, be sure to select them explicitly when you create your script deployment record.

To define the execution context:

1. Define a new script deployment record or edit an existing one. For more information, see the help topic Script Deployment.
2. In the script deployment record, click the **Context Filtering** subtab. By default, all execution contexts are selected except for Web Application and Web Store.
3. In the **Execution Context** field, select the contexts in which you want the script to execute. Cancel the selection of contexts in which the script should not execute.
4. Click **Save**.

Localization Context

You can define the localization context in which a client or user event script can execute. You can enable the script execution for specific localization contexts programmatically and in the UI.

Localization filtering allows you to execute a script based on the country of the active record or transaction. For a list of records that support localization context, see Records that Support Localization Context.

NetSuite automatically determines the localization context for records and transactions based on their values for country fields such as subsidiary and tax nexus. It is important to understand this determination before you set up localization context filtering for scripts. For more information, see the help topic Record Localization Context.

You can specify the execution order of localized client and user event scripts. A maximum of 10 localized/non-localized client scripts are supported. For more information, see The Scripted Records Page.

You can also search for localized scripts by using the N/query Module. For more information, see the help topic N/query Module.

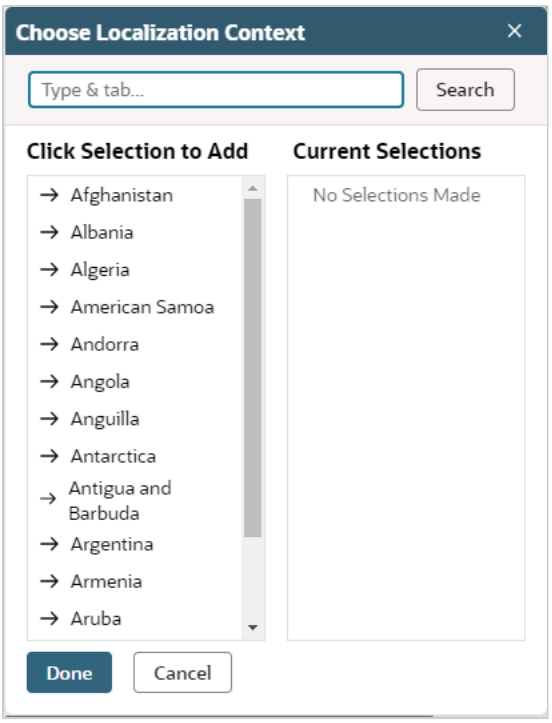
The following table shows how you can specify the localization context based on the script type.

Script Type	Defining Localization Context Filtering
SuiteScript 2.x Client Script Type	Complete the following steps to add localization context filtering to client scripts: 1. Use the localizationContextEnter(scriptContext) and localizationContextExit(scriptContext) entry points in your script. 2. Define the localization context on the Context Filtering subtab on the script deployment record.
SuiteScript 2.x User Event Script Type	Define the localization context on the Context Filtering subtab on the script deployment record only.

To define the localization context:

1. Define or edit your script deployment record. For more information, see the help topic Use the Script Deployment Record.
2. Click the **Context Filtering** subtab. All countries are selected by default.
3. Clear the **Select All** box.
4. Click on the arrow button beside the **Localization Context** field.

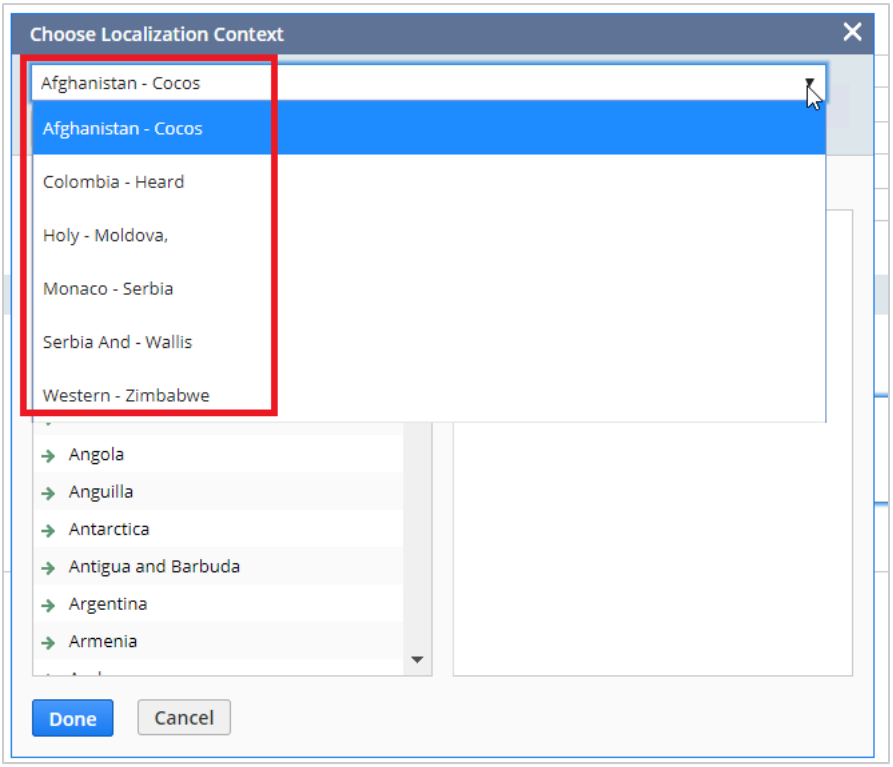
The **Choose Localization Context** popup window is displayed.



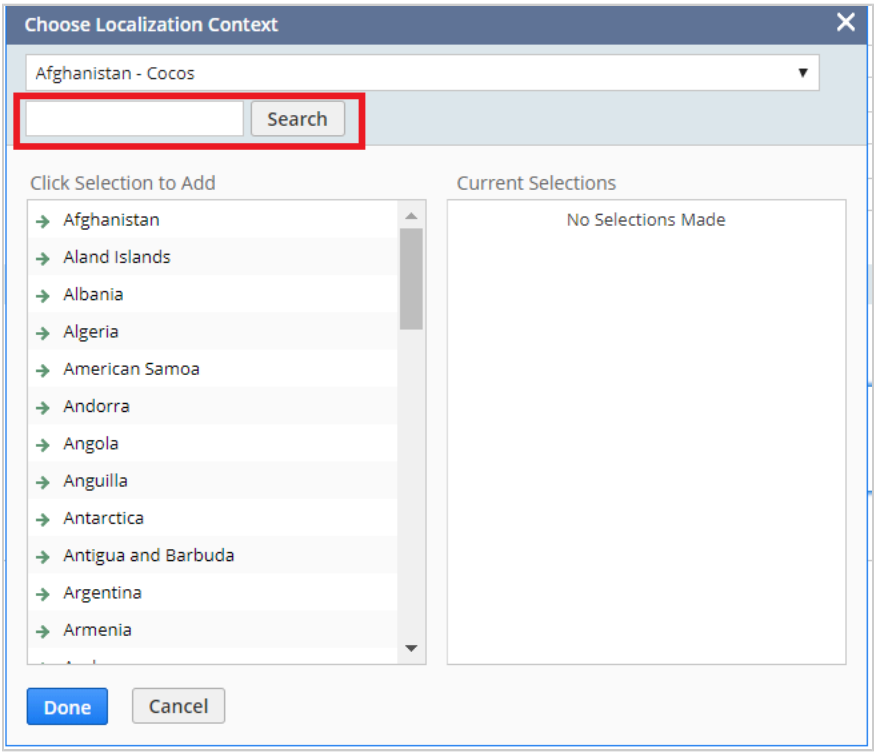
5. Select a country. The selected country is automatically displayed under Current Selections.
6. Click **Done**.

To avoid scrolling through a long list of countries, you can do any of the following:

- The drop-down list on top of the popup window displays the pagination of the countries that are listed alphabetically. Select the pagination based on the first letter of the country that you want to specify and make your selection.



- Enter the country in the **Search** field and select it.



Records that Support Localization Context

Transactions	Items	Entities	Others
Advanced Intercompany Journal Entry (advintercompanyjournalentry)	Assembly Item BOM (assemblyitembom)	Contact (contact)	Tax Type (taxtype)
Assembly Build (assemblybuild)	Description Item (descriptionitem)	Customer (customer)	Tax Code (salestaxitem)
Assembly Unbuild (assemblyunbuild)	Discount Item (discountitem)	Employee (employee)	–
Bin Transfer (bintransfer)	Download Item (downloaditem)	Generic Resource (genericresource)	
Bin Worksheet (binworksheet)	Gift Certificate Item (giftcertificateitem)	Lead (lead)	
Blanket Purchase Order (blanketpurchaseorder)	Item (item)	Other Name (othername)	
Cash Refund (cashrefund)	Item Group (itemgroup)	Partner (partner)	
Cash Sale (cashsale)	Kit Item (kititem)	Project (job)	
Check (check)	Lot Numbered Build/Assembly Item (lotnumberedassemblyitem)	Project Template (projecttemplate)	
Credit Card Charge (creditcardcharge)	Lot Numbered Inventory Item (lotnumberedinventoryitem)	Prospect (prospect)	
Credit Card Refund (creditcardrefund)	Markup Item (markupitem)	Tax Group (taxgroup)	
Credit Memo (creditmemo)	Non-Inventory Part (noninventoryitem)	Vendor (vendor)	
Customer Deposit (customerdeposit)	Other Charge Item (otherchargeitem)	–	
Customer Payment (customerpayment)	Payment Item (paymentitem)		
Customer Payment Authorization (customerpaymentauthorization)	Sales Tax Item (salestaxitem)		
Deposit (deposit)	Serialized Build/Assembly Item (serializedassemblyitem)		
Deposit Application (depositapplication)	Serialized Inventory Item (serializedinventoryitem)		
Customer Refund (customerrefund)	Service (serviceitem)		
Estimate (estimate)	Shipping Item (shipitem)		
Expense Report (expensereport)	Subscription Plan (subscriptionplan)		
Fulfillment Request (fulfillmentrequest)	Subtotal Item (subtotalitem)		
Intercompany Journal Entry (intercompanyjournalentry)	–		
Intercompany Transfer Order (intercompanytransferorder)			
Inventory Adjustment (inventoryadjustment)			
Inventory Cost Revaluation (inventorycostrevaluation)			
Inventory Count (inventorycount)			

Inventory Status Change (inventorystatuschange)			
Inventory Transfer (inventorytransfer)			
Invoice (invoice)			
Item Fulfillment (itemfulfillment)			
Item Receipt (itemreceipt)			
Journal Entry (journalentry)			
Opportunity (opportunity)			
Paycheck (paycheck)			
Paycheck Journal (paycheckjournal)			
Period End Journal (periodendjournal)			
Purchase Contract (purchasecontract)			
Purchase Order (purchaseorder)			
Requisition (purchaserequisition)			
Return Authorization (returnauthorization)			
Revenue Arrangement (revenuearrangement)			
Revenue Commitment (revenuecommitment)			
Revenue Commitment Reversal (revenuecommitmentreversal)			
Sales Order (salesorder)			
Statistical Journal Entry (statisticaljournalentry)			
Store Pickup Fulfillment (storepickupfulfillment)			
Transfer Order (transferorder)			
Vendor Bill (vendorbill)			
Vendor Credit (vendorcredit)			
Vendor Payment (vendorpayment)			
Vendor Return Authorization(vendorreturnauthorization)			
Work Order (workorder)			
Work Order Close (workorderclose)			
Work Order Completion (workordercompletion)			
Work Order Issue (workorderissue)			

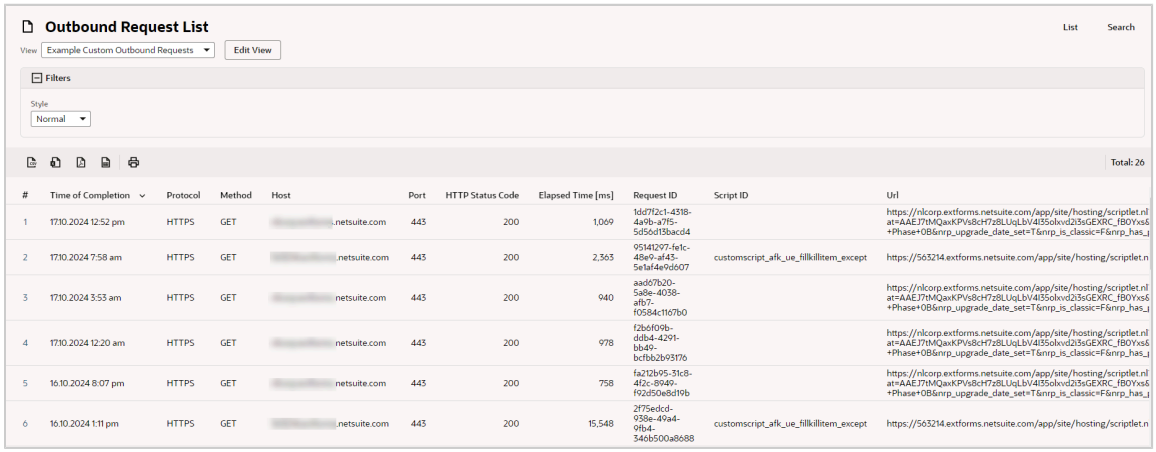
Reviewing Outbound HTTPS and SFTP Requests

Applies to: SuiteScript 2.x | SuiteCloud Developer

A log of outbound requests from NetSuite is available at Setup > Company > Communication > Outbound Request List > Search. The Outbound Requests log includes all outgoing HTTPS and SFTP requests made from your NetSuite account. Each SFTP request from SuiteScript is listed with a link to the specific script deployment that issued the request. The ability to view this type of log is important for auditing account activity. A review of this log can help you to identify requests that result in errors and requests that are not as efficient as they could be.

The Outbound Requests log displays data in a default format that includes the URL, the method, the result, and other useful details. You can modify this format by clicking Customize View. See the help topic Customizing List Views for more information.

The following figure shows a customized view in which the order of columns has been modified:



By default, the Outbound Request List search is accessible only to users with the administrator role. Administrators can provide access to additional users by assigning the Outbound Request permission, a List type permission, to other roles.

Users with the appropriate permissions also can run adhoc searches and create their own saved searches that return data on outbound requests. These capabilities are available at Setup > Company > Communication > Outbound Request List > Search.

Outbound Requests searches are also available from these menu options:

- Lists > Search > Saved Searches > New
- Reports > New Search
- Reports > Saved Searches > All Saved Searches > New

Working with the SuiteScript Records Browser

Applies to: SuiteScript 2.x | SuiteCloud Developer

To access the SuiteScript Records Browser, use the following link:

Go to the SuiteScript Records Browser

Important: When writing SuiteScript, you must use the IDs listed in the NetSuite Help Center. You can access IDs by viewing a NetSuite record's page source, but not all IDs in the source code are supported in SuiteScript. If you create a script that references an unsupported or undocumented ID, and NetSuite later changes the ID, your script may break.

Not all fields in the SuiteScript Records Browser can be set using SuiteScript. Some fields are read only. Check the NetSuite UI to see if a field can be set. Generally, if you can set a field in the UI, you can set it using SuiteScript. If you cannot set a field in the UI, you cannot set it using SuiteScript. However, you can still get the field's value using SuiteScript.

The SuiteScript Records Browser provides a summary of all records, fields, sublists, search joins, search filters, search columns, and record transformations that are supported in SuiteScript. Information about elements is displayed as a series of tables.

To find SuiteScript-supported records and IDs:

1. Open the SuiteScript Records Browser.

   Only records that officially support SuiteScript are listed in the SuiteScript Records Browser.
2. Click the record you want to reference in SuiteScript.

   All IDs currently supported for the record are listed.

The following table provides examples of objects and methods that use each type of ID:

ID Type	Object Examples	Method Examples
Field IDs	▪ serverWidget.Field ▪ record.Field ▪ record.Record ▪ currentRecord.CurrentRecord	▪ Record.getField(options) ▪ CurrentRecord.getField(options) ▪ Record.setValue(options) ▪ CurrentRecord.setValue(options) ▪ record.submitFields(options) ▪ search.lookupFields(options)
Sublist and sublist field IDs	▪ serverWidget.Sublist	▪ Record.commitLine(options) ▪ CurrentRecord.commitLine(options) ▪ Record.getCurrentSublistValue(options) ▪ CurrentRecord.getCurrentSublistValue(options) ▪ Record.insertLine(options) ▪ CurrentRecord.insertLine(options) ▪ Record.setSublistValue(options)
Search join, filter, and column IDs	▪ search.Search	▪ search.create(options)

ID Type	Object Examples	Method Examples
	▪ search.Filter ▪ search.Column ▪ search.Result ▪ search.ResultSet	▪ search.load(options) ▪ search.createFilter(options) ▪ search.createColumn(options)
Transformation IDs	▪ record.Record	▪ record.transform(options)

As you use the SuiteScript Records Browser, consider the following:

- You can use the Records Browser online, or you can download it. For more information, see the help topic Downloading the SuiteScript Records Browser.
- For information about governance applied to individual SuiteScript APIs, and about various SuiteScript script types, see SuiteScript Governance and Limits.
- Only the Records Browser for the most recent release of NetSuite is supported. Older versions may still be accessible, but they are not supported.
- Green highlighting indicates anything new in the current release.

For more details, see the following topics:

- Finding a Record or Subrecord
- SuiteScript Record Summary Overview
- Deleted Record Search

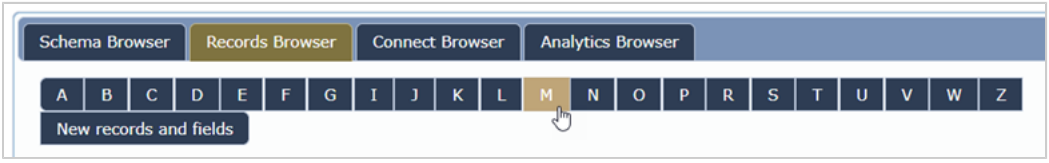
Finding a Record or Subrecord

Applies to: SuiteScript 2.x | SuiteCloud Developer

Use the A-Z index at the top of the SuiteScript Records Browser to find a record or subrecord.

To find a record or subrecord:

1. Click the appropriate letter at the top of the browser window.

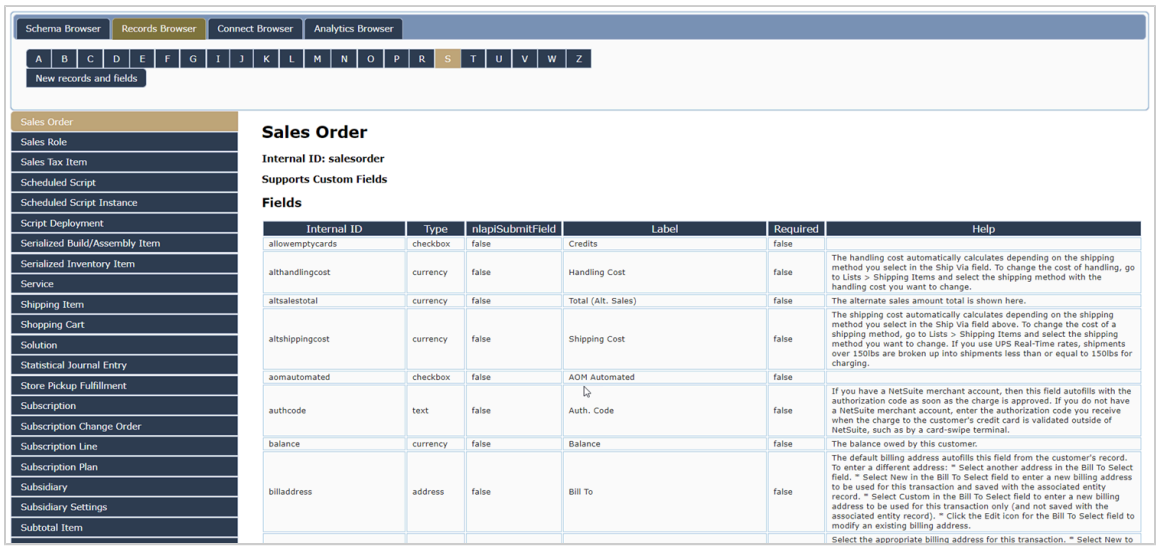


   The left pane shows record names starting with your selected letter. The center pane shows the details of the first record in the list.

2. In the left pane, click the name of the record you're interested in.

   The center pane shows the record details.

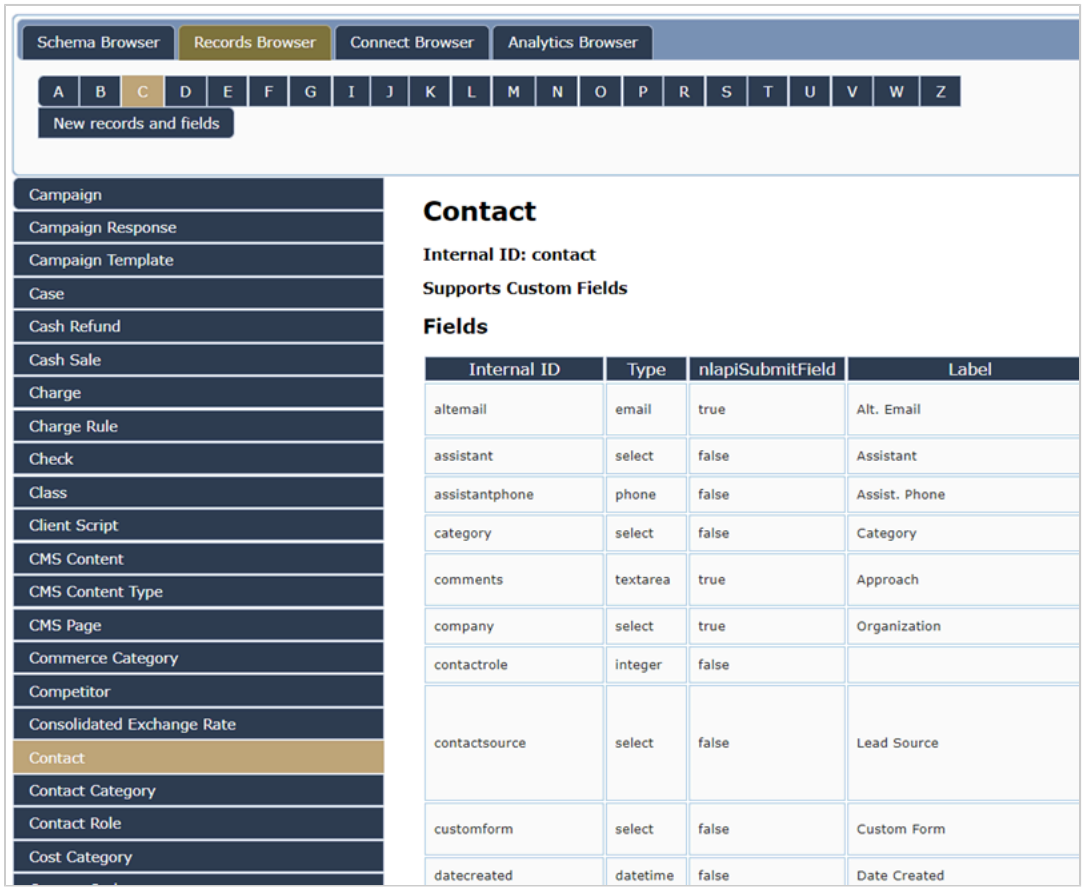
You can also get to a subrecord page from one of its parent records. Each subrecord summary field on a record page includes a link to the subrecord page. For example, on the Sales Order page of the Records Browser, the billingaddress and shippingaddress fields include links to the address subrecord page. Click the **summary** link in the **Type** column to go to the subrecord page.



SuiteScript Record Summary Overview

Applies to: SuiteScript 2.x | SuiteCloud Developer

The SuiteScript Records Browser includes a record summary for each record exposed to SuiteScript. For example, the following screenshot shows the record summary for a customer record:



For more information about the record summary, see Information Available in the SuiteScript Record Summary.

You can check whether the record you're currently viewing is also supported in SOAP web services or the SuiteAnalytics Connect NetSuite.com data source. Click the SOAP Schema Browser tab or the Connect Browser tab at the top of the page.

If the record is supported in SOAP web services or SuiteAnalytics Connect, you're directed to the corresponding page in the SOAP Schema Browser or SuiteAnalytics Connect Browser. Otherwise, you're directed to the first page of the respective browser. To compare record types support across SuiteScript, SOAP web services, and SuiteAnalytics Connect, see the help topic SuiteCloud Supported Records.

SuiteCloud Supported Records.

Information Available in the SuiteScript Record Summary

Applies to: SuiteScript 2.x | SuiteCloud Developer

In the SuiteScript Records Browser, the pages include the following designations for applicable records:

- **Search Only**: Indicates that the record supports only search actions in SuiteScript.
- **Supports Deleted Record Search**: Indicates that a record can be used as a filter for a deleted record search. For more information, see Deleted Record Search.

For each record you see a series of tables that include the following information:

- **Fields**: The record fields
- **Sublists**: A table representing each supported sublist
- **Tabs**: A list of tabs available on the record in the NetSuite UI
- **Search Joins**: A list of other searches you can access when searching this record
- **Search Filters**: Fields in the record that you can use as search criteria
- **Search Columns**: Fields you can include in search results
- **Transform Types**: A list of records that this record can be transformed into using record.transform(options)

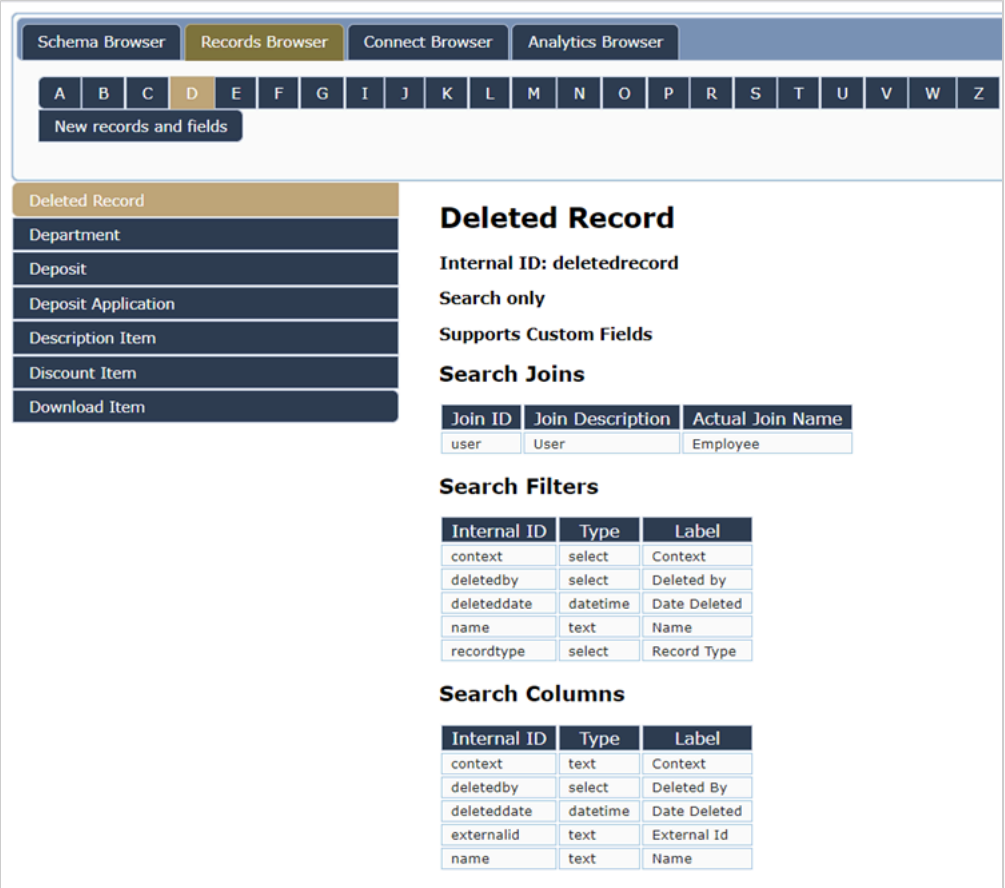
The following table describes some of the column names used in these tables.

Column Label	Description
Internal ID	The internal ID of the field.
Type	The data type of the field.
nlapiSubmitField	A boolean value indicating whether the field supports the record.submitFields(options) method, which lets you perform inline editing. This column name comes from the SuiteScript 1.0 API for inline editing (nlapiSubmitField). For more information, see the help topic Using Inline Editing.
Label	The label for the field as shown in the user interface.
Help	Additional details about working with the field.

Deleted Record Search

Applies to: SuiteScript 2.x | SuiteCloud Developer

The Records Browser includes a Deleted Record page, listed under **D**. This page lists the columns and filters available for deleted record searches in SuiteScript.



Check a record's page in the Records Browser to see if it can be used as a filter for deleted record searches. Records that can be used as filters have a **Supports Deleted Record Search** designation on their pages. For information about running deleted record searches in the UI, see the help topic Searching for Deleted Records.

The retrieval of deleted records in SOAP web services is a bit different. You can use the getDeleted operation instead of running a search. For a complete list of supported record types for this operation, refer to the `DeletedRecordType` enumeration in the coreTypes spreadsheet.

Creating Script Parameters Overview

Applies to: SuiteScript 2.x | SuiteCloud Developer

In the context of SuiteScript, script parameters are similar to custom fields; they are not considered to be parameters that are passed between JavaScript functions. Script parameters share characteristics with custom fields created through point-and-click customization. Script parameters are configurable by administrators and the users of your Suite App, and are accessible programmatically through SuiteScript. Script parameters are defined on the Parameters subtab of the Script record page.

Warning: Do not include confidential information in script parameters. Information saved in script parameters can be indexed by search engines and therefore be viewable by the public. This means, for example, that the information could be found in Google searches.

You should create script parameters in the following situations:

- You want part of your script to be configurable, either through script deployment or by the users of your SuiteApp. You do not need to create script parameters if your script is not designed to be configurable.
- You need to parameterize a script that was deployed multiple times. This approach makes it more convenient to customize the behavior of the script for each deployment.
- You want to configure a scheduled script. You're able to do this by specifying configuration parameters as arguments to task.create(options).

The advantages of using script parameters include:

- With deployment-specific parameters, you can configure script behavior without writing code. These parameters are useful when administrators deploy scripts that were installed as part of a bundle. The parameters let administrators to control or modify the script without knowing anything about the code. Deployment-specific parameters are similar to property or configuration files that some applications use to modify behavior at runtime.
- Script parameters let you to modify script behavior for troubleshooting without changing code, which is often expensive and not feasible (for example, if the original script author is unavailable).
- Script parameters provide flexibility to handle various inputs based on context. For example, consider a situation in which one script is deployed to 50 different records, but requires slightly different behavior for each record. You could hard-code and deploy 50 different scripts, but they may be difficult to maintain because the code isn't configurable, code might be duplicated unnecessarily, and changes in business requirements likely require code changes.

For more information, see the following help topics:

- Creating Script Parameters
- Referencing Script Parameters
- Script Parameter Preferences
- Creating a Custom Field

Creating Script Parameters

Note: This topic applies to all versions of SuiteScript.

Use the following steps to create script parameters. If you are unsure how to create a script record, see the help topic Creating a Script Record.

To create a script parameter:

1. Go to Customization > Scripting > Scripts.
2. Beside the script you want to add a parameter to, click **Edit**.
3. Click the **Parameters** tab, and click **New Parameter**.
4. In the **Label** field, type the name of the parameter (custom field) as it will appear in the UI after the script is deployed.
5. In the **ID** field, type a custom ID for the script parameter.

   Script parameter IDs must be in lowercase and contain no spaces. They also cannot exceed 30 characters.

   You can leave the ID field blank to use a system-generated ID. However, it is a best practice to create your own custom ID for script parameters. Doing so will help avoid naming conflicts if you later decide to package your script.
6. In the **Type** field, select the type of the script parameter (for example, Hyperlink, Date, Free-Form Text, or Check Box).

   For more information about field types, see the help topic Field Type Descriptions for Custom Fields.
7. (Optional) If the parameter type is List/Record, in the **List/Record** field, specify the list or record.

   If you define a saved search as a List/Record script parameter, only saved searches that are public will appear in the List/Record field. For more information about working with searches using SuiteScript, see the help topic N/search Module.
8. In the **Preference** field, select the set of preferences that you want to use for the parameter.

   Depending on the value of this field, the parameter's default value is based on the values set on either the General Preferences page (for a field value of Company), Set Preferences page (for a field value of User), or the portlet setup page (for a field value of Portlet). If you do not specify a preference, the parameter is considered to be a deployment script parameter, and its value is defined on the script deployment record.

   For more information, see Script Parameter Preferences.
9. (Optional) Use the **Display**, **Validation & Defaulting**, **Sourcing & Filtering**, **Access**, and **Translation** tabs to define additional values for the parameter.

   For information about defining these values, see the following help topics:

   - Setting Display Options for Custom Fields
   - Setting Validation and Defaulting Properties
   - Setting Sourcing Criteria
   - Setting Filtering Criteria
   - Restricting Access to Custom Fields
   - Adding Translations for Custom Fields
10. Click **Save** to save the parameter.

11. On the script record, click **Save** to save the script record.

Referencing Script Parameters

Applies to: SuiteCloud Developer | SuiteScript 2.x

You can use the Script.getParameter(options) method to reference script parameters that you create. This method is included in the N/runtime module, and you can use other methods in this module to work with scripts and script objects. For example, use runtime.getCurrentScript() to get a runtime.Script object representing your script. For more information, see the help topic N/runtime Module.

Before you can reference script parameters in your script, you must create them using the NetSuite UI. To learn how, see Creating Script Parameters. Remember the ID you used (or the ID that was generated automatically for you) when creating a script parameter. Use this ID in your script to get the script parameter's value.

The following example from a Suitelet obtains the value of a script parameter called `custscript_mycheckbox`:

```
// Add a script parameter called Check Box Required
var myField = Form.addField({
    id: 'custscript_mycheckbox',
    label: 'Check Box Required',
    type: serverWidget.FieldType.CHECKBOX
});

// Obtain an object that represents the current script
var myScript = runtime.getCurrentScript();

// Obtain the value of the Check Box Required script parameter
var scriptParameterValue = myScript.getParameter({
    name: 'custscript_mycheckbox'
});
```

You cannot write to a script parameter using SuiteScript. You can read from these fields, but not write to them. You can pass a value to a script parameter outside of the UI only when you call task.create(options) to schedule a script.

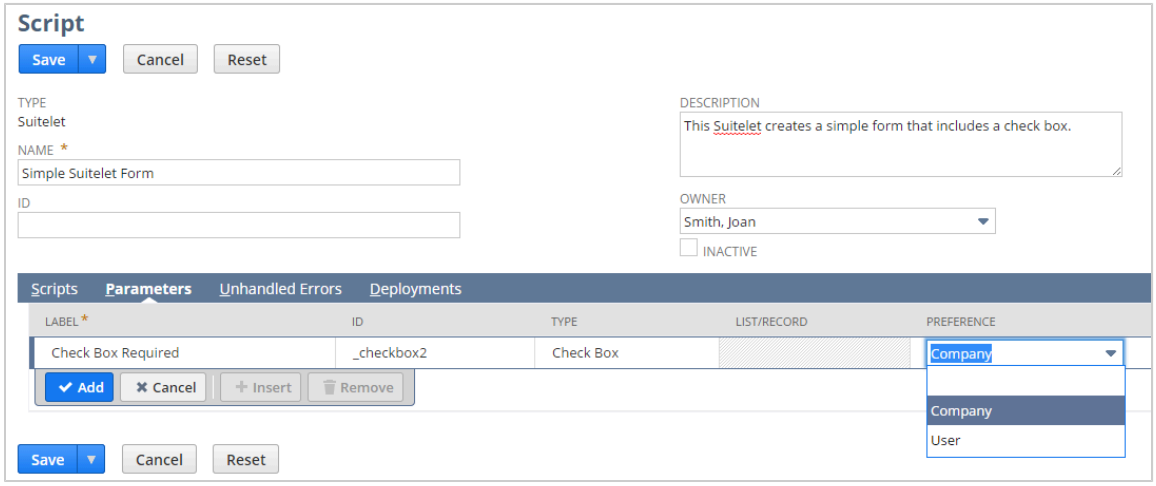
For a complete example working with script parameters in a SuiteCloud project, see the help topic Disable Tax Fields.

Script Parameter Preferences

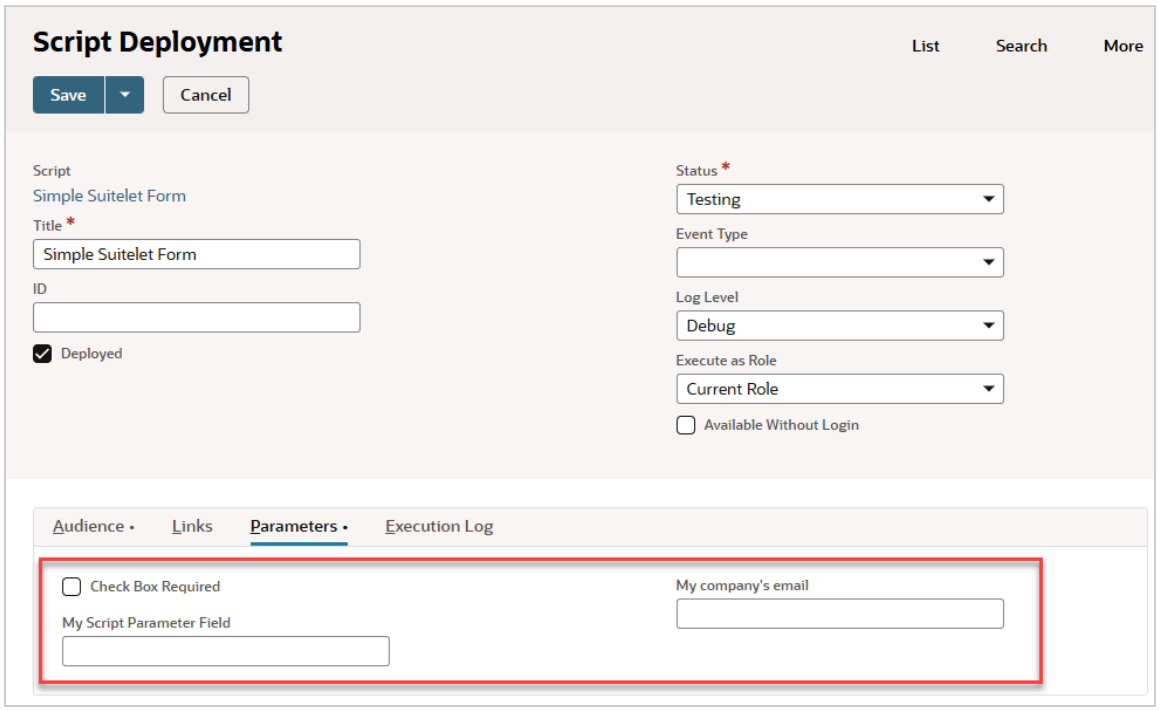
Applies to: SuiteScript 2.x | SuiteCloud Developer

When you use script parameters, you can specify a preference type. Available preference types are Company and User or you can choose not to set a preference.

- **Company:** For Company preference, the parameter's value comes from the value set in Setup > Company > General Preferences on the Custom Preferences subtab. For an example, see Setting Script Parameter Preferences Example.
- **User:** For User Preference, the parameter's value comes from the value set in Home > Set Preference on the Custom Preferences subtab. This lets end users override the company default script behavior and set their own default value. End users can change script parameter values without modifying the script or its deployments.



If you don't set a preference, the script parameter is considered a "deployment" script parameter by default. In this case, you define the value of the script parameter on the Parameters subtab of the Script Deployment record.

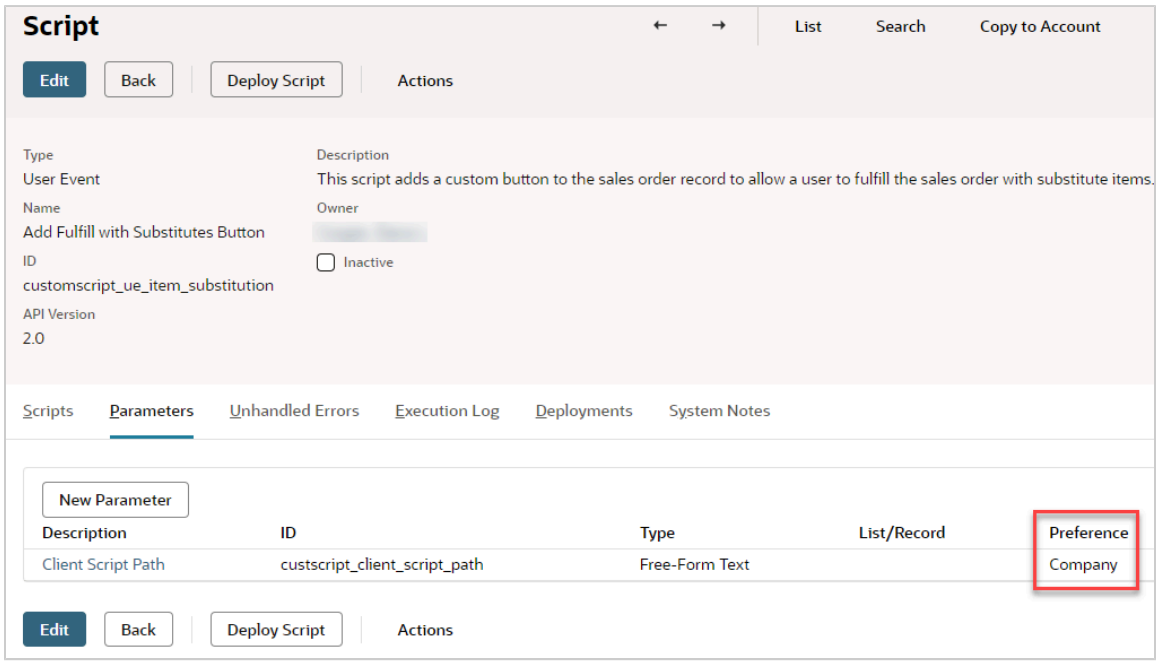


Note that users who install a bundled script that uses preferences can override the default behavior of the script and customize the script to their specific business needs. Setting preferences eliminates having to manipulate the script code or the script deployment. For more information, see the help topics SuiteBundler Overview and Script Parameter Preference Updates in Bundles.

Setting Script Parameter Preferences Example

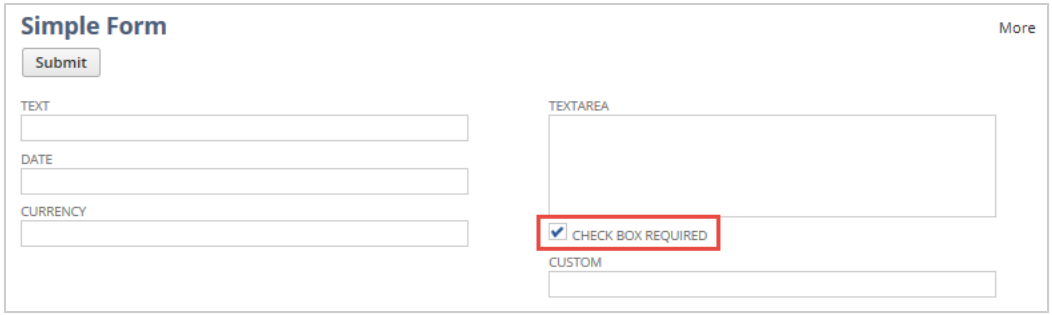
Applies to: SuiteScript 2.x | SuiteCloud Developer

In this example, the Suitelet script includes a parameter called **Check Box Required** (with the internal ID *custscript_checkboxtest2*), which is set to the Company preference.



By going to Setup > Company > General Preferences on the Custom Preferences subtab, administrators can set the default value of this parameter for the entire company. In this example, the value of the **Check Box Required** script parameter is set to T (True, or box checked).

When the Suitelet that contains this box is deployed, **Check Box Required** is checked.



If the **Check Box Required** parameter had been set to F (False, or box not checked), the box would have been cleared on the form when the Suitelet was deployed.

Note: To create a script parameter, see Creating Script Parameters. For information about accessing script parameter values, see Referencing Script Parameters.

Script Parameter Preference Updates in Bundles

Applies to: SuiteScript 2.x | SuiteCloud Developer

Script parameters that have a user or company preference set are not updated in target accounts when the bundle is updated. Script parameters without a preference are part of the script deployment and are updated in target accounts based on the bundle object preference:

- With Update Deployments, script deployment parameters in target accounts are updated to match the source account.
- With Do Not Update Deployments, script deployment parameters in target accounts remain unchanged.

Set the preference to Company or User if target account users need to change parameter values. Users can change parameter values as needed, and these changes won't be overwritten on bundle update even if the related bundle object preference is set to Update Deployments.

To prevent changes, set the bundle object preference to Do Not Update Deployments for parameters without a preference.

For more information, see the help topic Bundle Object Preferences.

SuiteScript IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

IDs are used for several different things including permissions, features, names, tasks, and buttons.

- Permission Names and IDs
- Feature Names and IDs
- Preference Names and IDs
- Task IDs
- Button IDs

Permission Names and IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following table provides permission names and IDs associated with each NetSuite feature. You can use the permission ID to return the permission levels that have been specified in your account by calling User.getPermission(options) in the N/runtime Module.

The following link provides access to a Microsoft Excel worksheet listing the usage of most NetSuite permissions: NetSuitePermissionsUsage.xls. You can use this list to understand the implications of assigning a specific permission, or to find the permission required to provide access to a specific task or page. For more information, see the help topic Permissions Documentation.

Permission ID	Permission Name	Feature	Valid Levels
ADMI_ACCOUNTING	Accounting Management	Accounting	None, Full
ADMI_ACCOUNTINGBOOK	Accounting Book	Accounting	None, View, Create, Edit, Full
ADMI_ACCOUNTINGLIST	Accounting Lists	Accounting	None, View, Create, Edit, Full
ADMI_ACCTPERIODS	Manage Accounting Periods	Accounting Periods	None, View, Full
ADMI_ACCTSETUP	Set Up Accounting	Accounting	None, Full
ADMI_ACCTSETUP	Accounting Preferences	Accounting	None, Full
ADMI_ADMINDOCSEU	Admindocs EU	—	None, Full
ADMI_ADMINDOCSNA	Admindocs NA	—	None, Full
ADMI_ADMINDOCSOTHER	Admindocs OTHER	—	None, Full
ADMI_ACH	Set Up ACH Processing	ACH Processing	None, Full
ADMI_ADVANCED_ORDER_MANAGEMENT	Advanced Order Management	Advanced Order Management	None, Full
ADMI_ADVANCED_TEMPLATES	Advanced PDF/HTML Templates	Advanced Printing	None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_ALLOW_JS_HTML_UPLOAD	Allow JS / HTML Uploads	Documents	None, Full
ADMI_ALLOWNONGLCHANGES	Allow Non G/L Changes	Accounting Periods	None, Full
ADMI_ANALYTICS	Analytics Administrator	—	None, Full
ADMI_APP_DEPLOYMENT	SuiteApp Deployment	SuiteApp	None, Full
ADMI_APPDEFPKG	App Definitions and Packages	—	None, Full
ADMI_APPPUBLISHER	Application Publishers	SSP Applications	None, Full
ADMI_AUDITLOGIN	View Login Audit Trail	Login Audit Trail	None, Full
ADMI_BACKUPEXPORT	Backup Your Data	CSV Export	None, Full
ADMI_BALANCE_TRX_BY_SEGMENTS	Balance Transactions by Segments	Balancing Segments	None, View, Create
ADMI_BANK_CONNECTIVITY_CONFIG	Bank Connectivity Plug-In Configuration	—	None, Full
ADMI_BILLINGINFO	Billing Information	Allow Multiple Users	None, Full
ADMI_BLCGA	Balance Location Costing Group Accounts	Group Average Costing	None, Full
ADMI_BUNDLER	SuiteBundler	SuiteBundler	None, Full
ADMI_BUNDLER	SuiteApp Marketplace	SuiteBundler	None, Full
ADMI_BUNDLERAUDITTRAIL	SuiteBundler Audit Trail	SuiteBundler	None, Full
ADMI_BUNDLERMANUP	SuiteBundler Upgrade Install Base	SuiteBundler	None, Full
ADMI_CAMPAIGNEMAIL	Set Up Campaign Email Addresses	Marketing Automation	None, Full
ADMI_CAMPAIGNSETUP	Setup Campaigns	Marketing Automation	None, Full
ADMI_CASEALERT	Case Alerts	Customer Support and Service	None, View, Full
ADMI_CASEFORM	Online Case Form	Customer Support and Service	None, Full
ADMI_CASEISSUE	Support Case Issue	Customer Support and Service	None, Full
ADMI_CASEORIGIN	Support Case Origin	Customer Support and Service	None, Full
ADMI_CASEPRIORITY	Support Case Priority	Customer Support and Service	None, Full
ADMI_CASERULE	Support Case Territory Rule	Customer Support and Service	None, View, Create, Edit, Full
ADMI_CASESTATUS	Support Case Status	Customer Support and Service	None, Full
ADMI_CASETERRITORY	Support Case Territory	Customer Support and Service	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_CASETYPE	Support Case Type	Customer Support and Service	None, Full
ADMI_CENTERLINK	Custom Center Link	Custom Centers	None, Full
ADMI_CERTIFICATES	Certificate Management	—	None, View, Create, Edit, Full
ADMI_CLASSESTOLOCS	Convert Classes to Locations	Locations	None, Full
ADMI_CLASSSEGMENTMAPPING	Class Segment Mapping	Classes	None, Full
ADMI_CLASSSEGMENTMAPPING	Class Mapping	Multi Book Version 2	—
ADMI_CLOSEPERIOD	Lock Transactions	Accounting Periods	None, Full
ADMI_COMMERCECATEGORY	Commerce Categories	—	None, Full
ADMI_COMMISSIONSETUP	Commission Feature Setup	Commissions	None, Full
ADMI_COMPANY	Company Information	Company Setup	None, Full
ADMI_CONVERTCLASSES	Convert Classes to Departments	Departments	None, Full
ADMI_CONVERTLEAD	Lead Conversion Mapping	Sales Force Automation	None, Full
ADMI_COPYPROJECTTASK	Copy Project Tasks	Project Management	None, Full
ADMI_CREATEJOBSFROMSALESTRANS	Create Jobs from Sales Transactions	Project Management	None, Full
ADMI_CREDITCARD	Credit Card Processing	Credit Card Payments	None, Full
ADMI_CRMLIST	CRM Lists	Opportunities	None, View, Create, Edit, Full
ADMI_CROSSCHARGE	Manage Cross Charge Automation	Intercompany Framework	View, None, Full
ADMI_CSVIMPORTPREF	Set Up CSV Preferences	Accounting	None, Full
ADMI_CUSTADDRESSFORM	Custom Address Form	Custom Forms	None, View, Create, Edit, Full
ADMI_CUSTBODYFIELD	Custom Body Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTCATEGORY	Custom Center Categories	Custom Centers	None, View, Create, Edit, Full
ADMI_CUSTCENTER	Custom Centers	Custom Centers	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_CUSTCOLUMNFIELD	Custom Column Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTEMAILLAYOUT	Custom HTML Layouts	Custom PDF/HTML Templates	None, View, Create, Edit, Full
ADMI_CUSTENTITYFIELD	Custom Entity Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTENTRYFORM	Custom Entry Forms	Custom Forms	None, View, Create, Edit, Full
ADMI_CUSTEVENTFIELD	Custom Event Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTFIELD	Custom Fields	SuiteFlow Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTFIELDTAB	Custom Subtabs	Custom Subtabs	None, View, Create, Edit, Full
ADMI_CUSTFORM	Custom Transaction Forms	Custom Forms	None, View, Create, Edit, Full
ADMI_CUSTITEMFIELD	Custom Item Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTITEMNUMBERFIELD	Custom Item Number Fields	Inventory	None, View, Create, Edit, Full
ADMI_CUSTLAYOUT	Custom PDF Layouts	Custom PDF/HTML Templates	None, View, Create, Edit, Full
ADMI_CUSTLIST	Custom Lists	Custom Lists	None, View, Create, Edit, Full
ADMI_CUSTOMERFORM	Online Customer Form	Sales Force Automation	None, View, Create, Edit, Full
ADMI_CUSTOMERRULE	Sales Territory Rule	Sales Force Automation	None, View, Create, Edit, Full
ADMI_CUSTOMER_SEGMENTS	Customer Segments Manager	—	None, Full
ADMI_CUSTOMIZEDFIELDLEVELHELP	Customize Field Level Help	SuiteBuilder Customized Field Level Help	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_CUSTOMSCRIPT	SuiteScript	Client SuiteScript	None, View, Create, Edit, Full
ADMI_CUSTOMSUBLIST	Custom Sublists	Custom Sublists	None, View, Create, Edit, Full
ADMI_CUSTOTHERFIELD	Other Custom Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTRECORD	Custom Record Types	Custom Records	None, View, Create, Edit, Full
ADMI_CUSTRECORDFORM	Online Custom Record Form	Custom Forms	None, View, Create, Edit, Full
ADMI_CUSTSECTION	Custom Center Tabs	Custom Records	None, View, Create, Edit, Full
ADMI_CUSTTASKS	Custom Center Links	Custom Records	None, View, Create, Edit, Full
ADMI_CUSTTRANFIELD	Custom Transaction Fields	Custom Fields	None, View, Create, Edit, Full
ADMI_CUSTTRANSACTION	Custom Transaction Types	Custom Transactions	None, View, Create, Edit, Full
ADMI_DELETEDRECORD	Deleted Records	Search for Deleted Records	None, Full
ADMI_DEPTSEGMENTMAPPING	Department Segment Mapping	Departments	None, Full
ADMI_DEPTSEGMENTMAPPING	Department Mapping	Multi Book Version 2	None, Full
ADMI_DEVICE_ID	Device ID Management	Suite Commerce InStore	None, Full
ADMI_DIRECTINVOICEPAYMENTSETUP	Direct Invoice Payment Setup	—	None, View, Create, Edit, Full
ADMI_DOMAINS	Set Up Domains	SuiteCommerce	None, Full
ADMI_DUPLICATESETUP	Duplicate Detection Setup	Duplicate Detection & Merge	None, Full
ADMI_EMPLCATEGORY	Publish Employee List	SuiteCommerce	None, Full
ADMI_EMPLOYEE_EXPENSE_SOURCE	Employee Expense Sources	Expense Reports	None, View, Create, Edit, Full
ADMI_EMPLOYEECENTERPUBLISHING	Employee Center Publishing	—	None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_EMPLOYEELIST	Other Lists	Accounting	None, View, Create, Edit, Full
ADMI_ENABLEFEATURES	Enable Features	Enable Company Features	None, Full
ADMI_ENTITYACCOUNTMAPPING	Entity Account Mapping	Accounting	None, Full
ADMI_ENTITYSTATUS	Customer Status	Customer Relationship Management	None, Full
ADMI_ESCALATIONRULE	Escalation Assignment Rule	Customer Support and Service	None, View, Create, Edit, Full
ADMI_ESCALATIONTERRITORY	Escalation Assignment	Customer Support and Service	None, View, Create, Edit, Full
ADMI_EXPENSEREPORTPOLICY	Expense Report Policies	Expense Reports	None, View, Full
ADMI_EXPORTIIF	Export as IIF	Accounting	None, Full
ADMI_FFTEXCEPTIONREASON	Fulfillment Exception Reason	—	None, View, Create, Edit, Full
ADMI_FINANCIALINSTITUTION	Financial Institution Records	—	None, Full
ADMI_FINCHARGEPREF	Finance Charge Preferences	A/R	None, Full
ADMI_GAINLOSSACCTMAPPING	Foreign Currency Variance Mapping	Accounting	None, Full
ADMI_GLOBALACCOUNTMAPPING	Global Account Mapping	Accounting	None, Full
ADMI_HTMLFORMULA	Create HTML Formulas in Search	—	None, View, Create, Edit, Full
ADMI_IMPORTCSVFILE	Import CSV File	CSV Import	None, Full
ADMI_IMPORTOVERRIDESSTRIG	Control SuiteScript and Workflow Triggers per CSV Import		None, Full
ADMI_IMPORTXML	Set Up ADP Payroll	Import ADP Payroll	None, Full
ADMI_INTEGRAPP	Integration Application	Integration	None, Full
ADMI_ISSUESETUP	Issue Setup	Issue Management	None, Full
ADMI_ISSUESHOWSTOPPER	Mark Issue As Showstopper	Issue Management	None, Full
ADMI_ITEMACCOUNTMAPPING	Item Account Mapping	Accounting	None, Full
ADMI_KERNEL	Core Administration Permissions	Core Administration Permissions	None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_KEYS	Key management	—	None, View, Create, Edit, Full
ADMI_KNOWLEDGEBASE	Publish Knowledge Base	Knowledge Base	None, View, Create, Edit, Full
ADMI_KPIREPORT	KPI Scorecards	KPI Scorecards	None, Full
ADMI_LOCATIONCOSTINGGROUP	Location Costing Group	Item Record Management	None, Full
ADMI_LOCSEGMENTMAPPING	Location Segment Mapping	Custom Segments	None, Full
ADMI_LOCSEGMENTMAPPING	Location Mapping	Multi Book Version 2	None, Full
ADMI_LOGIN_OAUTH	Log in using Access Tokens	TBA	None, Full
ADMI_LOGIN_OAUTH2	Log in using OAuth 2.0 Access Tokens	OAuth 2.0	None, Full
ADMI_MANAGECUSTOMSEGMENTS	Custom Segments	Custom Segments	None, View, Create, Edit, Full
ADMI_MANAGE_OAUTH2	OAuth 2.0 Authorized Applications Management	OAuth 2.0	None, Full
ADMI_MANAGE_OAUTH_TOKENS	Access Token Management	Token-based Authentication	None, Full
ADMI_MANAGE_OWN_OAUTH_TOKENS	User Access Tokens	Token-based Authentication	None, Full
ADMI_MANAGE_RESTRICTIONS	Manage Custom Restrictions	Advanced Employee Permissions	None, View, Edit, Full, Create
ADMI_MANAGEPERMISSIONS	Manage Custom Permissions	Allow Multiple Users	None, Full
ADMI_MANAGEROLES	Bulk Manage Roles	Show Role Differences	None, Full
ADMI_MANAGEUSERS	Manage Users	Managing Users	None, Full
ADMI_MANUFACTURING	Manufacturing Preferences	Assembly Items	None, Full
ADMI_MHLEVEL	Merchandise Hierarchy Level	—	None, Full
ADMI_MHNODE	Merchandise Hierarchy Node	—	None, Full
ADMI_MHVERSION	Merchandise Hierarchy Version	—	None, Full
ADMI_MIGRATEREVARRNGANDPLAN	Migrate Revenue Arrangements and Plans	Advanced Revenue Management	None, Full
ADMI_MOBILE_ACCESS	Mobile Device Access	—	None, Full
ADMI_MCP_SERVER	MCP Server Connection	—	None, Full
ADMI_NSASOIDCPROVIDER	OIDC Provider Setup		None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_NUMBERING	Auto-Generated Numbers		None, View, Edit, Full
ADMI_OIDC	OpenID Connect (OIDC) Single Sign-On	—	None, Full
ADMI_OIDCSETUP	Set Up OpenID Connect (OIDC) Single Sign-On	—	None, Full
ADMI_OPENIDSSO	OpenID Single Sign-on	—	None, Full
ADMI_OPENIDSSOSETUP	Set Up OpenID Single Sign-on	—	None, Full
ADMI_ORDERALLOCATIONSTRATEGY	Order Allocation Strategy	—	None, View, Create, Edit, Full
ADMI_ORDERPROMISING	Order Promising	—	None, View, Full
ADMI_OUTLOOKINTEGRATION	Outlook Integration 2.0	Outlook Integration	None, Full
ADMI_OUTLOOKINTEGRATION_V3	Outlook Integration 3.0	Outlook Integration	None, Full
ADMI_PARTNERCONTRIBUTION	Partner Contribution	—	None, Full
ADMI_PAYMENT_LINK_SETUP	Set Up Payment Link		None, Full
ADMI_PAYROLL	Set Up Payroll	Payroll	None, Full
ADMI_PENDINGBOOKJOURNAL	Allow Pending Book Journal Entry	—	None, Full
ADMI_PERIODCLOSING	Period Closing Management	—	None, Full
ADMI_PERIODOVERRIDE	Override Period Restrictions	—	None, Full
ADMI_PI_REMOVAL_CREATE	Remove Personal Information Create	—	None, Full
ADMI_PI_REMOVAL_RUN	Remove Personal Information Run	—	None, Full
ADMI_PROJECT_ACCOUNTING_SETUP	Project Profitability Setup	Project Management	None, Full
ADMI_PROJECT_ACCOUNTING_SETUP	Project Profitability	Advanced Project Profitability	None, Full
ADMI_PROMPTS	Prompts	AI	None, View, Create, Edit, Full
ADMI_PROVISION	Provisioning	NLCORP feature	None, View, Full
ADMI_REPOGROUPS	Financial Statement Sections	Accounting	None, Full
ADMI_REPOLAYOUTS	Financial Statement Layouts	Accounting	None, Full
ADMI_RESTWEBSERVICES	REST Web Services	REST Web Services	None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_REVIEW_CUSTOM_GL_RUNS	Review Custom GL plug-in executions	Custom GL Lines Plug-in	None, Full
ADMI_SALESCHANNEL	Sales Channel	—	None, View, Edit, Full, Create
ADMI_SALESTERRITORY	Sales Territory	Sales Force Automation	None, View, Create, Edit, Full
ADMI_SAMLSSO	SAML Single Sign-on	SAML Single Sign-on	None, Full
ADMI_SAMLSSOSETUP	Set Up SAML Single Sign-on	SAML Single Sign-on	None, Full
ADMI_SAVEDASHBOARD	Publish Dashboards	Publishing Dashboards	None, Full
ADMI_SETUPCOMPANY	Set Up Company	Company Preferences	None, View, Full
ADMI_SETUPIMAGERESIZE	Set Up Image Resizing	SuiteCommerce Advanced	None, Full
ADMI_SETUPYEARSTATUS	Set Up Year Status	Accounting	None, Full
ADMI_SFASETUP	Sales Force Automation Setup	Sales Force Automation	None, Full
ADMI_SITEMANAGEMENT	Web Site Management	Web Site	None, Full
ADMI_STATETAXIMPORT	Import State Sales Tax	Accounting	None, Full
ADMI_STORESEARCH	Site Search	—	None, Full
ADMI_STORESETUP	Set Up Web Site	Web Site	None, Full
ADMI_SUBLIST	Custom Sublist	Allow Multiple Users	None, View, Create, Edit, Full
ADMI_SUBSIDIARYHIERARCHYMOD	Subsidiary Hierarchy Modification	—	None, View, Edit
ADMI_SUBSIDIARYSETTINGSMANAGER	Subsidiary Settings Manager	—	None, View, Edit
ADMI_SUITE_OAX_CONNECTOR	NSAW Connector Administrator		None, Full
ADMI_SUITEANALYTICSCONNECT	SuiteAnalytics Connect	SuiteAnalytics Connect	None, Full
ADMI_SUITEAPP_MANAGEMENT	SuiteApp Management	SuiteApp Control Center	None, Full
ADMI_SUITECOMMERCEANALYTICS	SuiteCommerce Analytics		None, View, Full
ADMI_SUITESIGNON	SuiteSignOn	Outbound Single Sign-on	None, Full
ADMI_SUITE_OAX_CONNECTOR	OAX Connector Administrator	NetSuite Analytics Warehouse	None, Full

Permission ID	Permission Name	Feature	Valid Levels
ADMI_SUPPLYALLOCATIONSETUP	Supply Allocation Setup	—	None, View, Create, Edit, Full
ADMI_SUPPORTSETUP	Support Setup	Customer Support and Service	None, Full
ADMI_SWAPPRICES	Swap Prices Between Price Levels	Multiple Prices	None, Full
ADMI_TAXMIGRATION	SuiteTax Migration	—	None, Full
ADMI_TIMEMODIFICATION	Bulk Time Entry Modification	—	None, Full
ADMI_TAXPERIODS	Manage Tax Reporting Periods	Accounting	None, Full
ADMI_TEAMSELLINGCONTRIBUTION	Team Selling Contribution	Team Selling	None, Full
ADMI_TELEPHONY_SETUP	Telephony Integration	Telephony	None, Full
ADMI_TRAN_ACCOUNTING_RULES	Transaction Accounting Rules	—	None, Full
ADMI_TRANSITEMTXT	Translation	Multi-Language	None, Full
ADMI_TRANSLATION	Manage Translation	Multi-Language	None, Full
ADMI_TSTDRV_MASTER	Testdrive Masters	NLCORP feature	None, Full
ADMI_TWOFACTORAUTH	Two-Factor Authentication	Two-Factor Authentication	None, Full
ADMI_TWOFACTORAUTHBASE	Two-Factor Authentication base	Two-Factor Authentication	None, Full
ADMI_UNCATSITEITEMS	Uncategorized Presentation Items	Web Store	None, Full
ADMI_UPDATEPRICES	Update Prices	Item Record Management	None, Full
ADMI_UPSELLSETUP	Upsell Setup	Upsell Manager	None, Full
ADMI_WEBSERVICES	SOAP Web Services	SuiteTalk	None, Full
ADMI_WEBSERVICES	Web Services	Internal Web Services	None, Full
ADMI_WEBSERVICESLOG	View SOAP Web Services Logs	SuiteTalk	None, Full
ADMI_WEBSERVICESLOG	View Web Services Logs	Internal Web Services	None, Full
ADMI_WEBSERVICESSETUP	Set Up SOAP Web Services	SuiteTalk	None, Full
ADMI_WEBSERVICESSETUP	Set Up Web Services	Internal Web Services	None, Full
ADMI_WORKFLOW	Workflow	SuiteFlow	None, Full
ADMINDOCS	Admindocs	NLCORP feature	None, Full
LIST_ACCOUNT	Accounts	Accounting	None, View, Create, Edit, Full
LIST_ACH	Automated Clearing House	Payment Instruments	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_ALLGOVERNMENTISSUEDIDS	Advanced Government-Issued IDs	Advanced Government-Issued ID Tracking	None, View, Create, Edit, Full
LIST_ALLOCSCHEDULE	Allocation Schedules	Expense Allocation	None, View, Create, Edit, Full
LIST_AMORTIZATION	Amortization Schedules	Amortization	None, View, Create, Edit, Full
LIST_BASICGOVERNMENTISSUEDIDS	Basic Government-Issued IDs	Basic Government-Issued ID Tracking	None, View, Create, Edit, Full
LIST_BIG_SEARCH	Persist Search	Search	None, Create
LIST_BILLINBOUNDSHIPMENT	Bill Inbound Shipment	Inbound Shipment Management	None, View, Create, Edit, Full
LIST_BILLINGSCHEDULE	Billing Schedules	SuiteBilling	None, View, Create, Edit, Full
LIST_BILLOFDISTRIBUTION	Bill Of Distribution	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_BILLOFMATERIALSINQUIRY	Bill Of Materials Inquiry	Assembly Items	None, View, Create, Edit, Full
LIST_BOM	Bill of Materials	Assembly Items	None, View, Create, Edit, Full
LIST_BILLCAPTURE	Scanned Vendor Bills	—	None, View, Create, Edit, Full
LIST_BIN	Bins	Accounting	None, View, Create, Edit, Full
LIST_BONUS	Bonus	—	View, None, Create, Full, Edit
LIST_BONUSTYPE	Bonus Types	—	View, None, Create, Full, Edit
LIST_CALENDAR	Calendar	Calendar Preferences	None, View, Create, Edit, Full
LIST_CALL	Phone Calls	Phone Calls	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_CAMPAIGN	Marketing Campaigns	Marketing Automation	None, View, Create, Edit, Full
LIST_CAMPAIGNHISTORY	Campaign History	Marketing Automation	None, View, Create, Edit, Full
LIST_CARDHOLDERAUTHENTICATION	Cardholder Authentications	Credit Card Payments	None, View, Edit, Full, Create
LIST_CARDHOLDERAUTHEVENT	Cardholder Authentication Events	Credit Card Payments	None, View, Edit, Full, Create
LIST_CASE	Cases	Customer Support and Service	None, View, Create, Edit, Full
LIST_CASE_DUPLICATES	Duplicate Case Management		None, Full
LIST_CATEGORY	Expense Categories	Expense Reports	None, View, Create, Edit, Full
LIST_CERTIFICATES	Certificate Access	—	None, View, Create, Edit, Full
LIST_CHECKITEMAVAILABILITY	Check Item Availability	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_CLASS	Classes	Classes	None, View, Create, Edit, Full
LIST_COLORTHEME	Color Themes	SuiteCommerce	None, View, Create, Edit, Full
LIST_COMMISSIONRULES	Employee Commission Schedules/Plans	Employee Commissions	None, View, Create, Edit, Full
LIST_COMPANY	Companies	Customer Relationship Management	None, View, Create, Edit, Full
LIST_COMPETITOR	Competitors	Sales Force Automation	None, View, Create, Edit, Full
LIST_COMPONENTWHEREUSEDINQUIRY	Component Where Used	Assembly Items	None, View, Create, Edit, Full
LIST_CONTACT	Contacts	Contacts	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_CONTACTROLE	Contact Roles	Sales Force Automation	None, View, Create, Edit, Full
LIST_CONTACTSUBSIDIARYRELATION	Contact-Subsidiary Relationship	—	None, View
LIST_CONVERTLEAD	Lead Conversion	—	View, None, Create, Full, Edit
LIST_COSTEDBOMINQUIRY	Costed Bill Of Materials Inquiry	Assembly Items	None, View, Create, Edit, Full
LIST_CRMGROUP	CRM Groups	Customer Relationship Management	None, View, Create, Edit, Full
LIST_CRMMESSAGE	Track Messages	Customer Relationship Management	None, View, Create, Edit, Full
LIST_CRMTEMPLATE	Marketing Template	Marketing Automation	None, View, Create, Edit, Full
LIST_CURRENCY	Currency	Multiple Currencies	None, View, Create, Edit, Full
LIST_CUSTJOB	Customers	Prospects and Contacts	None, View, Create, Edit, Full
LIST_CUSTPROFILE	Customer Profile	Customer Profile	None, View, Create, Edit, Full
LIST_CUSTRECORDENTRY	Custom Record Entries	Custom Records	None, View, Create, Edit, Full
LIST_DEPARTMENT	Departments	Departments	None, View, Create, Edit, Full
LIST_DISTRIBUTIONNETWORK	Distribution Network	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_EARLIEST_AVAILABILITY	Earliest Availability	Supply Allocation	None, View, Create, Edit, Full
LIST_EMAILTEMPLATE	Email Template	Email Template	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_EMPLOYEE	Employees	Employees	None, View, Create, Edit, Full
LIST_EMPLOYEE_ACCESS	Employee Access Tab	Advanced Employee Permissions	None, View, Create, Edit, Full
LIST_EMPLOYEE_ ADMINISTRATION	Employee Administration	Advanced Employee Permissions	None, View, Create, Edit, Full
LIST_EMPLOYEE_ CONFIDENTIAL	Employee Confidential	Advanced Employee Permissions	None, View, Create, Edit, Full
LIST_EMPLOYEE_PUBLIC	Employee Public	Advanced Employee Permissions	None, View, Create, Edit, Full
LIST_EMPLOYEE_RECORD	Employee Record	—	View, Create, Edit, Full
LIST_EMPLOYEE_SELF	Employee Self	Advanced Employee Permissions	View, Create, Edit, Full
LIST_EMPLOYEECHANGETYPE	Employee Change Request Type	—	None, View, Create, Edit, Full
LIST_EMPLOYEECHANGEREAS ON	Employee Change Reason	Effective Dating	None, View, Create, Edit, Full
LIST_EMPLOYEECHANGEREQ UEST	Employee Change Request	Employee Change Requests	None, View, Create, Edit, Full
LIST_EMPLOYEECHANGETYPE	Employee Change Request Type	Employee Change Requests	None, View, Create, Edit, Full
LIST_EMPLOYEEEFFECTIVEDA TING	Employee Effective Dating	Effective Dating	None, View, Create, Edit, Full
LIST_EMPLOYEESEPARATION	Termination Reasons	Termination Reason Tracking	None, View, Create, Edit, Full
LIST_EMPLOYEESSN	Employee Social Security Numbers	Employees	None, View, Full
LIST_ENTITY_DUPLICATES	Duplicate Entity Management	Duplicate Detection & Merge	None, View, Full
LIST_EVENT	Events	Events	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_EXPENSEAMORTIZATION RULE	Expense Amortization Rule	—	None, View, Create, Edit, Full
LIST_EXPENSEPLAN	Expense Amortization Plan	—	None, View, Create, Edit, Full
LIST_EXPORT	Export Lists	Search Result Export	None, Create
LIST_FAIRVALUEDIMENSION	Fair Value Dimension	Revenue Recognition	None, View, Create, Edit, Full
LIST_FAIRVALUEFORMULA	Fair Value Formula	Revenue Recognition	None, View, Create, Edit, Full
LIST_FAIRVALUEPRICE	Fair Value Price	Revenue Recognition	None, View, Create, Edit, Full
LIST_FAXMESSAGE	Fax Messages	Communications	None, View, Create, Edit, Full
LIST_FAXTEMPLATE	Fax Template	Mail Merge	None, View, Create, Edit, Full
LIST_FILECABINET	Documents and Files	File Cabinet	None, View, Create, Edit, Full
LIST_FINANCIAL_EXCEPTION_ MGMT	Financial Exception Management	—	View
LIST_FIND	Perform Search	Order Management	None, View, Full
LIST_FINHISTORY	Financial History		None, View, Create, Edit, Full
LIST_FISCALCALENDAR	Fiscal Calendars	Accounting	None, View, Create, Edit, Full
LIST_GENERAL_TOKEN	General Token	Payment Instruments	None, View, Create, Edit, Full
LIST_GENERICRESOURCE	Generic Resources	Project Management	None, View, Create, Edit, Full
LIST_GIFT_CERTIFICATE	Gift Certificate	-	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_GLLINESAUDITLOG	Custom GL Lines Plug-in Audit Log	Custom GL Lines Plug-in	None, View, Create, Edit, Full
LIST_GLLINESAUDITLOGSEG	Custom GL Lines Plug-in Audit Log (Segments)	Custom GL Lines Plug-in	None, View, Create, Edit, Full
LIST_GLOBALINVTRELATIONSHIP	Global Inventory Relationship	—	None, View, Create, Edit, Full
LIST_GOVERNMENTISSUEDIDTYPE	Government-Issued ID Types	Advanced Government-Issued ID Tracking	None, View, Create, Edit, Full
LIST_HCMJOB	HCMJob Management	—	None, View, Create, Edit, Full
LIST_HCMPOSITION	Positions	—	None, View, Create, Edit, Full
LIST_HISTORY	Notes Tab	Communications	None, View, Create, Edit, Full
LIST_IMPORTED_EMPLOYEE_EXPENSE	Imported Employee Expenses	Expense Reports	View, None
LIST_INBOUNDSHIPMENT	Inbound Shipment	Inbound Shipment Management	None, View, Create, Edit, Full
LIST_INFOCATEGORY	Store Content Categories	Web Store	None, View, Create, Edit, Full
LIST_INFOITEM	Store Content Items	SuiteCommerce	None, View, Create, Edit, Full
LIST_INFOITEMFORM	Publish Forms	SuiteCommerce	None, View, Create, Edit, Full
LIST_INTEGRAPP	Integration Applications	SuiteTalk	None, View, Create, Edit, Full
LIST_INTERNALPUBLISH	Internal Publisher		None, View, Create, Edit, Full
LIST_INVCOSTTEMPLATE	Inventory Cost Template	Multi-Book Accounting	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_INVENTORYSTATUS	Inventory Status	Inventory Management	None, View, Create, Edit, Full
LIST_ISSUE	Issues	Issue Management	None, View, Create, Edit, Full
LIST_ITEM	Items	Item Record Management	None, View, Create, Edit, Full
LIST_ITEM_COLLECTION	Item Collection	—	None, View, Create, Edit, Full
LIST_ITEMDEMANDPLAN	Item Demand Plan	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_ITEMREVENUECATEGORY	Item Revenue Category	Revenue Recognition	None, View, Create, Edit, Full
LIST_ITEMPROCESSFAMILY	Item Process Family	Warehouse Management	None, View, Create, Edit, Full
LIST_ITEMPROCESSGROUP	Item Process Group	Warehouse Management	None, View, Create, Edit, Full
LIST_ITEM_REVISION	Item Revisions	Item Record Management	None, View, Create, Edit, Full
LIST_ITEMSUPPLYPLAN	Item Supply Plan	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_ITEMTEMPLATE	Item Templates		None, View, Create, Edit, Full
LIST_JOB	Jobs	Projects	None, View, Create, Edit, Full
LIST_JOBREQUISITION	HCMJob Requisitions	Job Requisitions	None, View, Create, Edit, Full
LIST_KEYS	Key access	—	None, View, Create, Edit, Full
LIST_KUDOS	Kudos	Kudos	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_KNOWLEDGEBASE	Knowledge Base	Knowledge Base	None, View, Create, Edit, Full
LIST_LABORCOSTING	Labor Costing	Labor Costing	View, None, Full, Create, Edit
LIST_LOCATION	Locations	Locations	None, View, Create, Edit, Full
LIST_MAILMERGE	Mail Merge	Mail Merge	None, View, Create, Edit, Full
LIST_MAILMESSAGE	Letter Messages	Communications	None, View, Create, Edit, Full
LIST_MAILTEMPLATE	Letter Template	Mail Merge	None, View, Create, Edit, Full
LIST_MASSUPDATES	Mass Updates	Mass Update	None, View, Create, Edit, Full
LIST_MATERIALREQUIREMENTSPLAN	Material Requirements Planning	Material Requirements Planning	View, None, Full, Create, Edit
LIST_MEDIAITEMFOLDER	Media Folders	File Cabinet	None, View, Create, Edit, Full
LIST_MEMDOC	Memorized Transactions	Transactions	None, View, Create, Edit, Full
LIST_MESSAGE_UNRESTRICTED	Messages for Analytics and REST	Email	None, View
LIST_MFGCOSTTEMPLATE	Manufacturing Cost Template	Inventory Management	None, View, Create, Edit, Full
LIST_MFGROUTING	Manufacturing Routing	Inventory Management	None, View, Create, Edit, Full
LIST_NEWSITEM	News Items	—	None, View, Create, Edit, Full
LIST_NOTIFICATION	Notifications	—	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_ONBOARDING_PLAN	Onboarding Plan	—	None, View, Create, Edit, Full
LIST_ORDER_REALLOCATION	Commit Orders	Advanced Inventory Management	None, View, Create, Edit, Full
LIST_ORDERMANAGEDASHBOARD	Order Management Dashboard	Sales Channel Allocation	None, View, Create, Edit, Full
LIST_ORGANIZATIONVALUE	Organizational Value	Kudos	None, View, Create, Edit, Full
LIST_OTHERNAME	Other Names	Sales Force Automation	None, View, Create, Edit, Full
LIST_OUTBOUNDREQUEST	Outbound Request	—	None, View, Create, Edit, Full
LIST_OVERTIMEPOLICY	Overtime Policies		None, View, Create, Edit, Full
LIST_PA_RECORDS	Product Analytics Records	—	None, View, Create, Edit, Full
LIST_PARTNER	Partners	Partner Relationship Management	None, View, Create, Edit, Full
LIST_PARTNERCOMMISSNRULES	Partner Commission Schedules/ Plans	Partner Commissions/Royalties	None, View, Create, Edit, Full
LIST_PAYCHECK	Paychecks	Payroll	None, View, Create, Edit, Full
LIST_PAYMENT_CARD	Payment Card	Payment Instruments	None, View, Create, Edit, Full
LIST_PAYMENT_CARD_TOKEN	Payment Card Token	Payment Instruments	None, View, Create, Edit, Full
LIST_PAYMENT_INSTRUMENTS	Payment Instruments	Payment Instruments	None, View, Create, Edit, Full
LIST_PAYMETH	Payment Methods	Order Management	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_PAYROLLITEM	Payroll Items	Payroll	None, View, Create, Edit, Full
LIST_PDFMESSAGE	PDF Messages	Communications	None, View, Create, Edit, Full
LIST_PDFTEMPLATE	PDF Template	Mail Merge	None, View, Create, Edit, Full
LIST_PHASEDPROCESS	Phased Processes		None, View, Create, Edit, Full
LIST_PICKDECOMPOSITION	Units for Pick Decomposition	—	None, View, Create, Edit, Full
LIST_PICKSTRATEGY	Pick Strategy	Warehouse Management	None, View, Create, Edit, Full
LIST_PICKTASK	Pick Task	Warehouse Management	None, View, Create, Edit, Full
LIST_PLANNEDREVENUE	Planned Revenue	Revenue Recognition	None, View, Create, Edit, Full
LIST_PLANNEDSTANDARDCOST	Planned Standard Cost	Item Record Management	None, View, Create, Edit, Full
LIST_PRESCATEGORY	Presentation Categories	SuiteCommerce	None, View, Create, Edit, Full
LIST_PRICEBOOK	Price Books	—	None, View, Create, Edit, Full
LIST_PRICEPLAN	Price Plans	—	None, View, Create, Edit, Full
LIST_PROJECT_BUDGET	Project Budget	Advanced Project Budgets	None, View, Create, Edit, Full
LIST_PROJECTREVENUERULE	Project Revenue Rules	Project Management	None, View, Create, Edit, Full
LIST_PROJECTTASK	Project Tasks	Project Management	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_PROJECTTEMPLATE	Project Templates	Project Management	None, View, Create, Edit, Full
LIST_PROMOTIONCODE	Promotion	Sales Force Automation	None, View, Create, Edit, Full
LIST_PUBLISHSEARCH	Publish Search	Publishing Search Results	None, View, Create, Edit, Full
LIST_QUANTITYPRICINGSCHEDULE	Quantity pricing Schedules	Quantity Pricing	None, View, Create, Edit, Full
LIST_REALLOCATE_ORDER_ITEM	Reallocate Order Item	Supply Allocation	View, None, Full, Create, Edit
LIST_RECOGNITIONEVENTTYPE	Custom Recognition Event Type	Advanced Revenue Recognition Or Advanced Expense Management	None, View, Create, Edit, Full
LIST_RECORDCUSTFIELD	Record Custom Field		None, View, Create, Edit, Full
LIST_RELATEDITEMS	Related Items	Web Site	None, View, Create, Edit, Full
LIST_RESOURCE	Resource	Project Management	None, View, Create, Edit, Full
LIST_RESOURCEGROUP	Resource Groups	Project Management	None, View, Create, Edit, Full
LIST_REVENUEELEMENT	Revenue Element	Revenue Recognition	None, View, Create, Edit, Full
LIST_REVENUEPLAN	Revenue Recognition Plan	Revenue Recognition	None, View, Create, Edit, Full
LIST_REVENUERECOGNITIONRULE	Revenue Recognition Rule	Revenue Recognition	None, View, Create, Edit, Full
LIST_REVRECSCHEDULE	Revenue Recognition Schedules	Revenue Recognition	None, View, Create, Edit, Full
LIST_REVRECTREATMENT	Recognition Treatment	—	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_REVRECTREATMENTRULE	Recognition Treatment Rule	—	None, View, Create, Edit, Full
LIST_REVRECFIELDMAPPING	Revenue Recognition Field Mapping	Revenue Recognition	None, View, Create, Edit, Full
LIST_REVRECVSOE	Revenue Management VSOE	VSOE	None, View, Create, Edit, Full
LIST_RSRCALLOCATION	Resource Allocations	Project Management	None, View, Create, Edit, Full
LIST_RSRCALLOCATIONAPPRV	Resource Allocation Approval	Project Management	None, View, Create, Edit, Full
LIST_RSSFEED	Publish RSS Feeds	—	None, View, Create, Edit, Full
LIST_SAASMETRIC	SaaS Metric	—	None, View, Create, Edit, Full
LIST_SALESCAMPAIGN	Sales Campaigns	Sales Campaigns	None, View, Create, Edit, Full
LIST_SALESROLE	Sales Roles	Team Selling	None, View, Create, Edit, Full
LIST_SCHEDULEMASSUPDATES	Schedule Mass Updates	—	None, View, Create, Edit, Full
LIST_SCSNAPSHOT	Supply Chain Snapshot List	—	None, View, Create, Edit, Full
LIST_SENTEMAIL	Sent Email	Sent Email List	None, View, Create, Edit, Full
LIST_SHIPITEM	Shipping Items	Accounting	None, View, Create, Edit, Full
LIST_SHIPPARTPACKAGE	Shipping Partner Package	Order Management	None, View, Create, Edit, Full
LIST_SHIPPARTREGISTRATION	Shipping Partner Registration	Order Management	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_SHIPPARTSHIPMENT	Shipping Partner Shipment	Order Management	None, View, Create, Edit, Full
LIST_SHORTCUT	Shortcuts	—	None, View, Create, Edit, Full
LIST_SITEEMAILTEMPLATE	Web Store Email Template	SuiteCommerce	None, View, Create, Edit, Full
LIST_STANDARDCOSTVERSION	Standard Cost Version	Item Record Management	None, View, Create, Edit, Full
LIST_STORECATEGORY	Store Categories	Web Store	None, View, Create, Edit, Full
LIST_STOREITEMLISTLA	Item/Category Layouts	Web Store	None, View, Create, Edit, Full
LIST_STORETAB	Store Tabs	Web Store	None, View, Create, Edit, Full
LIST_SUBSCRIPTION	Subscriptions	Advanced Subscription Billing	None, View, Create, Edit, Full
LIST_SUBSCRIPTIONCHANGEORDER	Subscription Change Orders	Advanced Subscription Billing	None, View, Create, Edit, Full
LIST_SUBSCRIPTIONPLAN	Subscription Plan	Advanced Subscription Billing	None, View, Create, Edit, Full
LIST_SUBSIDIARY	Subsidiaries	Subsidiaries	None, View, Create, Edit, Full
LIST_SUPPLY_REALLOCATION	Allocate Orders	—	None, View, Create, Edit, Full
LIST_SYSTEMNOTES	System Notes for Analytics and REST	System Notes	None, View
LIST_SYSTEMEMAILTEMPLATE	System Email Template	Customer Relationship Management	None, View, Create, Edit, Full
LIST_TALENT_ADMINISTRATION	Talent Administration	—	View, Full
LIST_TASK	Tasks	Customer Relationship Management	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_TAXDETAILSTAB	Tax Details Tab	Tax Overhauling	None, View, Edit, Full
LIST_TAXENGINESELECTION	Subsidiary Tax Registrations Tab	Tax Overhauling	None, View, Edit, Full
LIST_TAXITEM	Tax Records	Accounting	None, View, Create, Edit, Full
LIST_TAXSCHEDULE	Tax Schedules	Advanced Taxes	None, View, Create, Edit, Full
LIST_TEGATAACCOUNT	Tegata Accounts	Accounting	None, View, Create, Edit, Full
LIST_TEMPLATE_CATEGORY	Template Categories	Email Marketing Campaigns	None, View, Create, Edit, Full
LIST_TIMECODE	Time Codes	Overtime	None, View, Create, Edit, Full
LIST_TIMEOFFADMIN	Time-Off Administration	Time-Off Management	None, View, Create, Edit, Full
LIST_TRANNUMBERAUDITLOG	Access to transaction numbering audit log	Transactions	None, View, Create, Edit, Full
LIST_UNDELIVEREDEMAIL	Undelivered Emails	—	None, View
LIST_UNIT	Units	Accounting	None, View, Create, Edit, Full
LIST_UPSELL	Upsell Assistant	Upsell Manager	None, View, Create, Edit, Full
LIST_UPSELLWIZARD	Upsell Wizard	Upsell Manager	None, View, Create, Edit, Full
LIST_USAGE	Usage	Advanced Subscription Billing	None, View, Create, Edit, Full
LIST_VENDOR	Vendors	Accounting	None, View, Create, Edit, Full
LIST_VENDOR_ACH	Vendor Automated Clearing House	—	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
LIST_WBS	Work Breakdown Structure	Advanced Project Budgets	None, View, Create, Edit, Full
LIST_WEBSITE	Website (External) publisher	Web Site	None, View, Create, Edit, Full
LIST_WORKASSIGNMENT	Work Assignments	—	View, None, Full, Create, Edit
LIST_WORKCALENDAR	Work Calendar	Project Management	None, View, Create, Edit, Full
LIST_WORKPLACE	Workplaces	Payroll	None, View, Create, Edit, Full
LIST_ZONE	Zone	Warehouse Management	None, View, Create, Edit, Full
REGT_ACCTPAY	Accounts Payable Register	A/P	None, View, Create, Edit, Full
REGT_ACCTREC	Accounts Receivable Register	A/R	None, View, Create, Edit, Full
REGT_BANK	Bank Account Registers	Accounting	None, View, Create, Edit, Full
REGT_COGS	Cost of Goods Sold Registers	Accounting	None, View, Create, Edit, Full
REGT_CREDCARD	Credit Card Registers	Accounting	None, View, Create, Edit, Full
REGT_DEFEREXPENSE	Deferred Expense Registers	Amortization	None, View, Create, Edit, Full
REGT_DEFERREVENUE	Deferred Revenue Registers	Revenue Recognition	None, View, Create, Edit, Full
REGT_EQUITY	Equity Registers	Accounting	None, View, Create, Edit, Full
REGT_EXPENSE	Expense Registers	Accounting	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
REGT_FIXEDASSET	Fixed Asset Registers	Accounting	None, View, Create, Edit, Full
REGT_INCOME	Income Registers	Accounting	None, View, Create, Edit, Full
REGT_LONGTERMLIAB	Long Term Liability Registers	Accounting	None, View, Create, Edit, Full
REGT_NONPOSTING	Non Posting Registers	Accounting	None, View, Create, Edit, Full
REGT_OTHASSET	Other Asset Registers	Accounting	None, View, Create, Edit, Full
REGT_OTHCURRASSET	Other Current Asset Registers	Accounting	None, View, Create, Edit, Full
REGT_OTHCURRLIAB	Other Current Liability Registers	Accounting	None, View, Create, Edit, Full
REGT_OTHEXPENSE	Other Expense Registers	Accounting	None, View, Create, Edit, Full
REGT_OTHINCOME	Other Income Registers	Accounting	None, View, Create, Edit, Full
REGT_PAYROLL	Run Payroll	Payroll	None, View, Create, Edit, Full
REGT_STAT	Statistical Account Registers	Statistical Accounting	None, View, Create, Edit, Full
REGT_UNBILLEDREC	Unbilled Receivable Registers	Revenue Commitments	None, View, Create, Edit, Full
REPO_1099	Form 1099 - Federal Miscellaneous Income	A/P	None, View, Create, Edit, Full
REPO_940	Form 940 - Employer's Annual Federal Unemployment Tax Return	Payroll	None, View
REPO_941	Form 941 - Employer's Quarterly Federal Tax Return	Payroll	None, View
REPO_ACCOUNTDETAIL	Account Detail	Accounting	None, View
REPO_AMORTIZATION	Amortization Reports	Amortization	None, View

Permission ID	Permission Name	Feature	Valid Levels
REPO_ANALYTICS	SuiteAnalytics Workbook	SuiteAnalytics Workbook	None, Edit
REPO_AP	Accounts Payable	A/P	None, View
REPO_AR	Accounts Receivable	A/R	None, View
REPO_AUTHPARTNERCOMMISSION	Partner Authorized Commission Reports	Partner Commissions/Royalties	None, View
REPO_BALANCESHEET	Balance Sheet	Accounting	None, View
REPO_BOOKINGS	Sales Order Reports	Sales Force Automation	None, View
REPO_BUDGET	Budget	Accounting	None, View
REPO_CASHFLOW	Cash Flow Statement	Accounting	None, View
REPO_COMMISSION	Commission Reports	Employee Commissions	None, View
REPO_CONSOLIDATED_REPORTING	Consolidated Reporting	—	None, View
REPO_CUSTOMIZATION	Report Customization	Report Customization	None, View
REPO_DEFERREDEXPENSE	Deferred Expense Reports	—	None, View
REPO_FINANCIALS	Financial Statements	Accounting	None, View
REPO_GL	General Ledger	Accounting	None, View
REPO_GRANT_ACCESS	Granting access to Reports		None, View, Create, Edit, Full
REPO_GSTSUMMARY	GST Summary Report		None, View
REPO_INTEGRATION	Integration	SuiteTalk	None, View
REPO_INVENTORY	Inventory	Inventory	None, View
REPO_ISSUE	Issue Reports	Issue Management	None, View
REPO_MARKETING	Marketing Campaign Reports	Marketing Automation	None, View
REPO_NONPOSTING	Sales Order Transaction Report	Order Management	None, View
REPO_PANDL	Income Statement	Accounting	None, View
REPO_PARTNERCOMMISSION	Partner Commission Reports	Partner Commissions/Royalties	None, View
REPO_PAYCHECKDETAIL	Payroll Check Register	Payroll	None, View
REPO_PAYROLL	Payroll Summary & Detail Reports	Payroll	None, View
REPO_PAYROLLHIDEFINEMPINFO	Hide Employee Information on Financial Reports	Payroll	None, View
REPO_PAYROLLHOURSEARNING	Payroll Hours & Earnings	Payroll	None, View
REPO_PAYROLLJOURNAL	Payroll Journal Report	Payroll	None, View
REPO_PAYROLLLIAB	Payroll Liability Report	Payroll	None, View

Permission ID	Permission Name	Feature	Valid Levels
REPO_PAYROLLSTATEWITHHOLD	Payroll State Withholding	Payroll	None, View
REPO_PAYROLLW2	Form W-2 - Wage and Tax Statement	Payroll	None, View, Create, Edit, Full
REPO_PERIODENDFINANCIALS	Period End Financial Statements	—	None, View
REPO_PROJECT_ACCOUNTING	Project Accounting	Accounting	None, View
REPO_PSTSUMMARY	PST Summary Report		None, View
REPO_PURCHASEORDER	Purchase Order Reports	Purchase Orders	None, View
REPO_PURCHASES	Purchases	Accounting	None, View
REPO_QUOTA	Quota Reports	Sales Force Automation	None, View
REPO_RECONCILE	Reconcile Reporting	Accounting	None, View
REPO_REMINDEREMPLOYEE	Employee Reminders	Employees	None, View
REPO_RETURNAUTH	Return Authorization Reports	Return Authorizations	None, View
REPO_REVREC	Revenue Recognition Reports	Revenue Recognition	None, View
REPO_RSRCALLOCATION	Resource Allocation Reports	Project Management	None, View
REPO_SALES	Sales	Sales Force Automation	None, View
REPO_SALESORDER	Sales Order Fulfillment Reports	Sales Orders	None, View
REPO_SALES_PARTNER	Sales By Partner	Partner Relationship Management	None, View
REPO_SALES_PROMO	Sales By Promotion Code	Sales Force Automation	None, View
REPO_SALES_PROMO	Sales By Promotion	Sales Force Automation	None, View
REPO_SCHEDULE	Report Scheduling	Reports	None, Full
REPO_SNAPSHOTCASE	Support Case Snapshot/ Reminders	Support	None, View
REPO_SNAPSHOTLEAD	Lead Snapshot/Reminders	Business	None, View
REPO_SFA	Sales Force Automation	Customer Relationship Management	None, View
REPO_SUPPORT	Support	Customer Support and Service	None, View
REPO_TAX	Tax	Accounting	None, View
REPO_TAXREPORTS	Tax Reports	Accounting	None, View
REPO_TIME	Time Tracking	Time Tracking	None, View
REPO_TRAN	Transaction Detail	Transactions	None, View
REPO_TRIALBALANCE	Trial Balance	Accounting	None, View
REPO_UNBILLED	Accounts Receivable Un-Billed	Accounting	None, View

Permission ID	Permission Name	Feature	Valid Levels
REPO_W4	Form W4 - Employee's Withholding Allowance Certificate	Payroll	None, View
REPO_WEBSITE	Web Site Report	Web Site	None, View
REPO_WEBSTORE	Web Store Report	Web Store	None, View
REPO_WORKFORCEANALYTICS	Workforce Analytics	Workforce Analytics	View
TRAN_ADJUSTMENTJOURNAL	Currency Adjustment Journal	Accounting	None, View, Create, Edit, Full
TRAN_ALLOCSCHEDULE	Create Allocation Schedules	Expense Allocation	None, View, Create, Edit, Full
TRAN_AMENDW4	Amend W-4	Employees	None, View, Create, Edit, Full
TRAN_APPROVECOMMISSN	Employee Commission Transaction Approval	Employee Commissions	None, View, Create, Edit, Full
TRAN_APPROVEDD	Approve Direct Deposit	Direct Deposit	None, View, Create, Edit, Full
TRAN_APPROVEEFT	Approve EFT	Electronic Funds Transfer	None, View, Create, Edit, Full
TRAN_APPROVEPARTNERCOMM	Partner Commission Transaction Approval	Partner Commissions/Royalties	None, View, Create, Edit, Full
TRAN_AUDIT	Audit Trail	Transactions	None, View, Create, Edit, Full
TRAN_AUTO_CASH	Automated Cash Application	Accounting	None, Full
TRAN_BALANCEOVERVIEW	Balance Overview	Intercompany Framework	None, View
TRAN_BALJRNAL	Balancing Journals	—	None, View, Create, Full
TRAN_BINTRNFR	Bin Transfer	Bin Management	None, View, Create, Edit, Full
TRAN_BINWKSHT	Bin Putaway Worksheet	Bin Management	None, View, Create, Edit, Full
TRAN_BLANKORD	Blanket Purchase Order	Purchasing and Receiving	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_BLANKORDAPPRV	Blanket Purchase Order Approval	Purchasing and Receiving	None, View, Create, Edit, Full
TRAN_BUDGET	Set Up Budgets	Accounting	None, View, Create, Edit, Full
TRAN_BUILD	Build Assemblies	Assembly Items	None, View, Create, Edit, Full
TRAN_CARDCHRG	Credit Card	Accounting	None, View, Create, Edit, Full
TRAN_CARDHOLDERAUTHENTICATION	Cardholder Authentication	Credit Card Payments	None, View, Edit, Full, Create
TRAN_CARDHOLDERAUTHEVENT	Cardholder Authentication Event	Credit Card Payments	None, View, Edit, Full, Create
TRAN_CARDRFND	Credit Card Refund	Accounting	None, View, Create, Edit, Full
TRAN_CASHRFND	Cash Sale Refund	Accounting	None, View, Create, Edit, Full
TRAN_CASHSALE	Cash Sale	Accounting	None, View, Create, Edit, Full
TRAN_CHARGE	Charge	Project Management	None, View, Create, Edit, Full
TRAN_CHARGERULE	Charge Rule	Project Management	None, View, Create, Edit, Full
TRAN_CHECK	Check	Accounting	None, View, Create, Edit, Full
TRAN_CLEARHOLD	Override Payment Hold	Order Management	None, View, Create, Edit, Full
TRAN_COMMISSN	Employee Commission Transaction	Employee Commissions	None, View, Create, Edit, Full
TRAN_COMMITPAYROLL	Commit Payroll	Payroll	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_COPY_BUDGET	Copy Budgets	Accounting	None, View, Create, Edit, Full
TRAN_CREATEINVCOUNT	Create Inventory Counts	Inventory Count	None, View, Create, Edit, Full
TRAN_CUSTAUTH	Customer Payment Authorization	Customer Payment Authorizations	None, View, Create, Edit, Full
TRAN_CUSTCHRG	Statement Charge	A/R	None, View, Create, Edit, Full
TRAN_CUSTCRED	Credit Memo	A/R	None, View, Create, Edit, Full
TRAN_CUSTDEP	Customer Deposit	A/R	None, View, Create, Edit, Full
TRAN_CUSTINVC	Invoice	A/R	None, View, Create, Edit, Full
TRAN_CUSTINVCAPPRV	Invoice Approval	Order Management	None, View, Create, Edit, Full
TRAN_CUSTPYMT	Customer Payment	A/R	None, View, Create, Edit, Full
TRAN_CUSTRFND	Customer Refund	A/R	None, View, Create, Edit, Full
TRAN_DEPAPPL	Deposit Application	A/R	None, View, Create, Edit, Full
TRAN_DEPOSIT	Deposit	Accounting	None, View, Create, Edit, Full
TRAN_EDITBANKINGINFO	Personal Banking Information	—	None, View, Full
TRAN_EDITPROFILE	Edit Profile	Employees	None, View, Create, Edit, Full
TRAN_ESTIMATE	Estimate	Estimates	None, View, Create, Edit, Full
TRAN_ESTIMATEDCOSTOVERRIDE	Override Estimated Cost on Transactions		None, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_EXPREPT	Expense Report	Expense Reports	None, View, Create, Edit, Full
TRAN_FFTREQ	Fulfillment Request	Order Management	None, View, Create, Edit, Full
TRAN_FINCHRG	Finance Charge	A/R	None, View, Create, Edit, Full
TRAN_FIND	Find Transaction	Transactions	None, View, Create, Edit, Full
TRAN_FORECAST	Edit Forecast	Sales Force Automation	None, View, Create, Edit, Full
TRAN_FXREVAL	Currency Revaluation	Multiple Currencies	None, View, Create, Edit, Full
TRAN_GST_REFUND	Process GST Refund	Accounting	None, View, Create, Edit, Full
TRAN_IMPORTOLBFILE	Import Online Banking (QIF) File	Accounting	None, View, Create, Edit, Full
TRAN_INTERCOADJ	Intercompany Adjustments	Accounting	None, View, Create, Edit, Full
TRAN_INVADJST	Adjust Inventory	Inventory	None, View, Create, Edit, Full
TRAN_INVCOUNT	Count Inventory	Advanced Inventory Management	None, View, Create, Edit, Full
TRAN_INVDISTR	Distribute Inventory	Multi-Location Inventory	None, View, Create, Edit, Full
TRAN_INVREVAL	Revalue Inventory Cost	Inventory	None, View, Create, Edit, Full
TRAN_INVTRNFR	Transfer Inventory	Multi-Location Inventory	None, View, Create, Edit, Full
TRAN_INVWKSHT	Adjust Inventory Worksheet	Inventory	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_ITEMRCPT	Item Receipt	Advanced Receiving	None, View, Create, Edit, Full
TRAN_ITEMSHIP	Item Fulfillment	Advanced Shipping	None, View, Create, Edit, Full
TRAN_JOURNAL	Make Journal Entry	Accounting	None, View, Create, Edit, Full
TRAN_JOURNALAPPRV	Journal Approval	Accounting	None, View, Create, Edit, Full
TRAN_LIABPYMT	Payroll Liability Payments	Payroll	None, View, Create, Edit, Full
TRAN_MANAGEPAYROLL	Manage Payroll	Payroll	None, View, Create, Edit, Full
TRAN_MATCHING_RULES	Matching Rules for Online Banking	Accounting	None, View, Create, Edit, Full
TRAN_MGRFORECAST	Edit Manager Forecast	Sales Force Automation	None, View, Create, Edit, Full
TRAN_NETTINGSETTLEMENTAPPRV	Netting Settlement Approval	Intercompany Framework	None, Edit
TRAN_NETTSTLM	Netting Settlement	Intercompany Framework	None, View, Create, Edit, Full
TRAN_OPENBAL	Enter Opening Balances	Accounting	None, View, Create, Edit, Full
TRAN_OPPRTNTY	Opportunity	Opportunities	None, View, Create, Edit, Full
TRAN_ORDERRESERVATION	Order Reservation	—	None, View, Edit, Full, Create
TRAN_ORDRESVAPPRV	Approve Order Reservation		None, View, Create, Edit, Full
TRAN_OWNTRNSF	Ownership Transfer	—	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_PARTNERCOMMISSN	Partner Commission Transaction	Partner Commissions/Royalties	None, View, Create, Edit, Full
TRAN_PAYCHECK	Individual Paycheck	Payroll	None, View, Create, Edit, Full
TRAN_PAYMENTAUDIT	Access Payment Audit Log	Order Management	None, View, Create, Edit, Full
TRAN_PAYMENTEVENT	View Payment Events	Credit Card Payments	None, View, Create, Edit, Full
TRAN_PAYMENTRESULTPREVIEW	View Payment Result Previews	Credit Card Payments	None, View, Edit, Full, Create
TRAN_PAYROLLRUN	Process Payroll	Payroll	None, View, Create, Edit, Full
TRAN_PCHKJRNL	Paycheck Journal	Employees	None, View, Create, Edit, Full
TRAN_PEJRNL	Period End Journals	—	None, View, Create, Edit, Full
TRAN_POSTPERIODS	Posting Period on Transactions	Accounting	None, View, Create, Edit, Full
TRAN_POSTVENDORBILLVARIANCE	Post Vendor Bill Variances	Vendors	None, View, Create, Edit, Full
TRAN_PRICELIST	Generate Price Lists	Item Record Management	None, View, Create, Edit, Full
TRAN_PRINTSHIPMENTDOCS	Print Shipment Documents	Shipping Partners	None, View, Create, Edit, Full
TRAN_PROJECT_IC_CHARGE_REQUEST	Project Intercompany Cross Charge Request	Project Intercompany Cross Charge Request	None, View, Create, Edit, Full
TRAN_PURCHCON	Purchase Contract	Vendors	None, View, Create, Edit, Full
TRAN_PURCHCONAPPRV	Purchase Contract Approval	Vendors	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_PURCHORD	Purchase Order	Purchase Orders	None, View, Create, Edit, Full
TRAN_PURCHORDBILL	Bill Purchase Orders	Advanced Receiving	None, View, Create, Edit, Full
TRAN_PURCHORDRECEIVE	Receive Order	Purchase Orders	None, View, Create, Edit, Full
TRAN_PURCHREQ	Requisition	Purchasing and Receiving	None, View, Create, Edit, Full
TRAN_PURCHREQAPPRV	Requisition Approval	Purchasing and Receiving	None, View, Create, Edit, Full
TRAN_QUOTA	Establish Quotas	Sales Force Automation	None, View, Create, Edit, Full
TRAN_RECOG_GIFTCERT_INCOME	Recognize Gift Certificate Income	Accounting	None, View, Create, Edit, Full
TRAN_RECONCILE	Reconcile	Accounting	None, View, Create, Edit, Full
TRAN_REVARRNG	Revenue Arrangement	Revenue Recognition	None, View, Create, Edit, Full
TRAN_REVARRNGAPPRV	Revenue Arrangement Approval	Revenue Recognition	None, View, Create, Edit, Full
TRAN_REVCOMM	Revenue Commitment	Revenue Commitments	None, View, Create, Edit, Full
TRAN_REVCOMRV	Revenue Commitment Reversal	Revenue Commitments	None, View, Create, Edit, Full
TRAN_REVCONTR	Revenue Contracts	Revenue Recognition	None, View, Create, Edit, Full
TRAN_RFQ	Request For Quote	Purchasing and Receiving	None, View, Create, Edit, Full
TRAN_RTNAUTH	Return Authorization	Return Authorizations	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_RTNAUTHAPPRV	Return Auth. Approval	Return Authorizations	None, View, Create, Edit, Full
TRAN_RTNAUTHCREDIT	Refund Returns	Order Management	None, View, Create, Edit, Full
TRAN_RTNAUTHRECEIVE	Receive Returns	Order Management	None, View, Create, Edit, Full
TRAN_RTNAUTHREVERSEREVCOMMIT	Generate Revenue Commitment Reversals	Revenue Commitments	None, View, Create, Edit, Full
TRAN_SALESORD	Sales Order	Sales Orders	None, View, Create, Edit, Full
TRAN_SALESORDAPPRV	Sales Order Approval	Sales Orders	None, View, Create, Edit, Full
TRAN_SALESORDCOMMITREVENUE	Generate Revenue Commitment	Revenue Commitments	None, View, Create, Edit, Full
TRAN_SALESORDFULFILL	Fulfill Orders	Order Management	None, View, Create, Edit, Full
TRAN_SALESORDINVOICE	Invoice Sales Orders	Advanced Shipping	None, View, Create, Edit, Full
TRAN_SALESORDREVENUECONTRACT	Generate Single Order Revenue Contracts	Revenue Recognition	None, View, Create, Edit, Full
TRAN_STATEMENT	Generate Statements	A/R	None, View, Create, Edit, Full
TRAN_STATCHNG	Inventory Status Change	Inventory Status	None, View, Create, Edit, Full
TRAN_STATUSDD	Direct Deposit Status	Direct Deposit	None, View, Create, Edit, Full
TRAN_STATUSEFT	EFT Status	Electronic Funds Transfer	None, View, Create, Edit, Full
TRAN_STPICKUP	Store Pickup Fulfillment	SuiteCommerce InStore	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_SYSJRNL	System Journal	Accounting	None, View, Create, Edit, Full
TRAN_TAXLIAB	Tax Liability Payment	Accounting	None, View, Create, Edit, Full
TRAN_TAXPYMT	Pay Sales Tax	Accounting	None, View, Create, Edit, Full
TRAN_TEGPYBL	Tegata Payable	Accounting	None, View, Create, Edit, Full
TRAN_TEGRCVBL	Tegata Receivable	Accounting	None, View, Create, Edit, Full
TRAN_TIMEBILL	Track Time	Time Tracking	None, View, Create, Edit, Full
TRAN_TIMECALC	Calculate Time	Time Tracking	None, View, Create, Edit, Full
TRAN_TIMEPOST	Post Time	Project Management	None, View, Create, Edit, Full
TRAN_TIMER	Timer	Time Tracking	None, View, Create, Edit, Full
TRAN_TRANSFER	Transfer Funds	Accounting	None, View, Create, Edit, Full
TRAN_TRNFRORD	Transfer Order	Inventory Management	None, View, Create, Edit, Full
TRAN_TRNFRORDAPPRV	Transfer Order Approval	Inventory Management	None, View, Create, Edit, Full
TRAN_UNBUILD	Unbuild Assemblies	Assembly Items	None, View, Create, Edit, Full
TRAN_VENDAUTH	Vendor Return Authorization	Vendor Return Authorizations	None, View, Create, Edit, Full
TRAN_VENDAUTHAPPRV	Vendor Return Auth. Approval	Vendor Return Authorizations	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_VENDAUTHCREDIT	Credit Returns	Vendor Returns	None, View, Create, Edit, Full
TRAN_VENDAUTHRETURN	Vendor Returns	Vendor Return Authorizations	None, View, Create, Edit, Full
TRAN_VENDBILL	Bills	A/P	None, View, Create, Edit, Full
TRAN_VENDBILLAPPRV	Vendor Bill Approval	Vendor Bills	None, View, Create, Edit, Full
TRAN_VENDCRED	Enter Vendor Credits	A/P	None, View, Create, Edit, Full
TRAN_VENDPYMT	Pay Bills	A/P	None, View, Create, Edit, Full
TRAN_VENDPYMTAPPRV	Vendor Payment Approval	Vendor Bills	None, View, Create, Edit, Full
TRAN_VENDRFQ	Vendor Request For Quote	Purchasing and Receiving	None, View, Create, Edit, Full
TRAN_VPREPAPP	Vendor Prepayment Application	—	None, View, Create, Edit, Full
TRAN_VPREP	Vendor Prepayment	—	None, View, Create, Edit, Full
TRAN_WAVE	Wave	—	None, View, Create, Edit, Full
TRAN_WOCLOSE	Work Order Close	Inventory Management	None, View, Create, Edit, Full
TRAN_WOCOMPL	Work Order Completion	Inventory Management	None, View, Create, Edit, Full
TRAN_WOISSUE	Work Order Issue	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORD	Work Order	Work Orders	None, View, Create, Edit, Full

Permission ID	Permission Name	Feature	Valid Levels
TRAN_WORKORDBUILD	Build Work Orders	Work Orders	None, View, Create, Edit, Full
TRAN_WORKORDCLOSE	Close Work Orders	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORDCOMPLETE	Enter Completions	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORDISSUE	Issue Components	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORDMARKBUILT	Mark Work Orders Built	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORDMARKFIRMED	Mark Work Orders Firmed	Inventory Management	None, View, Create, Edit, Full
TRAN_WORKORDMARKRELEASED	Mark Work Orders Released	Inventory Management	None, View, Create, Edit, Full
TRAN_XCHGJRNL	Cross Charge Journal	Intercompany Framework	None, View
TRAN_YTDADJST	Enter Year-To-Date Payroll Adjustments	Payroll	None, View, Create, Edit, Full

Feature Names and IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following table provides the internal IDs and feature names for all NetSuite features. You can use the feature ID to see if a particular feature is enabled in your account by calling runtime.isFeatureInEffect(options) in the N/runtime Module.

Feature Internal ID	Feature Name
ACCOUNTING	Accounting
ACCOUNTINGPERIODS	Accounting Periods
ACTIVITYCODES	Activity Codes
ADDONS	Add-On Items
ADVANCEDBILLOFMATERIALS	Advanced Bill of Materials
ADVANCEDEMPLOYEEPERMISSIONS	Advanced Employee Permissions
ADVANCEDJOBS	Project Management
ADVANCEDNUMBERINGSEQUENCES	Advanced Numbering

Feature Internal ID	Feature Name
ADVANCEDPRINTING	Advanced PDF/HTML Templates
ADVANCEDPROCUREMENTAPPROVALS	Advanced Procurement Approvals
ADVANCEDPROJECTACCOUNTING	Advanced Project Profitability
ADVANCEDPROMOTIONS	Advanced Promotions
ADVANCEDREVENUERECOGNITION	Advanced Revenue Management
ADVANCEDREVENUERECOGNITIONAPP	Advanced Revenue Recognition SuiteApp
ADVANCEDSITECUST	Advanced Site Customization
ADVANCEDSITEMANAGEMENT	Site Management Tools
ADVBILLING	Advanced Billing
ADVBINSERIALLOTMGMT	Advanced Bin/Numbered Inventory Management
ADVFORECASTING	Advanced Forecasting
ADVINVENTORYMGMT	Advanced Inventory Management
ADVPARTNERACCESS	Advanced Partner Access
ADVRECEIVING	Advanced Receiving
ADVSHIPPING	Advanced Shipping
ADVSUBSCRIPTIONBILLING	Advanced Subscription Billing
ADVTAXENGINE	Advanced Taxes
ADVWEBREPORTS	Advanced Web Reports
ADVWEBSEARCH	Advanced Web Search
ALTSALESADVFORECAST	ALT_SALES Advanced Forecasting
ALTSALESAMOUNT	Alternate Sales Amount
AMORTIZATION	Amortization
APPROVALROUTING	Approval Routing
ARMINCONFIGMODE	Advanced Revenue Management In Configuration Mode
ARMREVENUEALLOCATION	Advanced Revenue Management (Revenue Allocation)
ASSEMBLIES	Assembly Items
ASYNCCUSTOMER	Asynchronous AfterSubmit Customer Processing
ASYNCSALESORDER	Asynchronous AfterSubmit Sales Order Processing
AUTOAPPLYPROMOTIONS	Auto-Apply Promotions
AUTOLOCATIONASSIGNMENT	Automatic Location Assignment
AVAILABLETOPROMISE	Available To Promise
BALANCING_SEGMENTS	Balancing Segments

Feature Internal ID	Feature Name
BARCODES	Bar Coding and Item Labels
BILLCAPTURE	Bill Capture
BILLINGACCOUNTS	Billing Accounts
BILLINGCLASSES	Per-Employee Billing Rates
BILLINGRATECARDS	Billing Rate Cards
BILLINGWORKCENTER	Billing Operations
BILLSCOSTS	Bill Costs To Customers
BINMANAGEMENT	Bin Management
BLANKETPURCHASEORDERS	Blanket Purchase Orders
BOXNET	Box Document Management
CAMPAIGNASSISTANT	Campaign Assistant
CAMPAIGNSUBSCRIPTIONS	Subscription Categories
CCTRACKING	Credit Card Payments
CENTRALIZEDPURCHASINGBILLING	Centralized Purchasing and Billing
CHARGEBASEDBILLING	Charge-Based Billing
CHECKOUTSUBDOMAIN	Customizable Checkout Subdomains
CLASSES	Classes
COMMERCECATEGORIES	Commerce Categories
COMMERCESEARCHANALYTICS	Commerce Search Analytics
COMMISSIONONCUSTOMFIELDS	Commission on Custom Fields
COMMISSIONS	Employee Commissions
COMMITPLUSOVERAGE	Commit Plus Overage
COMPENSATIONTRACKING	Compensation Tracking
CONSOLPAYMENTS	Consolidated Payments
CREATESUITEBUNDLES	Create bundles with SuiteBundler
CRM	Customer Relationship Management
CRMTIME	Time Tracking for CRM
CRM_TEMPLATE_CATEGORIES	CRM Template Categories
CROSSSUBSIDIARYFULFILLMENT	Intercompany Cross-Subsidiary Fulfillment
CUSTOMCODE	Client SuiteScript
CUSTOMERACCESS	Customer Access
CUSTOMGLLINES	Custom GL Lines

Feature Internal ID	Feature Name
CUSTOMRECORDS	Custom Records
CUSTOMSEGMENTS	Custom Segments
CUSTOMTRANSACTIONS	Custom Transactions
DEPARTMENTS	Departments
DISTRIBUTIONRESOURCEPLANNING	Distribution Resource Planning
DOCUMENTPUBLISHING	Document Publishing
DOCUMENTS	File Cabinet
DOWNLOADITEMS	Sell Downloadable Files
DROPSHIPMENTS	Drop Shipments & Special Orders
DUPLICATES	Duplicate Detection & Merge
DYNALLOCATION	Dynamic Allocation
EFFECTIVEDATING	Effective Dating
EMAILINTEGRATION	Capture Email Replies
EMPLOYEECENTERPUBLISHING	Employee Center Dashboard Publishing
EMPLOYEECHANGEREQUESTS	Employee Change Requests
EMPPERMS	Global Permissions
ENHANCEDINVENTORYLOCATION	Advanced Item Location Configuration
ENHANCEDPREMIERPAYROLL	Enhanced Premier Payroll
ESCALATIONRULES	Automated Case Escalation
ESTIMATES	Estimates
EXPENSEALLOCATION	Expense Allocation
EXPREPORTS	Expense Reports
EXTCRM	Online Forms
EXTREMELIST	Inline Editing
EXTSTORE	External Catalog Site (WSDK)
FCADVANCEDSECURITY	File Cabinet Advanced Security
FCEXPENSE	Enhanced File Security – Employee Expense Report Folders
FCEXPENSEMIGRATECONTROLLER	Expense Migration Scheduled From Start
FULFILLMENTREQUEST	Fulfillment Request
FXRATETYPE	Currency Exchange Rate Types
FXRATEUPDATES	Currency Exchange Rate Integration
GAINLOSSACCTMAPPING	Foreign currency variance mapping

Feature Internal ID	Feature Name
GIFTCERTIFICATES	Gift Certificates
GLAUDITNUMBERING	GL Audit Numbering
GRIDORDERMANAGEMENT	Grid Order Management
GROSSPROFIT	Gross Profit
GROUPAVERAGECOSTING	Group Average Costing
HELPDESK	Help Desk
HISTORICALMETRICS	Historical Metrics
HRANALYSIS	Workforce Analytics
HTML_FORMULAS_IN_SEARCH	HTML Formulas in Search
I18NTAXREPORTS	International Tax Reports
IC_FRAMEWORK_OR_AIM	IC Framework OR Intercompany Auto Elimination
INBOUNDCASEEMAIL	Email Case Capture
INBOUNDSHIPMENT	Inbound Shipment Management
INSTALLMENTS	Installments
INTELLIGENTRECOMMENDATIONS	Intelligent Recommendations
INTERCOMPANYAUTODROPSHIP	Automated Intercompany Drop Ship
INTERCOMPANYAUTOELIMINATION	Automated Intercompany Management
INTERCOMPANYELIMINATIONENGINE	IC Framework OR IC Netting OR Advanced Inter. Elimination
INTERCOMPANYFRAMEWORK	Intercompany Framework
INTERCOMPANYTIMEEXPENSE	Intercompany Time and Expense
INTERNATIONALPHONENUMBERS	Phone Number Formatting
INTRANET	Intranet
INTRANSITPAYMENTS	In-Transit Payments
INVENTORY	Inventory
INVENTORYCOUNT	Inventory Count
INVENTORYSTATUS	Inventory Status
INVOICEGROUP	Invoice Groups
IPADDRESSRULES	IP Address Rules
ISSUEDB	Issue Management
ITEMDEMANDPLANNING	Demand Planning
ITEMOPTIONS	Item Options
JOBCOSTING	Job Costing and Project Budgeting

Feature Internal ID	Feature Name
JOBMANAGEMENT	Job Management
JOBREQUISITION	Job Requisitions
JOBS	Projects
KILLINBOUNDSSO	Disable Inbound Single Sign-on
KNOWLEDGEBASE	Knowledge Base
KPIREPORTS	KPI Scorecards
KUDOS	Kudos
LANDEDCOST	Landed Cost
LEADMANAGEMENT	Lead Conversion
LOCATIONS	Locations
LOTNUMBEREDINVENTORY	Lot Tracking
MAILMERGE	Mail Merge
MARKETING	Marketing Automation
MATERIALREQUIREMENTSPLANNING	Material Requirements Planning
MATRIXITEMS	Matrix Items
MERCHANDISEHIERARCHY	Merchandise Hierarchy
MFGROUTING	Manufacturing Routing and Work Center
MFGWORKINPROCESS	Manufacturing Work In Process
MOBILEPUSHNTF	Mobile Push Notification
MOSS	EU One Stop Shop
MULTIBOOKMULTICURR	Multi-Book Accounting and Multiple Currencies
MULTICURRENCY	Multiple Currencies
MULTICURRENCYCUSTOMER	Multi-Currency Customers
MULTICURRENCYMERGE	Multi Currency Merge
MULTICURRENCYVENDOR	Multi-Currency Vendors
MULTILANGUAGE	Multi-Language
MULTILOCINVT	Multi-Location Inventory
MULTIPARTNER	Multi-Partner Management
MULTIPLEBUDGETS	Multiple Budgets
MULTIPLECALENDARS	Multiple Calendars
MULTISHIPTO	Multiple Shipping Routes
MULTISUBSIDIARYCUSTOMER	Multi Subsidiary Customer

Feature Internal ID	Feature Name
MULTIVENDOR	Multiple Vendors
MULTPRICE	Multiple Prices
NETSUITEAPPROVALSWORKFLOW	NetSuite Approvals Workflow
NOTALTSALESAMOUNT	Not ALT_SALES Amount
NOTMULTIPARTNER	Not Multipartner
NOTTEAMSELLING	Not Team Selling
NSASOIDCPROVIDER	NetSuite as OIDC Provider
NSAW_ADDITIONAL_INSTANCE	NetSuite Analytics Warehouse Additional Instance
NSAW_MULTI_INSTANCE_CONNECTOR	NSAW Multi-Instance Connector
OAUTH2	OAuth 2.0 Authentication
OIDC	OpenID Connect (OIDC) Single Sign-on
ONLINEORDERING	Online Ordering
OPENIDSSO	OpenID Single Sign-on
OPPORTUNITIES	Opportunities
OTHERSUBLISTFIELDS	Other Sublist Fields
OUTSOURCEDMFG	Outsourced Manufacturing
OVERTIME	Enhanced Timesheets Using Workforce Management Wage Rules
PARTNERACCESS	Partner Access
PARTNERCOMMISSIONS	Partner Commissions/Royalties
PAYABLES	A/P
PAYCHECKJOURNAL	Paycheck Journal
PAYMENTINSTRUMENTS	Payment Instruments
PAYMENTLINK	Payment Link
PAYPALINTEGRATION	PayPal Integration
PAYROLL	Payroll
PAYROLLSERVICE	Payroll Service
PERFORMANCEMANAGEMENT	Performance Management
PERIODENDJOURNALENTRIES	Period End Journal Entries
PERSONALIZED_CATALOG_VIEWS	Personalized Catalog Views
PICKPACKSHIP	Pick, Pack and Ship
PI_REMOVAL	Remove Personal Information
PLANNEDWORK	Planned Work

Feature Internal ID	Feature Name
PREPAYWITHDRAWDOWN	Prepay With Drawdown
PRM	Partner Relationship Management
PROJECTTASKMANAGER	Project Task Manager
PROMOCODES	Promotion Codes
PURCHASECARDDATA	Send Purchase Card Data
PURCHASECONTRACTS	Purchase Contracts
PURCHASEORDERS	Purchase Orders
PURCHASEREQS	Purchase Requests
QUANTITYPRICING	Quantity Pricing
RECEIVABLES	A/R
REQUIREDDEPOSITWORKFLOW	Required Deposit Workflow
REQUISITIONS	Requisitions
RESOURCEALLOCATIONAPPROVAL	Resource Allocation Approval Workflow
RESOURCEALLOCATIONCHART	Resource Allocation Chart
RESOURCEALLOCATIONS	Resource Allocations
RESOURCESKILLSETS	Resource Skill Sets
RESTWEBSERVICES	REST Web Services
RETURNAUTHS	Return Authorizations
REVENUECOMMITMENTS	Revenue Commitments
REVENUERECOGNITION	Revenue Recognition
REVRECSALESORDERFORECASTING	Sales Order Revenue Forecasting
REVRECVSOE	VSOE
RFQ	Request For Quote
RULEBASEDRECOGNITIONTREATMENT	Rule-Based Recognition Treatment
SAASMETRICS	SaaS Metrics
SALESCAMPAIGNS	Sales Campaigns
SALESCHANNELALLOCATION	Sales Channel Allocation
SALESORDERS	Sales Orders
SAMLSSO	SAML Single Sign-on
SDFCOPYTOACCOUNT	Copy To Account
SERIALIZEDINVENTORY	Serialized Inventory
SERVERSIDESCRIPTING	Server SuiteScript

Feature Internal ID	Feature Name
SERVICEPRINTEDCHECKS	Service Printed Checks and Stubs
SERVICEPRINTEDW2S	Service Printed W-2s and 1099s
SFA	Sales Force Automation
SFA_AND_NOTASA	SFA And Not ALT_SALES Amount
SHIPPINGLABELS	Shipping Label Integration
SHIPPINGPARTNERS	Shipping Partners
SITEBUILDER	Site Builder (Website)
SITEBUILDER_STORE	Site Builder (Web Store)
SITELOCATIONALIASES	Descriptive URLs
SOFTDESCRIPTORS	Credit Card Soft Descriptors
STACKABLEPROMOTIONS	SuitePromotions
STANDARDCOSTING	Standard Costing
STATACCOUNTING	Statistical Accounts
STOREPICKUP	Store Pickup
SUBSCRIPTIONBILLING	Subscription Billing
SUITE_OAX_CONNECTOR	NetSuite Analytics Warehouse
SUITEANALYTICSCONNECT	SuiteAnalytics Connect
SUITEAPPCONTROLCENTER	SuiteApp Control Center
SUITEAPPDEVELOPMENTFRAMEWORK	SuiteCloud Development Framework
SUITECOMMERCE	SuiteCommerce
SUITECOMMERCE_ADVANCED	SuiteCommerce Advanced
SUITECOMMERCE_IN_STORE	SuiteCommerce In-Store
SUITECOMMERCE_MY_ACCOUNT	SuiteCommerce MyAccount
SUITECUBE_ENTERPRISE	Cached Data in Datasets
SUITESIGNON	SuiteSignOn
SUITETAXENGINE	SuiteTax Engine
SUITETAXENGINEINDIA	India Localization SuiteTax Engine
SUITETAXREPORTS	SuiteTax Reports
SUITETAXREPORTSINDIA	India Localization SuiteTax Reports
SUPPLTAXCALC	Supplementary Tax Calculation
SUPPLYALLOCATION	Supply Allocation
SUPPLYCHAINCONTROLTOWER	Supply Chain Control Tower

Feature Internal ID	Feature Name
SUPPLYCHAINMANAGEMENT	Supply Chain Management
SUPPLYCHAINPREDICTEDRISKS	Supply Chain Predicted Risks
SUPPORT	Customer Support and Service
TABLEAU	Tableau Workbook Export
TAXAUDITFILES	Tax Audit Files
TAX_OVERHAULING	SuiteTax
TBA	Token-based Authentication
TEAMSELLING	Team Selling
TELEPHONY	Telephony Integration
TIMEBASEDPRICING	Time-Based Pricing
TIMEBASEDPRICINGSUITEAPP	Subscription Billing Enhanced UI SuiteApp
TIMETRACKING	Time Tracking
TRANDELETIONREASONCODE	Use Deletion Reason
UNITSOFMEASURE	Multiple Units of Measure
UPLIFTPRICING	Uplift Pricing
UPSELL	Upsell Manager
URLCOMPONENTALIASES	URL Component Aliases
USR	SuiteAnalytics Workbook
VENDORACCESS	Vendor Access
VENDORPREPAYMENTS	Vendor Prepayments
VENDORRETURNAUTHS	Vendor Return Authorizations
VENDOR_PAYMENT_INSTRUMENTS	Vendor Payment Instruments
WARRANTYANDREPAIRSMANAGEMENT	Warranty and Repairs Management
WBS	Advanced Project Budgets
WEBAPPLICATIONS	SuiteScript Server Pages
WEBDUPLICATEEMAILMANAGEMENT	Web Site Duplicate Email Management
WEBHOSTING	Host HTML Files
WEBSERVICESEXTERNAL	SOAP Web Services
WEBSITE	Web Site
WEEKLYTIMESHEETS	Weekly Timesheets
WEEKLYTIMESHEETSNEWUI	New Weekly Timesheets Interface
WITHHOLDINGTAX	Withholding Tax

Feature Internal ID	Feature Name
WMSSYSTEM	Warehouse Management
WORKFLOW	SuiteFlow
WORKORDERS	Work Orders

Preference Names and IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following topics list the internal IDs for all NetSuite preference configuration pages that support SuiteScript.

To interact with a configuration page, load the page using config.load(options). After the page loads, you can get or set configuration values using the returned record.Record object. Additionally, you can use User.getPreference(options) to get the values for **General Preferences** and **Accounting Preferences**.

NetSuite configuration preference IDs are grouped into the following categories:

- General Preferences
- Company Information Preferences
- User Preferences
- Accounting Preferences
- Accounting Periods
- Manufacturing Preferences
- Tax Setup
- Tax Periods

General Preferences

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the account preferences that can be found by going to Setup > Company > General Preferences.

The internal ID for the General Preferences page is **companypreferences**. All preference internal IDs are case-insensitive.

Preference UI Label	Preference Internal ID
Date Format	DATEFORMAT
Long Date Format	LONGDATEFORMAT
Time Format	TIMEFORMAT
Number Format	NUMBERFORMAT
Negative Number Format	NEGATIVE_NUMBER_FORMAT
Phone Number Format	PHONEFORMAT
First Day of Week	FIRSTDAYOFWEEK
Search Sorting	SEARCHSORTING

Preference UI Label	Preference Internal ID
Add Primary Contact to Bill To Address	CONTACTONBILLTO
Use Last Name First for Employees	LASTNAMEFIRST
Use Last Name First for Entities	LASTNAMEFIRSTENTITIES
Pre-Populate Contact Address	PREPOPULATECONTACTADDRESS
Show Employees as Contacts	SHOWEMPLOYEESASCONTACTS
Show Display Name with Item Codes	ITEMNUMBERING
Password Policy	PASSWORD_POLICY
Minimum Password Length	MINPASSWORDLENGTH
Password Expiration in Days	PASSWORDEXPIREDAYS
Idle Session Timeout in Minutes	IDLE_SESSION_TIMEOUT
Internal Web Site	INTERNALWEBSITE
Allow Free-Form States in Addresses	FREEFORMSTATES
Use State Abbreviations in Addresses	ABBREVIATESTATES
Company Logo Folder	COMPANYLOGOFOLDER
Default Role for New Customers	CUSTOMERROLE
Assign Tasks to Partners	ASSIGNTASKSTOPARTNERS
Auto Name Customers	AUTONAMECUSTOMERS
Calendar System	CALENDARSYSTEM
Default Customer Type	CUSTOMERTYPE
Customer Center Welcome Message	CUSTOMERWELCOMEMESSAGE
Show Help Link In Customer Center	CUSTOMERSHOWHELPLINK
Horizontal Labels	HORIZONTALLABEL
Screen Font	FONT
Landing Page	CUSTOMLANDINGPAGE
Number of rows in List segments	LISTSEGMENTSIZE
Maximum entries in Dropdown	MAXDROPDOWNSIZE
Log System Notes on Update Only	LOGSYSTEMNOTESONUPDATEONLY
Show Quick Add Row on Lists	SHOWQUICKADD
Show List When Only One Results	SHOWLISTONRESULT
Default Customer Type	CUSTOMERTYPE
Default Lead Type	DEFAULTLEADTYPE
Default Vendor Type	DFLTVENDORTYPE

Preference UI Label	Preference Internal ID
Default Partner Type	DFLTPARTNERTYPE
Auto Name Customers	AUTONAMECUSTOMERS
Auto Inactivate Contacts with Customers	INACTIVATE_CONTACTS
Show Individuals as Contacts	SHOWINDIVIDUALSASCONTACTS
Hide Attachment Folders	HIDEATTACHMENTFOLDERS
Assign Tasks to Partners	ASSIGNTASKSTOPARTNERS
Email Employee on Approvals	EMAILEMPLOYEEONAPPROVAL
Maintenance Complete Email Notification	EMAILMAINTENANCECOMPLETE
Show Reports in Grid	REPORTGRID
Collapse VSOE Column By Default On Sales Transactions	COLLAPSEVSOEFIELDSET
Time Selectors Use Fiscal Calendars Based on First Month	TIMESELECTORSUSEFIRSTFISCALMONTH
Web Site Hosting Files Always Available	HOSTING_FILES_PUBLIC
Show Transaction Numbering Setup	SHOW_TRAN_NUMBERING_SETUP
Asynchronous Job Plan Recalculation	SHOW_TRAN_NUMBERING_SETUP
Preferred Subcustomer Form	SUBCUSTOMERFORM

Company Information Preferences

Applies to: SuiteScript 2.x | SuiteCloud Developer

The Company Information preferences are available at Setup > Company > Company Information.

To interact with these preferences, use the `COMPANY_INFORMATION` value from the config.Type enum. All Company Information preference IDs are case-sensitive.

For details about working with the Company Information page in the UI, see the help topic Configuring Company Information.

Note: The Company Information page includes several preferences that are displayed as body fields. These fields are described in the following table.

Preference Label	Preference Internal ID
Company Name	companyname
Legal Name	legalname
Company Logo (Forms)	formlogo
Company Logo (Pages)	pagelogo
Display Logo Internally	displaylogointernally
Web Site	url
County/State/Province	state

Preference Label	Preference Internal ID
Country	country
Return Email Address	email
Fax	fax
Currency	basecurrency
Employer Identification Number (EIN)	employerid
SSN or TIN (Social Security Number, Tax ID Number)	taxid
First Fiscal Month	fiscalmonth
Time Zone	timezone
Account ID	companyid
Customer Center Login	customersurl
These three IDs represent text summaries of the data that exists on the three address subrecords.	mainaddress _text
	shippingaddress _text
	returnaddress _text

The Company Information page also includes gray boxes with the headings Address, Shipping Address, and Return Address. In the UI, you interact with these components by clicking the Edit link next to each box. Clicking any of these links displays a popup window that includes more fields.

To interact with these fields programmatically, you must use subrecord methods to access the fields associated with the Address, Shipping Address, and Return Address blocks. You must first instantiate the appropriate subrecord by using Record.getSubrecord(options). This method requires one argument: the ID of the subrecord. For the address subrecords on the Company Information page, use the following internal IDs:

- mainaddress
- shipaddress
- returnaddress

After you instantiate the subrecord, you can interact with any of the preferences shown in the following table. These preferences are handled as fields on each subrecord instance.

Preference Label	Preference Internal ID
Address1	addr1
Address2	addr2
Attention	attention
Addressee	addressee
Phone	addrphone
City	city
County/State/Province	state
Zip	zip

Preference Label	Preference Internal ID
Country	country

For an example, see the help topic Retrieving a Body Field Address Subrecord Example. For general information about subrecords, see the help topic SuiteScript 2.x Scripting Subrecords.

User Preferences

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the user preferences that can be found by going to Home > Set Preferences.

The internal ID for the user preferences page (which appears as the Set Preferences page in the UI) is **userpreferences**. All preference internal IDs are case-sensitive.

Be aware that the API for setting user preferences works the same as the UI in terms of setting a preference for a user's session or setting it permanently. In the UI if a user sets a preference, and the preference reverts back to a default setting on the user's next login, the same behavior is supported in SuiteScript.

Preference UI Label	Preference Internal ID
On the General tab	
Nickname	MESSAGE_NICKNAME
Signature	MESSAGE_SIGNATURE
Add Signature to Messages	MESSAGE_AUTOSIGNATURE
From Email Address	MESSAGE_EMAIL
Language	LANGUAGE
Search Sorting	SEARCHSORTING
Language of the Help Center	HELP_LANGUAGE
PDF Language	PDFLANGUAGE
Time Zone	TIMEZONE
First Day of the Week	FIRSTDAYOFWEEK
Date Format	DATEFORMAT
Long Date Format	LONGDATEFORMAT
Time Format	TIMEFORMAT
Number Format	NUMBERFORMAT
Negative Number Format	NEGATIVE_NUMBER_FORMAT
Phone Number Format	PHONEFORMAT
Auto Place Decimal	AUTOPLACE
CSV Column Delimiter	CSV_COLUMN_DELIMITER
CSV Decimal Delimiter	CSV_DECIMAL_DELIMITER

Preference UI Label	Preference Internal ID
Use Multicurrency Expense Reports	USE_MC_ON_EXPREPT
Download PDF Files	DOWNLOADPDFS
Address Mapping Type	MAPTYPE
Show Internal IDs	EXPOSEIDS
Only Show Last Subaccount	ONLYSHOWLASTSUBACCT
Only Show Last Subentity	ONLYSHOWLASTSUBENT
Only Show Last Subitem	ONLYSHOWLASTSUBITEM
Submit Warnings	SUBMITWARNINGS
Limit CC Field to Contacts & Employees	EMAILLIMITCC
Default Issue Email Notification	ISSUE_EMAIL_ME_WHEN
Notify Me Upon Issue Assignment	ISSUE_NOTIFY_UPON_ASSIGNMENT
Delay Loading of Sublists	DELAYLOADINGSUBLISTS
Number of Rows in List Segments	LISTSEGMENTSIZE
Maximum Entries in Dropdowns	MAXDROPDOWNSIZE
Type-Ahead on List Fields	TYPEAHEADSELECTS
Require Exact Match on Item Type-Ahead	ITEMEXACTMATCH
Show Quick Add Row on Lists	SHOWQUICKADD
Display Bounce Warning on Campaigns	CAMPAIGN_BOUNCE_WARNING
Prefer Native Select Fields Over NS Dropdowns in Internet Explorer	NATIVE_DROPDOWNS
Show App ID Field	SHOW_APPID_FIELD
Show ID Field on Sublists	SHOW_ID_FIELD
On the Appearance tab	
Color Theme	COLORTHEME
Screen Font	FONT
Compensate for Large Fonts	SYSTEMLARGEFONTS
Density Setting for Internet Explorer	MSIE_ZOOM_FACTOR
Register Look on Lists	REGISTERSTYLE
Only Show Field Boarders on Hover	SHOWFIELDBORDERONHOVER
Chart Theme	CHART_THEME
Chart Background	CHART_BACKGROUND
Landing Page	LANDINGPAGE
Show Portlet Hint	SHOWPORTLETHINT

Preference UI Label	Preference Internal ID
Set Customer Dashboard As Default View On Customer Record	DASHBOARD_DEFAULT_VIEW_FOR_CUSTOMER
Limit Entry Forms to Two Columns	LIMITTOTWOCOLUMNS
Expand Tabs on Entry Forms	UNLAYEREDTABS
Enable Rich Text Editing	RICHTEXTEDITOR
Default Rich Text Editor Font	EDITORFONT
Default Rich Text Editor Font Size	EDITORFONTSIZE
Display Default Them With Optimal Color Contrast	ACCESSIBILITY_HIGH_CONTRAST
On the Transactions tab	
Auto Fill Transactions	AUTOFILL
Alphabetize Items Regardless of Type	ALPHABETIZE_ITEMS
Duplicate Number Warnings	DUPLICATEWARNINGS
Inventory Level Warnings	STOCKWARNINGS
Customer Credit Limit Handling	CUSTCREDLIMHANDLING
Vendor Credit Limit Warnings	VENDCREDLIMWARNINGS
Print Using HTML	HTMLPRINTING
Transaction Email Attachment Format	TRANSACTION_ATTACHMENT_FORMAT
Horizontal Print Offset	HORZPRINTOFFSET
Vertical Print Offset	VERTPRINTOFFSET
On the Analytics tab	
Report by Period	REPORTBYPERIOD
Show Reports in Grid	REPORTGRID
Customize Font on Financial Reports	ENABLE_REPORT_FONT_CUSTOMIZATION
Print Company Logo	DISPLAYLOGO
Display Report Title on Screen	DISPLAYRPTTITLE
Display Report Description	DISPLAYRPTDESC
Calculate Forecasts as Weighted	FORECASTWEIGHTED
Default Bank Account	DEFAULT_BANKREG
Show Forecasts as Weighted	FORECASTWEIGHTED
Show List When Only One Result	SHOWLISTONERESULT
Quick Search Uses Keywords	KEYWORDSEARCH
Popup Search Uses Keywords	KEYWORDSEARCHPOPUP

Preference UI Label	Preference Internal ID
Include Inactives in Global & Quick Search	SEARCHINACTIVES
Popup Auto Suggest	POPUPAUTOSUGGEST
Global Search Auto Suggest	SEARCHAUTOSUGGEST
Global Search Sort by Name/ID	GLOBALSEARCHSORTBYNAME
Global Search Customer Prefix Includes Leads and Prospects	GLOBALSEARCHCUPREFIX
PDF Page Orientation	REPORTPDFORIENTATION
PDF Font Size	REPORTPDFFONTSIZE
CSV Export Character Encoding	CSVEXPORTENCODING
KPI Export Character Encoding	KPI_PERIOD_SPECIFIC_RATES
On the Activities tab	
Edit Activities from Calendar	EVENT_EDITFROMCALENDAR
Send Invitation Emails	EVENT_EMAILNOTIFICATION
Restrict Invitees to Employees	EVENT_INTERNALINVITEESONLY
Default Event Access Setting for New Events	EVENT_DEFAULTPUBLIC
Default Reminder Type	REMINDERTYPE
Default Reminder Time	REMINDERPERIOD
Play Audio with Popup Event Reminders	REMINDERPLAYWAVE
Default Priority for Tasks	DEFAULTTASKPRIORITY
Default New Tasks Public	TASK_DEFAULTPUBLIC
Default New Phone Calls Public	CALL_DEFAULTPUBLIC
Default Sync Category	DEFAULT_CONTACT_SYNC_CATEGORY
On the Alerts tab	
First Selection	EMAILALERT_AM
Second Selection	EMAILALERT_NOON
Third Selection	EMAILALERT_PM
Include links in HTML alerts	LINKS_EMAILALERT
Respect Quick Date Portlet Settings	USE_QUICKDATE_IN_ALERTS
E-Mail	EMAILALERT_EMAIL
Send an On-Demand Alert from this Role	ALERTONDEMAND
On the Restrict View tab	
Subsidiary	SUBSIDIARY
Include Sub-Subsidiaries	SUBSIDIARYSUBS

Preference UI Label	Preference Internal ID
Department	DEPARTMENT
Include Sub-Departments	DEPARTMENTSUBS
Include Unassigned	DEPARTMENTUNASSIGNED
Location	LOCATION
Include Sub-Locations	LOCATIONSUBS
Include Unassigned	LOCATIONUNASSIGNED
Class	CLASS
Include Sub-Classes	CLASSSUBS
Include Unassigned	CLASSUNASSIGNED
On the Telephony tab	
Telephony Option	TELEPHONY_OPTION
TAPI Device	TELEPHONYDEVICE
CTI URL	CTI_URL
Prefix to Dial Out	DIALOUTPREFIX

Accounting Preferences

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the account preferences that can be found by going to Setup > Accounting > Accounting Preferences. All preference internal IDs are case-insensitive.

The internal ID for the Accounting Preferences page is **accountingpreferences**.

Preference UI Label	Preference Internal ID
On the General tab	
Use Account Numbers	ACCOUNTNUMBERS
Use Legal Name In Account	ACCOUNTLEGALNAME
Show All Transaction Types In Reconciliation	RECONCILIATIONALLTRANTYPES
Expand Account Lists	ASSETCOGSITEMACCTS
Cash Basis Reporting	CASHBASIS
Aging Reports Use	AGEFROM
Void Transactions Using Reversing Journals	REVERSALVOIDING
Set Reversal Variance Date Equal To The Reversing Journal Date When Voided Transaction Is In A Closed Period	SETREVERSINGVARIANCEDATETOREVJES
Use Journal Entry Summarization On Intercompany Elimination	ELIMINATION_JE_SUMMARIZATION

Preference UI Label	Preference Internal ID
Require Approvals On Journal Entries	JOURNALAPPROVALS
Allow User Events On Bulk Journal Approval	BULK_JOURNAL_APPROVAL_EVENTS
Allow GL Custom Segment Deletion	CUSTOM_SEGMENT_DELETION
Enable Accounting Period Window	ALLOWFUTUREPERIODLOCK
Minimum Period Window Size	OPENCURFUTUREPERIODWINDOW
Allow Transaction Date Outside Of Posting Period	DATEPERIODMISMATCH
GL Audit Numbering Method	GLNUMBERINGBY
Default Posting Period When Transaction Date In Closed Period	DEFAULTPERIODIFCLOSED
Create and Edit Inventory Transactions Dated In Closed Periods	INVT_TRANS_CLOSED_PERIODS
Allow Quick Close Of Accounting Periods	ALLOW_PERIODS_QUICK_CLOSE
Apply Payments Through Top-Level Customers Only	PAYMENTSONLYFROMTOPPARENT
Show Only Open Transactions On Statements	OPENONLYSTMTS
Open Transactions On Statements	OPENONLYTRANSACTIONS
Customer Credit Limit Handling	CUSTCREDLIMHANDLING
Customer Credit Limit Includes Orders	CUSTCREDLIMORDERS
Days Overdue For Warning / Hold	CREDLIMDAYS
Include Tax For Term Discounts	TERMDISCOUNTSINCLUDETAX
Include Shipping For Term Discounts	TERMDISCOUNTSINCLUDESHIPPING
Default Vendor Payments To Be Printed	VENDPYMTTOPRINT
Vendor Credit Limit Warnings	VENDCREDLIMWARNINGS
Vendor Credit Limit Includes Orders	VENDCREDLIMORDERS
Use In-Transit Vendor Payments By Default	DEFAULTVENDORPAYMENTTYPE
Vendor In-Transit Payment Account	VENDORITPACCOUNT
Vendor Prepayment Account	VENDORPREPAYMENTACCOUNT
Auto-Apply Vendor Prepayments	AUTO_APPLY_VENDOR_PREPAYMENTS
Allow Bill Consolidation Of Purchase Orders With Different Terms	ALLOWBILLCONSOLFORMULTITERMPO
Make Departments Mandatory	DEPTMANDATORY
Make Classes Mandatory	CLASSMANDATORY
Make Locations Mandatory	LOCMANDATORY
Allow Per-Line Departments	DEPTSPERLINE
Allow Per-Line Locations	LOCSPERLINE

Preference UI Label	Preference Internal ID
Allow Per-Line Classes	CLASSESPERLINE
Always Allow Per-Line Classifications On Journals	CDLPERLINEONJE
Allow Non-Balancing Classifications On Journals	NONBALANCINGCDLONJE
Allow Empty Classifications On Journals	NULLCDLONJE
Allow Users to Modify Revenue Recognition Schedule	MODIFYREVRECTOTALAMOUNT
Prorate Revenue Recognition Dates For Partially Billed Sales Orders	PRORATEREVRECINVFROMSO
Create Revenue Journals In GL	REVRECJOURNALENTRYSUMMARIZATION
Default Revenue Recognition Journal Date To	REVRECJOURNALDATEDEFAULT
Allow Users To Modify VSOE Values on Transactions	MODIFYVSOEVALSONTRAN
Use System Percentage Of Completion For Schedules	USESYSCALCPCT4REVREC
Adv Billing: Use Sales Order Amount	CALCPCTCOMPFROMSALESORDERAMT
Allow Revenue Commitments In Advance Of Fulfillment	UNSHIPPEDREVENUECOMMITMENTS
Allow Revenue Commitment Reversals In Advance Of Item Receipt	UNRECEIVEDREVENUECOMMITMENTS
Default Deferred Revenue Reclassification Account	DEFAULTDEFERREDREVENUEACCOUNT
Default Foreign Currency Adjustment Revenue Account	DEFAULTFXREVENUEACCOUNT
Allow Users To Modify Amortization Schedule	MODIFYAMORTOTALAMOUNT
Default Amortization Journal Date To	AMORJOURNALDATEDEFAULT
Default Amortization Journal Entry Form	AMORTIZATIONJOURNALENTRYFORM
Intercompany Time	INTERCOMPANYTIME
Intercompany Expenses	INTERCOMPANYEXPENSE
Intercompany Expenses	INTERCOMPANYEXPENSE
Enable Budget With Elimination Subsidiaries	BUDGETINCLUDEELIMSUB
Require Approvals on Journal Entries	JOURNALAPPROVALS
Rate Provider	EXCHANGE_RATE_PROVIDER
Use Triangulation Calculation By NetSuite	EXCHANGE_RATE_PROVIDER
Maximum number of MLI locations	MAXLOCATIONS
On the Items/Transactions tab	
Purchase Discount Account	PURCHDISCACCT
External Inventory In Transit Account	EXTINTRINVTACCT
Sales Discount Account	SALESDISCACCT
Default Expense Account	EXPENSEACCOUNT

Preference UI Label	Preference Internal ID
Default Income Account	INCOMEACCOUNT
Default Receivables Account	ARACCOUNT
Default Cogs Account	COGSACCOUNT
Default Asset Account	ASSETACCOUNT
Default Payment Account	PAYMENTACCOUNT
Default Gain / Loss Account	GAINLOSSACCOUNT
Default Bill Quantity Variance Account	BILLPRICEVARIANCEACCT
Default Vendor Return Variance Account	VENDRETURNVARIANCEACCOUNT
Default Customer Return Variance Account	CUSTOMERRETURNVARIANCEACCOUNT
Default Production Quantity Variance Account	PRODQTYVARIANCEACCT
Default Production Price Variance Account	PRODPRICEVARIANCEACCT
Default Purchase Price Variance Account	PURCHASEPRICEVARIANCEACCT
Default Inventory Count Account	INVCOUNTACCOUNT
Anyone Can Set Item Accounts	EDITITEMACCOUNTS
Default Dropship Expense Account	DROPSHIPEXPENSEACCOUNT
Consolidate Jobs on Sales Transactions	CONSOLINVOICES
Maximum # of Quantity-based Price Levels	QTYPRICECOUNT
Allow Quantity Discounts per Price Level on Schedules	QTYPRICESCHEDULEMULTDISCOUNTS
Include Reimbursements in Sales and Forecast Reports	FORECASTINCLUDES_REIMB_EXP
Include Shipping in Sales and Forecast Reports	FORECASTINCLUDES_SHIPPING
Transaction Types to Exclude from Forecast Reports	FORECASTTRANTYPES
Transaction Types to Exclude from Sales Reports	SALESTRANTYPES
Default Fixed Date Charge Rule Stage	FFCRFIXEDDATE
Default Milestone Charge Rule Stage	FFCRMILESTONE
Default Project Progress Charge Rule Stage	FFCRPROGRESS
Default Time-Based Rule Stage	TIMECR
Scan Individual Items	SINGLEITEMBARCODING
Centralize Purchasing In A Single Location	CENTRALIZEDPURCHASING
Days Before Lot Expiration Warning	LOTEXPIRATIONWARNING
Require Bins on All Transactions Except Item Receipts	REQUIREBINSONTRANS
Use Preferred Bin on Item Receipts	USEPREFERREDBINONITEMRCPT
Include Landed Cost In Last Purchase Price	INCLUDELANDEDCOSTINLPP

Preference UI Label	Preference Internal ID
Inventory Costing Method	INVTCOSTMETHOD
Default Cost Category	DEFAULTCOSTCATEGORY
Average Cost Completion Unit Cost	AVGCOSTCOMPLETIONCOSTINGMETHOD
Use Intransit Value In Group Average Cost Calculations	USEINTRANSITINGACCALCULATIONS
Customers Can Pay Online	EXTERNALPAYMENTS
Get Authorization On Customer Center Sales Orders	AUTHORIZECUSTOMERCENTERORDERS
Use Card Security Code For Credit Card Transactions	CCSECURITYCODE
Allow Adjusted Expiration Date To Improve Recurring Payments	ALLOWADJUSTEDEXPIRATIONDATE
Enable “Sale” Payment Operations On A Sales Order By Automatically Creating A Customer Deposit	CREATE_DEPOSIT_ON_SALES_ORDER_SALE
Use Strict Rules For The Selection Of Payment Processing Profiles	STRICT_SELECTION_OF_PPP
Preserve Transactions When Payment Is On Hold	PRESERVE_TRAN_ON_HOLD_PAYMENT
Duplicate Number Warnings	DUPLICATEWARNINGS
Sort Reconcile By	RECONSORTCOL
Recalculate Estimated Cost on Creation of Linked Transactions	USELATESTCOSTESTIMATE
Matrix Item Name/Number Separator	SKUSEPARATOR
Gift Certificate Auth Code Generation	GIFTCERTAUTHCODEGENERATION
Enforce Minimum Quantity On Return Authorizations	ENFORCE_MIN_QUANTITY_RET_AUTH
On the Order Management tab	
Default Sales Order Status	DEFSALESORDSTATUS
Require Re-approval on Edit of Sales Order	REAPPROVESOONEDIT
Send Email Confirmation when Sales Order Canceled	EMAILCANCELORDER
Default Location for Sales Orders	DEFAULTSALESORDERLOCATION
Default Commit Option On Sales Order	DEFAULTSALESORDERCOMMITOPTION
Default Commit Option On Transfer Order	DEFAULTTRANSFERORDERCOMMITOPTION
Item Commitment Transaction Ordering	ITEMCOMMITMENTTRANSACTIONORDER
Perform Item Commitment After Transaction Entry	AUTOMATICITEMCOMMITMENT
Always Print Kit Items on Picking Tickets	PICKINGTICKETKITITEMS
Show Non-Inventory Items on Picking Tickets and Packing Slips	PICKINGTICKETNONINVT
Show Uncommitted Items on Picking Tickets	PICKINGTICKETUNCOMMITTED

Preference UI Label	Preference Internal ID
Name for Packed Status	NAMINGPACKED
Name for Picked Status	NAMINGPICKED
Name for Shipped Status	NAMINGSHIPPED
Show Additional Items on Packing Slips	SHOWADDLITEMSPACKSLIP
Show Drop Ship Items on Packing Slips	PACKINGSLIPDROPSHIP
Limit Status on Packing Slip Queue	PACKINGSLIPSTATUS
Fulfill Based on Commitment	FULFILLCOMMITTED
Default Items to Zero Received/Fulfilled	DEFAULTUNFULFILLED
Allow Overage on Assembly Builds	OVERBUILDS
Filter Bulk Fulfillment Page by Location	BULKFULFILLOCFILTERING
Send Order Fulfilled Confirmation Emails	ORDFULFILLCONFEMAIL
Use Web Site Template for Fulfillment Emails	ORDFULFILLUSESTORETEMPLATES
Update Transaction Date Upon Fulfillment Status Change	UPDATEITEMSHIPDATEONSTATUSCHANGE
Show Unfulfilled Items on Invoices	SHOWUNSHIPPEDITEMS
Invoice In Advance Of Fulfillment	UNSHIPPEDINVOICES
Convert Absolute Discounts to Percentage When Billing	CONVERTABSOLUTEDISCOUNTS
Base Invoice Date on Billing Schedule Date	INCOICEUSESCHEDULEDATE
Drop Ship P.O. Form	DROPSHIPTEMPLATE
Automatically Email Drop Ship P.O.s	EMAILDROPSHIPPOS
Queue Drop Ship P.O.s for Printing	PRINTDROPSHIPPOS
Automatically Fax Drop Ship P.O.s	FAXDROPSHIPPOS
Include Committed Quantities	DROPSHIPINCLUDECOMMITTED
Limit Vendor List on Items	LIMITITEMVENDORS
Update Drop Ship Order Quantities Automatically Prior To Shipment	KEEPDROPSHIPQUANTITIESINSYNC
Drop Ship FulFillment Quantity Validation	FULFILLDROPSHIPORDERFROMINVENTORY
Allow Both Mark Shipped Fulfillments And Receipts On A Drop Shipment Line	DROPSHIPANDRECEIVEINTOINVDROPSHIPPO
Update Special Order Quantities Automatically Prior To Shipment	KEEPSPECIALORDERQUANTITIESINSYNC
Allow Expenses On Purchases	POEXPENSES
Default Location for Purchase Orders	DEFAULTPURCHASEORDERLOCATION
Maximum {Purchase} Lines To Consolidate	MAXPURCHASES

Preference UI Label	Preference Internal ID
Allow Default Email On Purchase Orders Using SuiteFlow Approval Routing	ALLOW_DEFAULT_EMAIL_ON_PO
Default Number Of Requests For Quote Pricing Tiers	DEFAULT_RFQ_TIERS
Bill in Advance of Receipt	UNCRECEIVEDBILLS
Allow Overage on Item Receipts	OVERRECEIPTS
Default Receiving Exchange Rate	DEFAULTRECEIVINGEXCHANGERATE
Use Purchase Order Rate On Bills	USEPORATEONBILLS
Landed Cost Allocation Per Line	LANDEDCOSTPERLINEDEFAULT
Default Return Auth. Status	DEFRTNAUTHSTATUS
Refund in Advance of Return	UNCRECEIVEDRTNAUTHS
Restock Returned Items	RESTOCKRETURNS
Write-Off Account for Returns	WRITEOFFACCOUNT
Default Vendor Return Auth. Status	DEFVENDAUTHSTATUS
Credit in Advance of Vendor Return	UNRETURNEDVENDAUTHS
Default Transfer Order Status	DEFTRNFRORDSTATUS
Generate Transfer Orders In Supply Planning	GENERATETRNFRORDINPLANNING
Use Item Cost As Transfer Cost	ITEMCOSTASTRNFRORDCOST
Default Transfer Order Incoterms	DEFAULT_TRNFRORD_INCOTERM
Default Vendor Bill Status	DEFVENDBILLSTATUS
On the Time & Expenses tab	
Show Jobs Only for Time and Expense Entry	TIMEEXPENSEJOBONLY
Automatically Notify Supervisor	AUTONOTIFYSUPV
Expenses Billable by Default	DEFAULTEXPENSEBILLABLE
Items Billable by Default	DEFAULTITEMSBILLABLE
Combine Detail Items on Expense Reports	COMBINEEXPENSEITEMS
Copy Expense Memos to Invoices	COPYEXPENSEMEMOS
Default Advance To Apply Account For Expense Reports	DEFAULT_ADVANCE_ACCT_FOR_EXPREPT
Foreign Amount Change	EXP_REPORT_FOREIGN_AMOUNT_CHANGE
On the Approval Routing tab	
Expense Reports	CUSTOMAPPROVALEXPENSE
Purchase Orders	CUSTOMAPPROVALPURCHORD
Vendor Bills	CUSTOMAPPROVALVENDORBILL
Requisitions	CUSTOMAPPROVALPURCHREQ

Preference UI Label	Preference Internal ID
Invoices	CUSTOMAPPROVALCUSTINVC
Purchase Contracts	CUSTOMAPPROVALPURCHCON
Blanket Purchase Orders	CUSTOMAPPROVALBLANKORD
Journal Entries	CUSTOMAPPROVALJOURNAL

Accounting Periods

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the account preferences that can be found by going to Setup > Accounting > Manage G/L > Manage Accounting Periods.

The internal ID for the Accounting Periods page is **accountingperiods**. All preference internal IDs are case-insensitive.

Preference UI Label	Preference Internal ID
First Fiscal Month	fiscalmonth
Fiscal Year End	fiscalyear
Period Format	periodstyle
Year in Period Name	periodnameyear
One-Day Year-End Adjustment Period	lastday

Manufacturing Preferences

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the manufacturing preferences that can be found by going to Setup > Manufacturing > Manufacturing Preferences.

The internal ID for the Accounting Periods page is **manufacturingpreferences**. All preference internal IDs are case-insensitive.

Preference UI Label	Preference Internal ID
Default Unbuild Variance Account	UNBUILDVARIANCEACCOUNT
Default WIP Cost Variance Account	WIPVARIANCEACCT
Default Scrap Account	SCRAPACCT
Default WIP Account	WIPACCT
Average Cost Consumption Unit Cost	AVGCOSTSOMPLETIONSCOSTINGMETHOD
Allow Purchase Of Assembly Items	ALLOWASSEMBLYPURCHASE
Allow Editing of Legacy BOMs	ALLOWEDITINGOFMIGRATEDLEGACYBOMS

Preference UI Label	Preference Internal ID
Default Scheduling Method	DEFAULTSCHEDULINGMETHOD
Default Commit Option On Work Order	DEFAULTWORKORDERCOMMITOPTION
Manufacturing Issue Based On Commitment	BUILDCOMMITTED
Create Work Orders In Supply Planning	CREATEWORKORDERINPLANNING
Default Allocation Strategy on WorkOrder	DEFAULTWORKORDERALLOCATIONSTRATEGY
Allow Overage On Work Order Transactions	OVERBUILDS
Default Work Order Status	DEFAULTWORKORDERSTATUS
Check Completed Quantity In Prior Operations During Operation Completion	VALIDATEPREDECESSORCOMPLETEDQTY
Show Planned Capacity On Work Orders	SHOWPLANNEDCAPACITYONWORKORDERS
Automatically fill actual production start and end dates	FILL_ACTUAL_PRODUCTION_DATES

Tax Setup

Applies to: SuiteScript 2.x | SuiteCloud Developer

The internal ID for the Set Up Taxes page is **taxpreferences**. All preference internal IDs are case-insensitive.

Preference IDs for fields on this page vary according to the country of the nexus. Field internal IDs are suffixed with the nexus country code. The format for these preferences is <field internal id><nexus country code>. Be aware that different fields are available for different nexuses.

The table below shows some scriptable tax preference fields for a US nexus. Fields are suffixed with **us**, for the US nexus. This table is provided for example purposes.

Preference UI Label	Preference Internal ID
Customers Default to Taxable	defaulttaxableus
Charge out of District Sales Taxes	chargeoutofdistrictus
Per-Line Taxes on Transactions	perlinetaxesus
Charge Sales Tax on Store Orders	storeordertaxationus
Enable Tax Lookup on Sales Transactions	enabletaxlookupus

Field IDs in your account may be suffixed with a different nexus country code. And different fields may be available. You may be able to look up field IDs in the user interface, by going to Setup > Accounting > Set Up Taxes, and clicking on a nexus.

- To make field IDs available, go to Home > Set Preferences and ensure that the Show Internal IDs box is checked on the General subtab, Defaults area.
- Find the field in the NetSuite user interface and click the field label to display the field level help text. The field ID is displayed in the popup.

Tax Periods

Applies to: SuiteScript 2.x | SuiteCloud Developer

These are the tax preferences that can be found by going to Setup > Accounting > Taxes > Manage Tax Periods.

The internal ID for the Tax Periods page is **taxperiods**. All preference internal IDs are case-insensitive.

Preference UI Label	Preference Internal ID
First Fiscal Month	fiscalmonth
Fiscal Year End	fiscalyear
Period Format	periodstyle
Year in Period Name	periodnameyear

Task IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

Task IDs identify NetSuite pages. Each NetSuite page has a unique task ID.

You can reference task IDs when using the url.resolveTaskLink(options) and redirect.toTaskLink(options) methods.

To find a task ID for a NetSuite page:

1. Right-click in the NetSuite page, and select **View Page Source** from the context menu. Note that the exact command varies by browser.
2. Press Ctrl+F to open the search popup, and enter **taskId** in the search field. Note that the search mechanism varies by browser.
3. Press Enter, use the down arrow, or click **Find Next** until you reach the occurance of `taskId` as part of a line of code. For example: `"taskId":"LIST_EMPLOYEE_RECORD"` is included on a line of code on the Employees page.

Button IDs

Applies to: SuiteScript 2.x | SuiteCloud Developer

The following table lists the internal IDs for standard NetSuite buttons that support SuiteScript.

These buttons may appear as inline buttons or as actions in the Actions menu for some records. The following properties of the serverWidget.Button object can be used to customize buttons and actions:

- Button.label – Use this property to change the label of the button or action.
- Button.isHidden – Use this property to hide or show the button or action.

You cannot use SuiteScript to do the following:

- Change the display type of a button from an inline button to an action in the Actions menu, or from an action to an inline button.
- Add or remove a custom button to or from the Actions menu.

You can use point-and-click customization to change the display type of buttons and actions. For more information, see the help topic Configuring Buttons and Actions.

Important: Customizing the Save, Edit, Cancel, Back, and Reset buttons is not supported in SuiteScript or in point-and-click customization.

Button UI Label	Button Internal ID
Add Items	addmatrix
Accept	accept
Accept Payment	acceptpayment
Apply	apply
Approve	approve
Approve Return	approvereturn
Authorize Return	return
Auto Fill	autofill
Bill	bill
Bill Remaining	billremaining
Cancel Order	cancelorder
Cancel Return	cancelreturn
Clear Splits	clearsplits
Close	closeremaining
Convert	convertlead
Convert to Inventory	convertinvt
Convert to Lot Numbered Inventory	convertlot
Convert to Serialized Inventory	convertserial
Create Build	createbuild
Create Matrix	creatematrix
Credit	credit
Decline	decline
Delete	delete
Email	email
Fax	fax
Fulfill	process
Generate Price List	generatepricelist
Generate Statement	generatestatement
GL Impact	glimpact

Button UI Label	Button Internal ID
Go To Register	gotoregister
Grab	grab
Make Copy	makecopy
Make Payment	payment
Make Standalone Copy	makestandalonecopy
Memorize	memorize
Merge	merge
New	new
New Event Field	neweventfield
Next Bill	nextbill
Next Week	next
Prev Week	prev
Print	print
Print Bill of Materials	printbom
Print Label	printlabel
Print Labels	printlabels
Print Picking Ticket	printpicktick
Print Summary	depositsummary
Quick Accept	quickaccept
Recalc	recalc
Receive	receive
Refund	refund
Reject	reject
Renew	renewal
Reset	resetter
Revenue Commitment Reversal	revcomrv
Save As	submitas
Save & Bill	submitbill
Save & Convert	submitconvert
Save & Copy	submitcopy
Save & Edit	submitedit
Save & Email	saveemail

Button UI Label	Button Internal ID
Save & Fulfill	submitfulfill
Save & New	submitnew
Save & Next	submitnext
Save & Print	saveprint
Save & Print BOM	saveprintbom
Save & Print Label	saveandprintlabel
Save & Refund	submitrefund
Save & Same	submitsame
Save Baseline	savebaseline
Search	search
Show Activity	showactivity
Submit Invoice	submitinvoice
Tentative	tentative
Unbuild	createunbuild
Update Matrix	updatematrix
Update VSOE	updatevsoe
View All Transactions	viewalltransactions
Void	void
W4 Worksheet	w4data

Working with UI Objects

Applies to: SuiteScript 2.x | SuiteCloud Developer

You can use UI objects to build an assistant in NetSuite that has the same look-and-feel as other built-in NetSuite assistants.

For information about NetSuite Assistants, see Understanding NetSuite Assistants.

For information about working with UI objects in SuiteScript 2.x, see the help topic SuiteScript 2.x Working with UI Objects.

Understanding NetSuite Assistants

Applies to: SuiteScript 2.x | SuiteCloud Developer

In NetSuite, assistants are made of steps that users follow to complete a bigger task. Some assistants require steps to be done in order, while others let users complete steps in any order or skip some. In those assistants, the steps are provided only as guidelines for what users can do to finish the task.

The UI objects you use to build your assistant give it the same look and feel as built-in NetSuite assistants. For examples of these assistants, see these topics:

- SuiteBundler Assistant
- Import Assistant

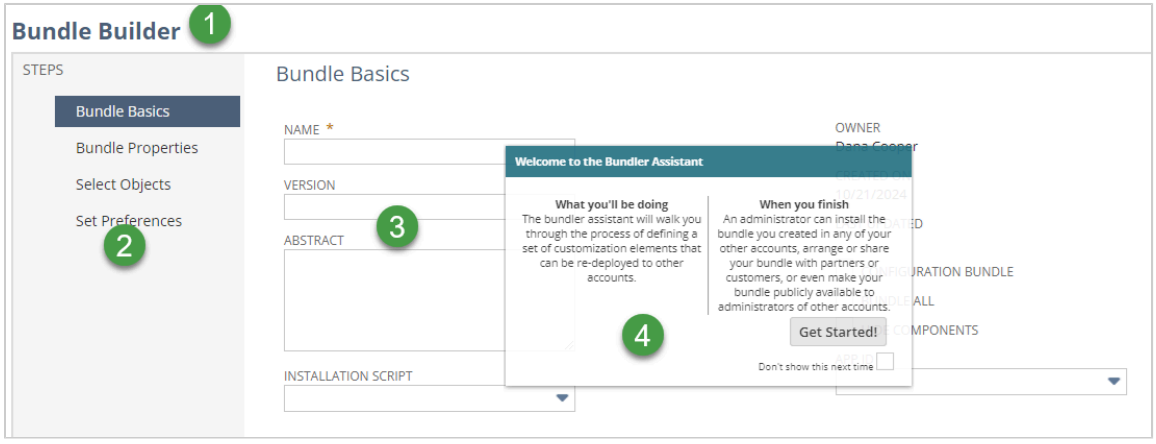
SuiteBundler Assistant

The SuiteBundler Assistant is a built-in NetSuite assistant tool that guides users through the steps to bundle custom NetSuite solutions for deployment to other accounts.

Note: To access the SuiteBundler Assistant, go to Customization > SuiteBundler > Create Bundle.

This figure shows Step 1 (page 1) of the SuiteBundler Assistant. Steps are in order and appear horizontally below the title.

You can build all the components shown here in your own custom assistant.



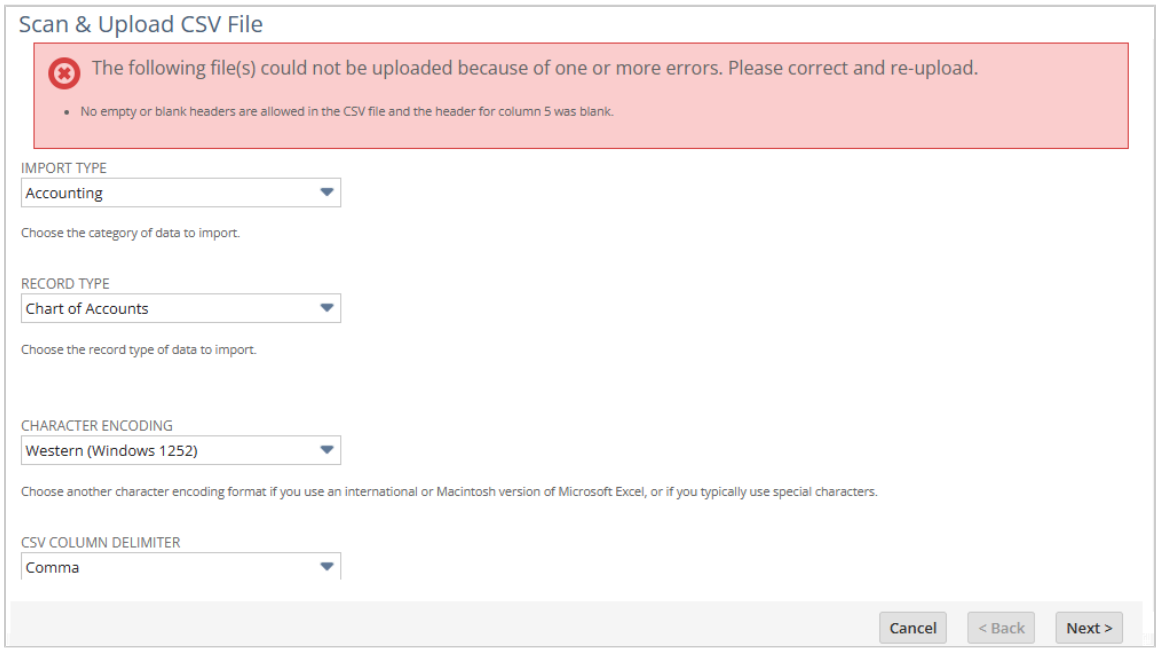
1	Assistant title
2	Steps positioned vertically
3	Fields in a field group
4	Splash screen

Import Assistant

The Import Assistant guides users through steps to import data into NetSuite.

Note: To access the Import Assistant, go to Setup > Import/Export > Import CSV Records.

The following figure shows how an error message appears in an assistant. Users can't move to the next step until the error is resolved. When you build custom assistants, you can throw errors that prevent users from moving to the next step.



Single Page Applications

A single page application (SPA) is a web application or site that loads one HTML page and updates it dynamically as you interact with it.

In NetSuite, you can build SPAs that match the NetSuite look and feel using the User Interface Framework (UIF), with full SuiteScript support and SuiteCloud Development Framework (SDF) deployment capabilities.

See the following help topics for information about SPAs in NetSuite:

Introduction to Single Page Applications

Single page applications (SPAs) are web applications used on many websites to provide a way to interact with the page without needing to reload it. When you interact with SPAs, most of the page stays the same and only specific sections are updated.

To learn more about SPAs and how to use them in NetSuite, see the following topics:

How a Single Page Application Works

SPAs load the application code (HTML, CSS, JavaScript) only one time, the first time you access the page. Because JavaScript handles all browser events, each new request consists only of the specific information needed to update the page.

In contrast, with multi-page applications, every time you request new information, the browser loads a whole new page instead of only updating the parts that changed.

SPAs convert data to HTML in the client (browser), instead of converting data in the server and sending it back to the client.

Advantages of Single Page Applications

The main advantages of SPAs are:

- More dynamic and seamless user experience. You get real-time updates without reloading the page or resubmitting the form.
- Faster loading and less bandwidth consumption compared to HTML.
- Easier to debug and maintain because the presentation and application are kept separate.
- Built-in NetSuite look and feel.

Single Page Applications in NetSuite

You can use SPAs instead of Suitelets to build a customized user interface in NetSuite.

To develop SPAs, you need the following NetSuite features:

- **NetSuite User Interface Framework (UIF)** – Gives you a library of UI components to build SPAs with the NetSuite look and feel. Access UIF through your account-specific URL: https://*<AccountID>*.app.netsuite.com/ui/apps/catalog.nl. For more information, see NetSuite User Interface Framework for Single Page Applications.
- **SuiteScript** – Lets you use client and server scripts, which you need to create an SPA. You can also load SuiteScript modules from the SPA's client and server scripts.
- **SuiteCloud Development Framework** – Lets you create SuiteApp projects, which you use to develop and deploy SPAs to target NetSuite accounts.

For information about the setup requirements for developing SPAs in NetSuite, see Single Page Application Creation And Development.

Single Page Application Terminology

To understand how single page applications work in NetSuite, you should be familiar with these terms.

- **NetSuite User Interface Framework (UIF)** – The NetSuite frontend framework for building user interfaces in JavaScript. This UI library loads by default when you run an SPA. You can use it in SPAs to get the NetSuite look and feel. UIF uses features such as JSX syntax and ECMAScript module management.
- **Single Page Application (SPA)** – A web application that dynamically updates the page when you interact with it, without fully reloading.

  In NetSuite, SPAs are built as part of an SDF SuiteApp project and includes these components:

  - **SPA client script** – This script runs the application in your browser. It loads the components and can also load SuiteScript modules and custom modules.
  - **SPA server script** – This script is the entry point for the application and has an API to modify the response (add scripts, perform validations, etc.). For example, the application can show an error if the user doesn't have a specific role.
  - **Assets folder** – This folder holds all the assets for the SPA (images, stylesheets, subfolders, etc.). You can access these assets through the SPA asset endpoint `/spa-app/<ApplicationID>/<SpaFolder>/assets/<path_to_asset>`, where `<path_to_asset>` is the file's relative path in the assets folder.
  - **SPA object definition** – The XML file that defines the SPA object in the SDF SuiteApp project. It sets the SPA name, URL, references to the SPA folder and script files. In SDF, SPAs are defined

as a singlepageapp SDF custom object. For more information, see the help topic Single Page Applications as XML Definitions.

- **SPA URL** – The link you use to run the SPA. It's made of up of:
  - **Base URL** – Where you access your NetSuite account.
  - **Application ID** – The SuiteApp application ID, made up of the publisher ID and project ID from when you created the SDF SuiteApp project for the SPA.
  - **Custom URL** – The custom URL you set in the SPA object definition XML file.

  Here's an example of an SPA URL: https: *<AccountID>*.app.netsuite.com/spa-app/*<ApplicationID>*/*<CustomURL>*.
- **SuiteApps** – Applications or bundles that extend NetSuite for specific industry and business needs. You use SuiteApp projects created with SDF to develop and deploy SPAs in NetSuite. For more information, see the help topic SuiteApp Projects.
- **SuiteCloud Development Framework (SDF)** – A development framework you use to create SuiteApps or customize NetSuite accounts using SuiteCloud projects. For information, see .
- **SuiteCloud projects** – Projects you create using SDF. There are two types: account customization projects and SuiteApp projects. For information, see the help topic SuiteCloud Project Types.

Components and Structure of Single Page Applications

In NetSuite, single page applications (SPAs) are built as part of a SuiteCloud Development Framework (SDF) SuiteApp project. You can build SPAs with the UI components available in the NetSuite UI Framework (UIF).

NetSuite UIF and SuiteApps use different ways to manage modules. NetSuite UIF uses ECMAScript Modules (ESM), whereas SuiteApps use RequireJS, which is based on the Asynchronous Module Definition (AMD). Because of this, you need to convert SPA source files before you can deploy a SuiteApp project with SPAs. The conversion process includes:

- Transpiling JSX syntax to regular JavaScript
- Converting `import` and `export` statements to `define` and `require` statements

You also need a specific folder structure to set up SPAs and make sure the conversion process works as expected. Each SPA in your SuiteApp project needs the following:

- A unique SPA folder
- Two required scripts, the SPA client script and SPA server script, stored inside the SPA folder
- An SPA object definition file that sets the SPA name, URL, and references to the SPA folder and script files

You can download sample SPA SuiteApps from the SuiteCloud Project Repository on Oracle Samples GitHub and use them as templates for your own SPAs. These samples use the required folder structure and include a file that handles (using Gulp) the build and bundle steps to convert the source files. You can use the `gulpfile` from the samples, or use any standard tool you prefer to transpile and convert the files. For more information, see Single Page Application Samples.

Read the following topics for more information:

- SuiteApp Project Structure for Single Page Applications – Describes the required folder structure for SPAs.
- Build Process for SuiteApp Projects with Single Page Applications – Describes how the `gulpfile` works and the supported build tasks.
- NetSuite User Interface Framework for Single Page Applications – Provides information about NetSuite UIF.

SuiteApp Project Structure for Single Page Applications

To use NetSuite UIF with single page applications (SPAs), you need a specific folder structure to convert the source files before you deploy the SuiteApp project. This folder structure lets you use JavaScript or TypeScript in your source files. During conversion, the SPA source files are transpiled and the converted files are saved in a different folder in the project directory.

You can find sample SuiteApp projects with SPAs in the SuiteCloud Project Repository on Oracle Samples GitHub. For more information, see Single Page Application Samples.

For information about SDF SuiteApp projects, see the help topics SuiteApp Projects and SuiteCloud Project Structure and File Components.

Project Structure for SDF SuiteApps with SPAs

SDF SuiteApp projects with SPAs need to follow this structure:

- `/src/SuiteApps/<ApplicationID>/<SpaFolder>` – This SPA folder holds the SPA server script, SPA client script, and the application's root component. It can also include the `assets` folder for static files.
- `/test` – Holds unit tests and test stubs.
- `/src/manifest.xml` and `/src/deploy.xml` – These files contain the SuiteApp configuration.
- `/src/Objects/custspa_xxx.xml` – The XML object definition file for the SPA SDF custom object.
- `/src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>` – This folder holds the transpiled and converted sources that get pushed to NetSuite when you deploy the SuiteApp. You can use the `gulpfile` from the sample SPA projects to run the build and bundle tasks that convert the SPA source files (see Build Process for SuiteApp Projects with Single Page Applications). Or, you use your own tool to transpile and convert the files, and save the converted files in this folder.

  > **Note:** All JavaScript files in the `src/FileCabinet/SuiteApps` folder must use the Asynchronous Module Definition (AMD) format.

If you have non-SPA script files in your SuiteApp project, take note of the following when using the `gulpfile` from the sample projects:

- If you have JavaScript and AMD script files, save them directly in the `/src/FileCabinet/SuiteApps/<ApplicationID>` folder. Keep them outside the SPA folder (if it already exists) because the `gulpfile` regenerates that folder each time you run a bundle task.
- Don't place AMD script files inside the `/src/SuiteApps/<ApplicationID>/<SpaFolder>`. The `gulpfile` assumes that all files in that folder are in ECMAScript Modules (ESM) format. If you include AMD scripts, the `gulpfile` treats them as ESM and tries to convert them to AMD, which causes errors.
- The `gulpfile` only processes files inside the SPA folders, anything outside is ignored. If you want to place scripts in the `/src/SuiteApps/<ApplicationID>` folder, you'll need to handle building, bundling, and moving those scripts to the `src/FileCabinet/SuiteApps/<ApplicationID>` folder for deployment.
- The `gulpfile` is designed specifically for converting SPA scripts. If you want to use it to convert non-SPA scripts in TypeScript and ESM, you'll need to handle any errors that may occur during this process.

For more information about the `gulpfile`, see Build Process for SuiteApp Projects with Single Page Applications.

The following shows a sample folder structure for an SPA SuiteApp after running the build and bundle tasks implemented in the `gulpfile`.

```
<ApplicationID>
├── build/
├── node_modules/
├── src/
│   ├── FileCabinet/
```

```
│         ├── SuiteApps/<ApplicationID>/
│               ├── <OtherScripts>/
│               ├── <SpaFolder>/
│                     ├── assets/
│                     ├── SpaClient.js
│                     └── SpaServer.js
│         └── Web Site Hosting Files/
│   ├── InstallationPreferences/
│   ├── Objects/
│         └── custspa_helloworld.xml
│   ├── SuiteApps/<ApplicationID>/
│         ├── <SpaFolder>/
│               ├── assets/
│               ├── HelloWorld.tsx
│               ├── SpaClient.tsx
│               └── SpaServer.tsx
│   ├── Translations/
│   ├── deploy.xml
│   └── manifest.xml
├── test/
├── gulpfile.mjs
├── jest.config.js
├── package.json
├── suitecloud.config.js
├── tsconfig.json
└── tsconfig.test.json
```

Build Process for SuiteApp Projects with Single Page Applications

Before you deploy SDF SuiteApp projects with single page applications (SPAs), you'll need to transpile and convert SPA source files first. You can use any standard tool to convert your sources, or try the build tools included in the sample SPA SuiteApp projects.

In the sample SPA projects, you'll find the build and bundle tasks for converting SPA sources in the `package.json` (under the `scripts` section) and in the `gulpfile`. You can run these tasks with Node Package Manager (npm):

- npm run build
- npm run bundle
- npm run clean
- npm run deploy
- npm run test

For information about how these tasks are used in the SPA creation process, see Single Page Application Creation And Development.

For information about the sample SPA SuiteApps, see Single Page Application Samples.

You can also set up source concatenation and minification in the `gulpfile` with these variables:

- `concatenateScripts` – Set the value to `true` to concatenate or bundle the sources for each entry point into a single file. This helps reduce the number of requests and loading time. Set the value to `false` to disable concatenation.
- `minifyScripts` – Set the value to `true` to remove unnecessary characters from the source code to decrease file size. This helps save bandwidth and improve web performance. Set the value to `false` to disable minification.

npm run build

This task uses the TypeScript compiler on the source files, builds them, and puts the result in a new `/build` directory.

The following source files are processed:

- SPA and backend script sources from `/src/SuiteApps/<ApplicationID>/<SpaFolder>`
- Unit test sources from `/test/unit`
- Configuration from `tsconfig.json`

Take note of the following:

- Both TypeScript and JavaScript sources are supported.
- SPA scripts in `/src/SuiteApps/<ApplicationID>/<SpaFolder>` need to be in JSX and ESM. The `gulpfile` is built for converting SPA scripts, and there's no guarantee that it will work for other script types.
- The process ignores files inside `/src/SuiteApps/<ApplicationID>` and outside `/src/SuiteApps/<ApplicationID>/<SpaFolder>`.
- If you have other AMD scripts that don't need to be converted, place them directly in the `/src/FileCabinet/SuiteApps/<ApplicationID>` folder.

npm run bundle

This task takes the source files from the `/build` folder, searches for entry points in SPA folders, and bundles all of the SPA dependencies. It uses these steps to find the entry points:

- Identify SPA folders by searching for directories that contain an `SpaServer` script
- In each SPA folder, it looks for these entry points:
  - `SpaClient` scripts
  - `SpaServer` scripts
  - Any script with the `@NScriptType` tag

This task changes all `import` and `export` statements in the source files to `define` and `require` statements. It also uses the `concatenateScripts` and `minifyScripts` options set in the `gulpfile`.

The results are saved in `/src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>` and they're ready to be deployed to the target NetSuite account. This task also copies all asset files from `/src/SuiteApps/<ApplicationID>/<SpaFolder>` to `/src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>`. Any file in the SPA folder that is not a JavaScript or a TypeScript file is treated as an asset.

npm run clean

This task cleans the `/build` and `/src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>` directories.

npm run deploy

This task bundles the project and runs `suitecloud project:deploy` to deploy the SuiteApp to the target NetSuite account.

npm run test

This task runs unit tests from the `/test/unit` folder using the Jest testing framework.

Understanding the Single Page Application Execution Process

The following steps summarize the single page application (SPA) execution process:

1. You can access the SPA by clicking a link or by typing the URL in your browser.
2. The SPA server script runs before any client-side code goes to the browser.

The SPA server script implements and exports the `initializeSpa` function, which receives a context object with an API that can modify the response by adding styles.

The system checks if the user is in the allowed audience set in the SPA definition, then runs the `initializeSpa` function from the SPA server script.

3. The SPA client script loads after the rest of the page elements (like the NetSuite center) finish loading.

   The SPA client script implements and exports the `run` function, which acts as the SPA's entry point and can return a promise or any other type of data.

   The SPA client script loads the root component to initialize the application.

   The SPA shows a loading icon until the `run` function returns. If it returns a promise, the icon stays until the promise is resolved or rejected.

4. The SPA content appears in the browser.

For information about creating SPAs, see Single Page Application Creation And Development.

NetSuite User Interface Framework for Single Page Applications

The NetSuite User Interface Framework (UIF) is a frontend framework and UI library for NetSuite.

To access NetSuite UIF, copy and paste this URL into your browser and replace *<AccountID>* with your NetSuite account ID: **https://***<AccountID>***.app.netsuite.com/ui/apps/catalog.nl**.

NetSuite UIF gives you a catalog of UI components and APIs to build custom UIs for your SPAs. You'll also find code samples and a sandbox to experiment with your code and see how each component renders. Other utilities like routing, state management, and internationalization are also available.

NetSuite UIF has its own documentation organized by feature, with basic tutorials and advanced guides. You'll find SPA SuiteApp documentation under Documentation > NetSuite Integration > SuiteApps.

Every NetSuite UIF release is published in the NPM repository with its TypeScript type declarations. Type declarations describe the classes, methods, properties, and APIs in NetSuite UIF. To use the type declarations, you must install the @netsuite-uif-types package.

Single Page Application Samples

Sample SuiteApp projects that use single page applications (SPA) are available in the SuiteCloud Project Repository on Oracle Samples GitHub (see spa-suiteapp-samples).

These samples use the SPA SuiteApp project template and incude a `gulpfile` for the build and bundle steps to convert source files. For information about the project structure for SuiteApps with SPAs, see Components and Structure of Single Page Applications.

You'll find these sample SPA SuiteApps in the repository:

- **Airport 360** – Sample 360-type SuiteApp that demonstrates the use of various NetSuite UI Framework (UIF) features such as routing, side menu, state management, asynchronous loading, and SuiteScript APIs.
- **Basics Routing** – Demonstrates the basic principles of routing.
- **Basics State Management** – Demonstrates the basic principles of state management.
- **Hello World JS** – Minimal Hello World application written in JavaScript.
- **Hello World TS** – Minimal Hello World application written in TypeScript.
- **Item 360** – Sample 360-type SuiteApp that demonstrates common SPA patterns and data handling using SuiteScript.

Single Page Application Creation And Development

Before you start building single page applications (SPAs), ensure that you've enabled the required features in your account. For more information, see Prerequisites for Single Page Applications.

In NetSuite, SPAs are built in SuiteApp projects using the SuiteCloud Development Framework (SDF). The NetSuite UI Framework (UIF) provides the UI components that you can use to build your SPAs.

The following steps provide an overview of the process for developing SPAs with NetSuite UIF:

1. Create the SDF SuiteApp project where you'll build your SPA. Alternatively, you can use a sample SPA SuiteApps as a starting point or add SPAs to an existing SuiteApp project.
2. Within the SuiteApp project, create the following for each SPA implementation:
   - A unique SPA folder
   - Two required scripts: SPA client script and SPA server script
   - An SPA object definition file
   - (Optional) An assets folder

   If you're using a sample project, ensure that you update the folder names, file names, and the contents for your project.
3. Build your application using the UI components and APIs from NetSuite UIF. You also can use SuiteScript modules in your SPA client and server scripts.
4. Convert your SPA source files and build the SuiteApp. Sample SPA projects include a `gulpfile` for the build and bundle steps. You can use this file or choose your own tool to transpile and convert the files.
5. Validate and deploy your SuiteApp project.
6. Open your single page application in NetSuite.

For more information, see the following topics:

- Creating a Single Page Application – Describes how to create an SDF SuiteApp project that includes SPAs.
- Single Page Applications as XML Definitions – Provides information about the XML definition for an SPA (singlepageapp) SDF custom object.
- Developing Single Page Applications – Describes how to develop the SPA client script and SPA server script.
- Troubleshooting Single Page Applications – Provides information about errors related to SPAs.

For information about the sample SPA SuiteApps, see Single Page Application Samples.

Prerequisites for Single Page Applications

These are the minimum requirements for developing single page applications in NetSuite:

- Must use SuiteCloud Development Framework (SDF) SuiteApp projects (see the help topic SuiteApp Projects)
- Must use SuiteScript 2.1

You can use the UI components and APIs from the NetSuite User Interface Framework (UIF) to build SPAs with NetSuite look and feel.

Before you can start developing SPAs, ensure that you've set up the following prerequisites in your account:

Required Features and Tools	Related Help Topic
Enable the following SuiteScript features: ■ Client SuiteScript ■ Server SuiteScript	Enabling SuiteScript
Enable the SuiteCloud Development Framework (SDF) feature and set up the requirements for SDF.	SuiteCloud Development Framework Setup
Enable the following authentication features: ■ Token-Based Authentication ■ OAuth 2.0	Enable the Token-based Authentication Feature Enable the OAuth 2.0 Feature
Install Node.js from Node.js Downloads (if you do not have it installed).	—
Set up your preferred development tool. You can choose from the following SuiteCloud Software Development Kit (SDK) tools: ■ SuiteCloud Plug-ins and Extensions: □ SuiteCloud Extension for Visual Studio Code □ SuiteCloud IDE Plug-in for WebStorm ■ SuiteCloud CLIs: □ SuiteCloud CLI for Java □ SuiteCloud CLI for Node.js	SuiteCloud Plug-ins and Extensions SuiteCloud CLIs

Creating a Single Page Application

This procedure describes how to create single page applications (SPAs) in an SDF SuiteApp project.

Before you start, make sure you've completed the prerequisite tasks and you've read and understood the concepts discussed in the following topics:

- Components and Structure of Single Page Applications – Describes the project folder structure required to implement SPAs.
- NetSuite User Interface Framework for Single Page Applications – Describes how to access and use NetSuite UIF to develop SPAs.
- Single Page Application Samples – Provides sample SPA SuiteApps, from basic SuiteApps that you can use as a template to more complex SuiteApps that demonstrate NetSuite UIF features and use SuiteScript to handle data.
- Prerequisites for Single Page Applications – Describes the prerequisite setup tasks that must be completed before working on SPAs.

Note: If you're creating an SPA with the SuiteCloud IDE Plug-in for WebStorm, note that the SPA object type is not listed in the New Custom Object window.

Likewise, script templates for the SPA client script and SPA server script are not available when you create a new SuiteScript file. You'll need to manually create the SPA object definition and SPA script files. Alternatively, you can start with a sample SPA SuiteApp or copy the structure from an existing SPA.

To create an SDF SuiteApp project with a single page application:

1. Create an SDF SuiteApp project. If you need guidance, check the help topic for your preferred NetSuite SDK tool. See References to SuiteCloud SDK Tools.

   Alternatively, you can start with a sample SPA SuiteApp or open an existing SuiteApp project where you want to add SPAs.

2. Set up the target NetSuite account for your project. If you need guidance, check the help topic for your preferred NetSuite SDK tool. See References to SuiteCloud SDK Tools.

3. In the `manifest.xml` file, add the SuiteScript Server (`SERVERSIDESCRIPTING`) feature dependency. You can copy the following code sample into your `manifest.xml` file.

```
<dependencies>
    <features>
        <feature required="true">SERVERSIDESCRIPTING</feature>
    </features>
</dependencies>
```

4. Create the following files and folders for each SPA implementation.

   If you're using a sample SuiteApp project as a template, ensure that you update the folder names, file names, and contents for your project.

   a. **SPA folder** (`src/SuiteApps/<ApplicationID>/<SpaFolder>`) – This folder is where you'll store the source files for your SPA. Each SPA in the SuiteApp project needs its own unique folder. Inside each SPA folder, you can organize your files into sub-folders. For guidance on creating and organizing the folder for your source files, see SuiteApp Project Structure for Single Page Applications.

   b. **SPA client script and SPA server script** – If you copied the folder from the sample project, the `SpaClient` and `SpaServer` scripts with the minimum content are already in the `src/SuiteApps/<ApplicationID>/<SpaFolder>` folder. You can use these scripts as a starting point, and change the file names if you prefer. For information about the script structure and examples of SuiteScript modules that you can add to these scripts, see Developing Single Page Applications.

      If you want to create these scripts manually, ensure that you save them inside the SPA folder.

   c. **SPA assets folder** (`src/SuiteApps/<ApplicationID>/<SpaFolder>/assets`) – This folder is optional and, if available, must be located inside the SPA folder. Use it to store assets like style sheets and images. Your SPA can access these assets through the asset endpoint `/spa-app/<ApplicationID>/<SpaFolder>/assets/<path_to_asset>`, where `<path_to_asset>` is the asset file's relative path in the assets folder.

      For example, an image located at `src/SuiteApps/com.netsuite.spa/helloworld/assets/logo.png` can be accessed using the endpoint: `/spa-app/com.netsuite.spa/helloworld/assets/logo.png`.

   d. **SPA object definition XML file** – You can copy the SPA object definition file from a sample project, or manually create a new XML file for your SPA object. Save this file in the `src/Objects` folder. If you're using the XML definition file from a sample project, be sure to update the file name and edit its contents (such as the script ID and paths) to match your project details.

      SPAs are defined in SDF as a singlepageapp SDF custom object. For more information, see the help topic Single Page Applications as XML Definitions.

      Follow these guidelines for the SPA object XML definition file:

      - Each SPA in the SuiteApp project needs its own SPA object definition XML file in the `Objects` folder.
      - The `scriptid` specified in the object definition must match the SPA object definition file name.

- The paths specified in the object definition are relative to the `src/FileCabinet/` folder. Take note of the following:
  - The target location for the SPA `folder` is inside `src/FileCabinet/SuiteApps/<ApplicationID>/` (where the transpiled files are saved after you run the build). In the XML definition, don't include `src/FileCabinet` in the path.
  - The path specified for the SPA client script (`clientscriptfile`) and SPA server script (`serverscriptfile`) must point to a location inside the SPA folder.
  - The assets folder is optional. If you add a path in the `assetsfolder` field of the SPA object definition, then you also need to have the assets folder in the SPA folder.

The basic XML definition for an SPA object in a SuiteApp looks like the following:

```
<singlepageapp scriptid="custspa_helloworld">
    <name>Hello World</name>
    <description>This is a sample SuiteApp that uses NetSuite UIF components.</description>
    <url>helloworld-suiteapp-spa</url>
    <folder>[/SuiteApps/com.netsuite.spa/helloworld/]</folder>
   <clientscriptfile>[/SuiteApps/com.netsuite.spa/helloworld/SpaClient.js]</clientscriptfile>
    <serverscriptfile>[/SuiteApps/com.netsuite.spa/helloworld/SpaServer.js]</serverscriptfile>
    <assetsfolder>[/SuiteApps/com.netsuite.spa/helloworld/assets/]</assetsfolder>
</singlepageapp>
```

For a complete example showing all the supported fields, see the help topic Single Page Application XML Definition Example.

5. Create the root component for your SPA and save the script inside the appropriate folder under `src/SuiteApps/<ApplicationID>/<SpaFolder>`.

   Go to NetSuite UIF for information about the components and APIs you can use to develop your SPA. To access NetSuite UIF, copy this URL into your browser and replace *<AccountID>* with your NetSuite account ID: **https://***<AccountID>***.app.netsuite.com/ui/apps/catalog.nl**.

   If you want to use the NetSuite UIF TypeScript type declarations, follow these steps to install the @netsuite-uif-types package:

   1. In the `package.json` file, add `@oracle/suitecloud-uif-types` and `typescript` to `devDependencies` and specify the appropriate version based on the NetSuite version that you are developing for.

      For example:

      ```
      "devDependencies": {
          "@oracle/suitecloud-uif-types": "^7.0.0",
          "typescript": "^5.2.0"
      }
      ```

   2. In `tsconfig.json`, add `@oracle/suitecloud-uif-types` as an additional type root.

      For example:

      ```
      "compilerOptions": {
          "typeRoots": {
              "node_modules/@types",
              "node_modules/@oracle/suitecloud-uif-types"
          }
      }
      ```

   3. Open a terminal or command line, and run `npm i` to install dependencies.

Important: After you define the root component, import it in the SPA client script and pass it as a parameter to the `setContent(rootComponent)` method. For more information, see Single Page Application Client Script.

At this point, your SuiteApp project folder structure looks like this:

```
com.netsuite.spa
├── src/
│   ├── FileCabinet/
│   │       ├── SuiteApps/com.netsuite.spa/
│   │       └── Web Hosting Files
│   ├── InstallationPreferences/
│   ├── Objects/
│   │       └── custspa_helloworld.xml
│   ├── SuiteApps/com.netsuite.spa/
│   │       └── helloworld/
│   │             ├── assets/
│   │             ├── HelloWorld.tsx
│   │             ├── SpaClient.tsx
│   │             └── SpaServer.tsx
│   ├── Translations/
│   ├── deploy.xml
│   └── manifest.xml
├── test/
├── gulpfile.mjs
├── jest.config.js
├── package.json
├── suitecloud.config.js
├── tsconfig.json
└── tsconfig.test.json
```

6. Build the SuiteApp. The build process transpiles the source files in `src/SuiteApps/<ApplicationID>/<SpaFolder>` and stores the converted files in `src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>`.

   You can use any standard tool to transpile and convert your SPA source files, or use the `gulpfile` from the sample SPA SuiteApp projects. For details about the tasks in the `gulpfile`, see Build Process for SuiteApp Projects with Single Page Applications.

   To use the `gulpfile`, open a terminal or command line, and run these NPM tasks from the `src` folder:

   - `npm i` – To install build dependencies.
   - npm run build – To build the project. This task runs the TypeScript compiler on the files in `src/SuiteApps/<ApplicationID>/<SpaFolder>` and saves the result in a new `build` folder.
   - npm run bundle – To bundle the SuiteApp. This task takes the transpiled sources from the `build` folder and bundles them together. The result is saved in `src/FileCabinet/SuiteApps/<ApplicationID>/<SpaFolder>` and is ready to be deployed to the target NetSuite account.

   After building the SuiteApp, your project folder structure looks like this:

```
com.netsuite.spa
├── build/
├── node_modules/
├── src/
│   ├── FileCabinet/
│   │       ├── SuiteApps/com.netsuite.spa/
│   │       │     └── helloworld
│   │       │           ├── assets/
│   │       │           ├── SpaClient.js
│   │       │           └── SpaServer.js
│   │       └── Web Hosting Files
│   ├── InstallationPreferences/
│   ├── Objects/
│   │       └── custspa_helloworld.xml
```

```
│   ├── SuiteApps/com.netsuite.spa/
│   │       └── helloworld/
│   │           ├── assets/
│   │           ├── HelloWorld.tsx
│   │           ├── SpaClient.tsx
│   │           └── SpaServer.tsx
│   ├── Translations/
│   ├── deploy.xml
│   └── manifest.xml
├── test/
├── gulpfile.mjs
├── jest.config.js
├── package.json
├── suitecloud.config.js
├── tsconfig.json
└── tsconfig.test.json
```

7. Validate and deploy the SuiteApp project. If you're using the `gulpfile`, run npm run deploy to bundle and deploy the SuiteApp to the target NetSuite account.

   If you prefer to deploy the SuiteApp from your NetSuite SDK tool, see References to SuiteCloud SDK Tools for guidance.
8. When the SuiteApp is successfully deployed, you can open the single page application in NetSuite.
   - To see the list of single page applications in your account, go to Customization > Scripting > Single Page Applications. To open the management page for an SPA, click the SPA name.
   - To open a single page application:
     - From the SPA list, click the link icon next to the SPA name.
     - From the SPA record, click the link in the URL field.

After setting up the SPA SuiteApp project, you can continue to develop and configure your SPA. For more information about these topics, see Developing Single Page Applications and Single Page Application Management.

References to SuiteCloud SDK Tools

SuiteCloud Software Development Kit (SDK) are tools that enable you to customize accounts through SuiteCloud Development Framework (SDF). If you need guidance on how to perform specific tasks using these tools, refer to the referenced help topics in the following table. These tasks are referenced in Creating a Single Page Application.

Task	General Information	SuiteCloud Extention for Visual Studio Code	SuiteCloud IDE Plug-in for WebStorm	SuiteCloud CLI for Java	SuiteCloud CLI for Node.js
Create a SuiteApp project	SuiteApp Projects	Creating a SuiteCloud Project in SuiteCloud Extension for Visual Studio Code	Creating a SuiteApp Project in SuiteCloud IDE Plug-in for WebStorm	createproject	project:create
Set up account	—	Setting Up NetSuite Accounts in SuiteCloud Extension for Visual Studio Code	SuiteCloud IDE Plug-in for WebStorm Account Setup	authenticate	account:setup
Validate a SuiteApp project	SuiteCloud Project Validation	Deploying a SuiteCloud Project to Your NetSuite Account with SuiteCloud Extension	Validating a SuiteCloud Project with SuiteCloud	validate	project:validate

Task	General Information	SuiteCloud Extention for Visual Studio Code	SuiteCloud IDE Plug-in for WebStorm	SuiteCloud CLI for Java	SuiteCloud CLI for Node.js
		for Visual Studio Code	IDE Plug-in for WebStorm		
Deploy a SuiteApp project	SuiteCloud Project Deployment Preparation	Deploying a SuiteCloud Project to Your NetSuite Account with SuiteCloud Extension for Visual Studio Code	Deploying a SuiteCloud Project to Your NetSuite Account with SuiteCloud IDE Plug-in for WebStorm	deploy	project:deploy

Developing Single Page Applications

Single page applications (SPAs) in NetSuite require the use of SuiteScript. SPAs use two specific script types: the SPA server script and the SPA client script.

The SPA server script is the entry point for the application and can run logic needed before rendering the page. The SPA client script loads components and can also load custom modules or third-party libraries. These scripts also let you load server-side and client-side SuiteScript modules.

For more information about the structure of these scripts and the APIs available, see the following topics:

- Single Page Application Client Script
- Single Page Application Server Script

For examples, see the following topics:

- SPA Client Script Basic Examples using SuiteScript modules
- SPA Server Script Basic Examples using SuiteScript modules

Note: All code samples in this section use `import`, so they need to be transpiled before deployment to NetSuite. For more information, see Components and Structure of Single Page Applications and Creating a Single Page Application.

For information about the execution process of SPAs, see Understanding the Single Page Application Execution Process.

Single Page Application Client Script

The SPA client script is a regular JavaScript file. It defines a module that must return an object with a `run(context)` member, which is the entry point function that is called when the SPA runs.

You can load SuiteScript modules from the SPA client script. For examples, see SPA Client Script Basic Examples using SuiteScript modules.

The SPA client script has the following structure:

1. It includes a `run(context)` entry point function in the `return` statement. This function can return any type of data, including a promise.

```
export const run = (context) => {
    // Add code here
};
```

2. The entry point function can receive an object that provides access to data or methods. In this case, the `context` object has the following properties:

- `baseUrl` – URL of the SPA.
- The `context` object also exposes two functions:
  - `context.setLayout(LayoutType)` – Sets the `LayoutType` property of the Shell component. This method accepts the following values for the `LayoutType` parameter:

Parameter	Type	Accepted Values	Description
`LayoutType`	string	application	Makes the content fill the whole height of the window. For example, it enables you to place some things on the bottom of the window.
		natural	Makes the content only as tall as it needs to be.

  - `context.setContent(rootComponent)` – Attaches the root component to the appropriate element in the NetSuite user interface.

Parameter	Type	Description
`rootComponent`	NetSuite UIF component	The root component for the SPA, represented as a NetSuite UIF component. The root component is defined in a separate script file.

In the following example, the layout type is set to `application` using the `setLayout(LayoutType)` method. The SPA client script imports the root component from the `HelloWorld.js` file. To set the page context, the `setContent(rootComponent)` method is used to render the HTML element of the `HelloWorld` component, which is defined in the `HelloWorld.js` file

```
import HelloWorld from './HelloWorld.js';

export const run = (context) => {
    context.setLayout('application'); // Make the application fill the entire viewport
    context.setContent(<HelloWorld />); // NetSuite UIF component
};
```

3. The `HelloWorld.js` file is a separate script file that defines the root component that initializes the SPA using NetSuite UIF.

```
import {ContentPanel, Heading} from '@uif-js/component';

export default function HelloWorld() {
   return (
      <ContentPanel horizontalAlignment={ContentPanel.HorizontalAlignment.CENTER} verticalAlignment={ContentPanel.Verti
calAlignment.CENTER} >
         <Heading>Hello World!</Heading>
      </ContentPanel>
   );
}
```

SPA Client Script Basic Examples using SuiteScript modules

The following code sample sends an email when the SPA is executed. It uses the N/email Module to create and send the email, and the N/runtime Module to get the internal ID of the current user and use it as the author of the email.

```
import email from 'N/email';
import runtime from 'N/runtime';

export const run = (context) => {
    email.send({
        author: runtime.getCurrentUser().id,
        recipients: 'test@myCompany.com',
```

```
        subject: 'SPA CS test',
        body: 'N/email and N/runtime modules work OK'
    });
};
```

The following example shows how to load an existing employee record using the N/record Module, and enter some value in the **Notes** field. Replace the value of the internal ID of the record with a valid value in your account.

```
import record from 'N/record';

export const run = (context) => {
    var employee = record.load({
        type : record.Type.EMPLOYEE,
        id : -5
    });
    employee.setValue({
        fieldId: 'comments',
        value: 'Note added using an SPA'
    });
    employee.save();
};
```

Single Page Application Server Script

The SPA server script file is the script file associated with the script record. It acts as a plug-in with an API that can modify the response (for example, add scripts or perform validations).

You can load most server-side SuiteScript modules from the SPA server script. For examples, see SPA Server Script Basic Examples using SuiteScript modules.

The SPA server script has the following structure:

1. It includes the two JSDoc tags, `@NApiVersion` and `@NScriptType`, declaring the appropriate version (`2.1`) and script type (`SpaServerScript`).

   ```
   /**
    * @NApiVersion 2.1
    * @NScriptType SpaServerScript
    */
   ```

2. It implements and exports an `initializeSpa(context)` entry point function. This function receives a context parameter, an object that contains the `context.addStyleSheet(options)` method.

   ```
   /**
    * @NApiVersion 2.1
    * @NScriptType SpaServerScript
    */
   export const initializeSpa = (context) => {};
   ```

3. The `context.addStyleSheet(options)` method adds a style sheet to the SPA. It returns `void`.

   This method has the following parameters:

Parameter	Type	Required/Optional	Description
`relativePath`	string	At least one is required (and not null).	The path to the CSS file relative to the assets folder (for example, "/main.css").
`url`	string	At least one is required (and not null).	The URL where the CSS file is located (for example, "http://sampleweb.com/css/main.css"). The URL member overrides any value provided in the `relativePath` member.

The following example shows how to use the `context.addStyleSheet(options)` method with both parameters specified. The `url` gets precedence over the `relativePath`.

```
/**
 * @NApiVersion 2.1
 * @NScriptType SpaServerScript
 */
import log from 'N/log';

export const initializeSpa = (context) => {
      context.addStyleSheet({
          relativePath: "/main.css",
          url: "http://sampleweb.com/css/main.css"});
      log.debug({
          title: 'Debug log server script',
          details: 'Server script added stylesheet'
      });
};
```

SPA Server Script Basic Examples using SuiteScript modules

The following example shows an SPA server script that loads the N/runtime Module to identify the current user and retrieve the user's email address. Then, it compares the current user email address with the specified email address in the code, and throws an error if the values don't match. When the error is thrown, a standard error message is shown in NetSuite with the specified message (for example, "Unauthorized user") and the SPA is not served.

```
/**
 * @NApiVersion 2.1
 * @NScriptType SpaServerScript
 */
import runtime from 'N/runtime';

export const initializeSpa = (context) => {
    var currentUser = runtime.getCurrentUser();
    var userEmail = currentUser.email;
    if (userEmail !== 'email@netsuite.com') {
        throw "Unauthorized user";
    }
    return;
};
```

The following sample adds a `custom.css` style sheet, located in the folder `<folder>/<assetsfolder>/css/`.

```
/**
 * @NApiVersion 2.1
 * @NScriptType SpaServerScript
 */

export const initializeSpa = (context) => {
    context.addStyleSheet({
        relativePath: "/css/custom.css"
    });
    return;
};
```

Troubleshooting Single Page Applications

The following topics provide additional information about errors related to single page applications (SPAs) and how to solve them:

- Validation and Deployment Errors – Describes errors that may occur when you validate and deploy an SDF SuiteApp project that contains an SPA object.
- Execution Errors – Describes errors that may occur when you run an SPA.

Validation and Deployment Errors

The following sections describe the errors that may occur when validating and deploying an SDF SuiteApp project that contains a single page application (SPA) object.

- Missing Fields and Field Values
- Incorrect Format
- Duplicate Values
- Invalid Values
- Values Do Not Match
- Reference File or Folder Does Not Exist

Missing Fields and Field Values

Error Message	Scenario	Solution
When the SuiteCloud project contains a singlepageapp, the manifest must define the SERVERSIDESCRIPTING feature as required.	The `SERVERSIDESCRIPTING` feature is missing in the manifest.xml file.	Add the `SERVERSIDESCRIPTING` feature to the manifest.
The object script ID must not be empty and must start with custspa.	The `scriptid` is empty or it does not start with `custspa`.	Specify a script ID in the SPA object definition file and ensure that it starts with the prefix `custspa`.
The object field name is missing.	The `name` field is missing.	Add the `name` field and specify a value for it.
The object field name must not be empty.	The `name` field is empty.	Specify a value for the `name` field.
The object field url is missing.	The `url` field is missing.	Add the `url` field and specify a value for it.
The object field url must not be empty.	The `url` field is empty.	Specify a unique value for `url`.
The object field folder is missing.	The `folder` field is missing.	Add the `folder` field and specify a value for it.
The object field folder must not be empty.	The `folder` field is empty.	Specify a unique value for `folder`.
The object field clientscriptfile is missing.	The `clientscriptfile` field is missing.	Add the `clientscriptfile` field and specify a value for it.
The object field serverscriptfile is missing.	The `serverscriptfile` field is missing.	Add the `serverscriptfile` field and specify a value for it.
The object field clientscriptfile must not be empty.	The value of the `clientscriptfile` field is missing.	Specify a value for the `clientscriptfile` field.

Error Message	Scenario	Solution
The object field serverscriptfile must not be empty.	The value of the `serverscriptfile` field is missing.	Specify a value for the `serverscriptfile` field.
The linkcategory field is missing for the (link) subrecord.	The `link` sublist element is included in the SPA object definition, but there is no `linkcategory` field inside it.	Add the `linkcategory` field and specify a value for it.
The linklabel field is missing for the (link) subrecord.	The `link` sublist element is included in the SPA object definition, but there is no `linklabel` field inside it.	Add the `linklabel` field and specify a value for it.
The linkcategory field for the (link) subrecord must not be empty.	The `linkcategory` field is empty.	Specify a value for the `linkcategory` field.
The linklabel field for the (link) subrecord must not be empty.	The `linklabel` field is empty.	Specify a value for the `linklabel` field.

Incorrect Format

Error Message	Scenario	Solution
The scriptid value must be lowercase alphanumeric without spaces or punctuation. Underscores are permitted.	The value for `scriptid` contains characters not allowed in the ID field.	Ensure that the value for `scriptid` contains only lowercase alphanumeric characters and underscores.
The script ID must not exceed 28 characters.	The value for `scriptid` has more than 28 characters.	Ensure that the value for `scriptid` is limited to 28 characters.
The name field contains more than the maximum number (1000) of characters allowed.	The value for `name` has more than 1000 characters.	Ensure that the value for `name` is limited to 1000 characters.
The description field contains more than the maximum number (1000) of characters allowed.	The value for `description` has more than 1000 characters.	Ensure that the value for `description` is limited to 1000 characters.
The url field contains more than the maximum number (1000) of characters allowed.	The value for `url` has more than 1000 characters.	Ensure that the value for `url` is limited to 1000 characters.
The URL cannot contain the following characters: ? ! * ' ( ) ; : @ & = + $, / ? # [ ] .	The value for `url` contains spaces or some of the following characters: ? ! * ' ( ) ; : @ & = + $, / ? # [ ] .	Ensure that the value for `url` does not contain spaces or any of the characters that are not allowed.
The {folder value} folder value referenced by the folder object field must start and end with the '/' character and must not contain the following characters: ?, *, <, >, |, ', \\ .	The value for `folder` is entered in the wrong format.	Ensure that the value for `folder` starts and ends with '/' and does not contain any of the following characters: ? * < > | ' \\.
The folder name, {folder name}, must not exceed 99 characters.	The folder name has more than 99 characters.	Ensure that the folder name is limited to 99 characters.
The filename [filename] must not exceed 199 characters.	The client script file name, including the file extension, has more than 199 characters.	Ensure that the file name is limited to 199 characters, including the file extension.
The filename [filename] must not exceed 199 characters.	The server script file name, including the file extension, has more than 199 characters.	Ensure that the file name is limited to 199 characters, including the file extension.

Error Message	Scenario	Solution
The file {folder value}/SpaClient referenced by the field clientscriptfile is an invalid file type. The file name can only contain the following file extension(s): .js. Change the file name reference and then try again.	The client script file does not have a file extension or has an incorrect file extension.	Ensure that the client script file is a JavaScript file (.js).
The file {folder value}/SpaServer referenced by the field serverscriptfile is an invalid file type. The file name can only contain the following file extension(s): .js. Change the file name reference and then try again.	The server script file does not have a file extension or has an incorrect file extension.	Ensure that the server script file is a JavaScript file (.js).
The {assetsfolder} folder value referenced by the assetsfolder object field must start and end with the '/' character and must not contain the following characters: ?, *, <, >, |, ', \\.	The value for `assetsfolder` is entered in the wrong format.	Ensure that the value for `assetsfolder` starts and ends with '/' and does not contain any of the following characters: ? * < > | ' \\.
The folder name, {<assets folder name>}, must not exceed 99 characters.	The assets folder name has more than 99 characters.	Ensure that the assets folder name is limited to 99 characters.
The linklabel field contains more than the maximum number (30) of characters allowed.	The `linklabel` field has more than 30 characters.	Ensure that the value for `linklabel` is limited to 30 characters.
The notifyemails field contains more than the maximum number (999) of characters allowed.	The value for `notifyemails` has more than 999 characters.	Ensure that the value for `notifyemails` is limited to 999 characters.

Duplicate Values

Error Message	Scenario	Solution
This URL already exists. Please enter a different one.	The value for `url` is not unique and already exists in the account.	Specify a unique value for `url`.
The URL <url> specified by the object field url is already referenced by another object. Select a different URL and try again.	More than one SPA object is using the specified URL. Each SPA must have a unique URL.	Specify a unique value for `url`.
The folder <folder> referenced by the object field folder is already referenced by another object. Select a different folder and try again.	The value for `folder` is not unique and already exists in the account.	Specify a unique value for `folder`.
Duplicate values {linkcategory value} were found in the 'links' sublist on the 'singlepageapp' record.	There is more than one `link` element with the same value in the `linkcategory` field. You cannot set more than one link for each location.	Remove the duplicate values for the `linkcategory` field.

Invalid Values

Error Message	Scenario	Solution
The object field executeas must not be {x}.	The value for executeas field does not correspond to any existing role.	Ensure that the value specified in executeas is an existing role. Use uppercase letters when entering a generic role value.

Error Message	Scenario	Solution
[ADMINISTRATOR] role cannot be used for the 'Execute As' field. Select a different value.	The value entered for the executeas field is ADMINISTRATOR. This value cannot be used for this field.	Specify a role other than ADMINISTRATOR in the executeas field.
The object field loglevel must not be {x}.	The value for loglevel is not valid.	Set loglevel to any of the following: DEBUG, AUDIT, ERROR, or EMERGENCY.
The object field audienceroles must not be {x}.	The value for audienceroles field does not correspond to any existing role.	Ensure that the value specified in audienceroles is an existing role. Use uppercase letters when entering a generic role value.
The audienceallroles field must be set to a valid Boolean value, 'T' or 'F'.	The value for audienceallroles is not valid.	Set audienceallroles to either T or F.
The object reference {executeas value/audienceroles value} is missing in the project and also not included in the dependencies list.	The custom role referenced in the executeas or audienceroles field does not exist in the project and is not included in the dependencies list.	Ensure that the custom role referenced in executeas or audienceroles exists in the SuiteApp project.
The linkcategory field for the (link) subrecord must not be {linkcategory value}.	The value in the linkcategory field is not valid.	Check the list of possible values for the center category in generic_centercategory. The linkcategory field also accepts references to centercategory custom type.
The {linkcategory value} object referenced by the linkcategory field in the (link) subrecord is missing.	The linkcategory field value references an object that is missing from the project.	Check that the custom centercategory object that you are referencing in the linkcategory field exists in your SuiteApp project.
The notifyuser field must be set to a valid Boolean value, 'T' or 'F'.	The value for notifyuser is not valid.	Set notifyuser to either T or F.
The notifyowner field must be set to a valid Boolean value, 'T' or 'F'.	The value for notifyowner is not valid.	Set notifyowner to either T or F.
The notifyadmins field must be set to a valid Boolean value, 'T' or 'F'.	The value for notifyadmins is not valid.	Set notifyadmins to either T or F.
The notifyemails field value is invalid. To include multiple addresses, separate them by commas.	The notifyemails field value is not a valid email address or is entered in the wrong format.	Ensure that the email address specified is valid and entered in the correct format. Use a comma to separate multiple email addresses.

Values Do Not Match

Error Message	Scenario	Solution
The script ID must match the file name (excluding the file extension).	The scriptid value specified in the object definition does not match the SPA object file name.	Specify the same value for the SPA object file name (custspa_*<name>*) and for the scriptid attribute.
The folder field in the {/SuiteApps/<publisherid2>.<projectid2>/<folder name>} reference is invalid.	The *<publisherid2>.<projectid2>* does not match the *<publisherid>.<projectid>* of the manifest.xml or deploy.xml file.	Check that all of the following values match:

Error Message	Scenario	Solution
		■ The values in the `publisherid` and `projectid` fields in the manifest.xml file ■ The value in the `path` tag under `files` in the deploy.xml file ■ The values used in the `folder` field in the SPA object definition file

Reference File or Folder Does Not Exist

The following errors are thrown when a path in the SPA object definition file is incorrect or when the reference file or folder does not exist in the SuiteApp project folder.

Folder and file paths specified in the SPA object definition are absolute and must point to the SuiteApp project folder `src/FileCabinet/SuiteApps/`*<publisherid>.<projectid>*`/` (where the transpiled and bundled source files of the SPA are located).

Error Message	Scenario	Solution
The SPA folder is not within the application folder. The folder field in the {/SuiteApps/<publisherid>.<projectid>/<folder name>} reference is invalid.	The folder path in the `folder` field points to a location outside of the `/SuiteApps/`*<publisherid>.<projectid>*`/` folder.	Ensure that the SPA folder is inside your local SuiteApp project folder `src/FileCabinet/SuiteApps/`*<publisherid>.<projectid>*`/`.
The folder {/SuiteApps/<publisherid>.<projectid>/<folder>} referenced by the object field folder is missing.	The SPA folder specified in the `folder` field does not exist in the SDF SuiteApp project folder.	Ensure that the SPA folder has been created in your local project.
The file {folder value}/SpaClient.js referenced by the object field clientscriptfile is missing.	The client script file specified in `clientscriptfile` does not exist in the SPA folder inside the SDF SuiteApp project.	Ensure that you have the SPA client script inside the SPA folder.
The file {folder value}/SpaServer.js referenced by the object field serverscriptfile is missing.	The server script specified in `serverscriptfile` does not exist in the SPA folder inside the SDF SuiteApp project.	Ensure that you have the SPA server script inside the SPA folder.
The folder {/SuiteApps/<publisherid>.<projectid>/<folder>/assets/} referenced by the object field assetsfolder is missing.	The assets folder specified in the `assetsfolder` field does not exist in the SDF SuiteApp project folder.	Ensure that the folder has been created in your local project and is inside the SPA folder.
The client script is not within the SPA folder.	The client script path points to a different folder than the SPA folder.	Check that the value in the `clientscriptfile` field uses the path specified in the `folder` field.
The server script is not within the SPA folder.	The server script path points to a different folder than the SPA folder.	Check that the value in the `serverscriptfile` field uses the path specified in the `folder` field.
The assets folder is not within the SPA folder.	The assets folder path points to a different folder than the SPA folder.	Check that the value in the `assetsfolder` field uses the path specified in the `folder` field.

Error Message	Scenario	Solution
/SuiteApps/<publisherid2>.<projectid2>/<folder name>/SpaClient.js is invalid. The clientscriptfile field must start with /SuiteApps/<publisherid>.<projectid>/, /Web Site Hosting Files/.	The client script path points to a folder in a different application.	Check that the value in the `clientscriptfile` field uses the correct value for *<publisherid>.<projectid>*.
/SuiteApps/<publisherid2>.<projectid2>/<folder name>/SpaServer.js is invalid. The serverscriptfile field must start with /SuiteApps/<publisherid>.<projectid>/, /Web Site Hosting Files/.	The server script path points to a folder in a different application.	Check that the value in the `serverscriptfile` field uses the correct value for *<publisherid>.<projectid>*.
The assetsfolder field in the {/SuiteApps/<publisherid2>.<projectid2>/<folder name>/assets} reference is invalid.	The assets folder path points to a folder in a different application.	Check that the value in the `assetsfolder` field uses the correct *<publisherid>.<projectid>* value.

Execution Errors

The following table describes the errors that may occur when executing your single page application (SPA).

Error Message	Description	Error Code
The requested SPA does not exist.	The SPA that you are trying to run does not exist in the NetSuite account. The SPA may have been deleted, or the URL entered for the SPA is incorrect.	—
The client script file cannot be loaded: the file is not active.	The client script file, the server script file, or the SPA folder is marked as inactive.	—
The client script file cannot be loaded: you do not have permission to access this file.	You do not have permission to access this file, or you do not have access to the SuiteApps folder (for example, the folder access is restricted).	—
An error occurred while executing the server script. You do not have privileges to view this page.	You accessed the SPA using a user or a role that does not have permission to view the SPA.	—

The client script file cannot be loaded: the script dependencies cannot be loaded.	The script tried to load a module that does not exist. For example:
	`define(['N/module'], function(module) {`
	`  return {`
	`  run: function (context) {`
	`  }`
	`  };`
	`});`
		`SCRIPT_FILE_DEPENDENCY_ERROR`

The entry point function cannot be called: the script does not have the specified function.	The script does not have the `run()` function. For example:
	`define([], function() {`
	`  return {`
	`  test: function (context) {`
	
	`  }`
	`  };`
	`});`
		`SCRIPT_DOES_NOT_HAVE_RUN_MEMBER`

The entry point function cannot be called: the function exists but cannot be called.	The `run()` function exists but is not well defined. For example, it is not defined as a function but as a variable: `define([], function() {`	`SCRIPT_MEMBER_NOT_CALLABLE`

Error Message	Description	Error Code

	<pre>2 return {
	3 run: 1
	4 };
	5 });</pre>	

The entry point function cannot be called: no define function found.	The script does not expose any module. The script must include one `define` function. You will also see this error if the return statement is missing or if other structure errors are present. For example: <pre>1 console.log('TEST');</pre>	`SCRIPT_DOES_NOT_EXPOSE_ANY_MODULE`

The entry point function throws an exception.	The script throws an error. For example:
	<pre>1 define([], function() {
	2 return {
	3 run: function (context) {
	4 throw 'spaclient.js Error';
	5 }
	6 };
	7 });</pre>
		`SCRIPT_ENTRY_POINT_THROWS_ERROR`

An error occurred while rendering the SPA.	An unexpected error occurred during the execution of the SPA. This error also appears if the SPA folder is marked private and the user running the SPA is not the owner.	—
Invalid call to addStyleSheet. No style sheet path or URL were provided.	The parameter object of the `addStyleSheet` method is missing both the `relativePath` and `url` members, or both members are set to null. At lease one of these parameters must be specified and not null.	—
An error occurred while executing the server script. {string thrown}	An error is thrown by the server script. For example: An `error occurred while executing the server script. Unauthorized user.`	—
404 error	The assets folder is included in the URL for the SPA, but it does not exist. For example, this error occurs if you try to access the following URL and the assets folder is null: `https://`*<AccountID>*`.app.netsuite.com/spa/`*<spaUrl>*`/assets/image.png`.	—

Single Page Application Management

You can manage these SPA settings from both the NetSuite UI and the SuiteCloud Development Framework (SDF):

- View and edit the details of the SPA
- Handle the roles used to run the SPA
- Specify the log level
- Specify the audience
- Create, edit, and delete links to the SPA
- Specify who receives error notifications
- View the logs
- Delete the SPA

- View System Notes

For more details, see the following topics:

- Managing a Single Page Application from SuiteCloud Development Framework
- Managing a Single Page Application from the NetSuite User Interface

Managing a Single Page Application from SuiteCloud Development Framework

With SuiteCloud Development Framework, you can manage the settings of single page applications (SPAs) by editing the fields on the SPA object definition XML file. These fields also correspond to fields on the **Basic Info** and **Configuration** tabs of the SPA record in the NetSuite UI.

When updating an SPA object in an SDF SuiteApp project, consider the following:

- From the SDF perspective, if you change the SPA ID, it's treated as a new SPA with the new ID and the SPA with the old ID is removed. You also need to update the URL and the SPA folder before you validate or deploy the project. Otherwise, uniqueness validation errors will be thrown.
- You cannot create new files or folders by only updating the XML definition. When you update the XML definition to include a new file or folder, only the references are updated. You must create new files and folders in the File Cabinet, or you can create them in your IDE and deploy them to your account. For example, if you type a file or folder name incorrectly in the XML definition, this incorrectly named file or folder is not created in the project. Instead, an error is thrown because the specified file or folder cannot be found.
- If you remove an optional field from your XML definition, the existing value is preserved and no change is applied to the field (except for links that are removed).
- For most optional fields, if you leave the value empty, it is reset to its default value. The exception is the Boolean field `notifyowner`, which has a default value of true (T). If you leave the value for this field empty, it is changed to false (F).
- Setting the `audienceallroles` to true (T) selects all internal roles only. You can use the `audienceroles` field to individually specify internal and external roles that can run the SPA. If `audienceallroles` is set to true, all internal roles will be included in the audience along with any specific external roles listed in `audienceroles`.

To update an SPA object, you must have the SuiteApp project files saved in your local environment (even if you don't change other files). Otherwise, when you redeploy the SuiteApp with the updated SPA object, it will fail. If you don't have these files saved locally, you can import the SPA object from a target NetSuite account into your SDF SuiteApp project and download the script files from the NetSuite File Cabinet.

To manage the settings of your SPA from SDF:

1. If you have the SuiteApp project files in your local environment, you can proceed directly to step 2. Otherwise, follow these steps to download the SuiteApp project files:
    1. Import the SPA object from a target NetSuite account to your local environment.

       The Download XML button is not available for SPA objects, but you can use SuiteCloud plug-ins and extensions and SuiteCloud CLI to import an SPA object from a NetSuite account. For more information, see the help topic Account Component Imports to SuiteCloud Projects.

       You can import an object from a specific SuiteApp by specifying the application ID. To filter your search, select `singlepageapp` as the object type and select script IDs that contain `custspa`.

2. Download the SPA folder, SPA client script file, and SPA server script file, or create them in your local environment. If the `assetsfolder` field has a value in the XML definition, you also need to create an assets folder inside the SPA folder. The assets folder is not downloaded if the folder is empty.

   You cannot import files from a NetSuite account into a SuiteApp project. You must manually download the SuiteApp project files from the NetSuite File Cabinet or create the files in your local SuiteApp project folder.

2. In the SPA object definition XML file, change the fields that you want to modify. For information about the XML definition for SPA objects, see the help topic Single Page Applications as XML Definitions. For information about fields and possible values in the SPA object definition, see the help topic singlepageapp.
3. Deploy the project to your account. For more information, see the help topic SuiteCloud Project Deployment Preparation.

> **Note:** You can apply installation preferences to SPAs. The SPA supports object locking, and hiding and locking of files. For more information about installation preferences and how to apply them, see the help topic Installation Preferences in the SuiteApp of SuiteCloud Project.

Managing a Single Page Application from the NetSuite User Interface

To see the list of single page applications (SPAs) in your NetSuite account, go to Customization > Scripting > Single Page Applications. Click the name of the SPA to see its details and configuration settings that you can edit. You must have the SuiteScript permission (Edit or Full) to edit the configuration settings of SPAs in your account.

From the SPA list, you can do the following:

- To open the management page for an SPA, click the SPA name.
- To open a single page application, click the link icon next to the SPA name.

> **Note:** SPAs that are part of SuiteApps installed from the SuiteApp Marketplace have a limited version of the SPA management page, where basic information is in read-only mode and only configuration fields are editable. For information about managing SPAs from installed SuiteApps, see the help topic Managing Single Page Applications Included in SuiteApps.

For developers working on SPAs, you can manage the SPA settings from these tabs on the SPA management page:

- **Basic Info** – Shows the details of the SPA, including the SPA name, SPA ID, name and ID of the SuiteApp that the SPA is part of, SPA URL, and the SPA sources (SPA folders and script files). Click the link icon next to the SPA source files to go to the location of the folder or file in the File Cabinet.
- **Configuration** – Includes the configuration fields that you can edit to change the settings of the SPA. For information about the changes that you can make on this tab, see Configuring Settings from the SPA Management Page.
- **Logs** – Shows the logs for the SPA. Click **Refresh** to reload the list and see any new logs after running the SPA.
- **System Notes** – Shows the changes for the SPA (creation and updates). The following details are shown: date/time, user, role of the user, type of action, object changed, and old and new values. The information about deletion actions is also retained.

Note: Updates to the values of following fields are not tracked by System Notes: **Audience**, **Center Links**, and **Error Notifications**.

Configuring Settings from the SPA Management Page

The following table shows the fields available on the Configuration tab of the SPA management page.

Field	Description	How to Configure
Execute As	Sets the permissions and restrictions to apply when running the SPA server script based on the selected role. By default, this field is set to Current Role. For more information, see Executing Scripts Using a Specific Role. **Note:** The SPA in the browser will always run based on the role of the currently logged-in user.	To change the role, click the edit icon to display the list of available roles. Select a role, and click **Save**.
Log Level	Filters the log entries shown on the Logs tab. The default log level is Debug. Other log level options include Audit, Error, and Emergency. For more information, see Setting Script Execution Log Levels.	To change the log level, click the edit icon to display the log level options. Select a value, and click **Save**.
Release Audience	Determines who can run the SPA. When the SPA is created, it runs only in the accounts of the roles selected in this field. If no role is specified, the SPA is accessible only to the script owner.	To change the audience, click **Set Up Audience** to see the current roles selected and other possible options. You can filter by role name. Audience roles are shown in two separate multiselect fields: **Internal Roles** and **External Roles**. ■ To make the SPA available to all internal roles, check the **All Internal Roles** box. ■ To add roles, check the boxes for each role on the left, then click the right arrow. ■ To remove roles, check the boxes for each role on the right, then click the left arrow. Click **Save** to save your selection.
Center Links	Lets you create, edit, or delete center links for the SPA. Center links are used to specify a menu path that lets users access the SPA directly from that location.	■ To create a center link, click **Add Link**. On the Center Link window: □ In the **Menu Location** field, select the center and tab where you want to add the SPA link. By default, the new link is added at the bottom of the list. If you want it before an existing element, select a value in the **Insert Before** field. □ In the **Label** field, enter the label for the SPA link. Click **Save**. Reload the page to see the new link in the menu. You can add only one link per menu location. ■ To edit a center link, click the edit icon, update the fields, and click **Save**.

Field	Description	How to Configure
		▪ To delete a center link, click the delete icon, then click **Delete Link**.
Error Notification	Indicates who is notified about errors. By default, the script owner is selected.	To set up who receives notifications, click **Set Up Notifications** and choose from these options: ▪ **Current User** – Notifies the user running the SPA. ▪ **Script Owner** – Notifies the user who owns the script. ▪ **Notify All Admins** – Notifies all users with the Administrator role. ▪ **Groups** – Notifies all users who are part of the selected group. To define new groups, go to Lists > Relationships > Groups > New. In the **Specific Email Addresses** field, enter a comma-separated list of email addresses for users who should be notified.

SuiteScript FAQ

Applies to: SuiteScript 2.x | APIs | SuiteScript 1.0 | SuiteCloud Developer

This FAQ covers several topics:

- General
- SDF and SuiteApps
- Code
- Additional tools to work with along with SuiteScript
- Errors
- Logs

For FAQ information related to SOAP web services, see the help topic SOAP Web Services Frequently Asked Questions (FAQ).

General

How often is the SuiteScript API documentation updated?

The SuiteScript API documentation is updated frequently and pushed to the NetSuite Help Center.

For the most current API documentation, see the help topic SuiteScript 2.x API Reference.

Which records are supported in SuiteScript?

You can find all information about SuiteScript-supported records, fields, tabs, sublists, search filters, and search columns in the NetSuite Help Center.

See the help topic SuiteScript Supported Records for information about supported records.

See the SuiteScript Records Browser for details related to SuiteScript-supported records.

Which sublists are supported in SuiteScript?

You can find all information about SuiteScript-supported records, fields, tabs, sublists, search filters, and search columns in the SuiteScript Records Browser.

See the SuiteScript Records Browser for details related to SuiteScript-supported sublists and their internal IDs.

What is the difference between SuiteScript 1.0 and SuiteScript 2.x?

SuiteScript 1.0 uses global functions while SuiteScript 2.x is modular. The functions in SuiteScript 1.0 are synchronous on client, while SuiteScript 2.x can use asynchronous processing. SuiteScript 2.x offers more APIs. New functionality is added to SuiteScript 2.x only.

See the SuiteScript 1.0 to SuiteScript 2.x API Map for more details on the differences.

A new version of SuiteScript, SuiteScript 2.1, is also available. This version is the latest minor version of SuiteScript. It extends SuiteScript 2.x by supporting additional ECMAScript language features and syntax. For more information, see the help topic SuiteScript 2.1.

What version of SuiteScript should I use?

Use SuiteScript 2.x for new scripts that you develop, and consider converting your SuiteScript 1.0 scripts to SuiteScript 2.0 or SuiteScript 2.1.

SuiteScript 1.0 isn't being updated with new features or enhancements. SuiteScript 1.0 scripts are still supported, but for new or revised scripts, use SuiteScript 2.x to get the latest features and enhancements.

A new version of SuiteScript, SuiteScript 2.1, is also available. This version is the latest minor version of SuiteScript. It extends SuiteScript 2.x by supporting additional ECMAScript language features and syntax. For more information, see the help topic SuiteScript 2.1.

Does SuiteScript support the use of multiple library files?

Yes. Multiple library files can be loaded onto the Script record page. The Script record page is where you set up your SuiteScript .js file and any required library files.

Also be aware that the system reads your library files in the order they appear on the Library Script File subtab on the Script record page. For example, if your first library file references the second library file, an error will be thrown, since the first library file is loaded before the second.

If you're unsure about creating a Script record or loading library files, go to the topic Creating a Script Record in the NetSuite Help Center.

What is the difference between Standard and Dynamic mode?

When a SuiteScript 2.x script creates, copies, loads, or transforms a record in standard mode, the record's body fields and sublist line items are not sourced, calculated, and validated until the record is saved. In dynamic mode, the record's body fields and sublist line items are sourced, calculated, and validated in real-time.

See the help topic SuiteScript 2.x Standard and Dynamic Modes for more details.

What are the best practices when using Standard and Dynamic mode?

Standard mode is usually faster, so use it for read-only operations.

Dynamic mode may be needed when doing write operations of fields on which other field depend.

See the help topic SuiteScript 2.x Standard and Dynamic Modes for more details.

How do I use require Configuration?

If you set up a valid `@NAmdConfig` JSDoc tag, SuiteScript implements the require configuration settings *before* loading dependencies.

See the help topic require Configuration for more details.

What is SuiteScript governance?

Governance checks scripts for issues like infinite loops and stops them if necessary.

NetSuite uses a governance model to optimize performance, based on usage units. If you exceed the allowed units, the script will be terminated.

See SuiteScript Governance and Limits for more details.

What is SuiteCloud Processors?

SuiteCloud Processors is the current system used to process scheduled scripts and map/reduce scripts. Before SuiteCloud Processors was introduced, scheduled scripts and map/reduce scripts were exclusively processed by scheduling queues.

See SuiteCloud Processors for more details.

Which is better to use: Scheduled script or Map Reduce script?

Use scheduled scripts for simple tasks that can't run at the same time. Map/Reduce script is better suited for processing massive data in parallel, since it is faster then Scheduled script.

See the help topics SuiteScript 2.x Scheduled Script Type and SuiteScript 2.x Map/Reduce Script Type for more details.

What version of ECMAScript does SuiteScript support?

SuiteScript 1.0 and SuiteScript 2.0 support ECMAScript 5.1.

SuiteScript 2.1 supports ECMAScript 6. This version is the latest minor version of SuiteScript. It extends SuiteScript 2.x by supporting additional ECMAScript language features and syntax. For more information, see the help topic SuiteScript 2.1.

Can a User Event script be used to trigger another User Event script?

Nested user event script execution is not supported.

See the help topic SuiteScript 2.x User Event Script Type for more details.

SDF and SuiteApps

What is SDF and how does it help me develop with SuiteApps?

SuiteCloud Development Framework (SDF) is a framework that lets you create SuiteApps from your local computer using an integrated development environment (IDE). Using SDF lets you manage client and server scripts as part of file-based customization projects. You can also use SDF with the SuiteCloud Software Developer Kit (SDK) as the best solution for creating a script as a part of an overall NetSuite customized solution packaged with other factors, such as custom records, custom forms, other scripts, or more. For more information about SDF, see the help topic SuiteCloud Development Framework,

Is there a special script type for installing SDF projects?

Yes, the SDF installation script type helps with tasks when developing a SuiteApp from SDF to your target account. You can use the SDF installation script type to perform setup, configuration, and data management tasks that otherwise would have to be completed by account administrators. For more information about the SDF installation script type, see the help topic SuiteScript 2.x SDF Installation Script Type.

Does SuiteScript have a module that I can use to see information about my SuiteApp?

Yes, the N/suiteAppInfo module provides information about your SuiteApp, including whether a specific SuiteApp is installed and a list of all SDF SuiteApps that are installed, along with the ID for the SDF SuiteApp that contains a specific script (for multiple scripts specified). For more information about the N/suiteappInfo module, see the help topic N/suiteAppInfo Module.

Code

How can I control the execution order of scripts deployed on a record?

To control script execution order, log in to NetSuite, go to Customization > Scripting > Scripted Records, select your record type, and drag the scripts up or down

How many client and user event scripts can I deploy in a record?

You can deploy as many scripts as you need, but be aware that too many scripts deployed on single record can cause performance issues.

How do I add a button in a User Event script that redirects to a Suitelet in SuiteScript 2.x?

In the `beforeLoad` callback, you get a form object on which any button can be added. That button can run an arbitrary function. The function must use `document.location = <url of a suitelet>` for redirect.

See the help topic SuiteScript 2.x User Event Script Type or SuiteScript 2.x Suitelet Script Type for more details.

Can I write to a script parameter field I have created in my code?

You can read script parameter field values, but you can't set them programmatically. The only time you can specify a value outside of the UI is when you call task.create(options).

```
var myTask = task.create({task.TaskType.SCHEDULED_SCRIPT});
myTask.scriptId = <id of your script>;
myTask.deploymentId = <id of your script deployment>;
myTask.params = <your params>;
myTask.submit();
```

See Creating Script Parameters Overview for more information about creating and working with script parameters.

How do I mass delete records using SuiteScript?

Currently, there's no way to mass delete records. You need to delete each record individually using record.delete(options).

Which certificate authorities (CAs) does NetSuite support?

NetSuite supports the same list of trusted third-party certificate authorities (CAs) as Mozilla Included CA Certificate List.

The endpoint you're connecting to must use a trusted CA, or the connection won't work. When connecting to an endpoint from NetSuite, make sure it provides a full certification chain, including intermediate certificates.

For a list of certificate authorities, see https://wiki.mozilla.org/CA/Included_Certificates.

Can I throw an alert in a Suitelet?

You can attach a client script by using the N/ui/message Module.

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget', 'N/ui/message'], function(serverWidget, message){
    function onRequest(context){
        let form = serverWidget.createForm('Alert Form');
        let msg = message.create({
            type:message.Type.WARNING,
            title:"Alert",
            message: "Something occurred",
            duration: 5000
        });
        form.addPageInitMessage(msg);
```

```
        context.response.writePage(form);
    }

    return{
        onRequest: onRequest
    }
});
```

N/ui/message Module and N/ui/dialog Module only work in client scripts.

Is there a change in creating subrecords in SuiteScript 2.x?

Yes, SuiteScript 2.x has changes in how you create subrecords.

See the help topic Subrecord Scripting in SuiteScript 2.x Compared With 1.0 for more details.

How do I optimize SuiteScript performance?

There are several guidelines that you can follow to optimize SuiteScript performance.

See Optimizing SuiteScript Performance for more details.

Should I use record-level or form-level scripts?

There are different forms used for different users. Use them when you need to run different scripts for different users. Record-level script is always used regardless of the user. Based on that, you can decide which type of script should be used.

See the help topic Record-Level and Form-Level Script Deployments for more details.

Where can I find the records and their scripts?

The Scripted Records page shows all records with associated user event or global client scripts.

See The Scripted Records Page for more details.

What is the difference between Map Reduce and Scheduled Script?

Use map/reduce scripts for handling large datasets that can be broken down into smaller, independent parts.

Scheduled scripts are server scripts that run sequentially on SuiteCloud Processors, whereas map/reduce scripts run in parallel.

See the help topic SuiteScript 2.x Script Types for more details.

Is DOM supported in SuiteScript?

You can't access the NetSuite UI directly using Document Object Model (DOM). Use SuiteScript APIs instead.

What are the required SuiteScript 2.x JSDoc Tags?

For entry point scripts, you need to include `NApiVersion` and `NScriptType` JSDoc tags. For more information about JSDoc tags, see the help topic: SuiteScript 2.x JSDoc Validation.

What are the stages required in a Map Reduce script?

You need to include at least `getInput` and one of map/reduce stages.

What does the script context seen in System Notes represent regarding the action on the record?

The script context shows where the operation that changed the field came from - it's the second-most nested scope in the stack of operations.

For example, if in the UI, the `pageInit` callback is used on a record to call `nlapiRequestURL` to trigger a Suitelet. This calls another `nlapiRequestURL` to trigger a RESTlet, which loads a record that triggers a user event that changes the field, then the context is RESTlet. The stack will be `UI/Client/Suitelet/Restlet/UserEvent`. RESTlet is the second most nested scope.

How do you reference Custom Modules in SuiteScript 2.x?

Add custom modules to your `define` callback dependencies by path.

What is the difference between scriptContext.newRecord and scriptContext.oldRecord in SuiteScript 2.x?

`scriptContext.oldRecord` stores the original record before any changes

`scriptContext.newRecord` stores the record which can be modified in the `beforeSubmit` callback, so you can save it differently than what's shown in the UI. For example, you can override a memo before the record is saved.

Additional tools to work with along with SuiteScript

What other tools are available to help me write my SuiteScript scripts?

You have several tools at your disposal, such as SDF, SDK, Records Catalog, and Records Browser.. For more information about SDF and SDK, see the help topic SuiteCloud Development Framework. For more information about the Records Catalog and the Records Browser, see the help topics The Records Catalog and SuiteScript Records Browser. For more information about additional SuiteCloud tools, see the help topic SuiteCloud Reference Tools.

Errors

You can find SuiteScript error descriptions in SuiteAnswers. Type the error in the search box and click **Search**.

How does 'Record Has Been Changed' error get triggered in SuiteScript execution?

This error occurs when someone else (or another script) changes the record at the same time.

See the SuiteAnswers article Prevent losing data because of the error "Record has been changed" for more details.

What is the 'You do not have privileges to view this page' error message on the script deployments page?

Users will get this error if any of the **Select All** box on the Audience subtab on the Script Deployment page isn't checked. This applies to both internal and external users, even those accessing the Suitelet without logging in.

See the help topic Errors Related to the Available Without Login URL for more details.

What is the 'You are not allowed to navigate directly to this page' error message when accessing Suitelet?

Users will get this error if the Suitelet's **Status** isn't set to Released, even if they're accessing it externally.

See the help topic Errors Related to the Available Without Login URL for more details.

Why am I getting an 'SSS_USAGE_LIMIT_EXCEEDED ID' error?

You'll get this error if your script uses too many usage points in one go. ii it happens twice, the script will stop running.

For more information about usage limits, see the help topic SuiteScript Governance and Limits

Keep an eye on your script usage, as there are limits in place. For more information, see Monitoring Script Usage.

Why am I getting an 'SSS_REQUEST_LIMIT_EXCEEDED' error?

You'll get this error if your server script or application takes too long to run.

For more information about usage limits, see the help topic SuiteScript Governance and Limits

Why am I getting 'SSS_INVALID_SUBLIST_OPERATION' error?

This error happens when you try to set a value on a line that doesn't exist, or if the sublist isn't editable.

See the SuiteAnswers article SuiteScript Error SSS_INVALID_SUBLIST_OPERATION for more details.

Logs

How long will the script execution logs be available on my account?

You can find script execution logs in two places:

- **Execution Log** subtab on Script and Script Deployment records – Shares log storage with other customers on the same database. To prevent excessive logging, there's a storage limit for script execution logs on each NetSuite database instance. If the log storage limit is reached on a server, all logs for all customers on that server will be deleted. You might have logs older than 30 days if the volume is low, but if it's high, you might only have logs from the past week. To keep important log information, consider storing it in custom records.
- **Script Execution Log** page – The execution logs at Customization > Scripting > Script Execution Logs stores logs for up to 30 days, no matter how many logs there are. This page can only show up to 10,000 log results at a time. If you have more than 10,000 logs, you can filter the page to see the ones you need. Use the **Filters** section at the top of the page to narrow down the logs by date, script, or log level. Keep in mind that you can't do a full search or customize the view on this page.

Can I use execution logs in a client script?

Yes, you can use execution logs in client script. For SuiteScript 1.0, use `nlapiLogExecution` (see the help topic SuiteScript 1.0 Documentation). For SuiteScript 2.x, use N/log Module.